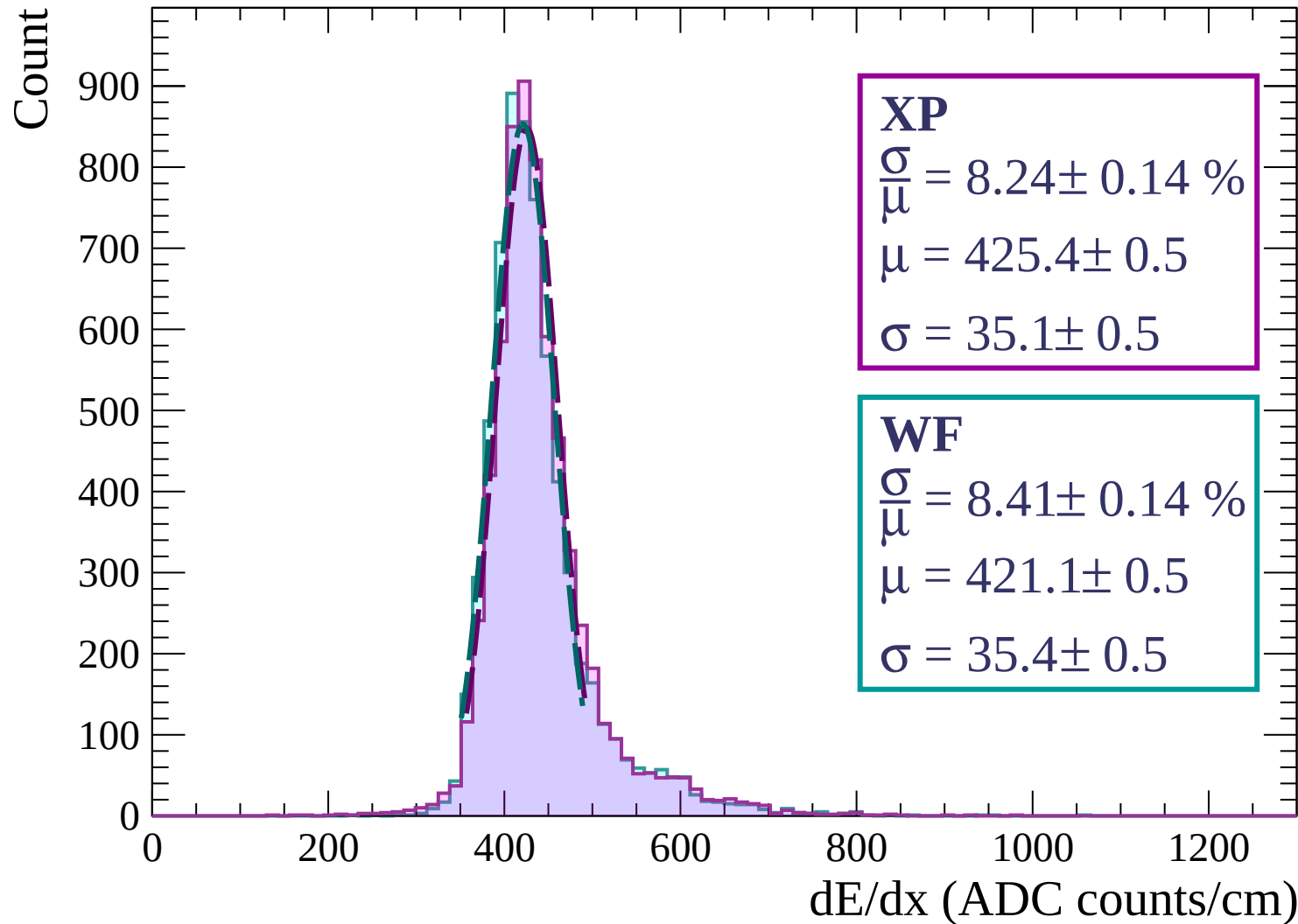
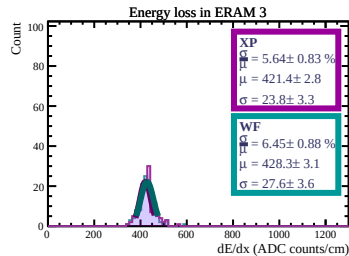
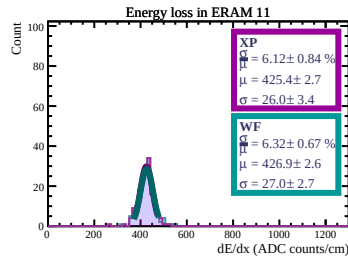
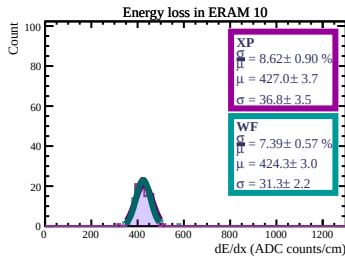
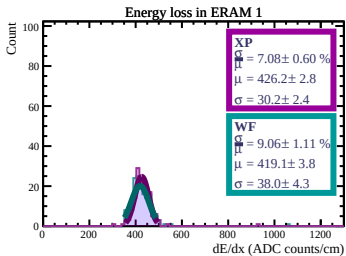
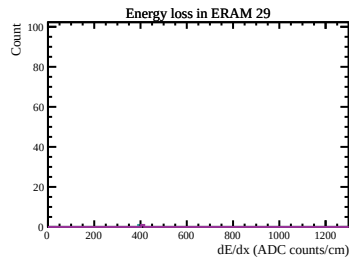
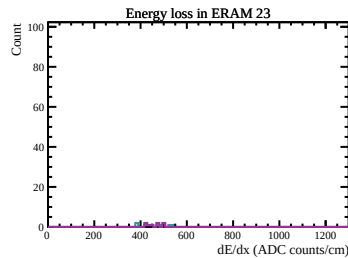
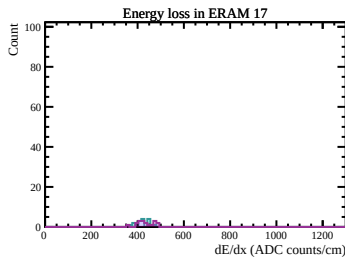
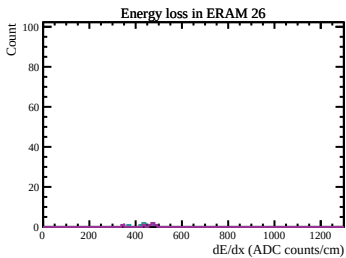
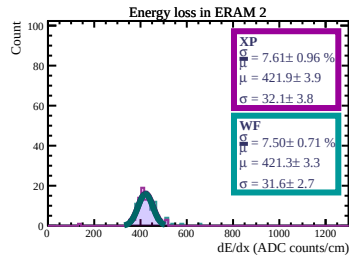
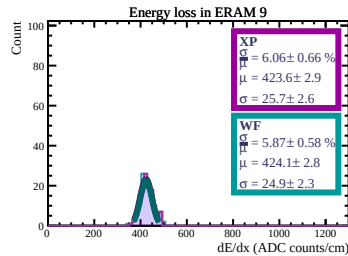
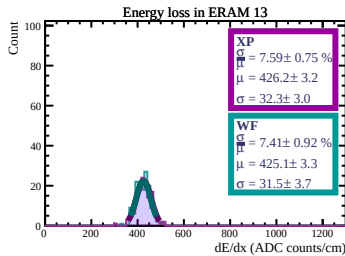
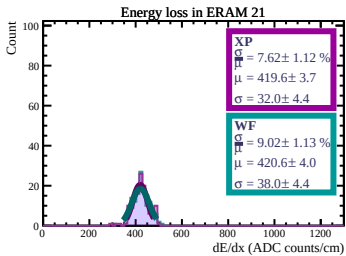
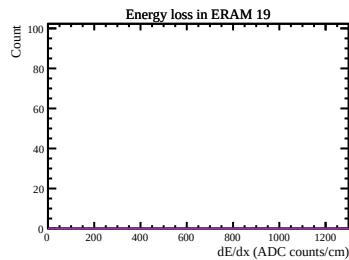
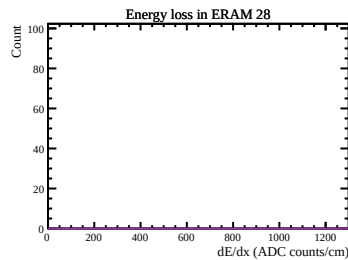
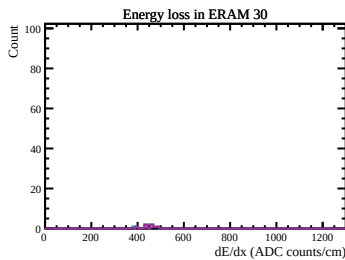
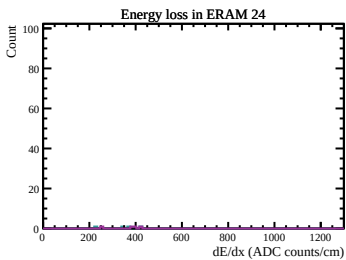
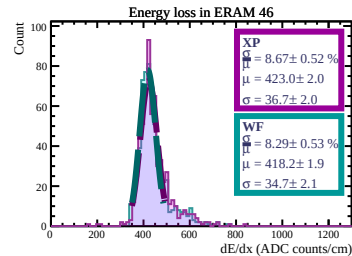
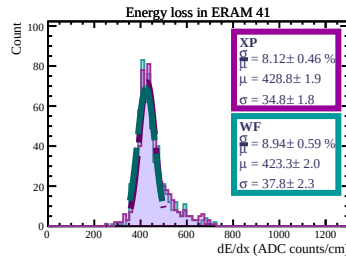
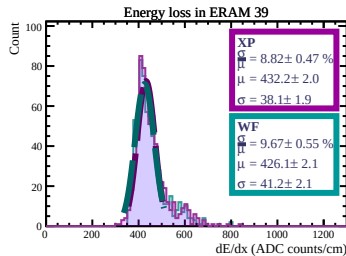
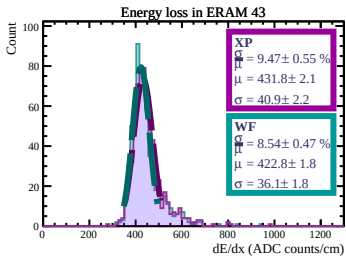
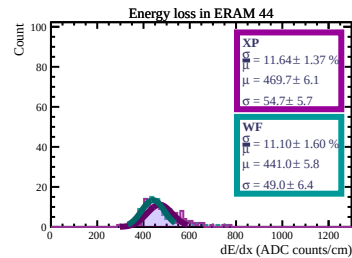
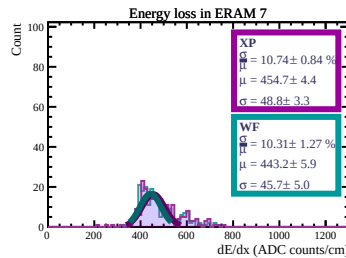
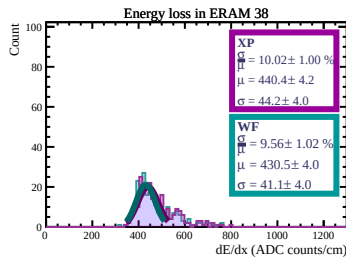
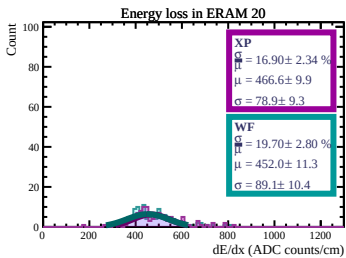
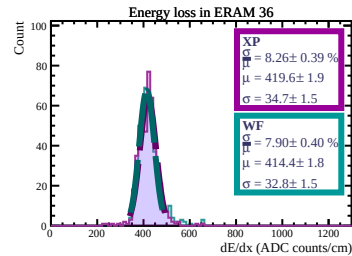
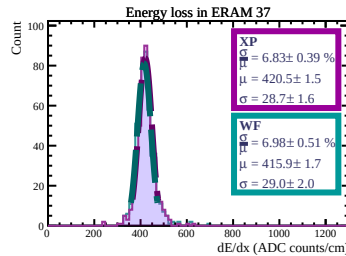
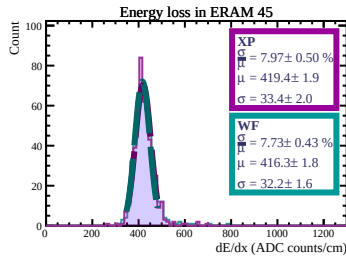
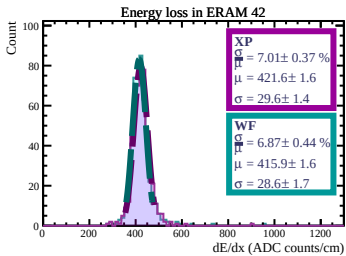
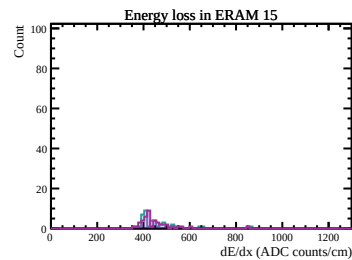
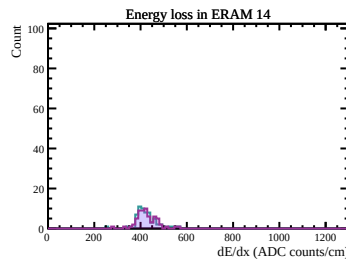
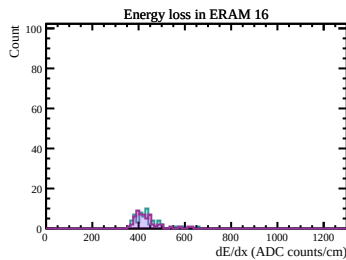
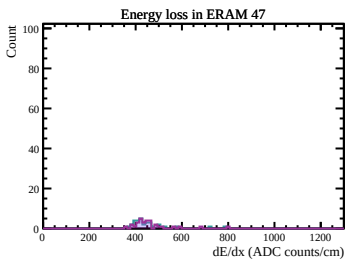
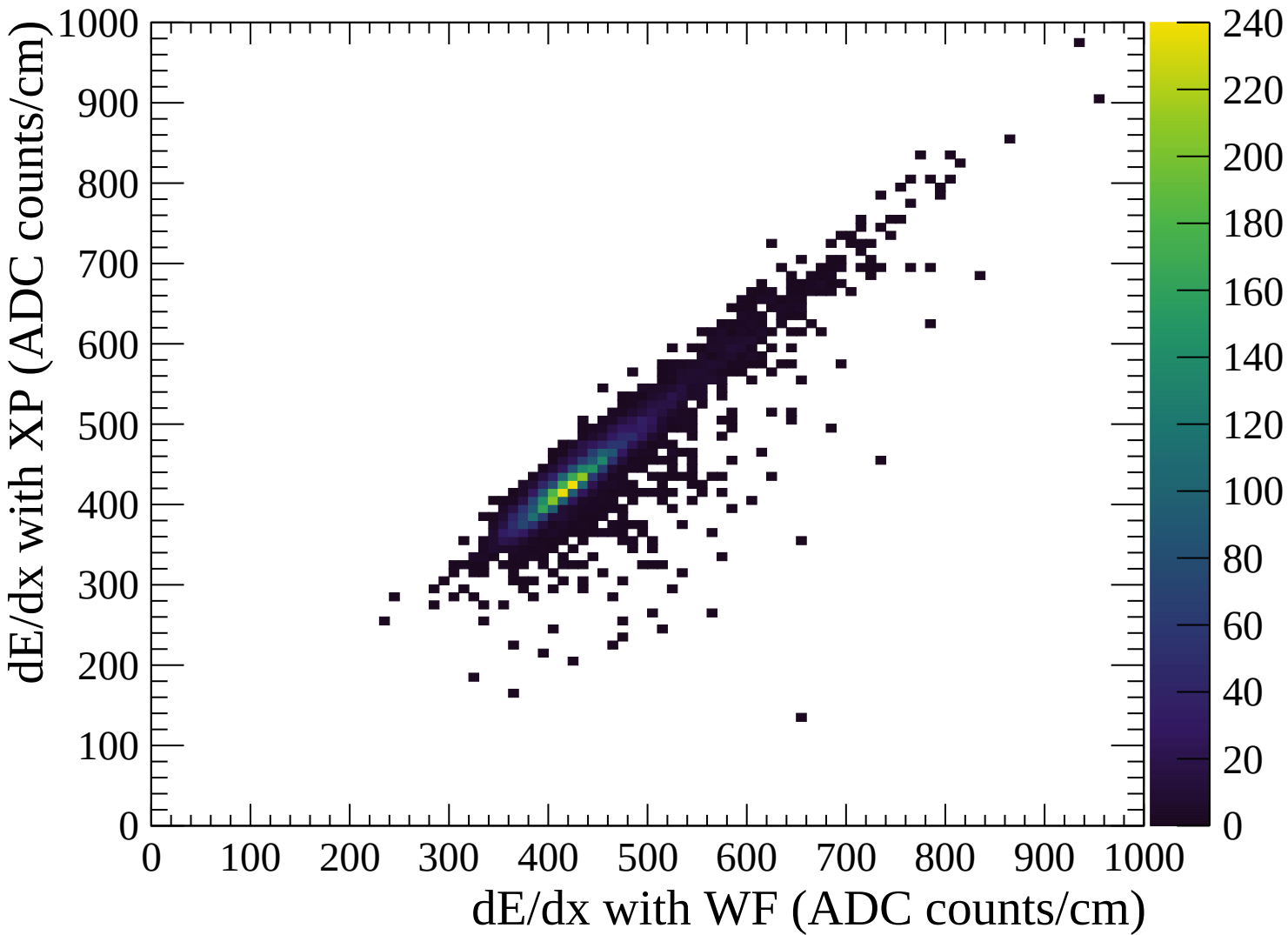
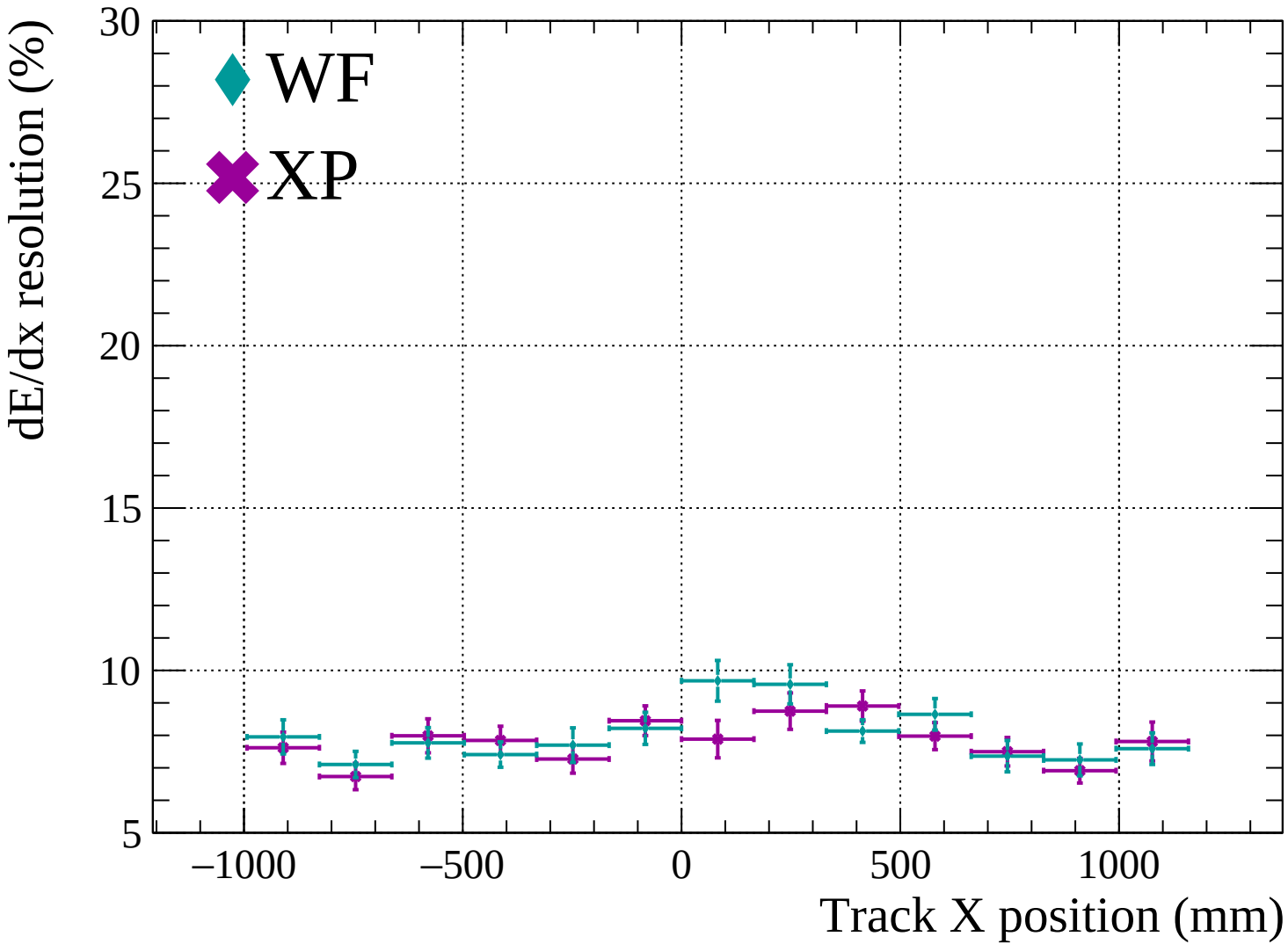


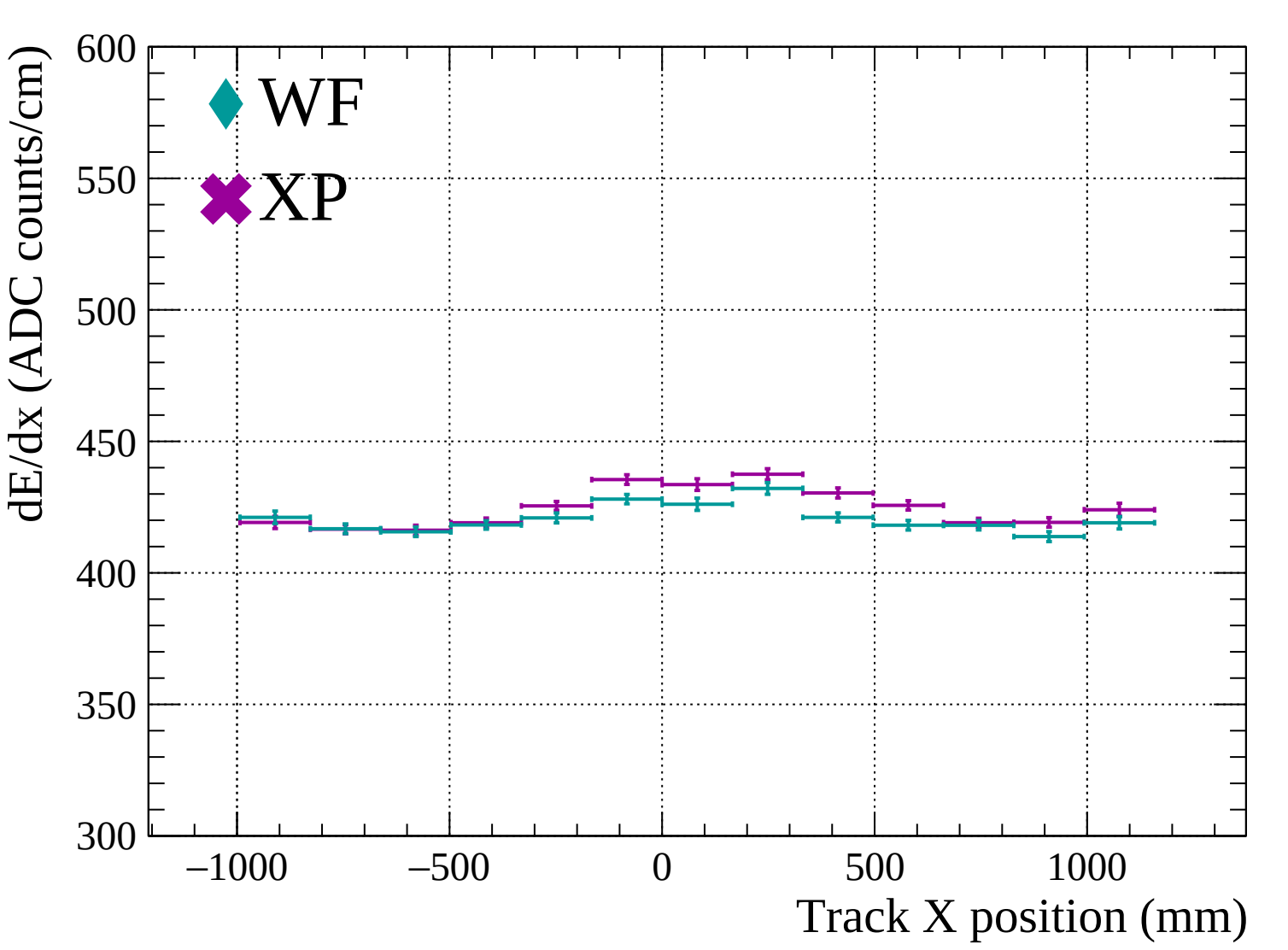


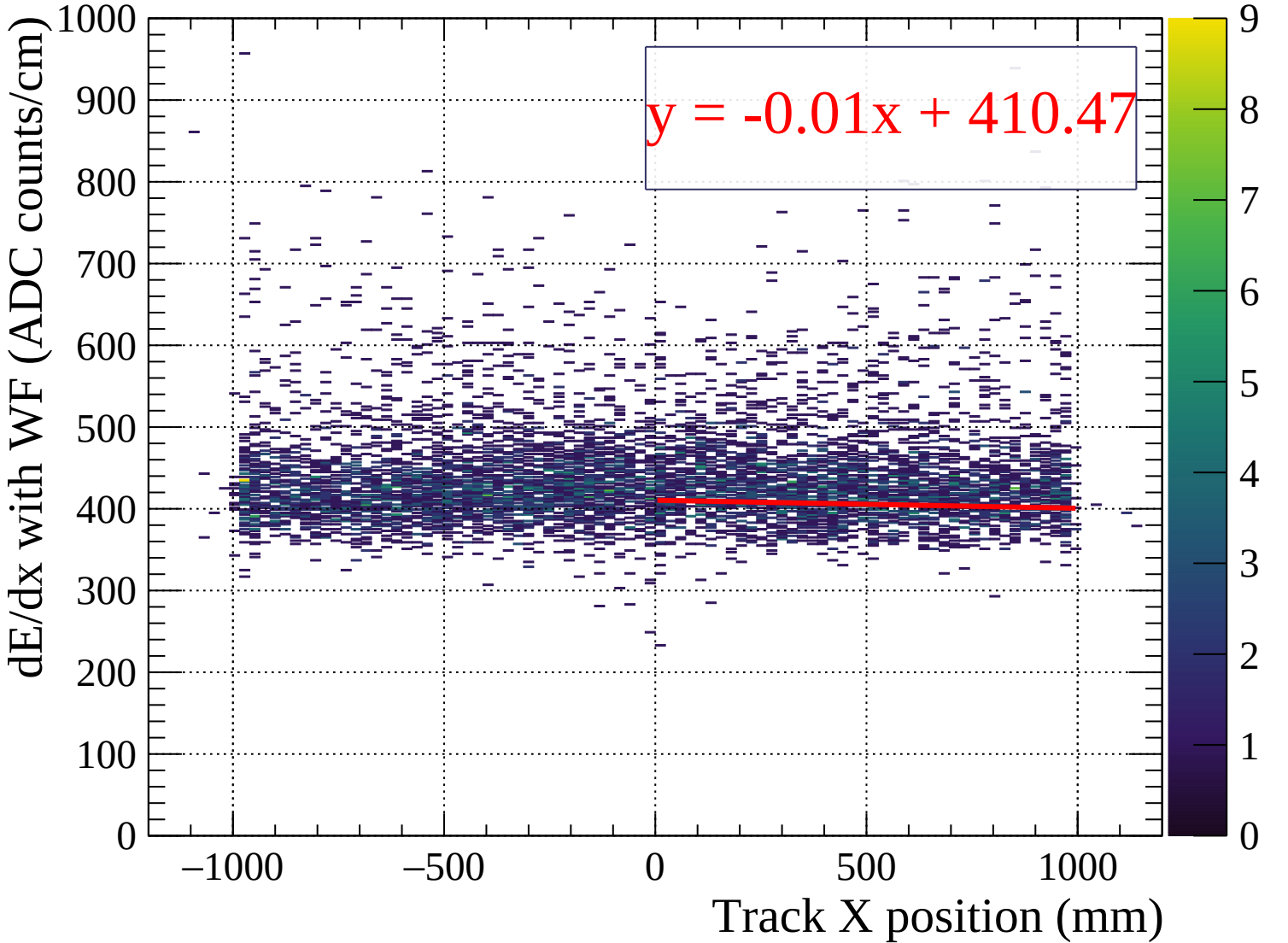


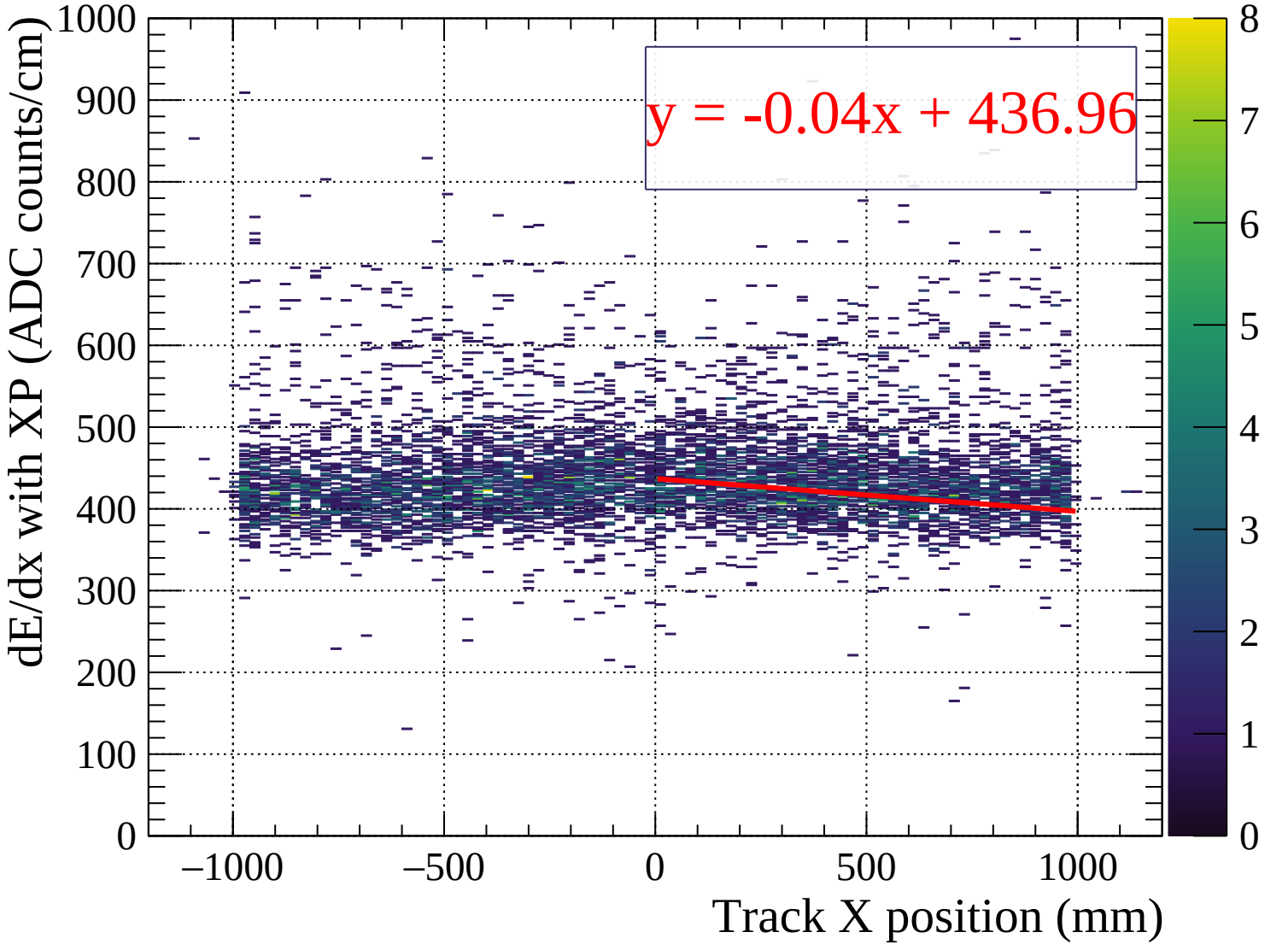


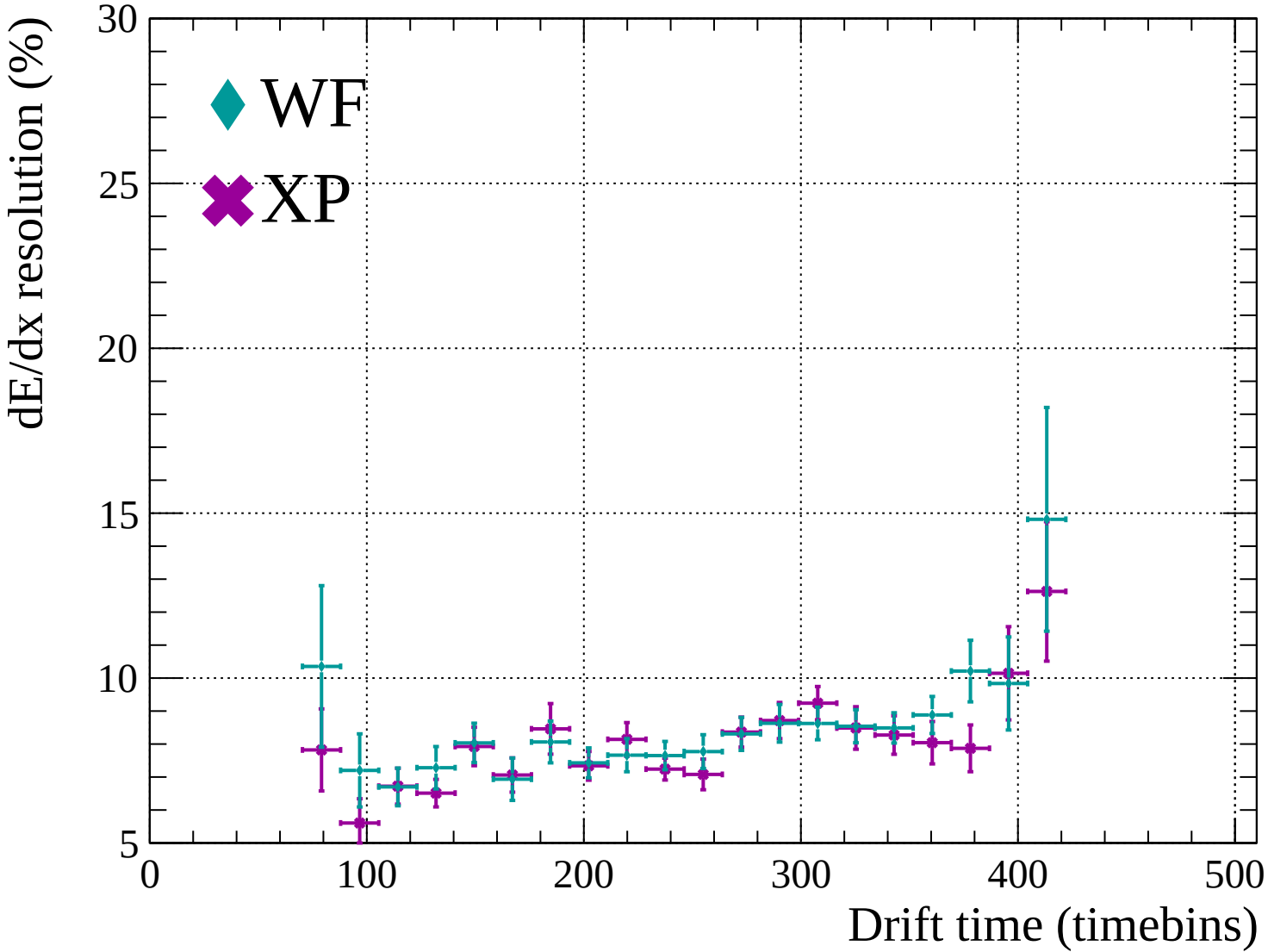


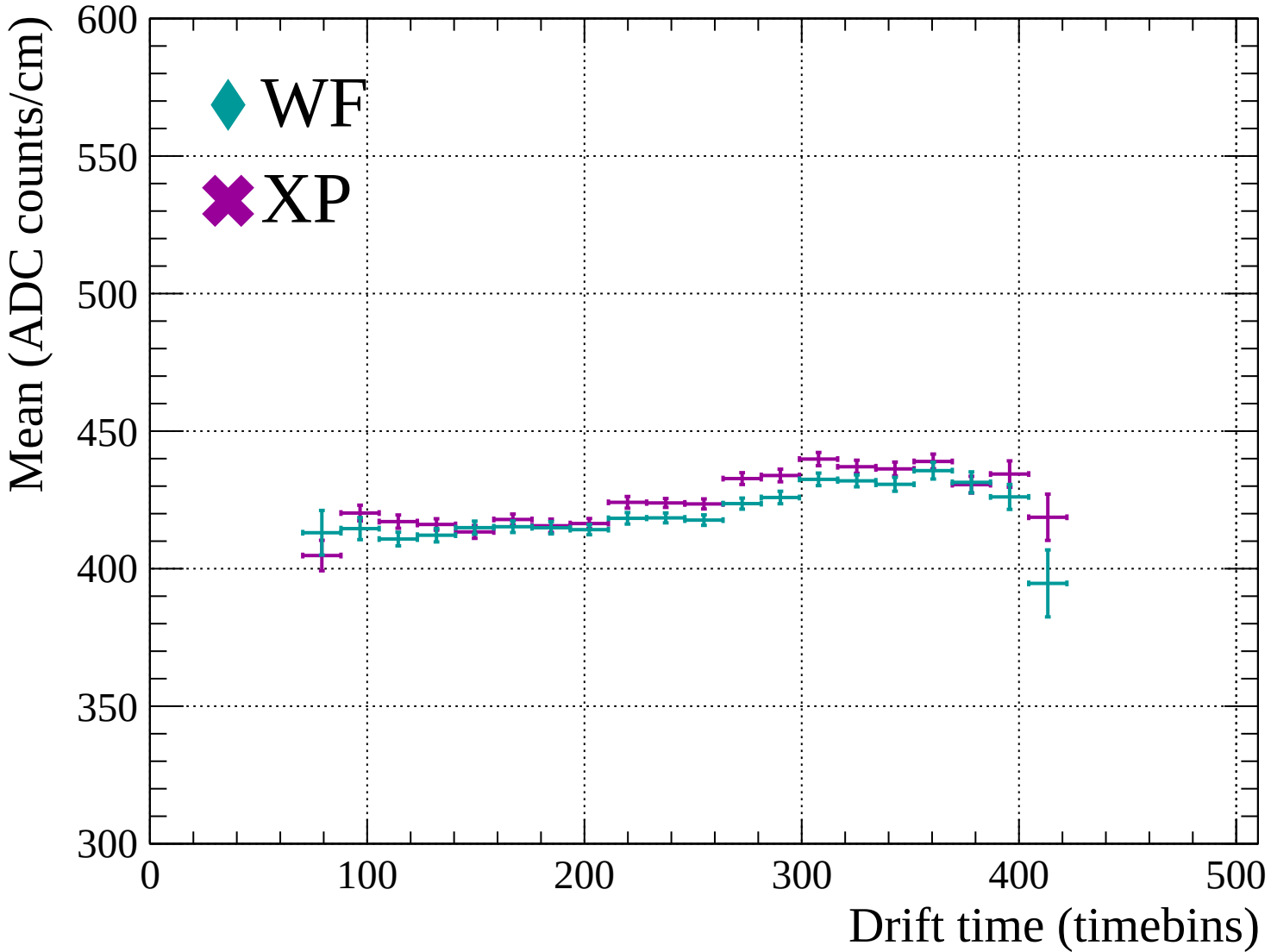


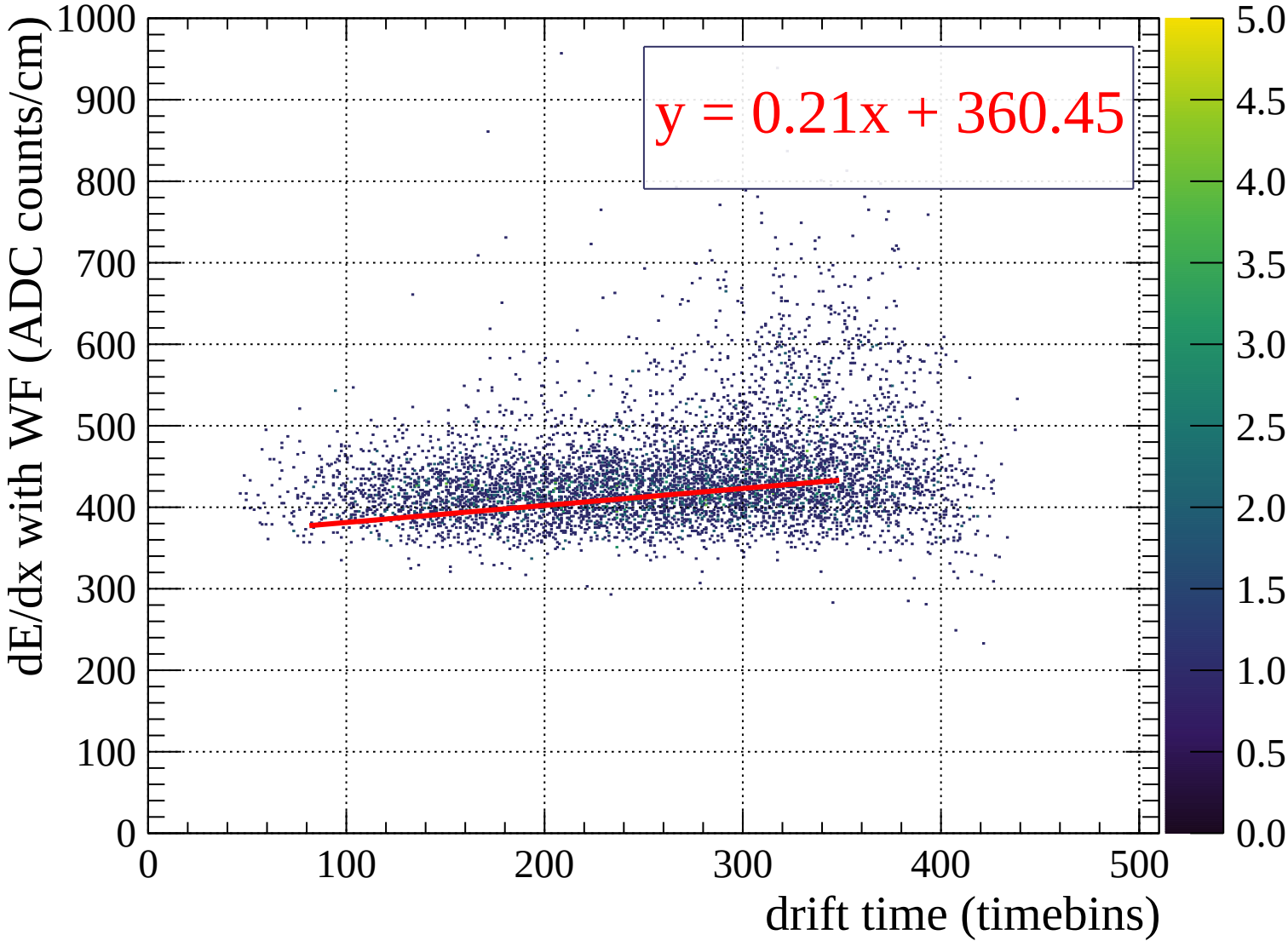


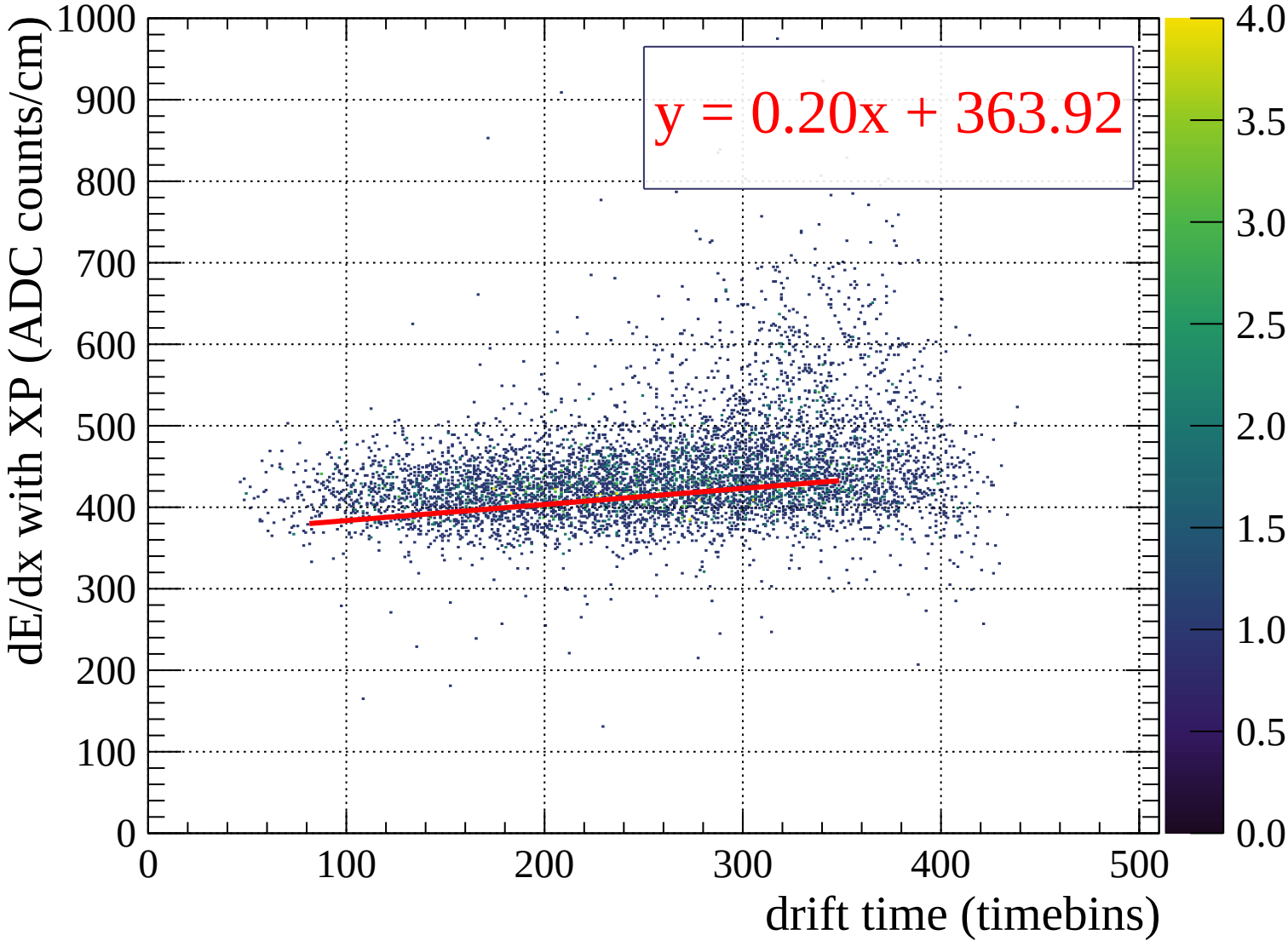


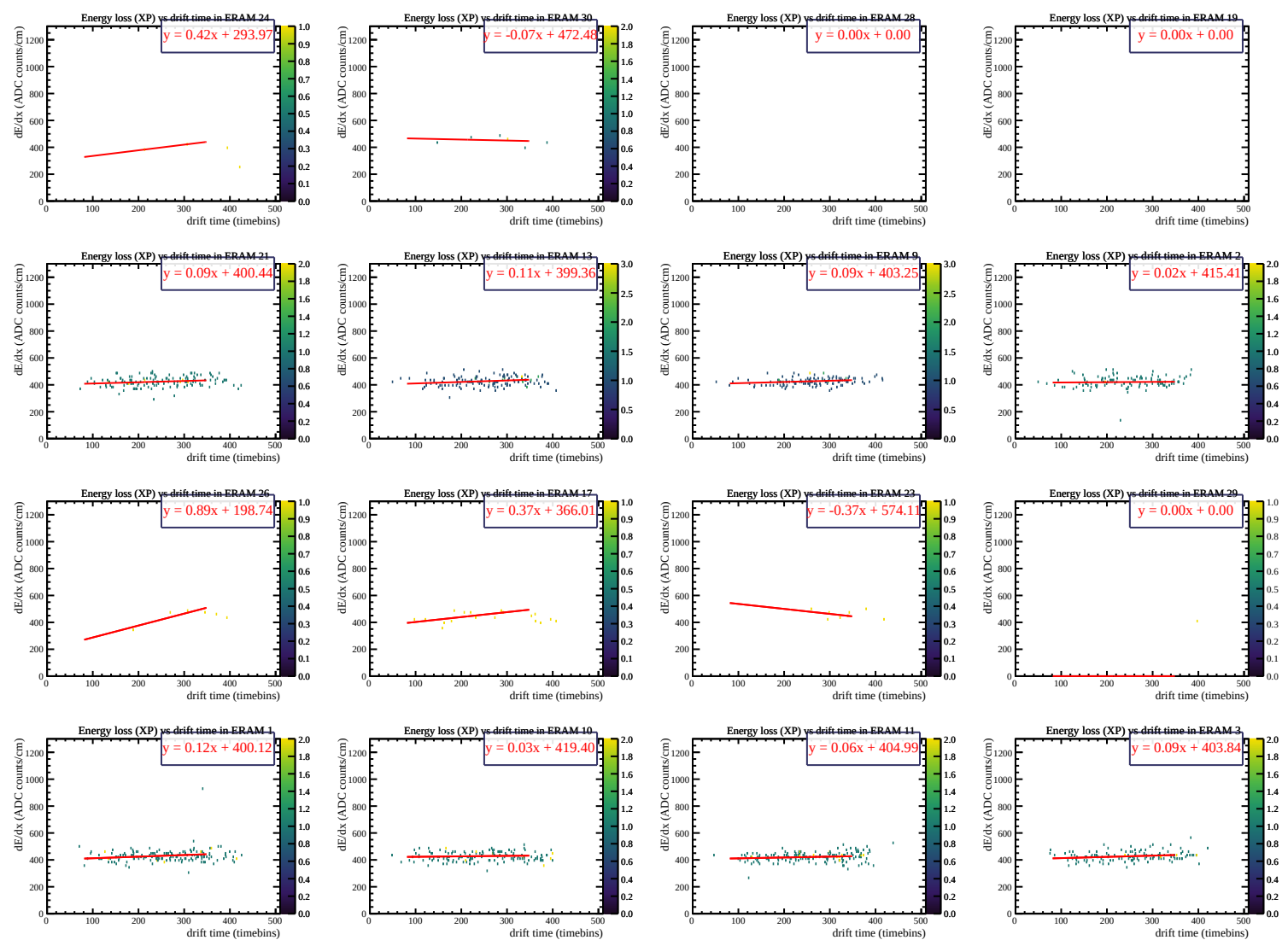


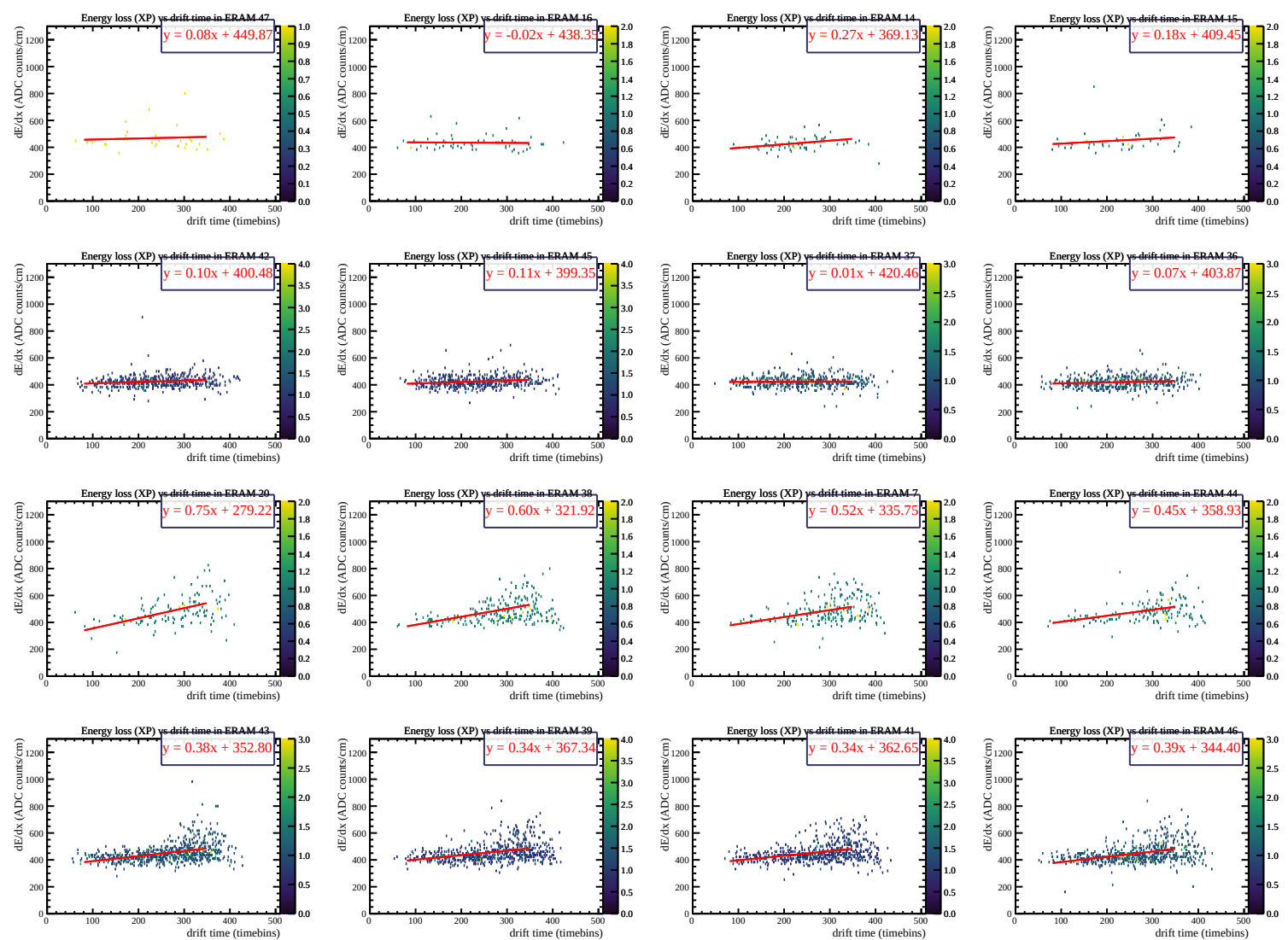


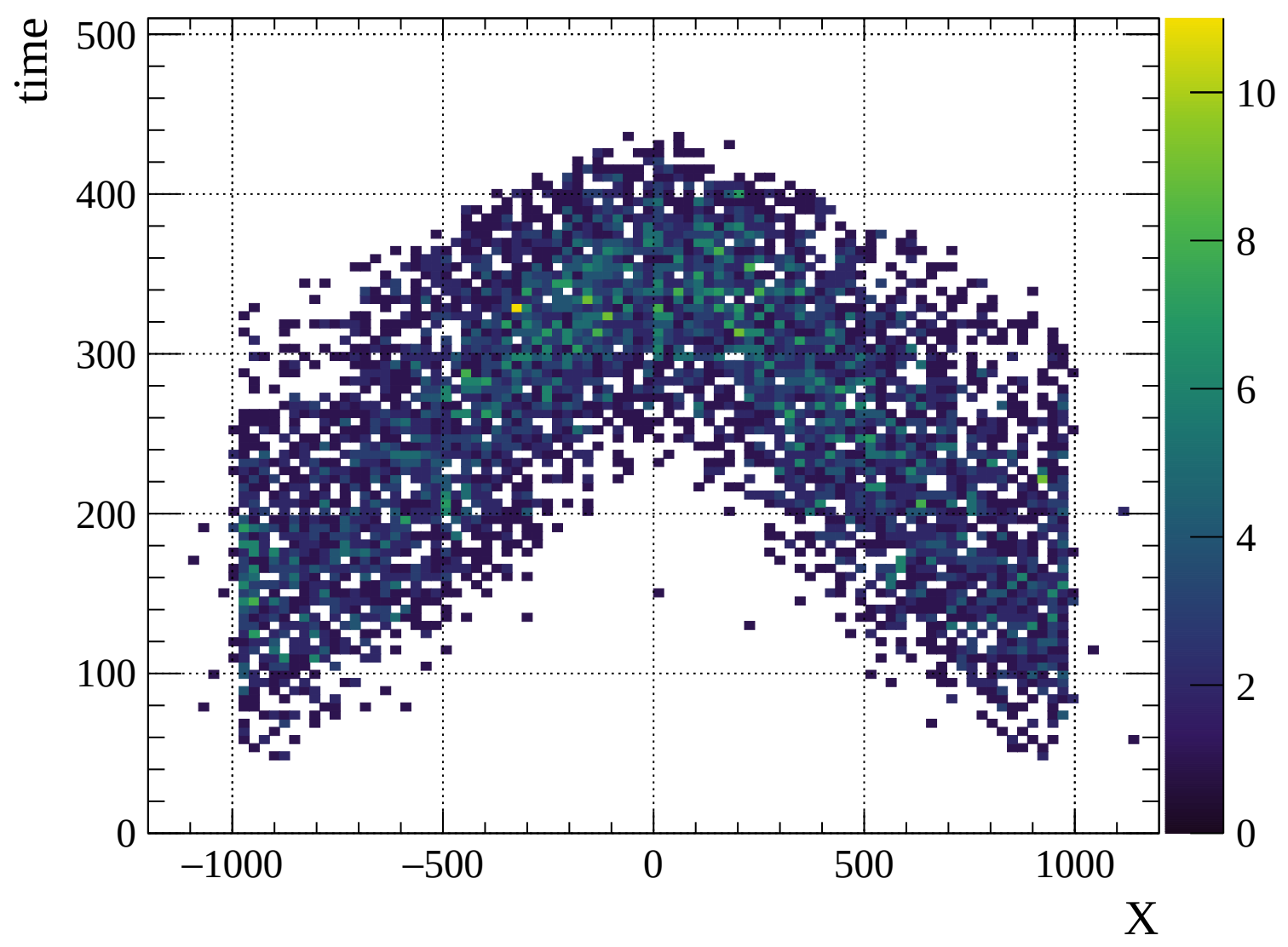


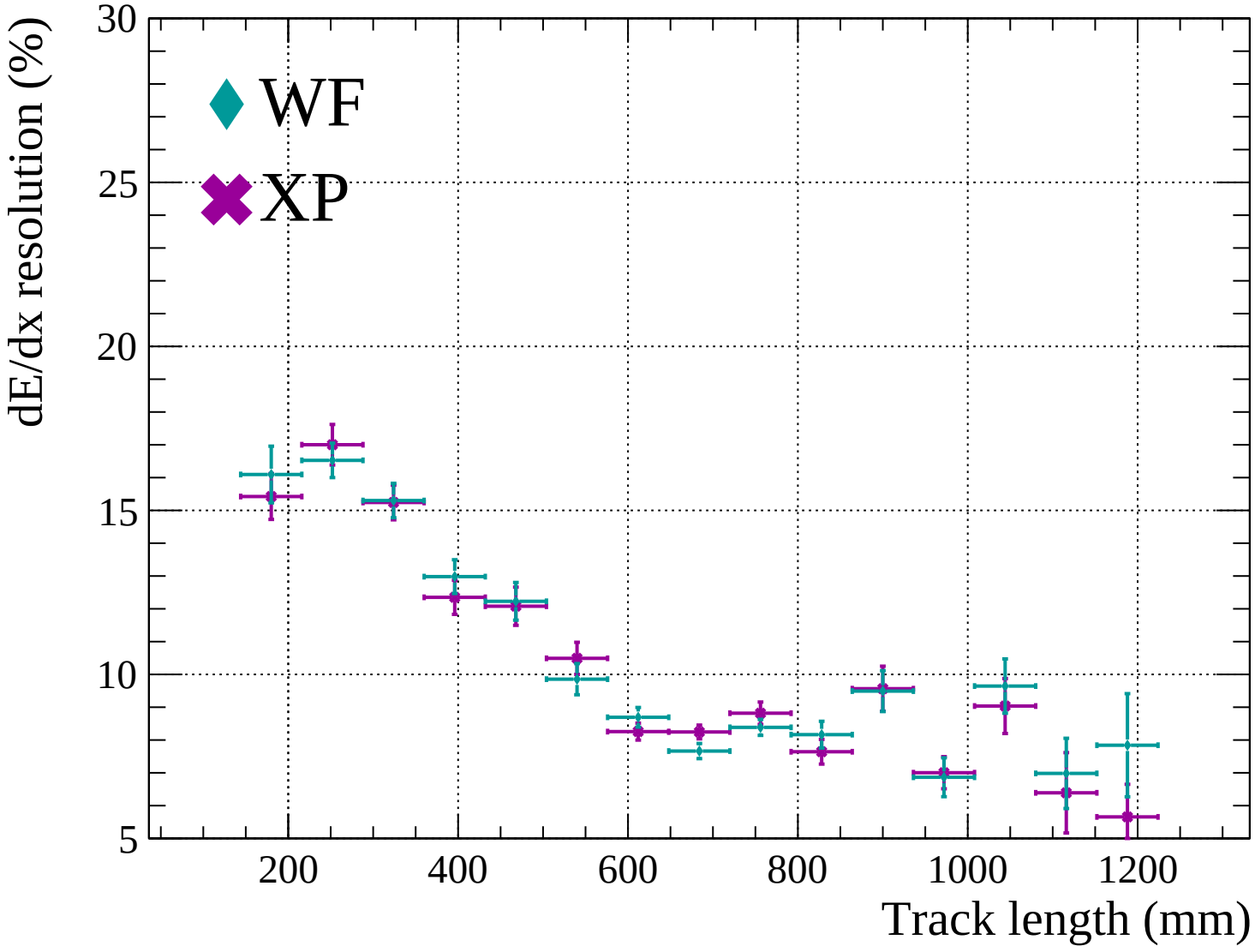


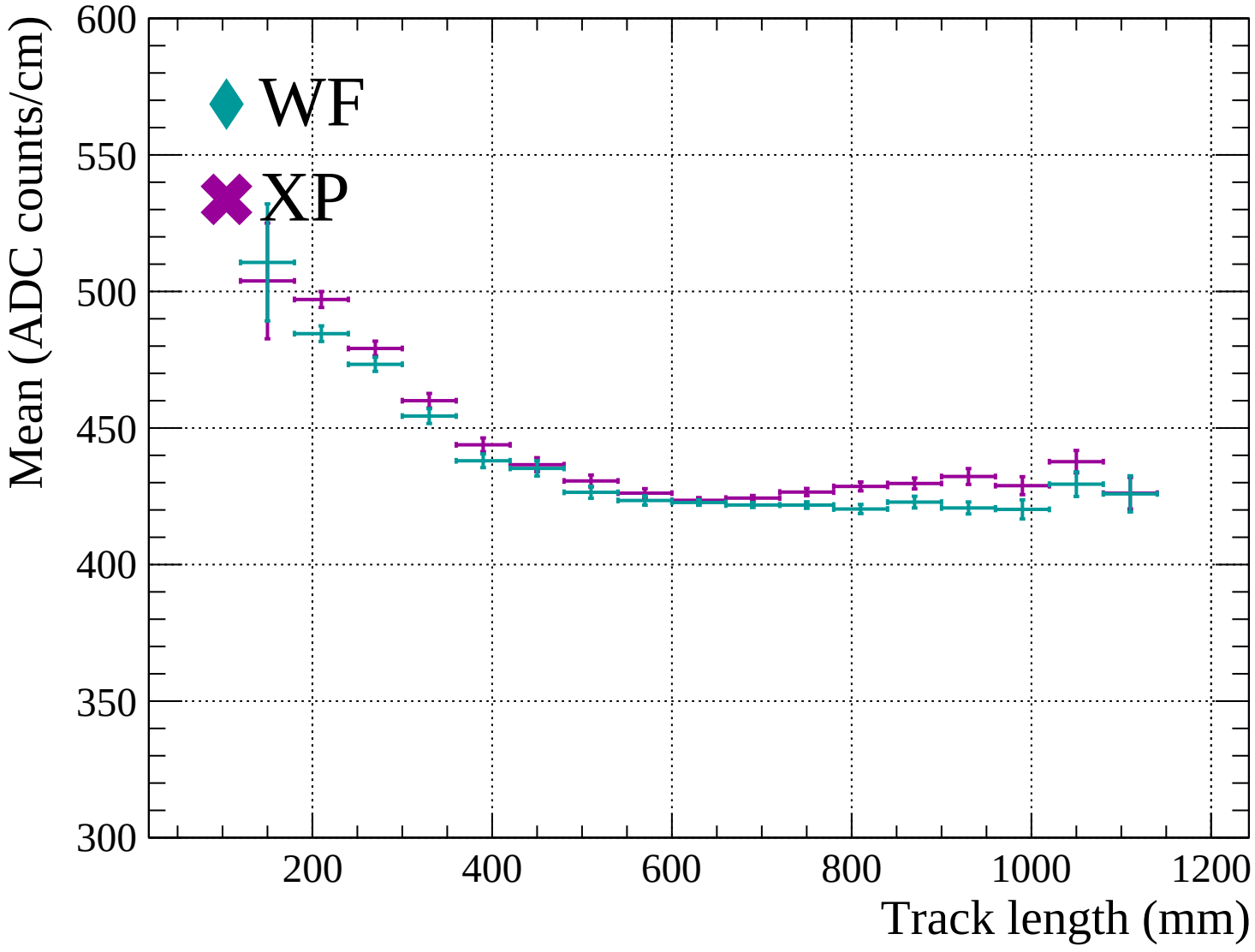


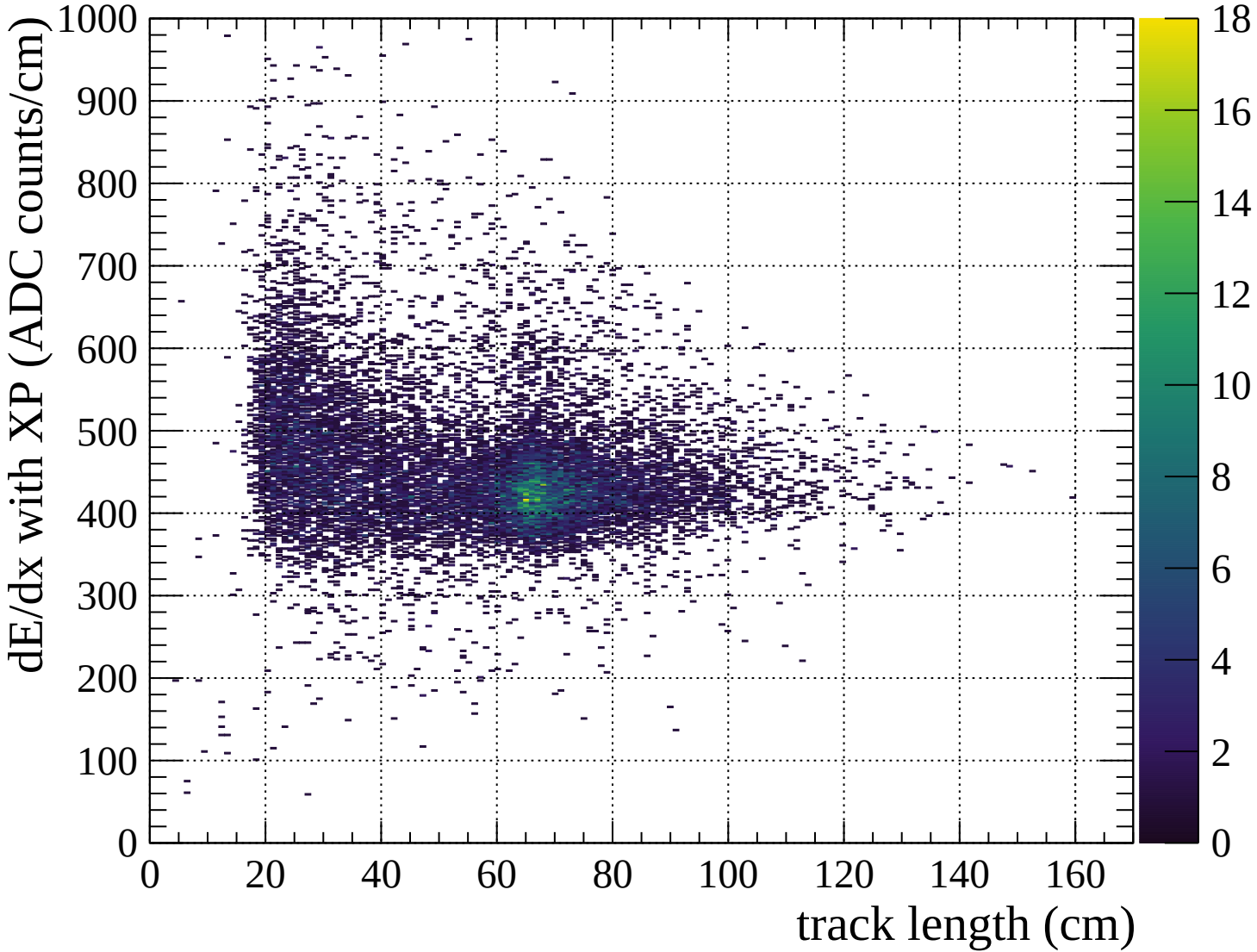


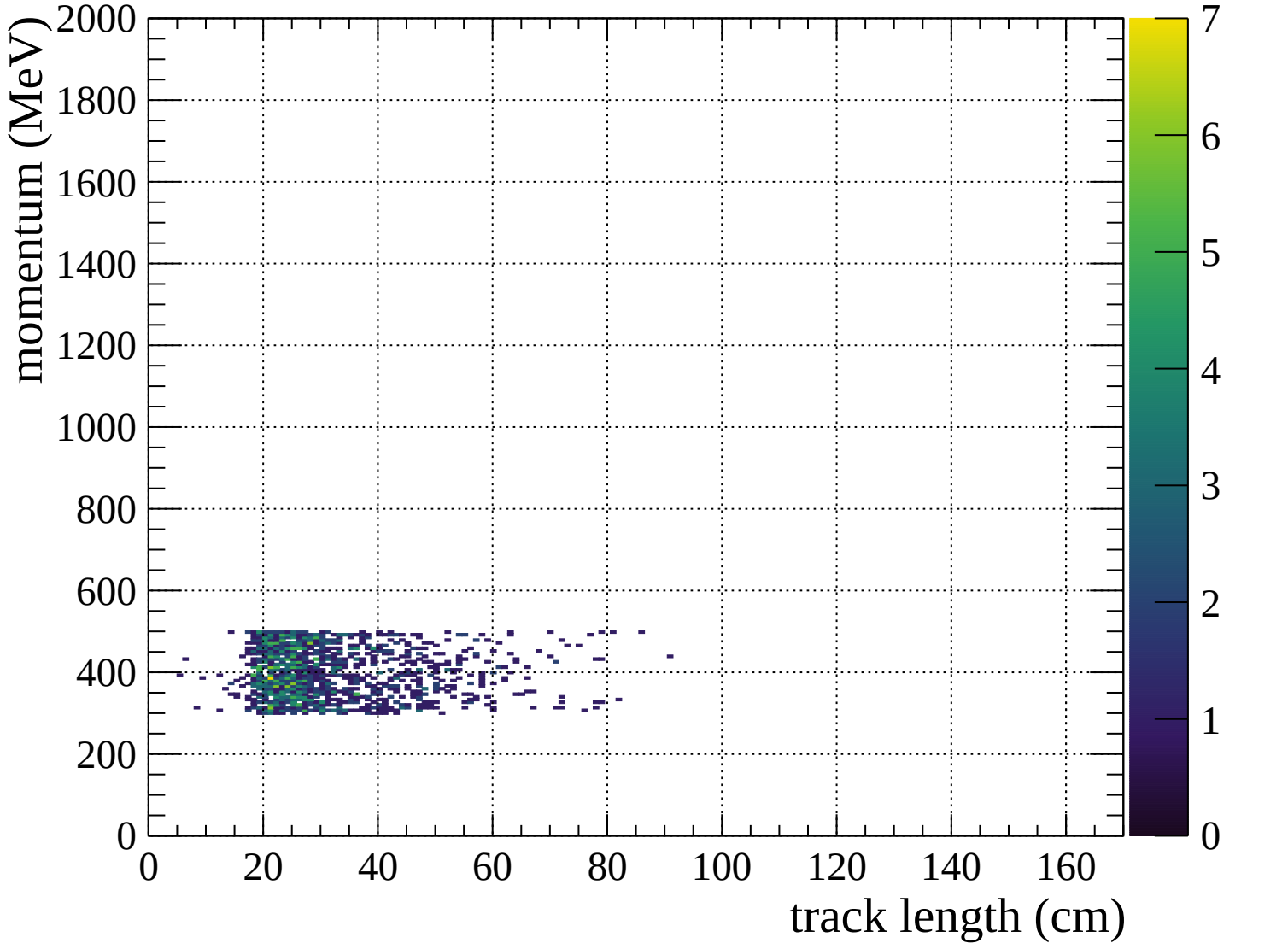


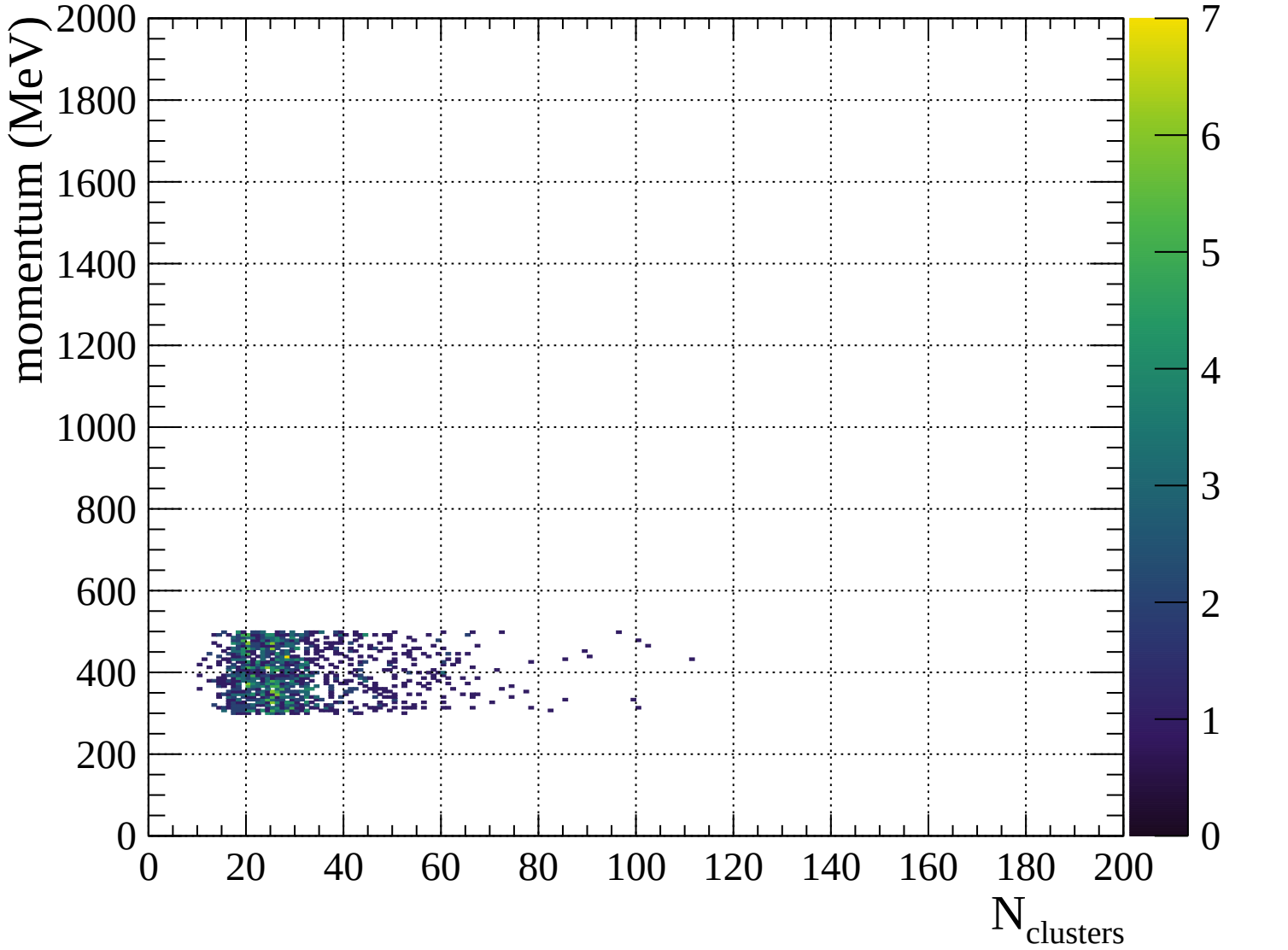


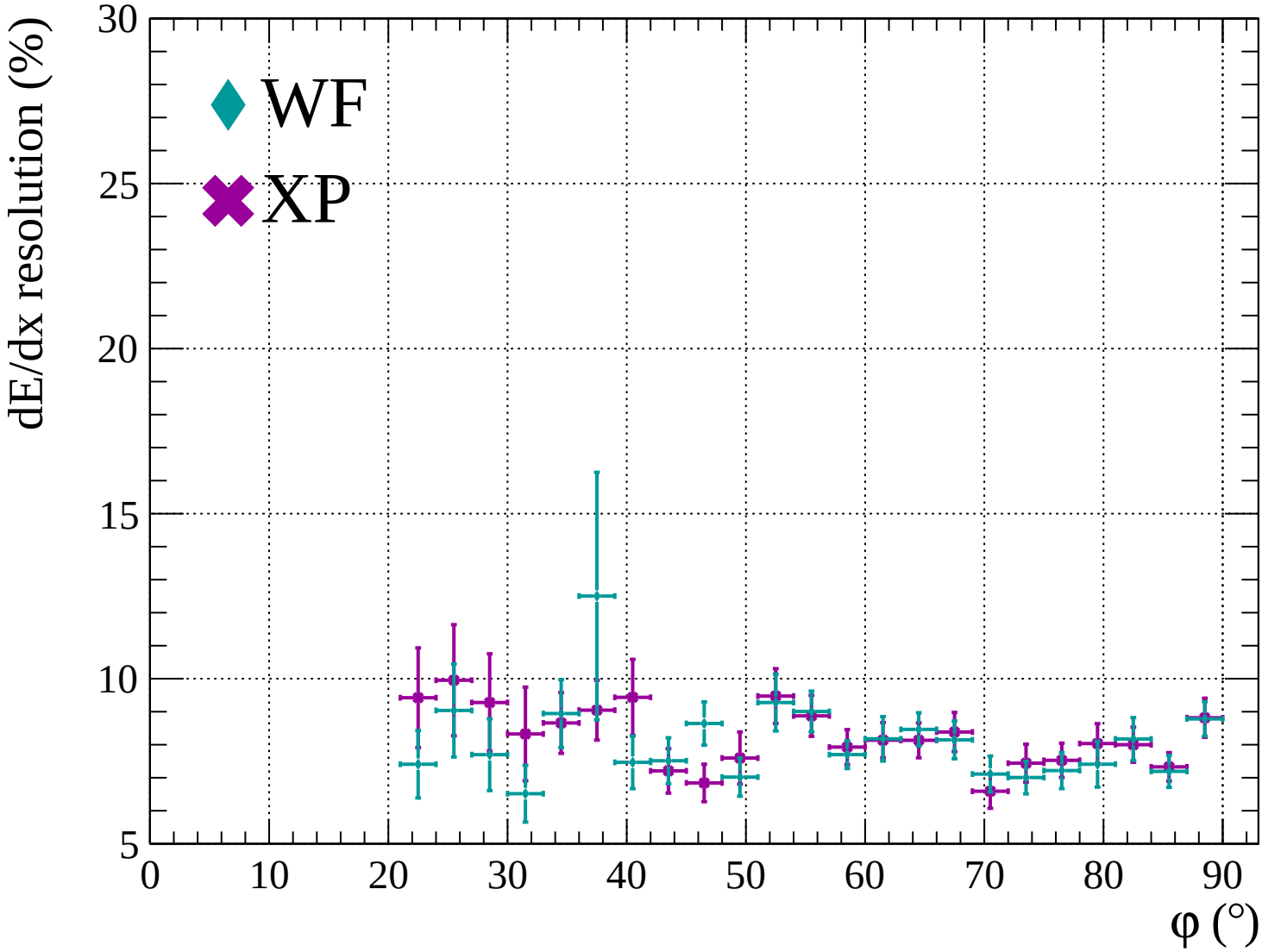


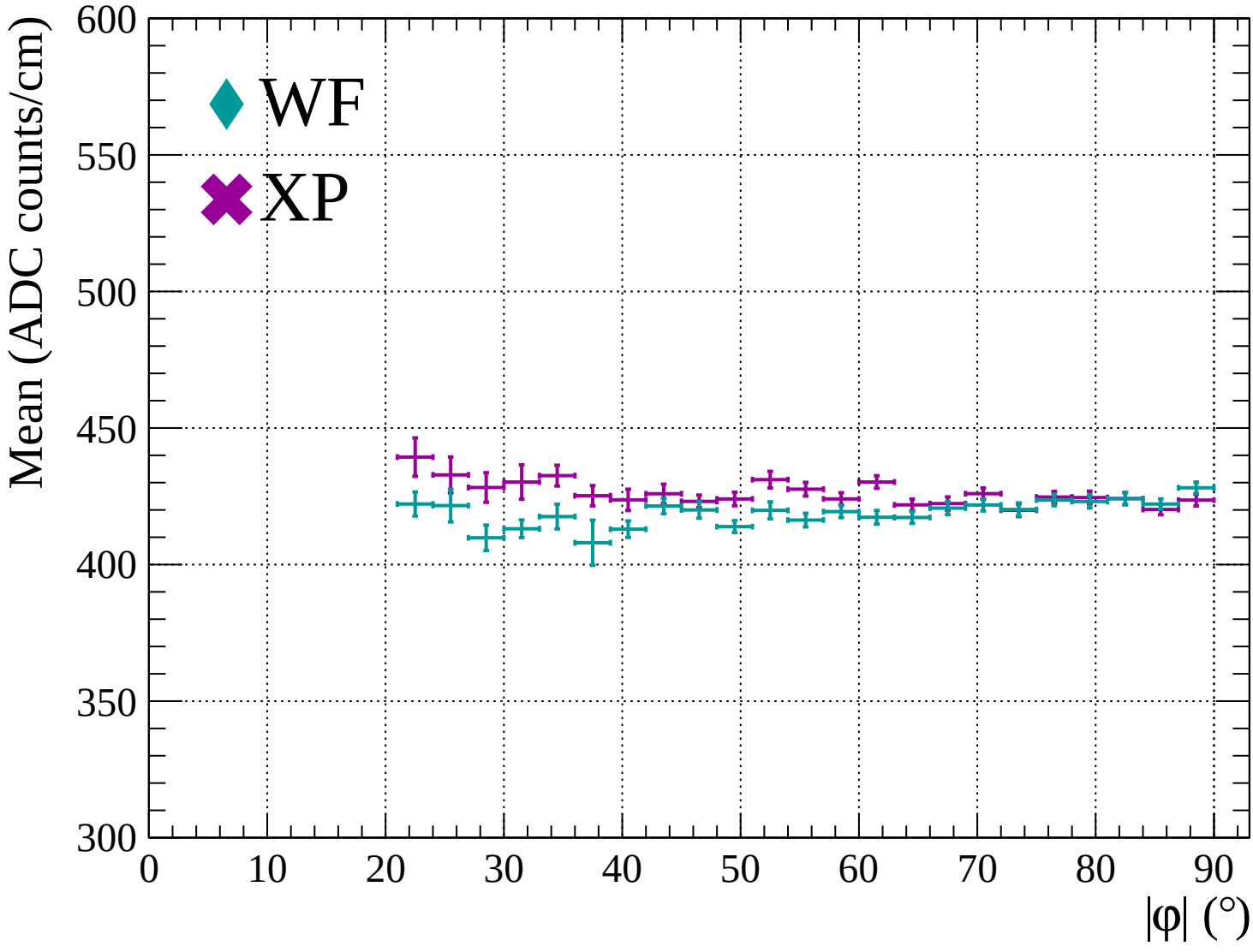


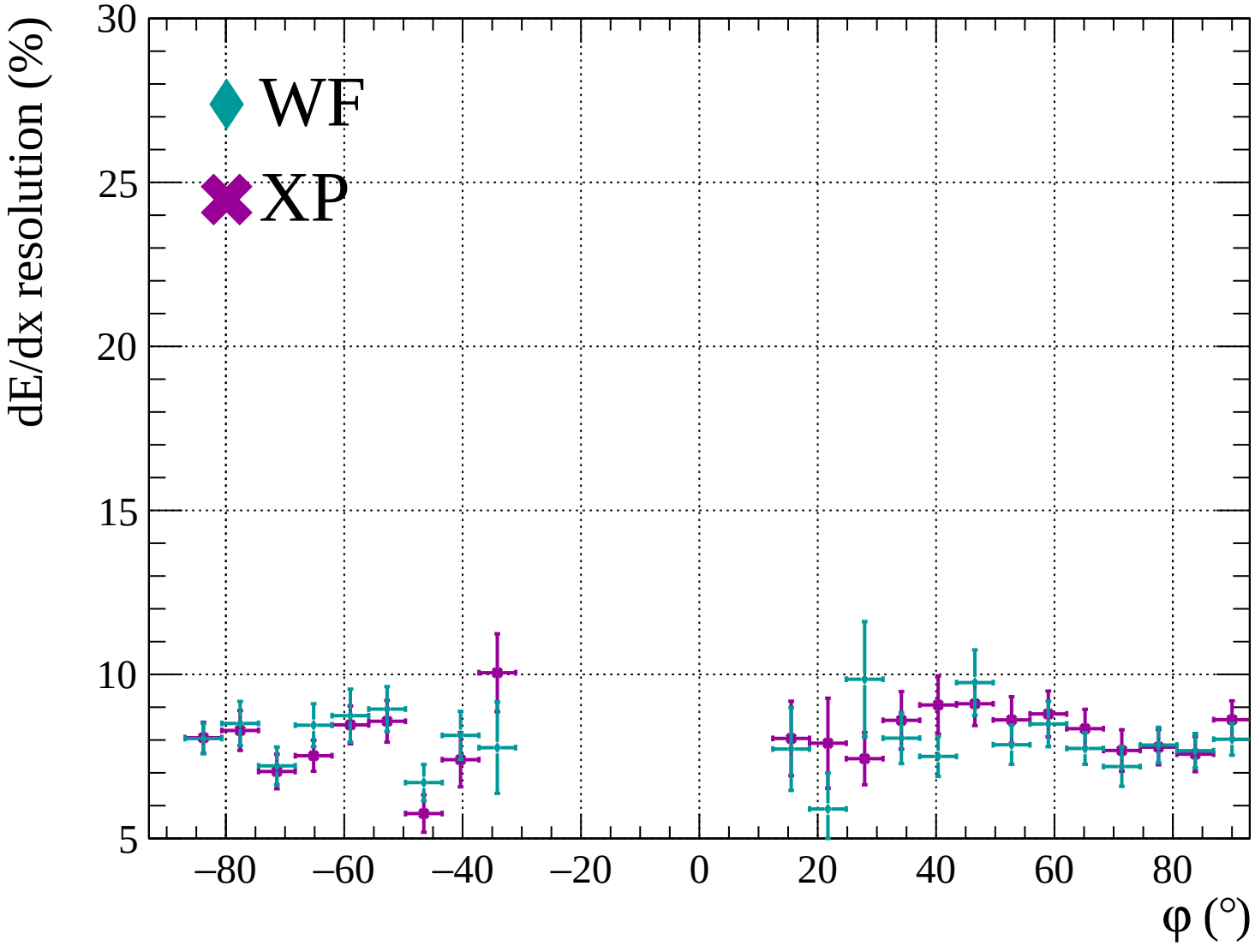


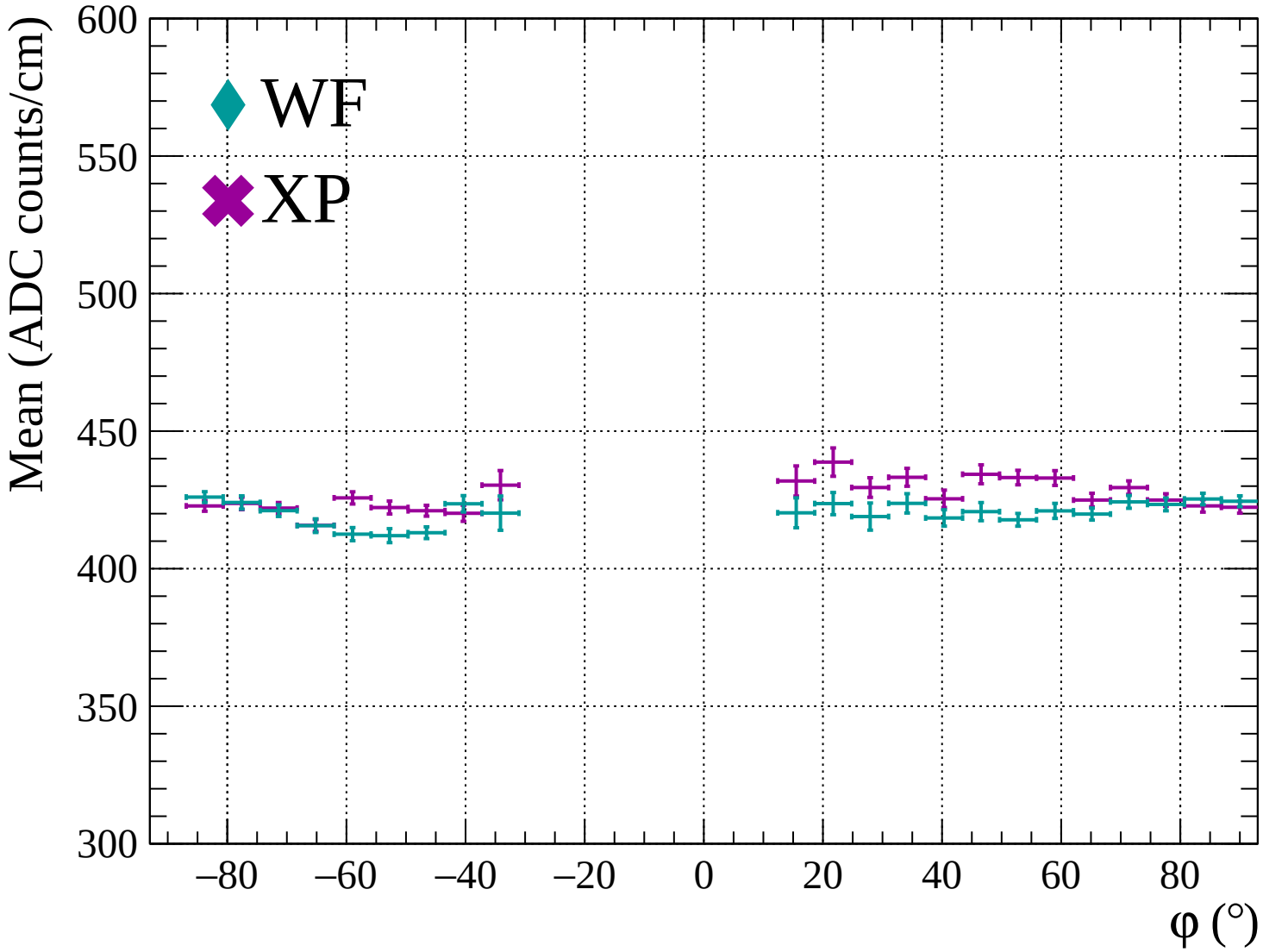


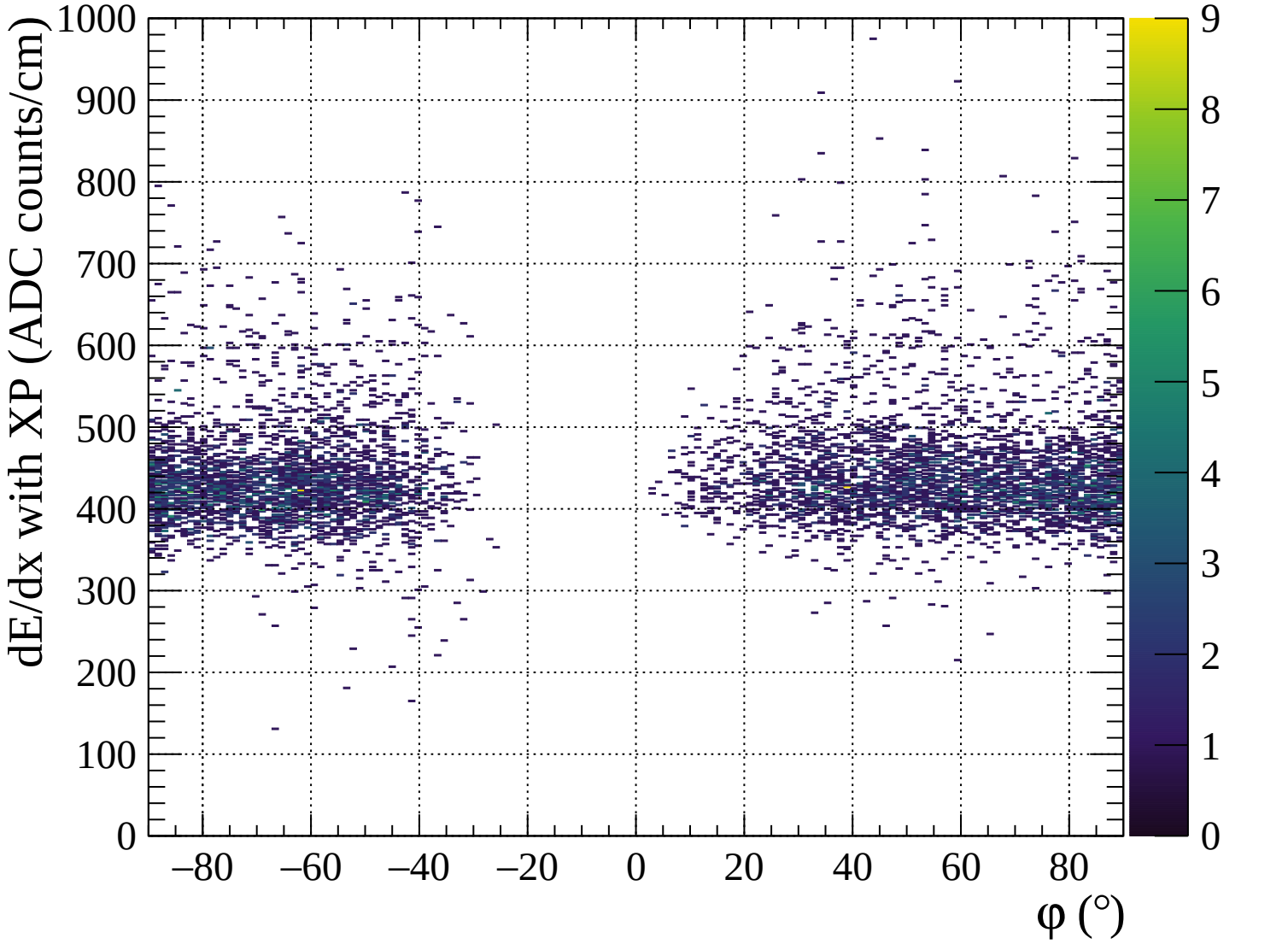


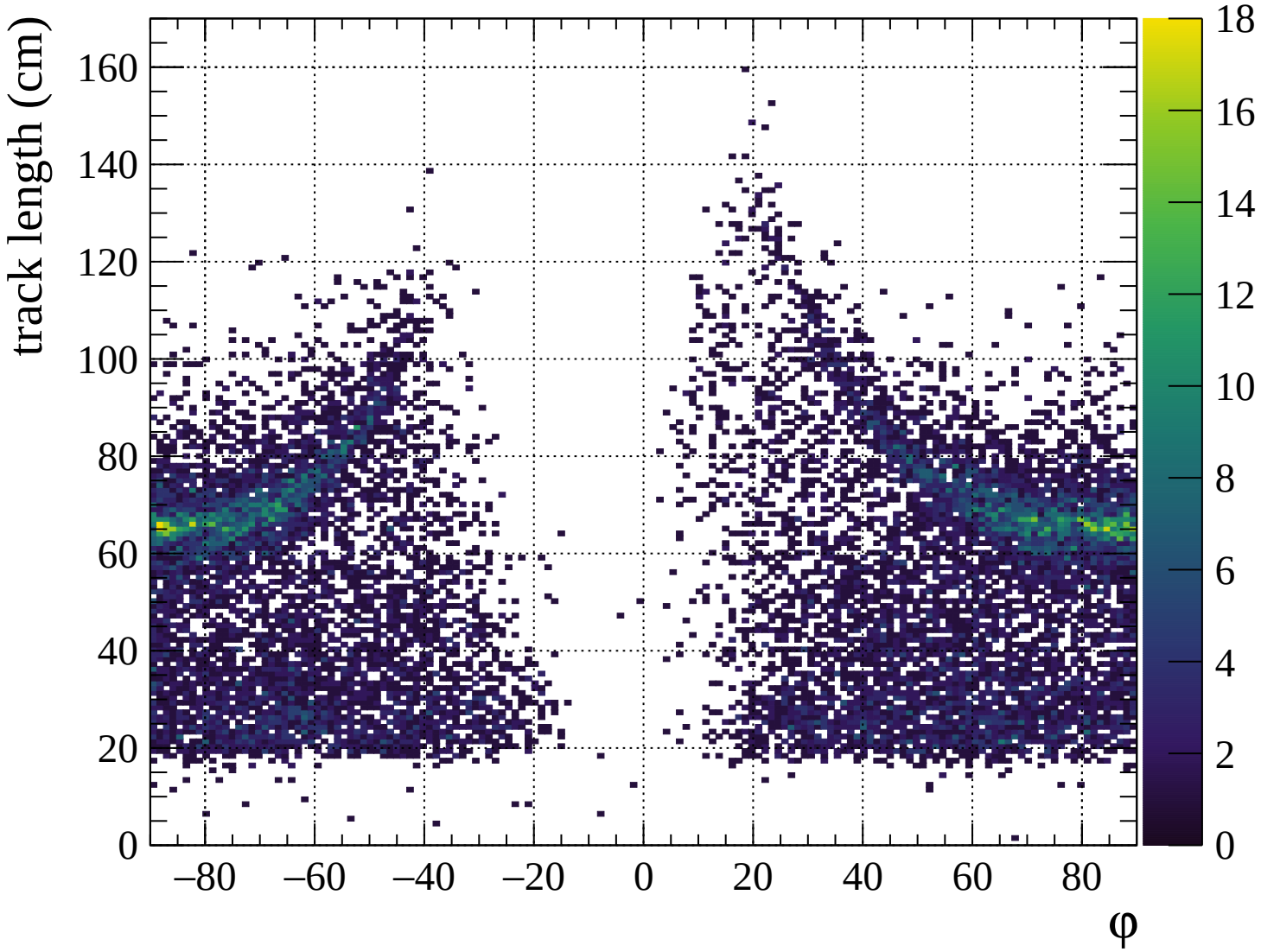


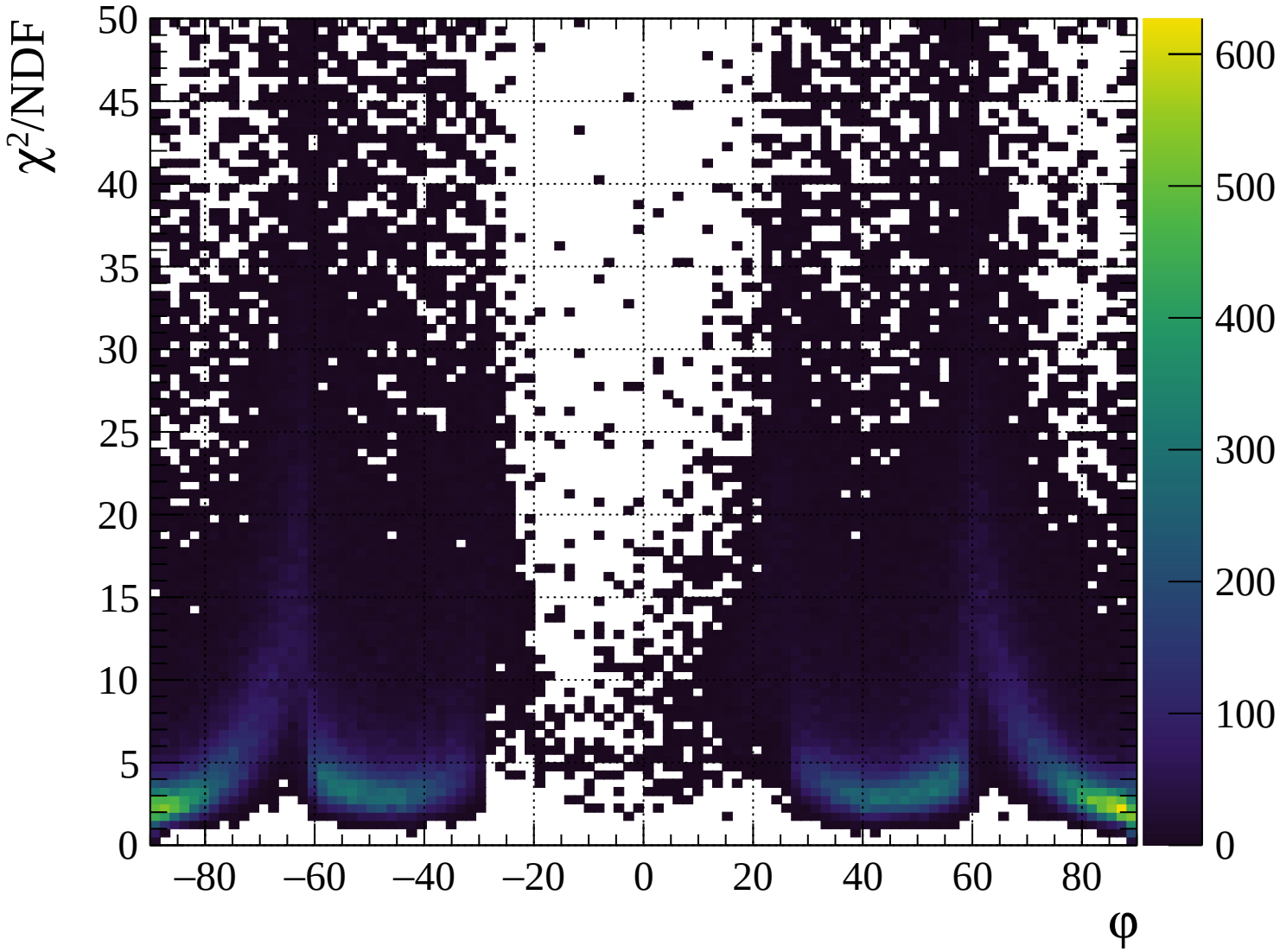


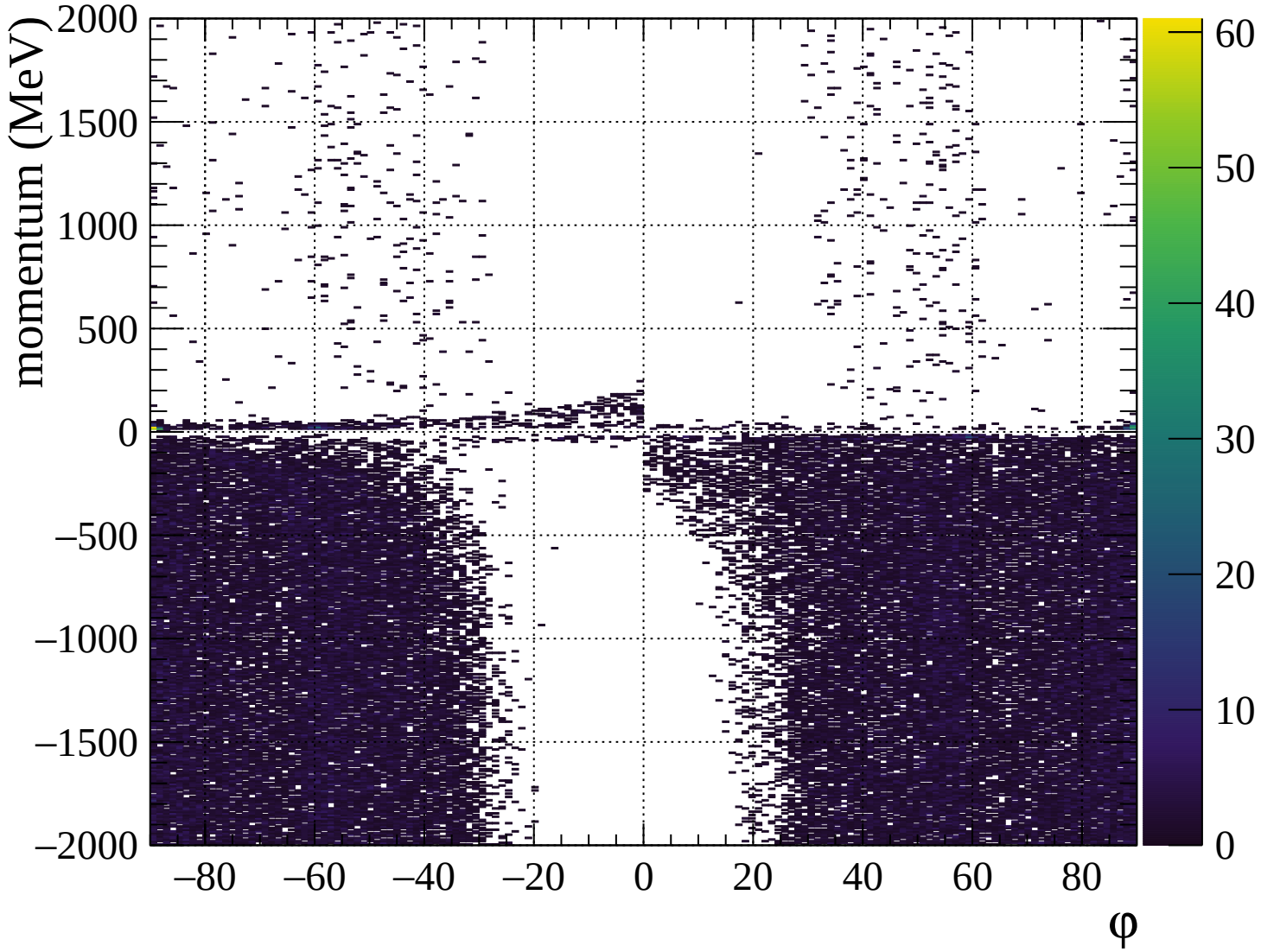


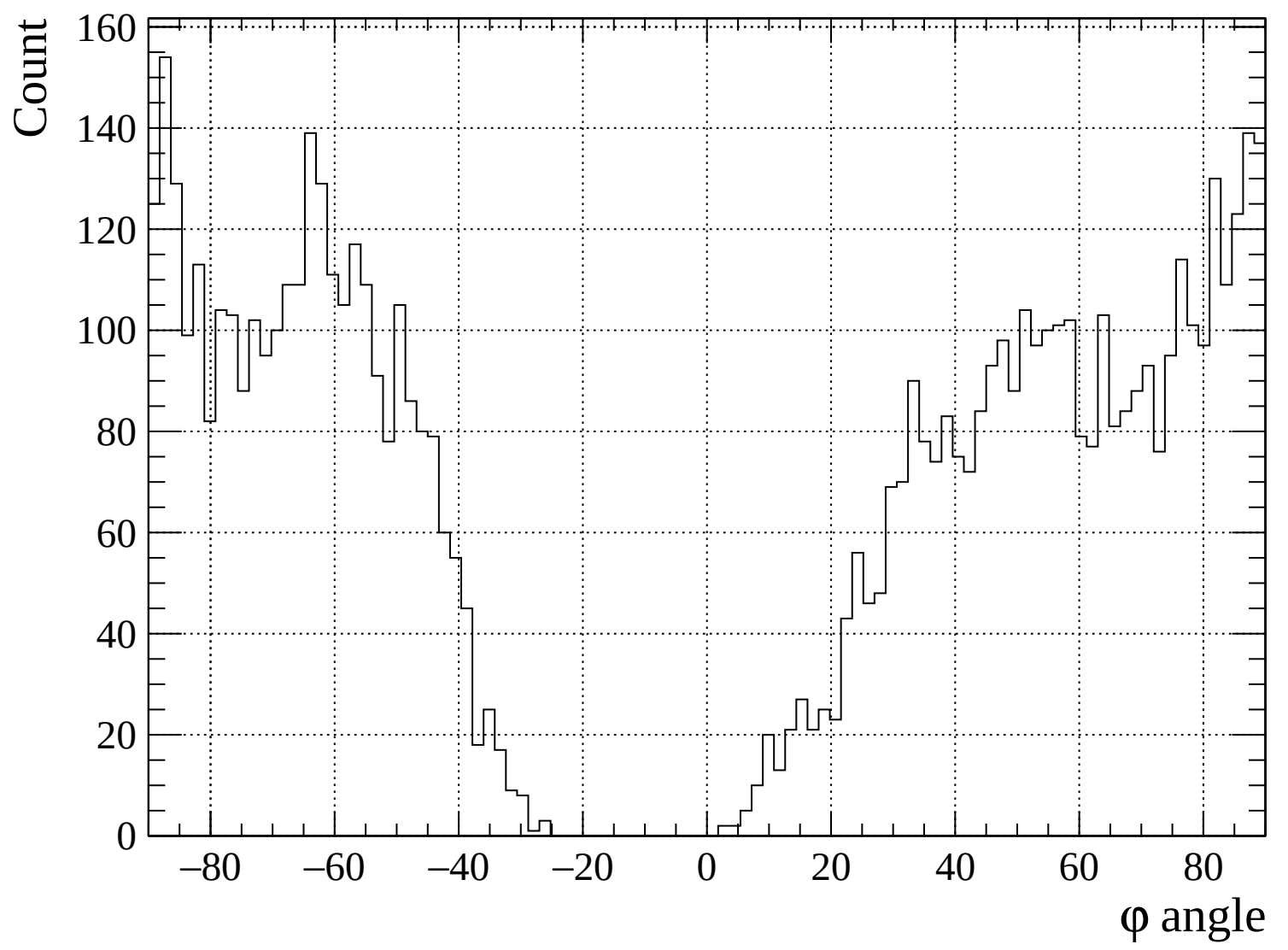


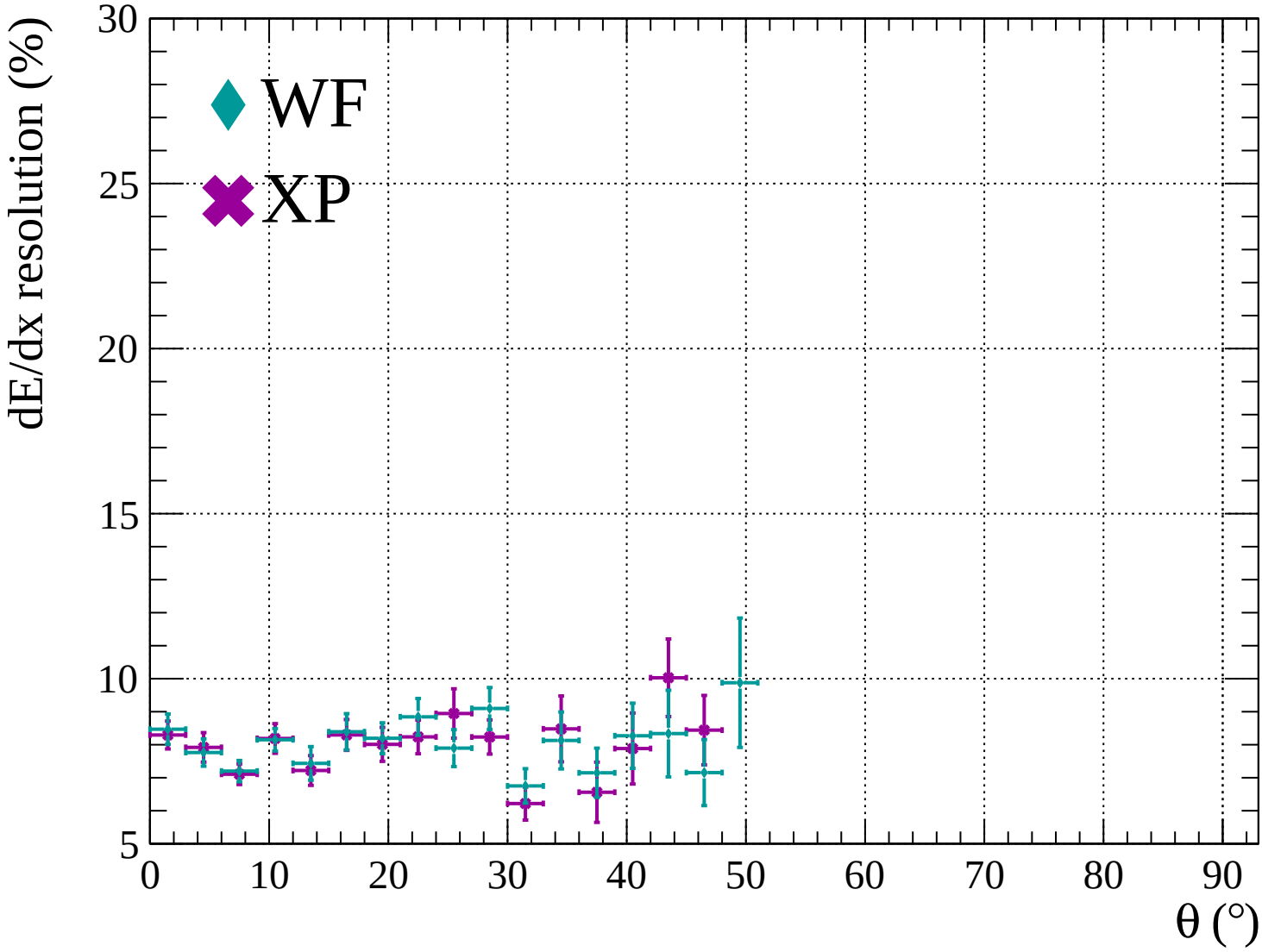


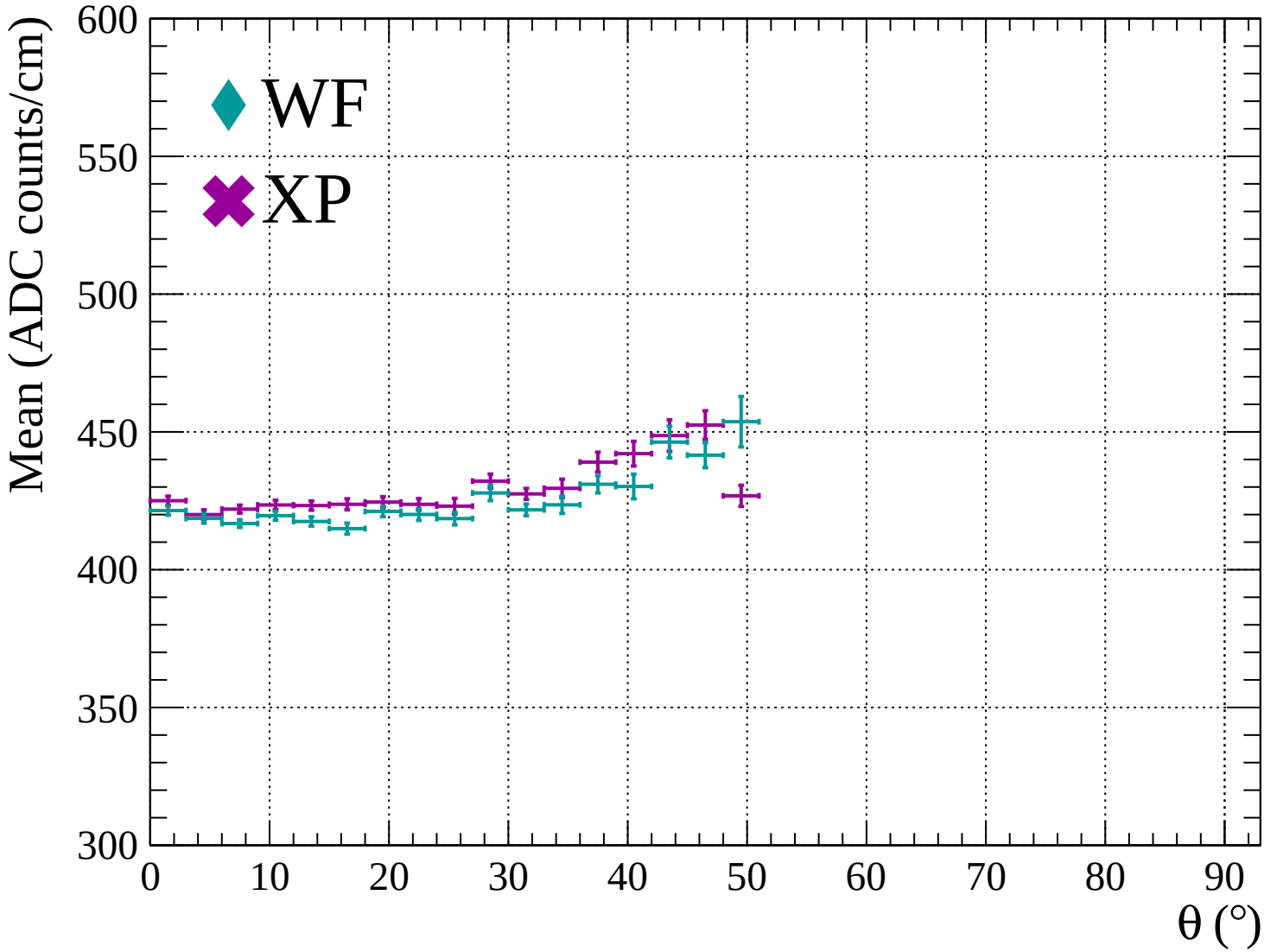


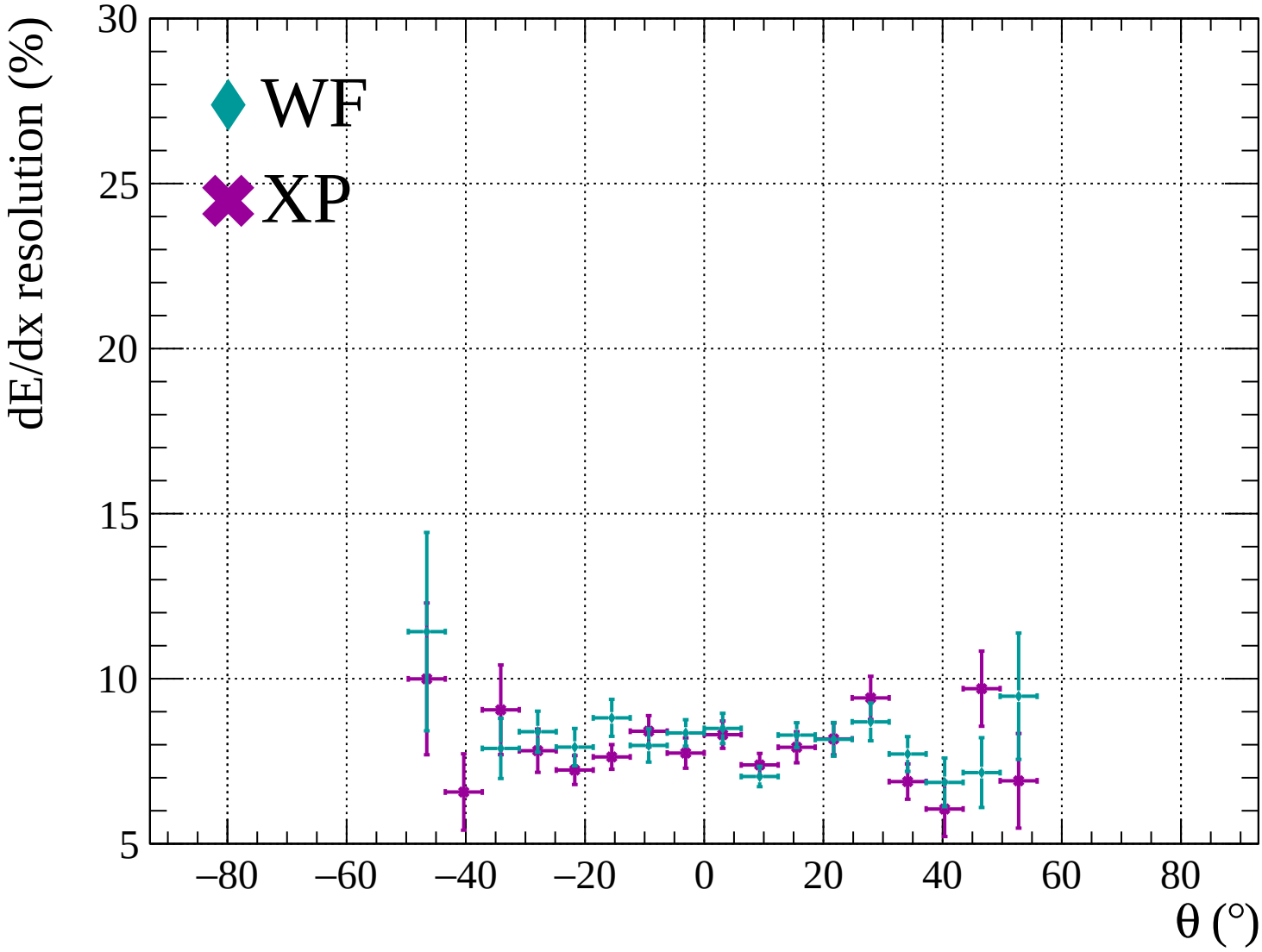


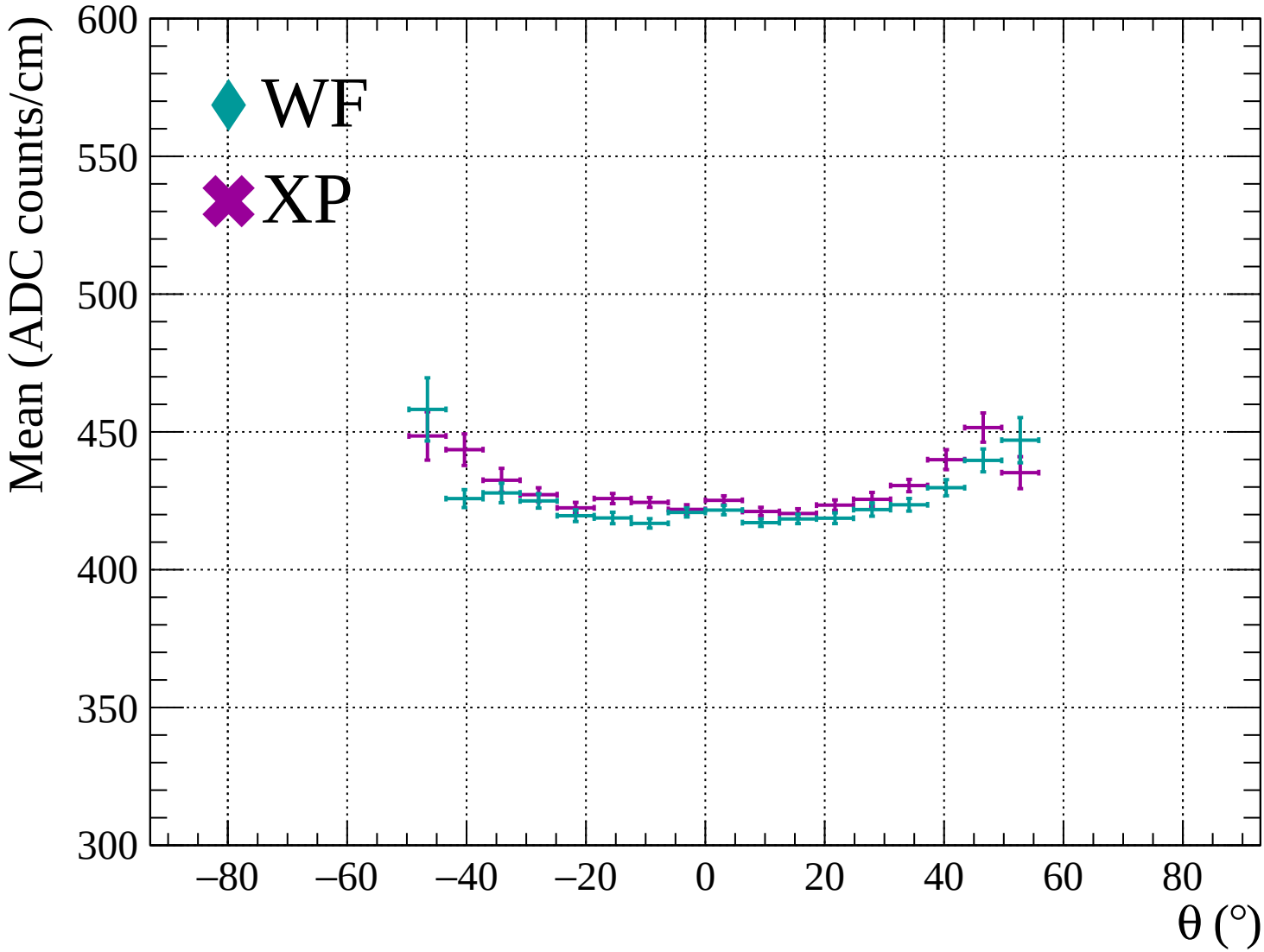


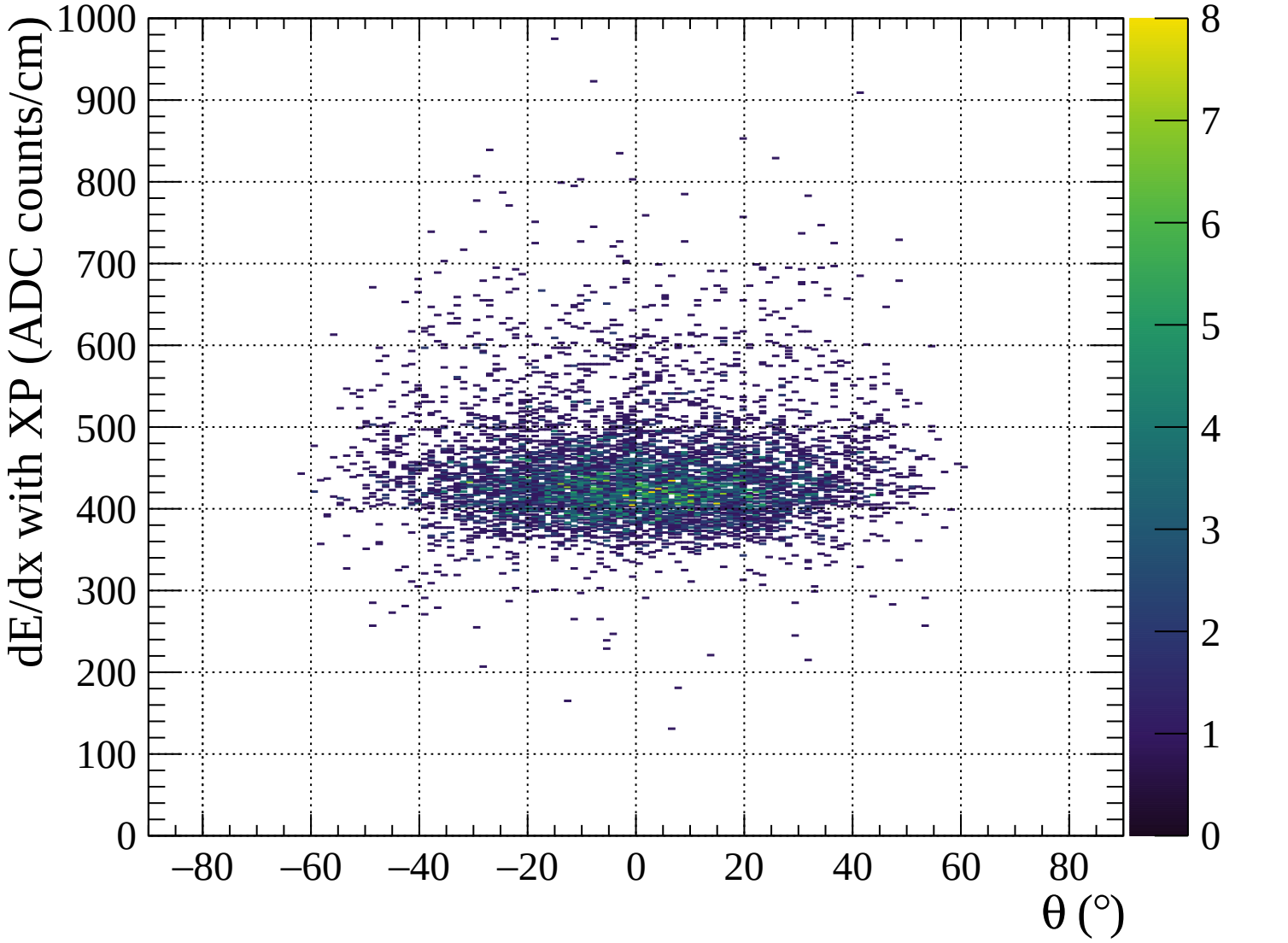


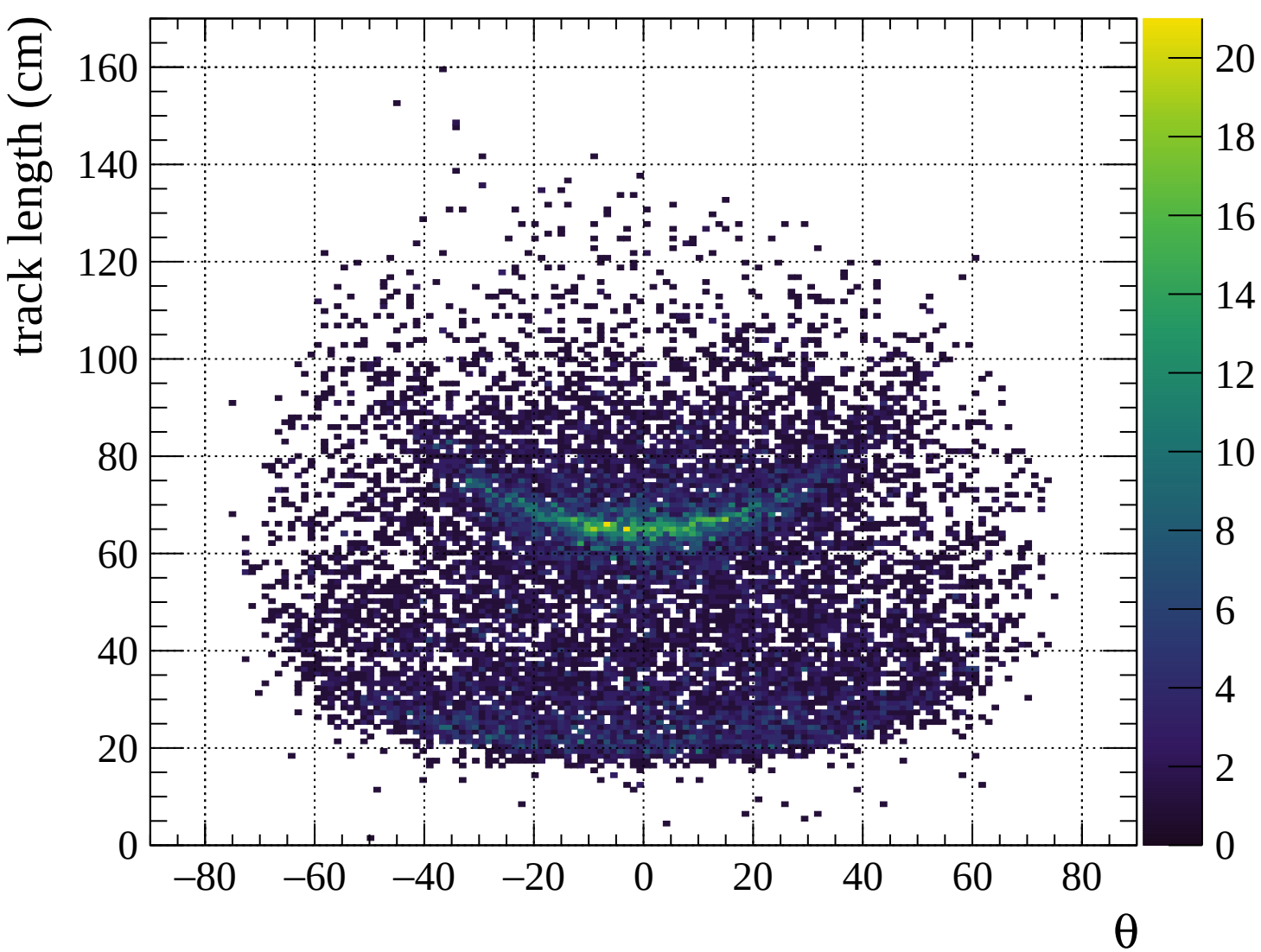


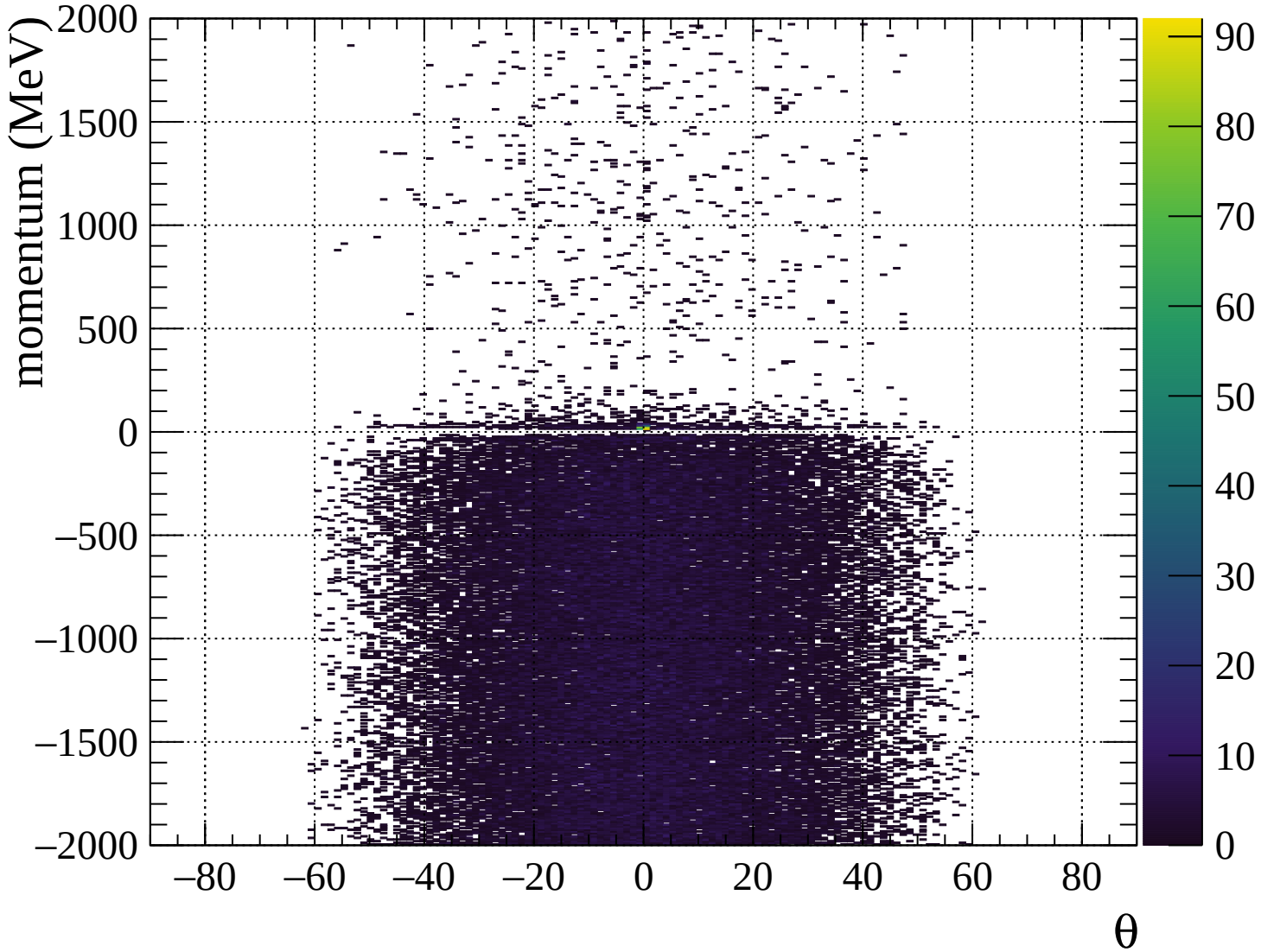


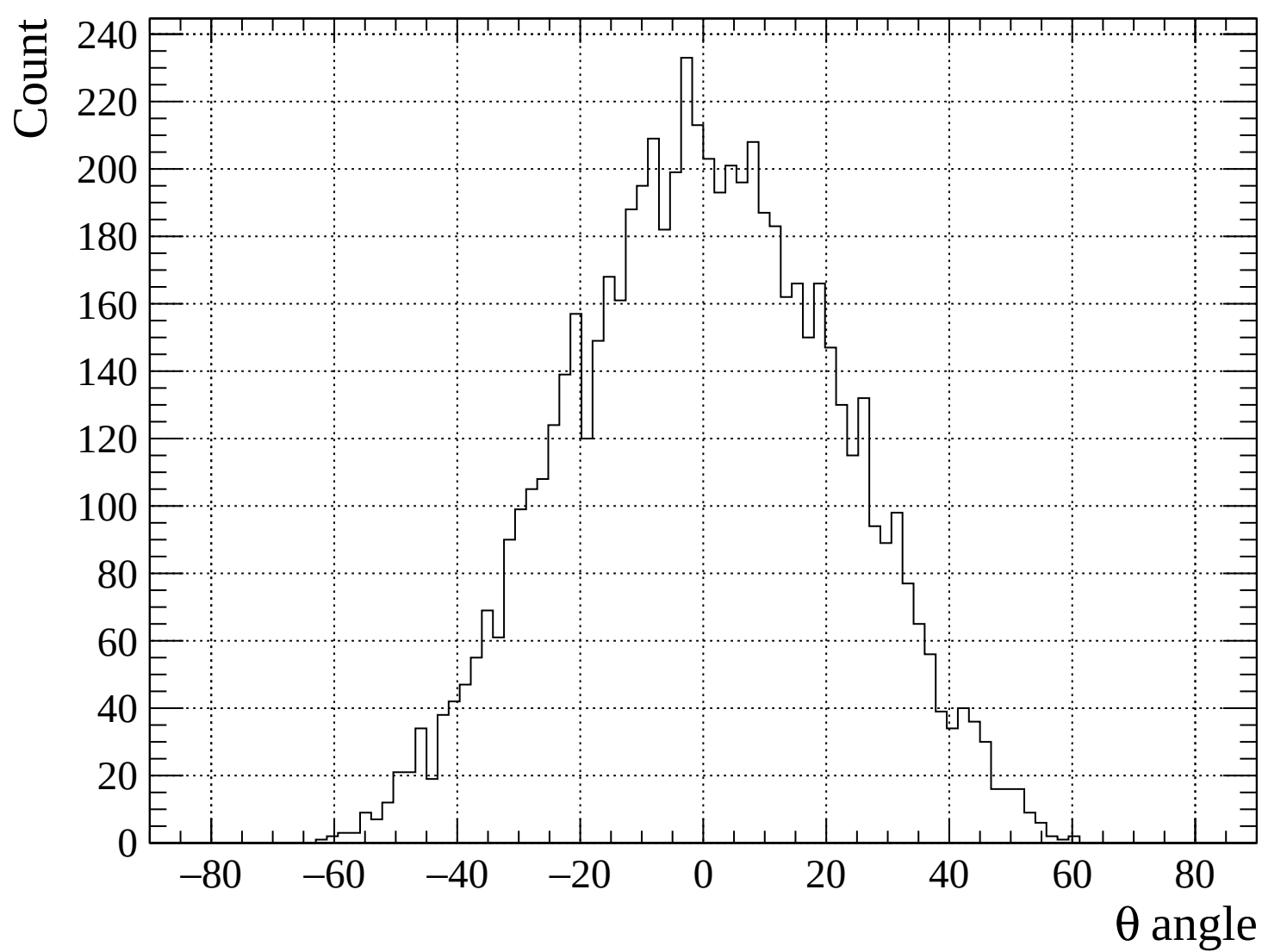


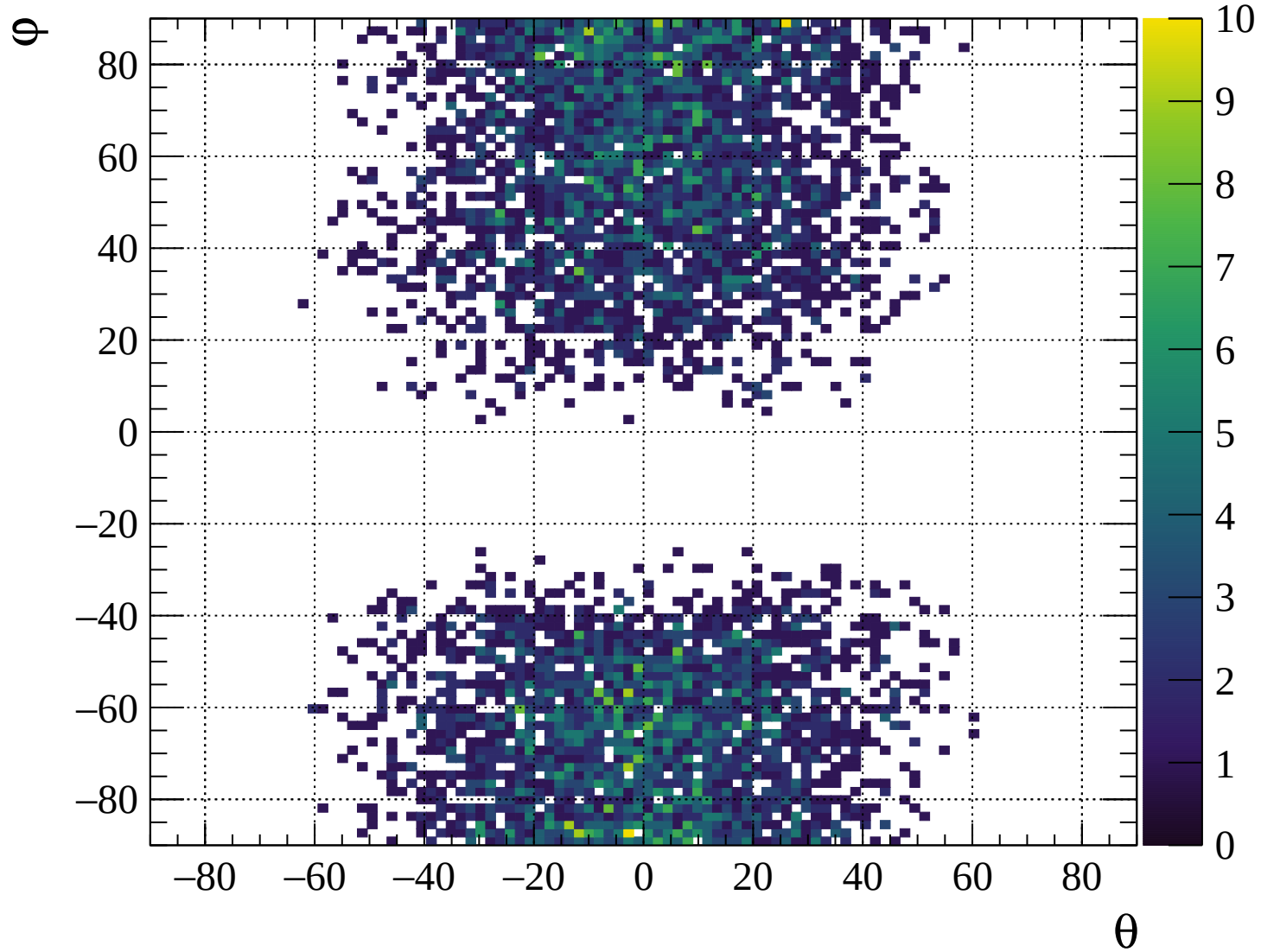


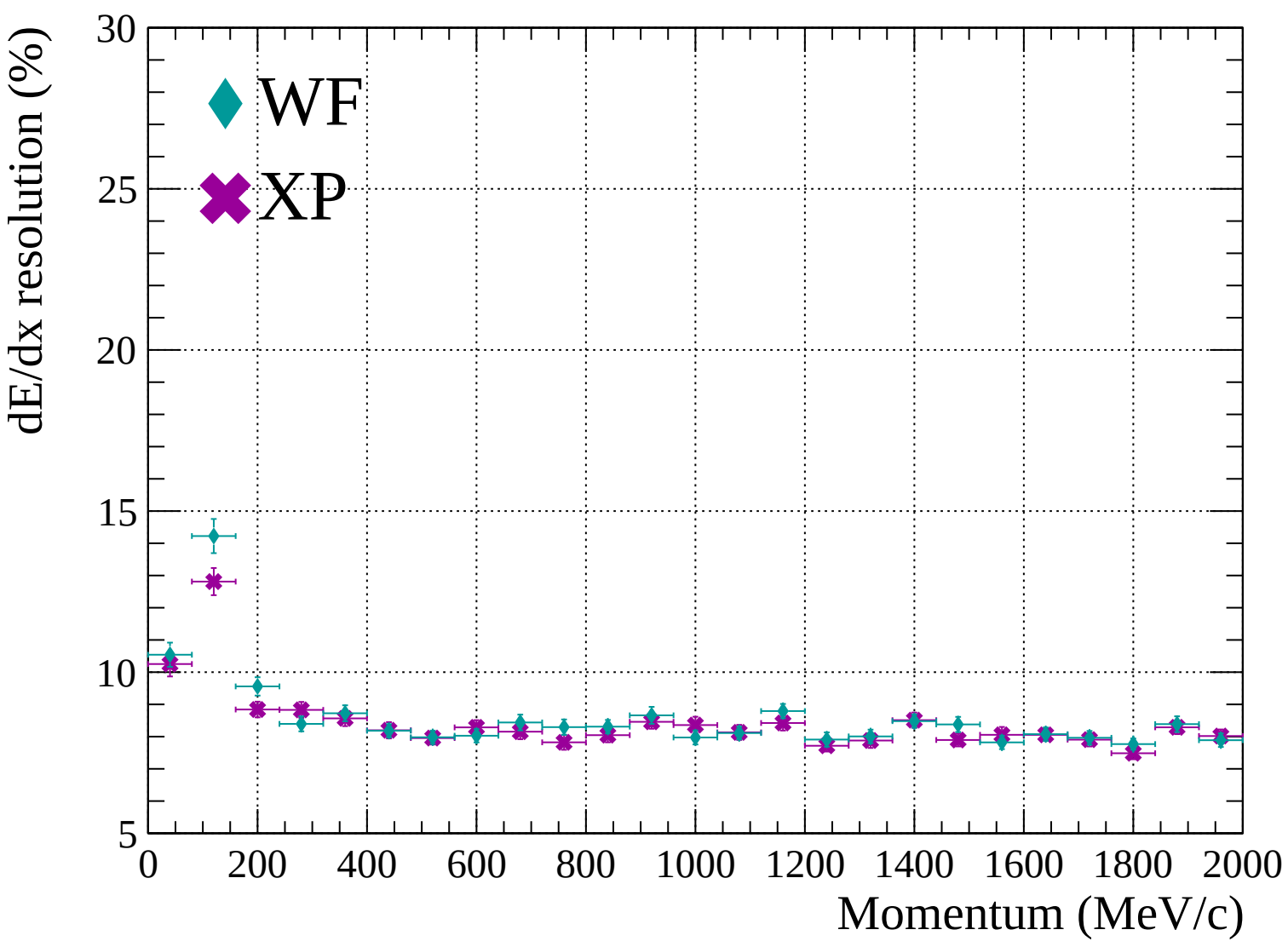


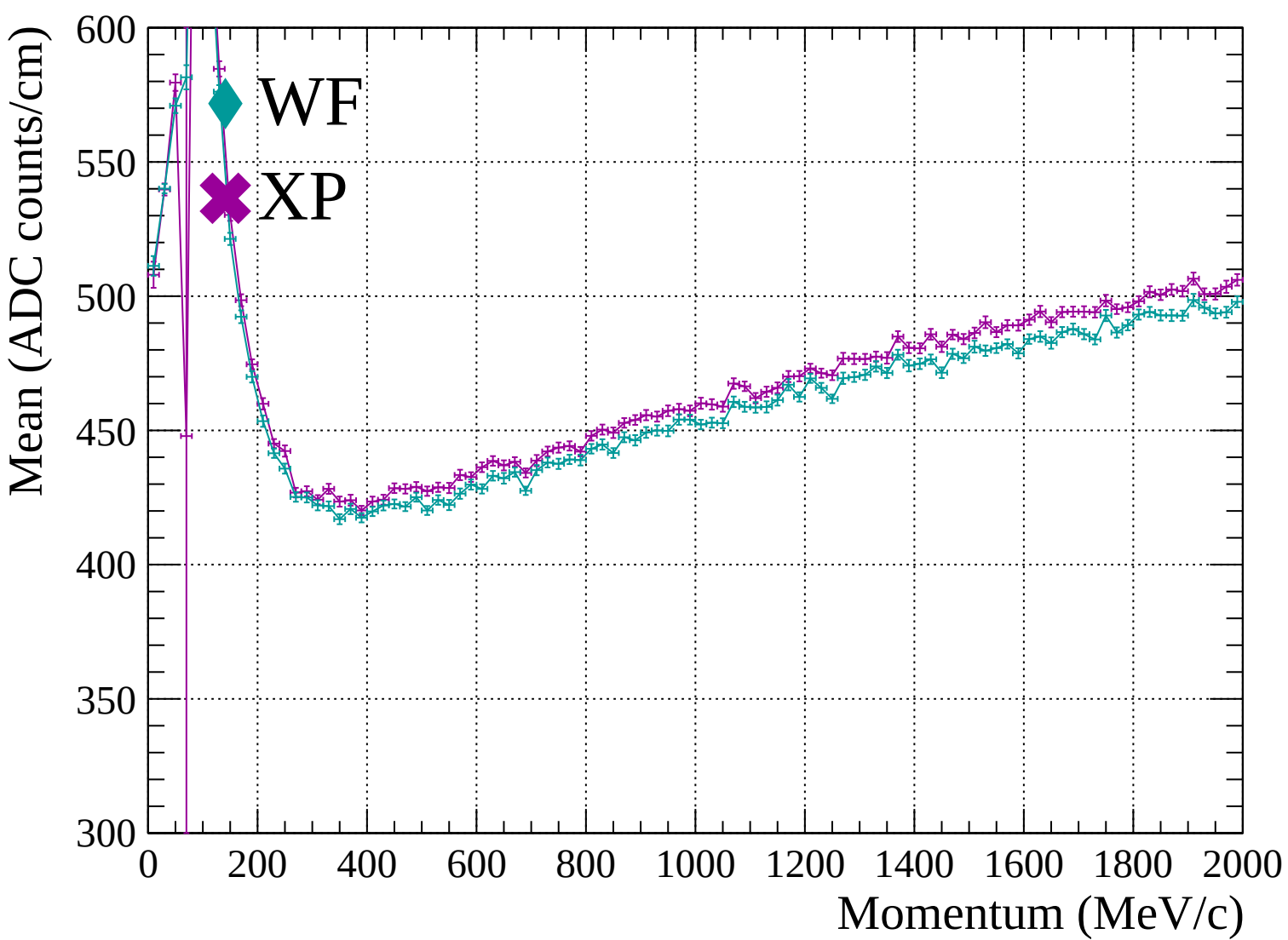


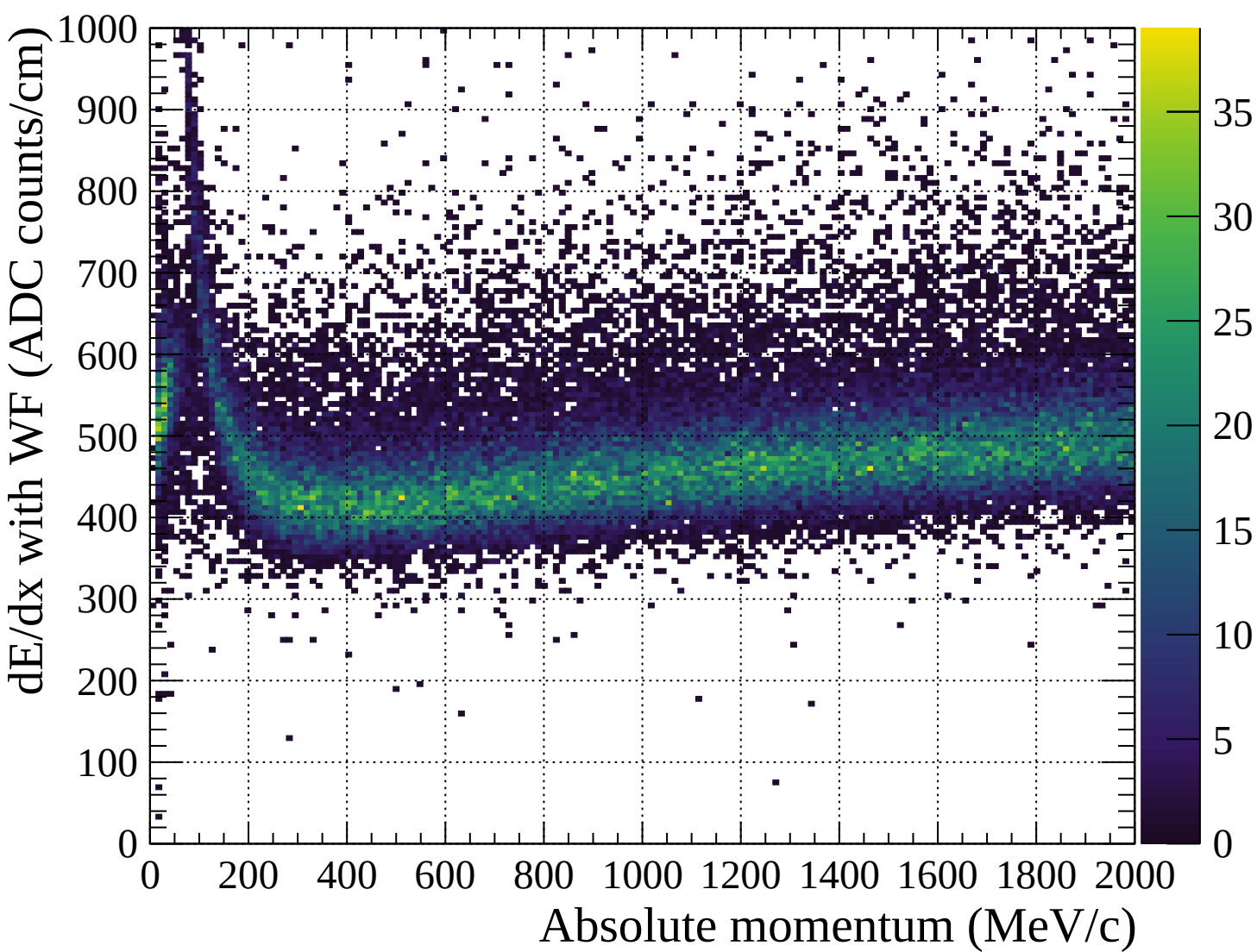


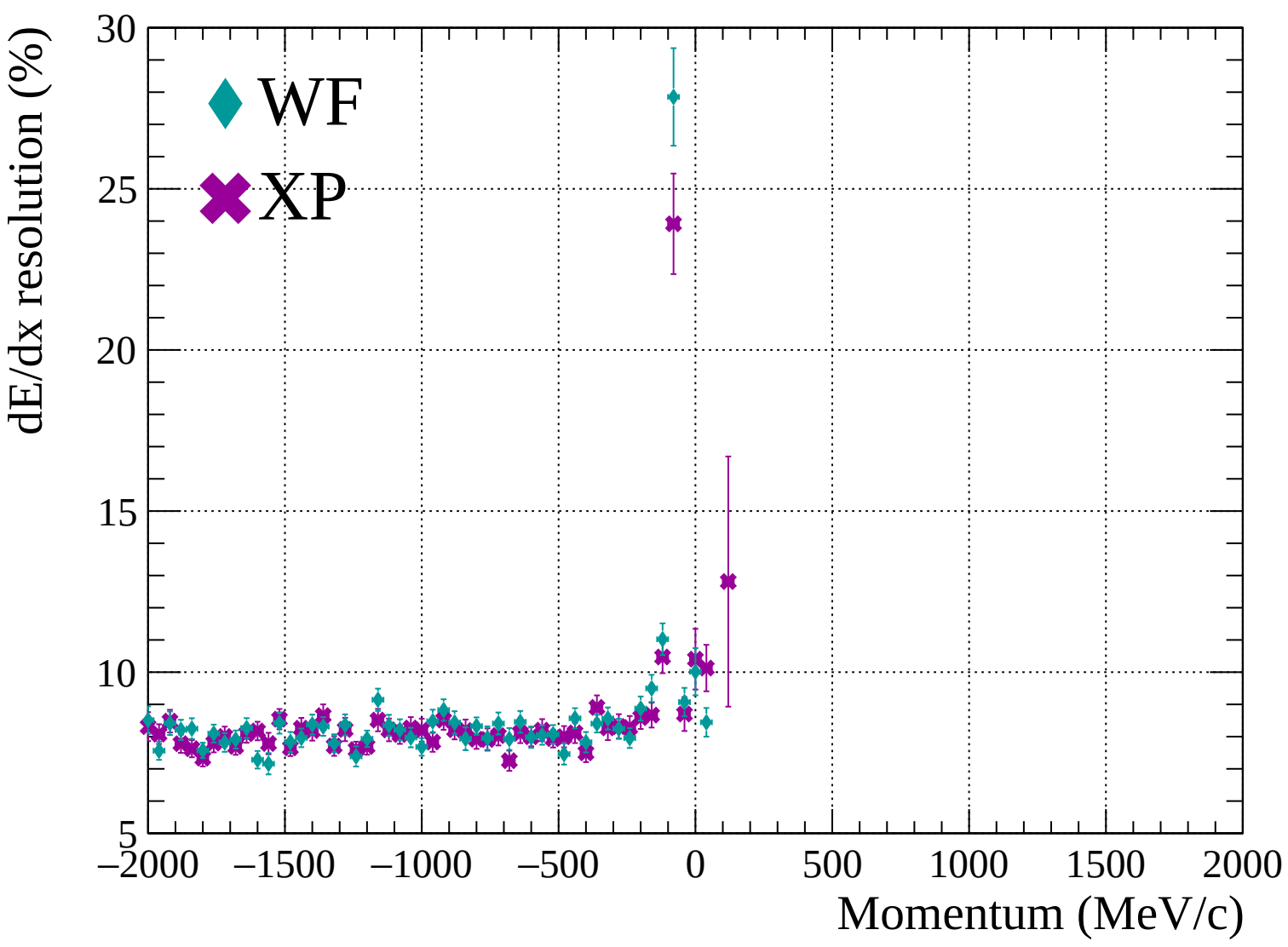


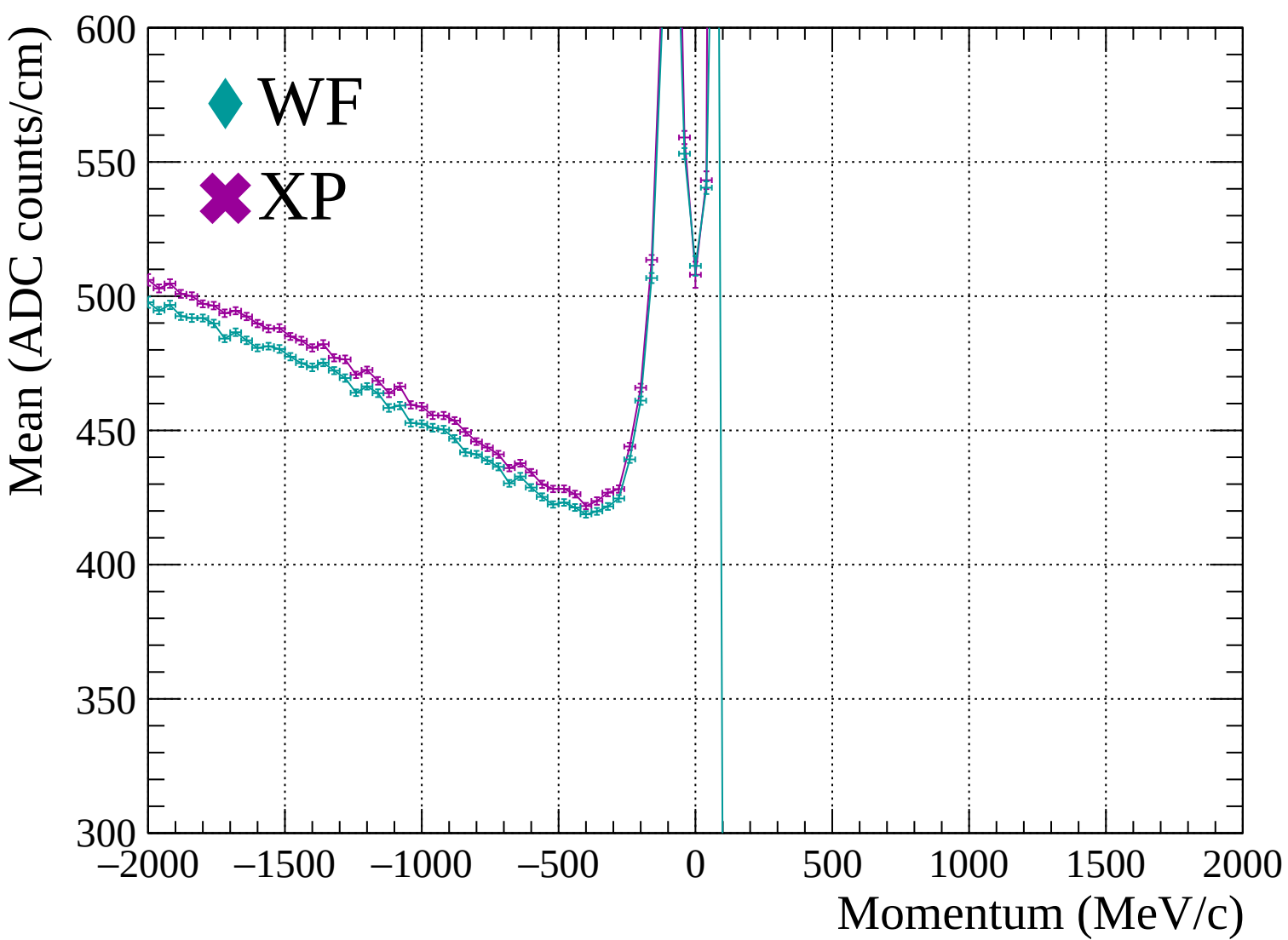


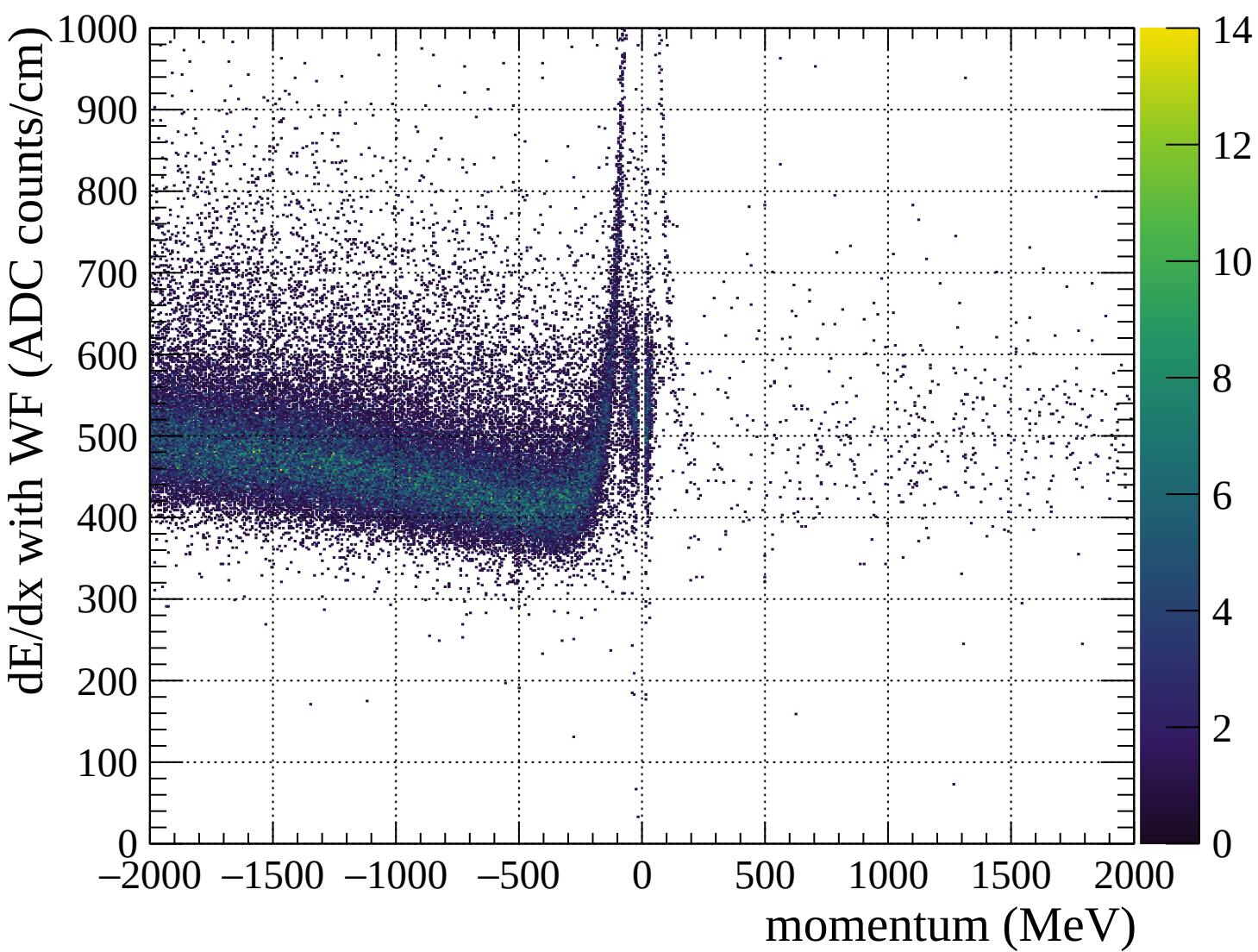


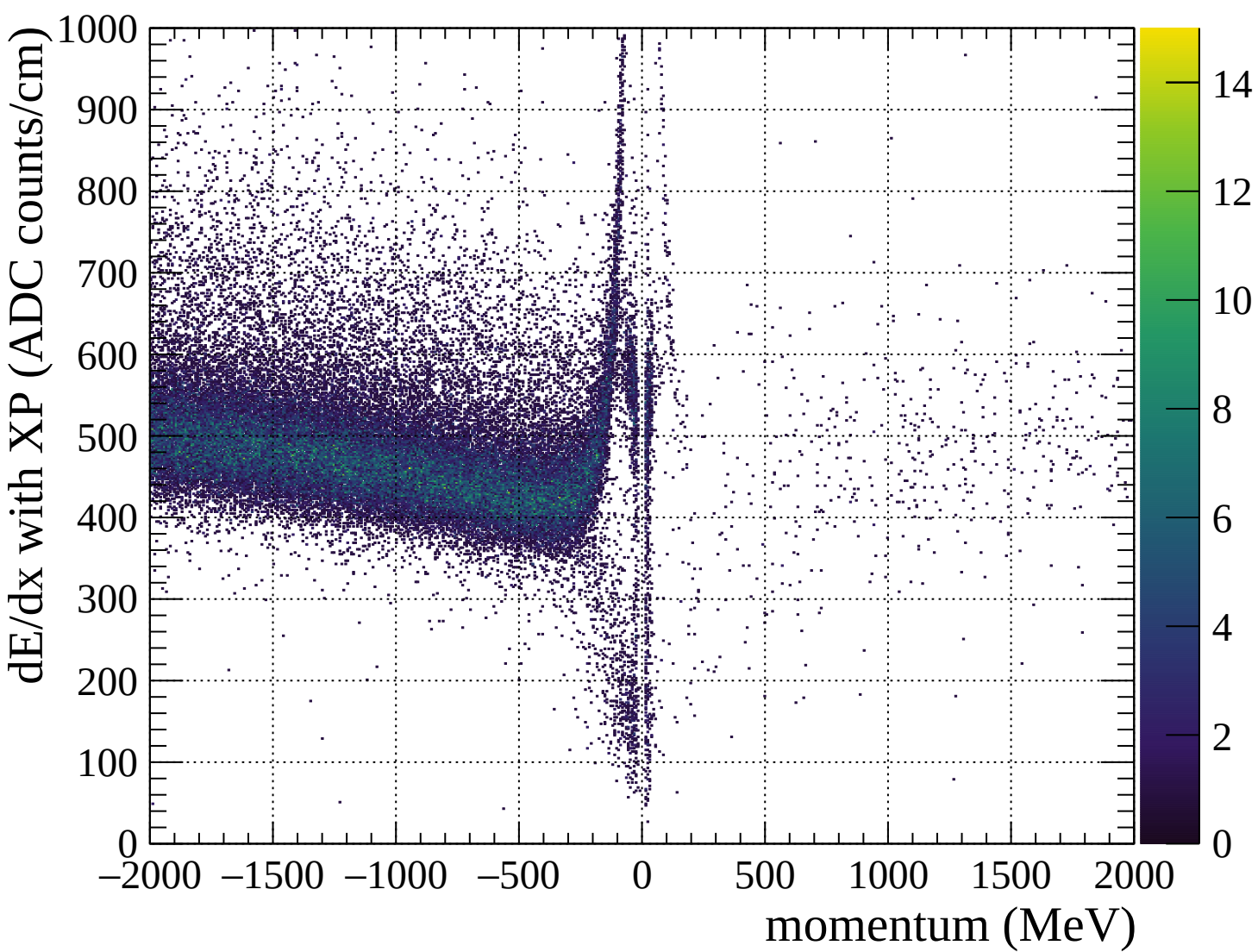


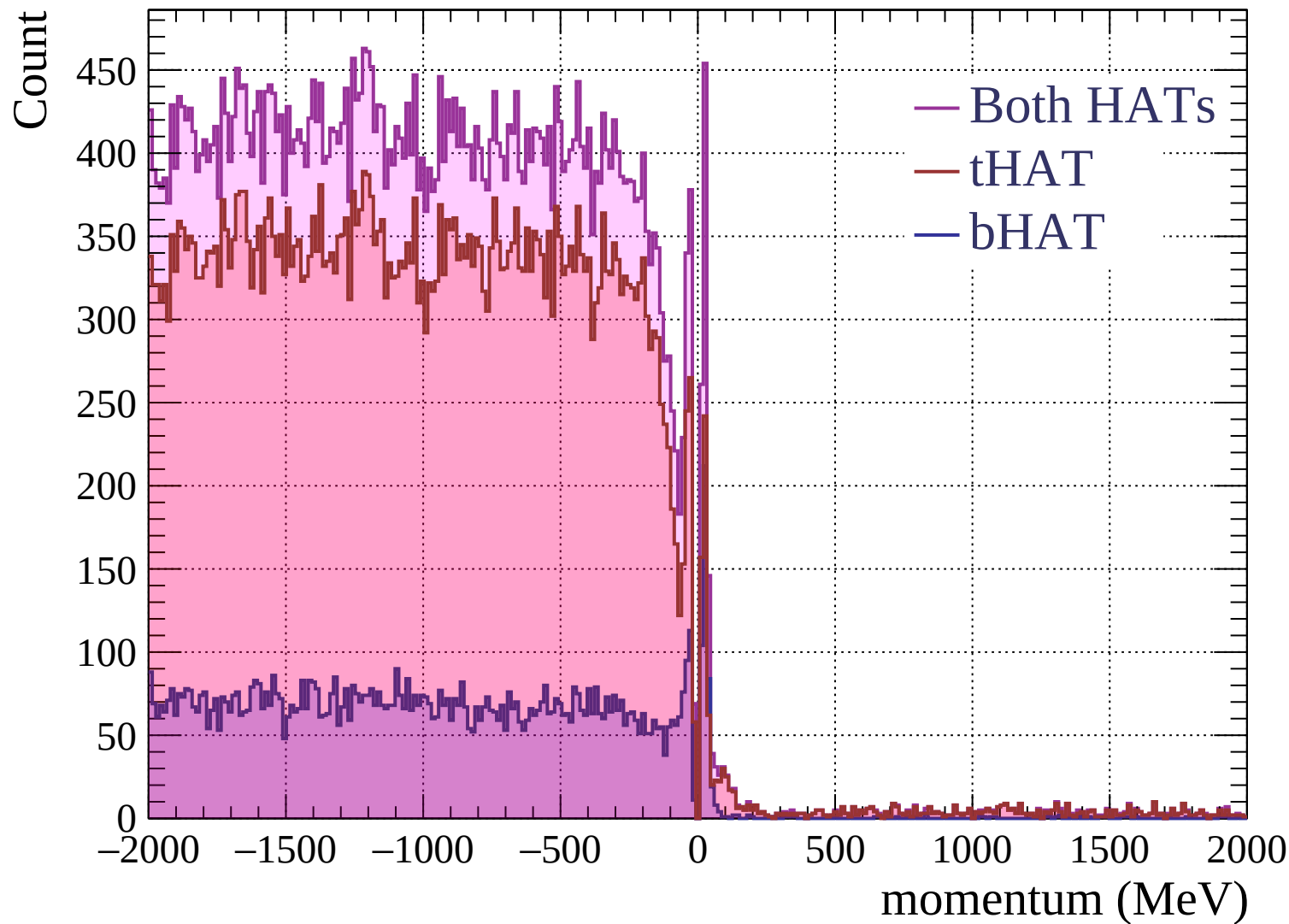


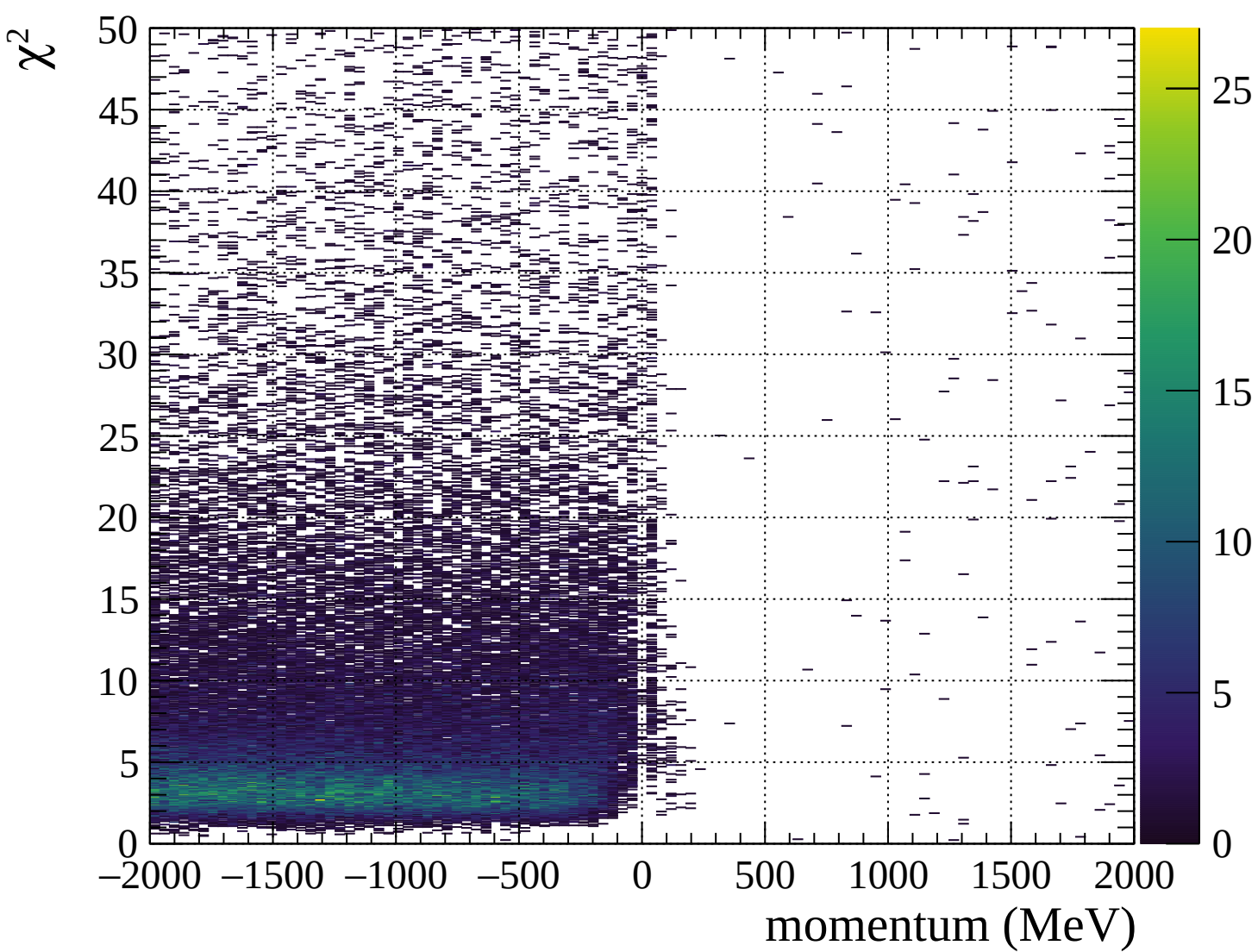




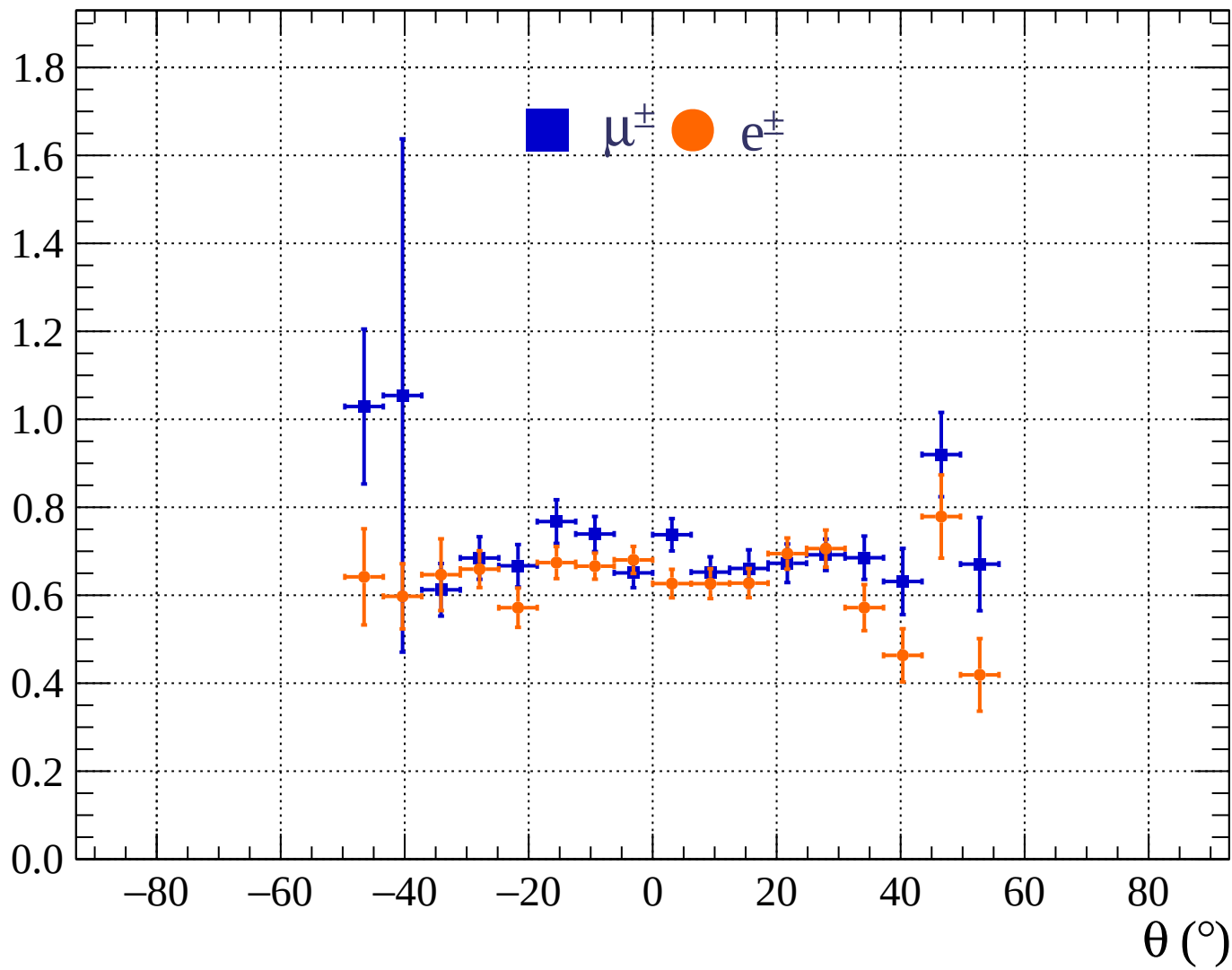




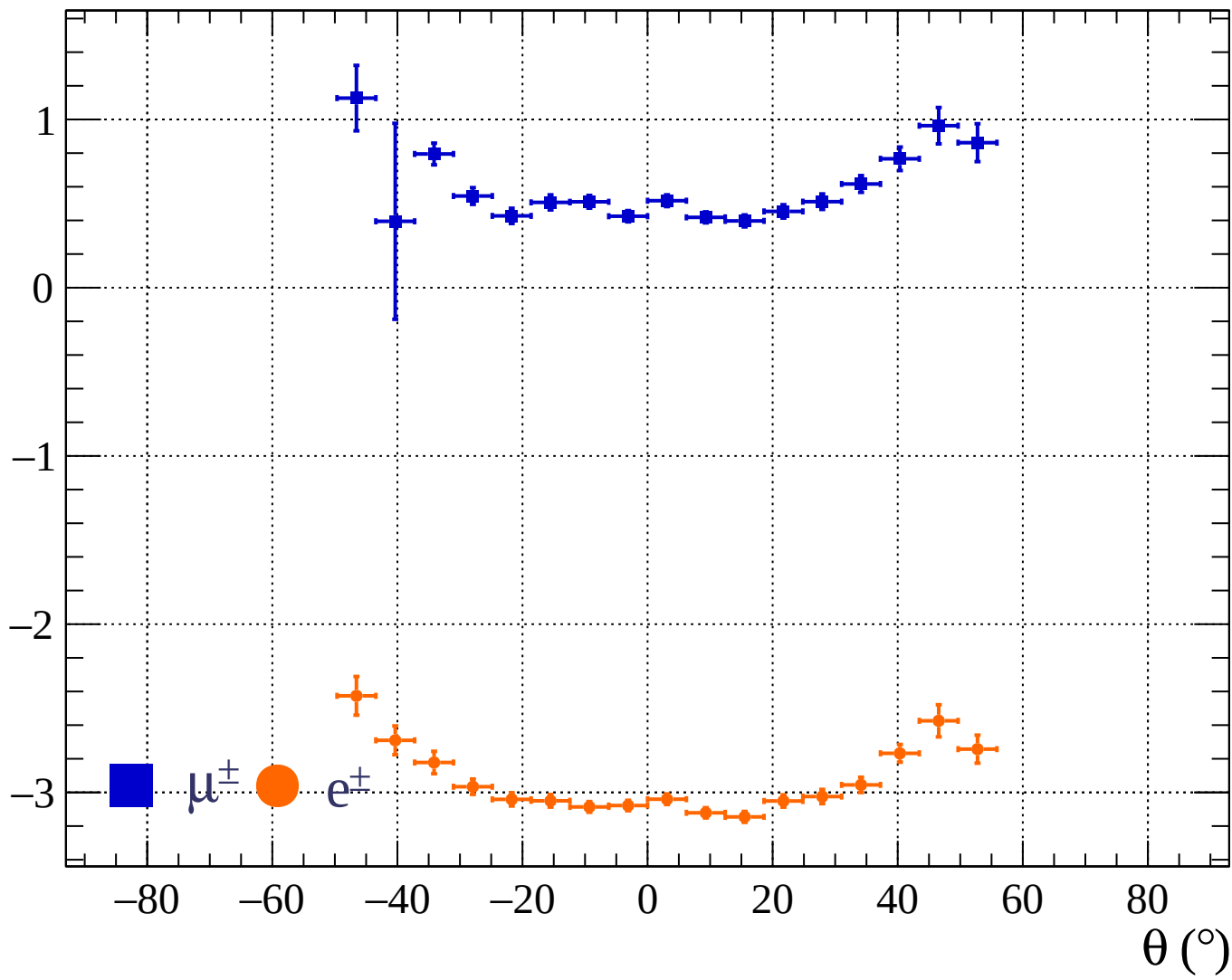




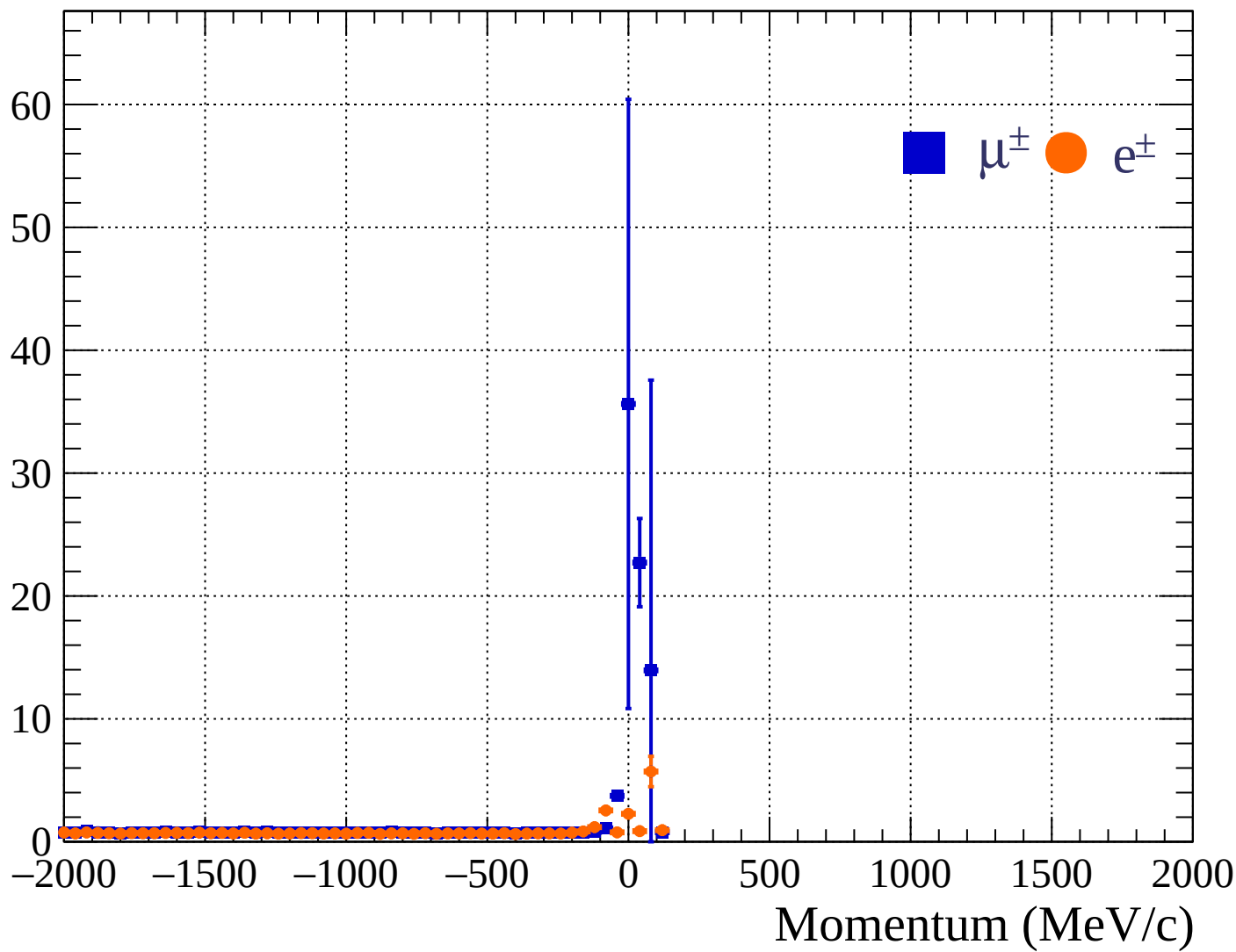
Pulls standard deviation



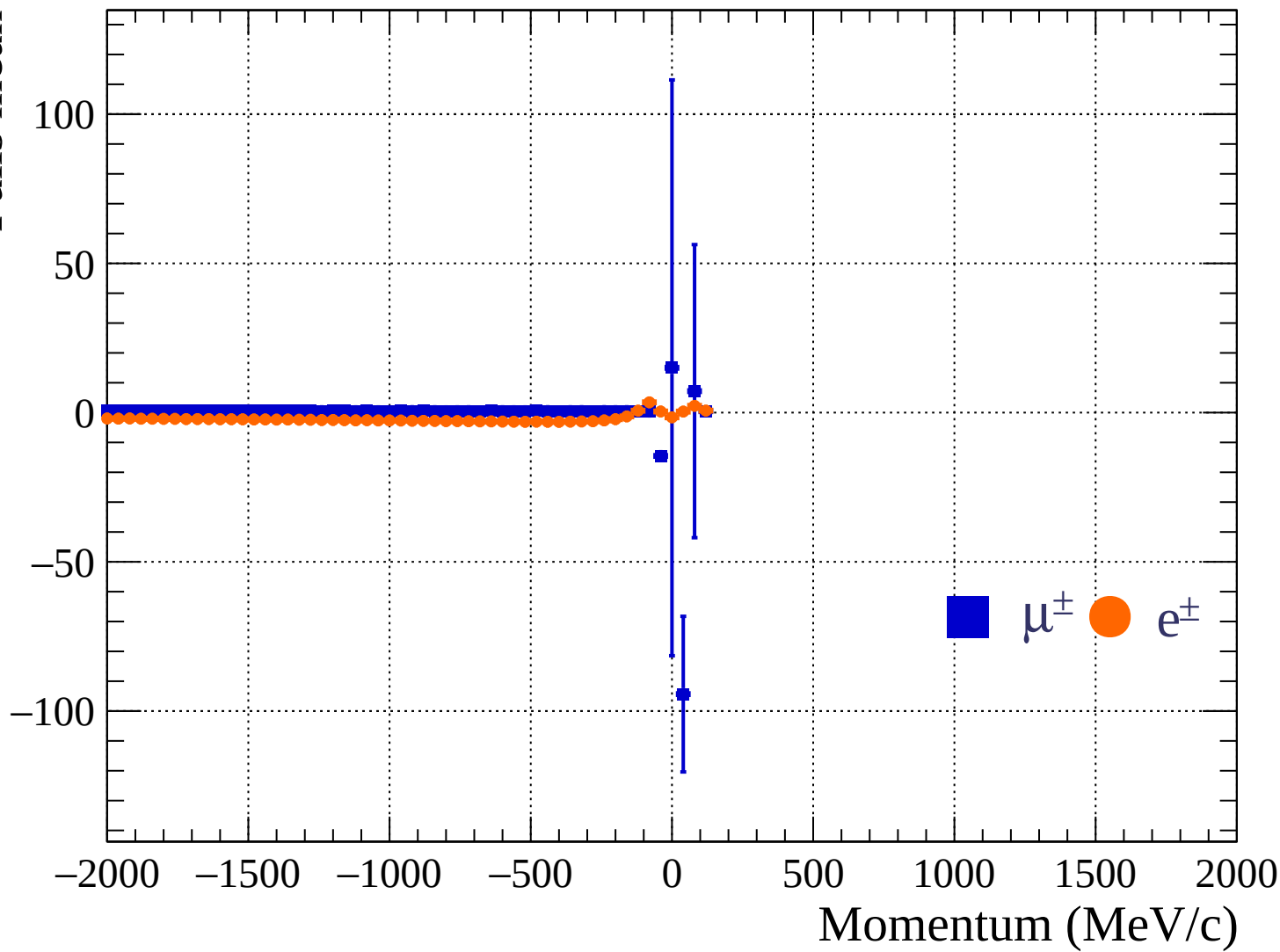
Pulls mean

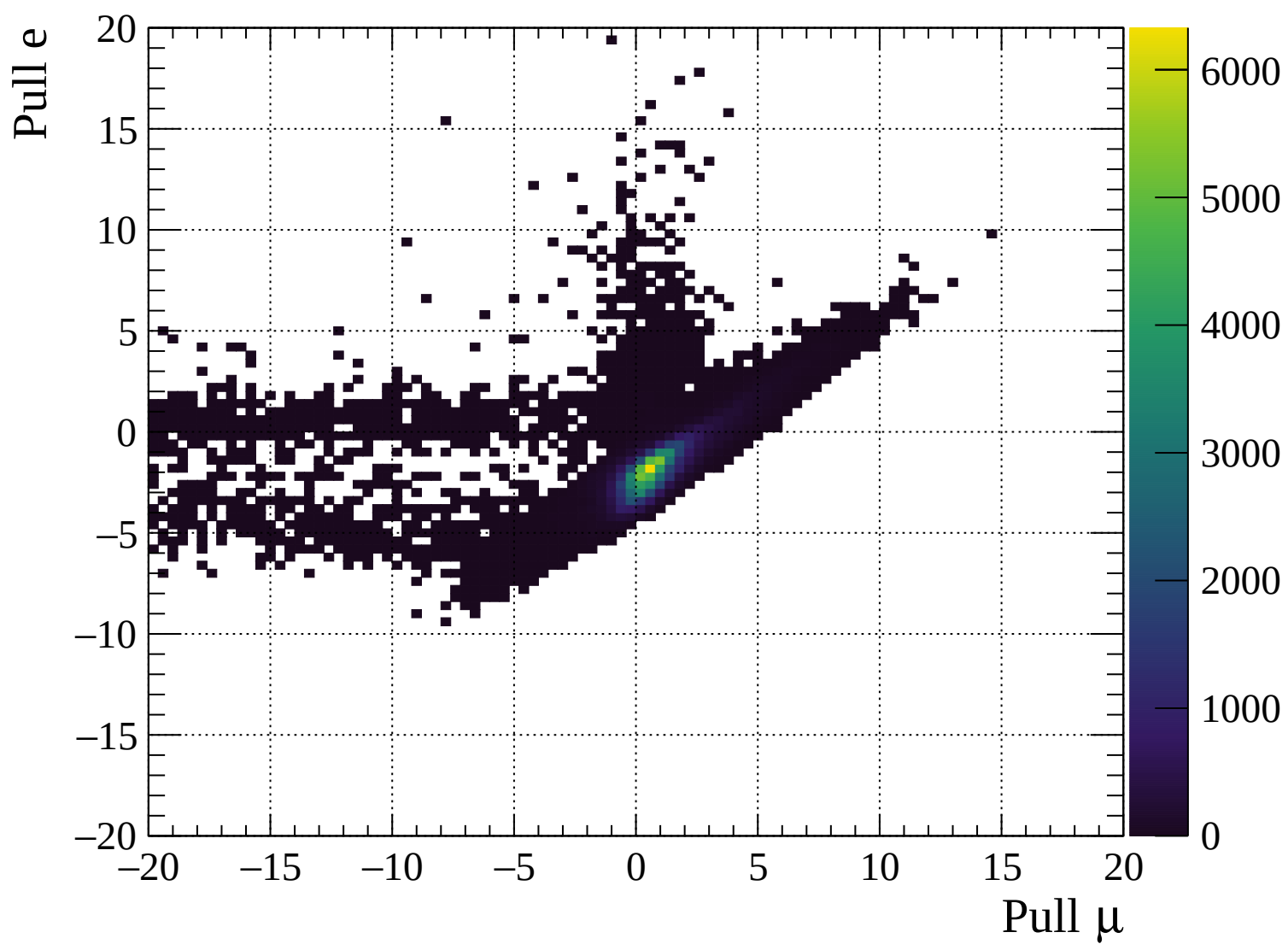


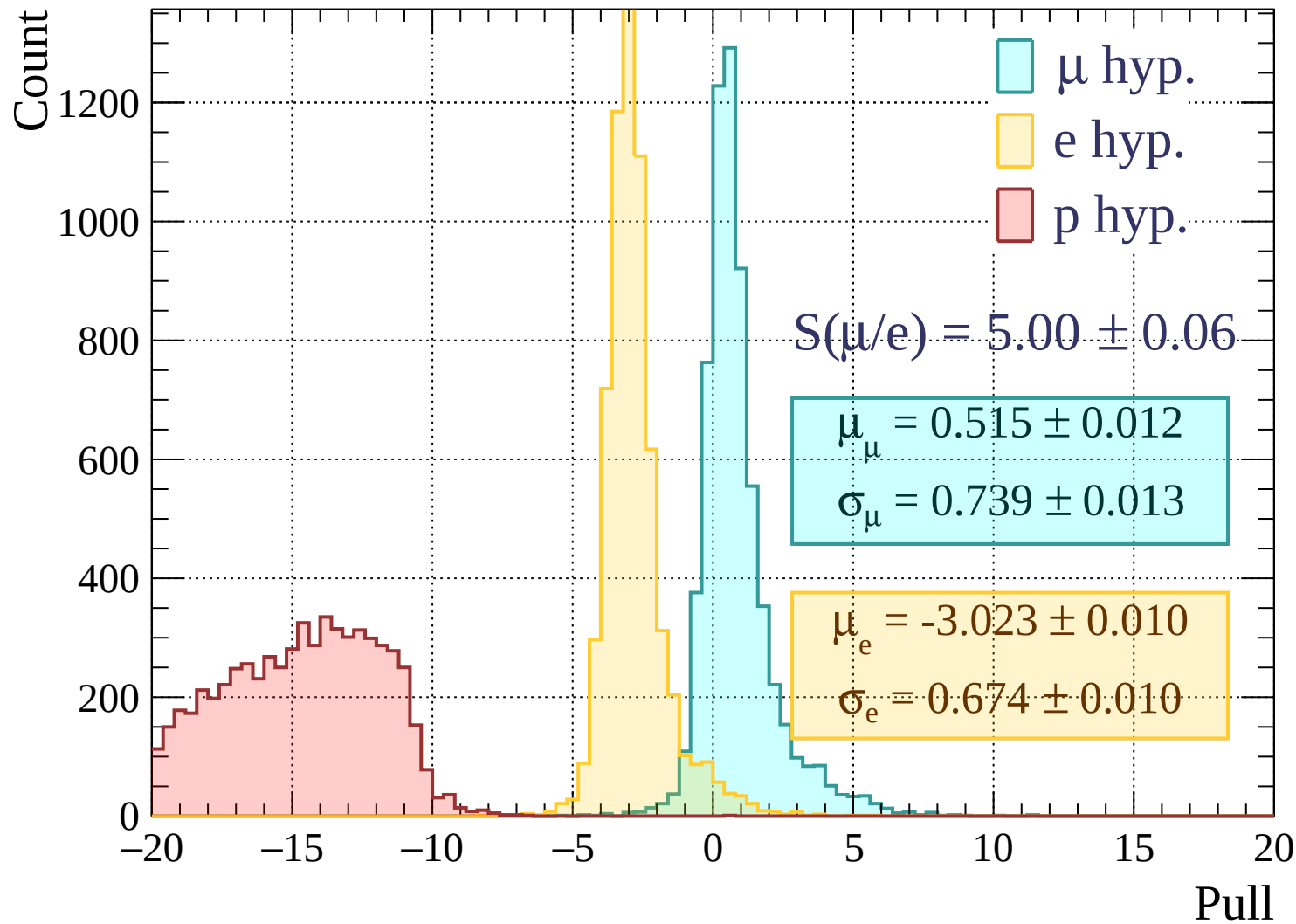
Pulls standard deviation

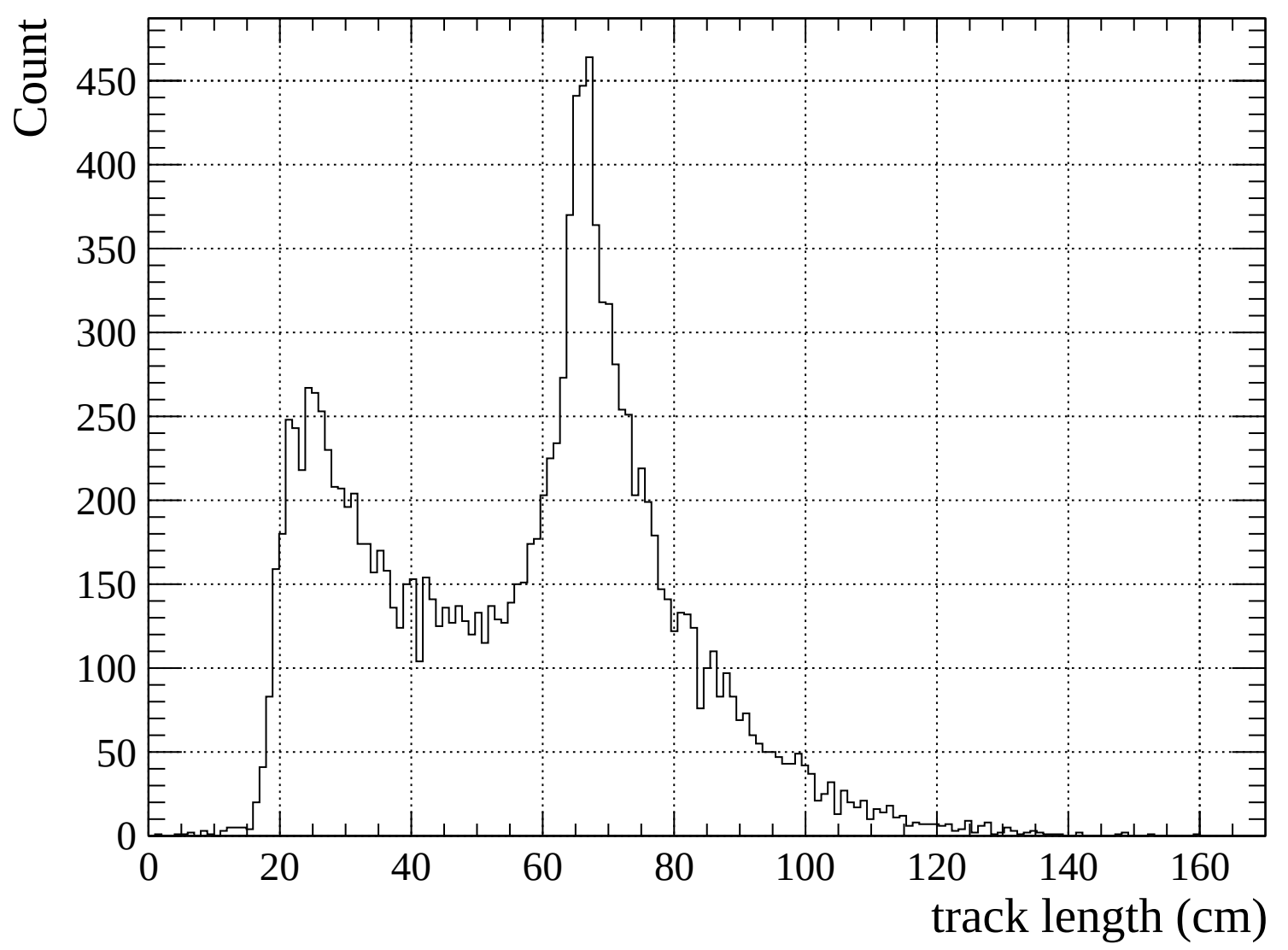


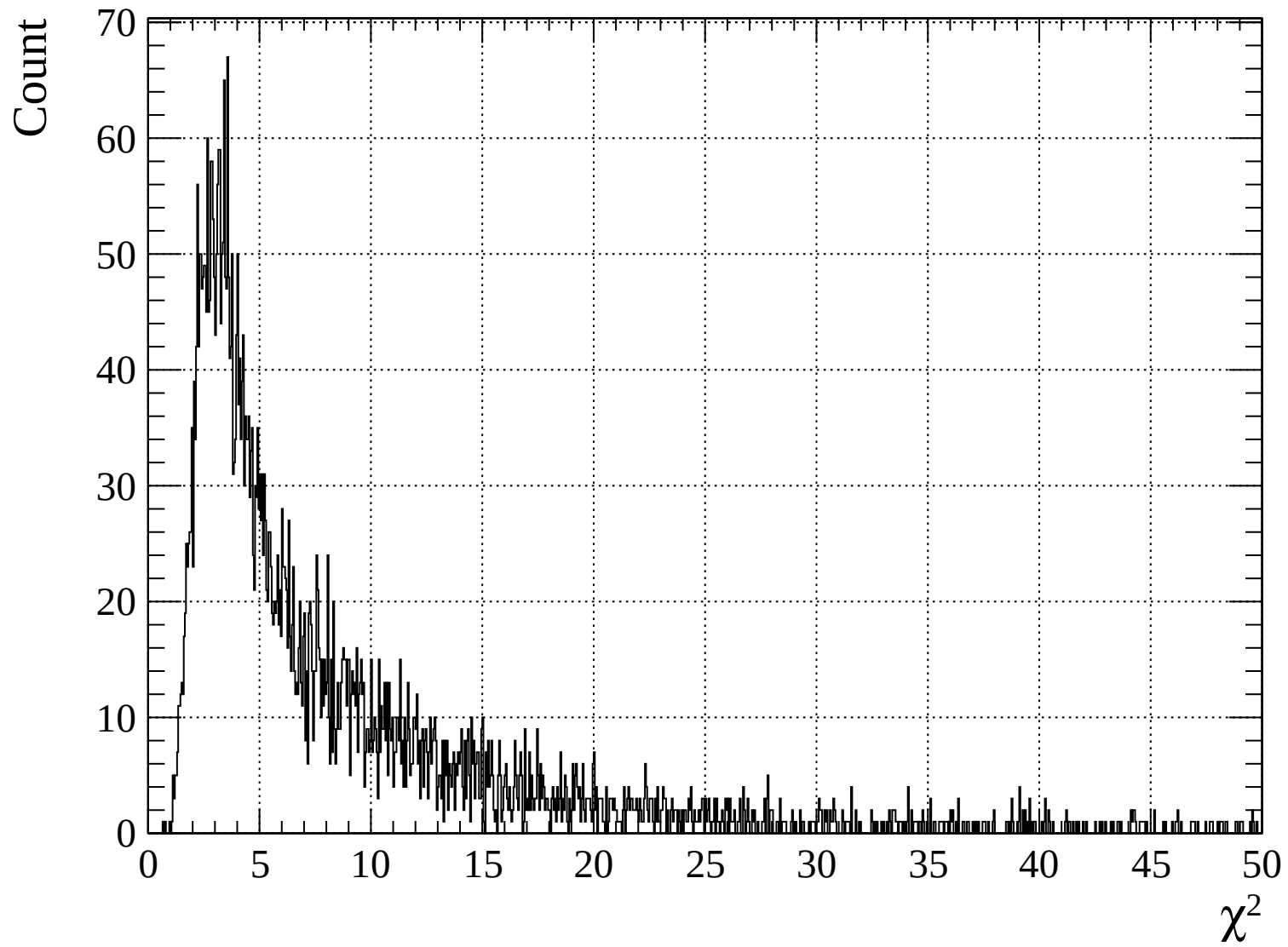
Pulls mean

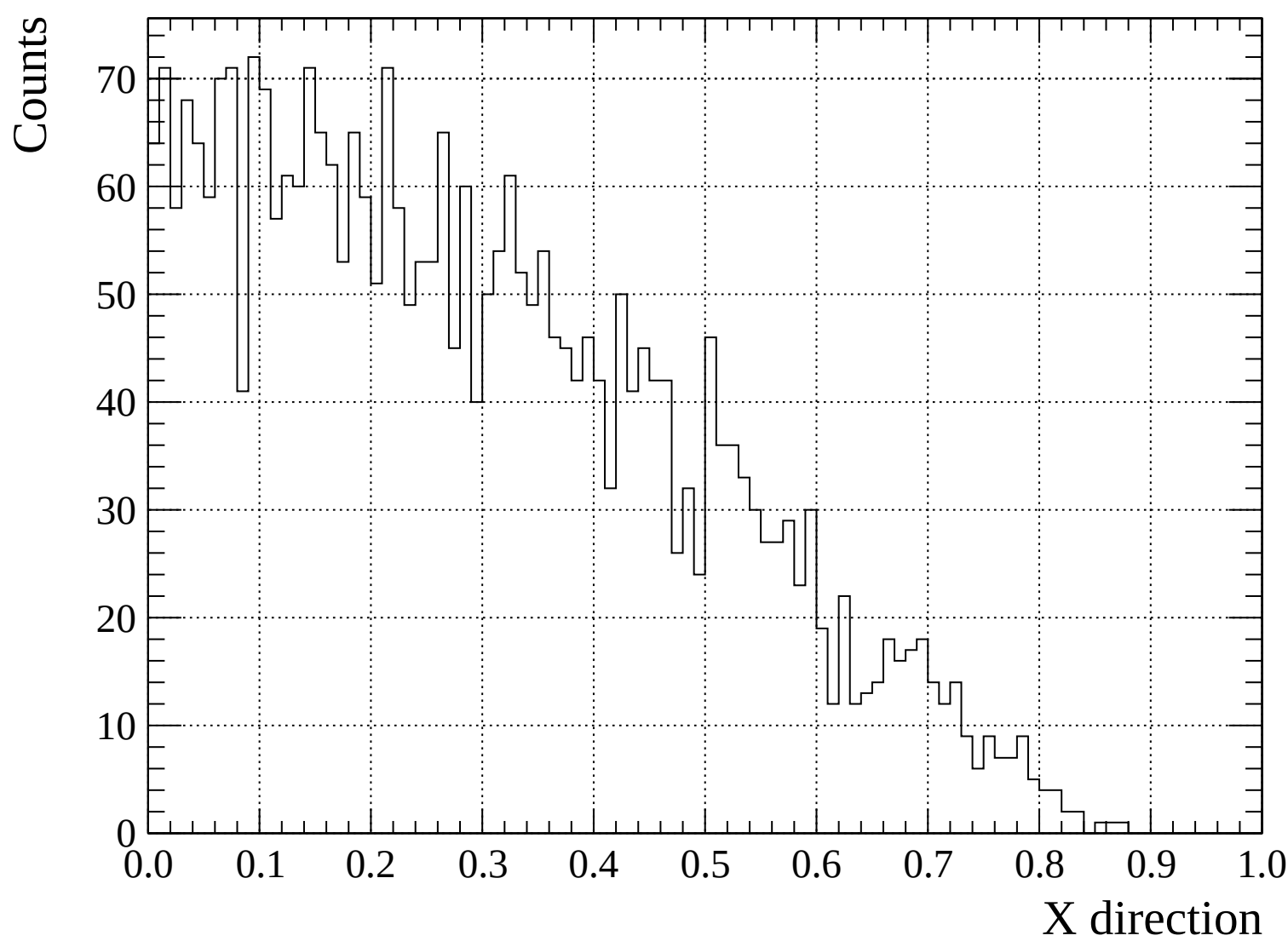


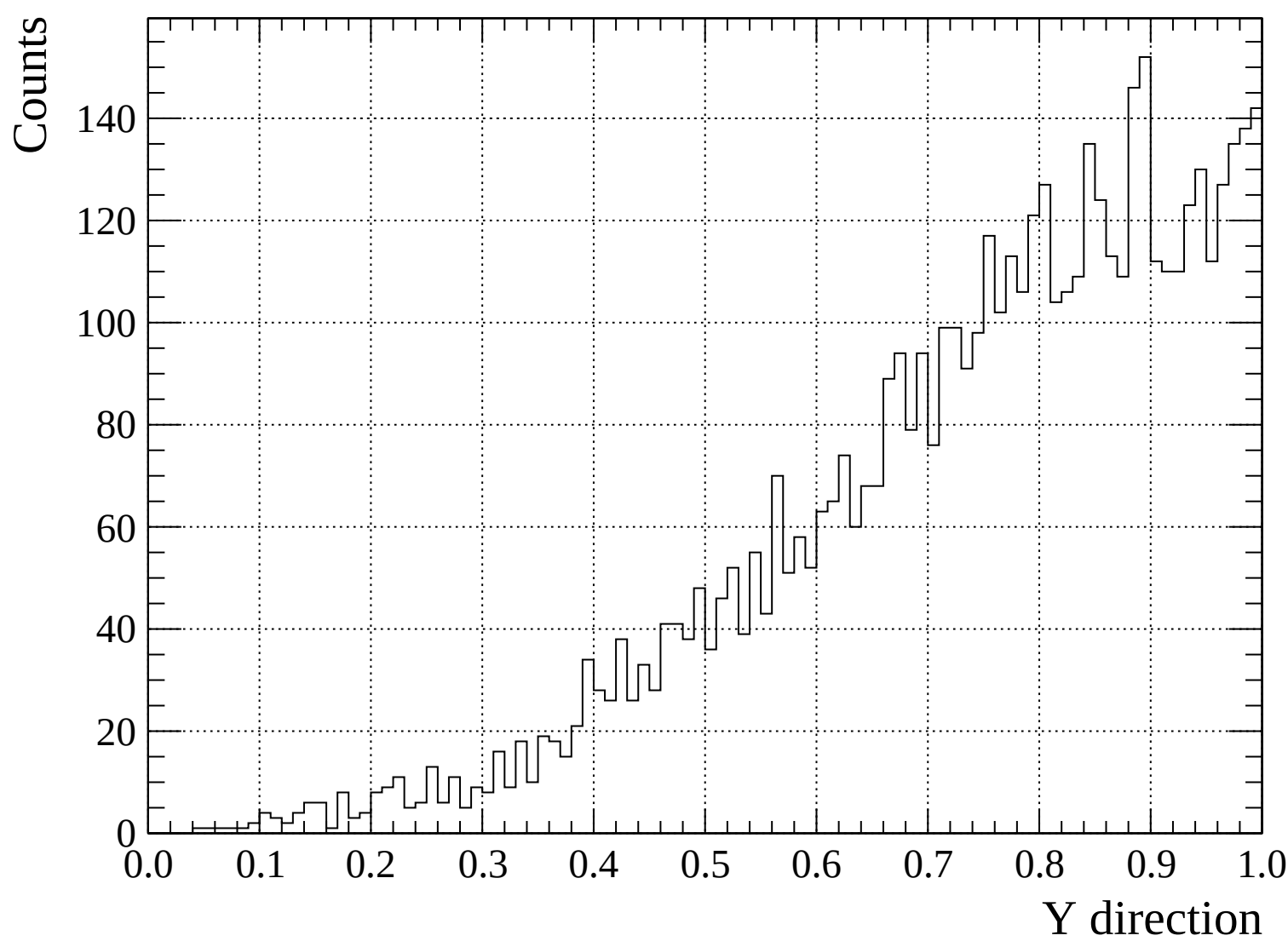


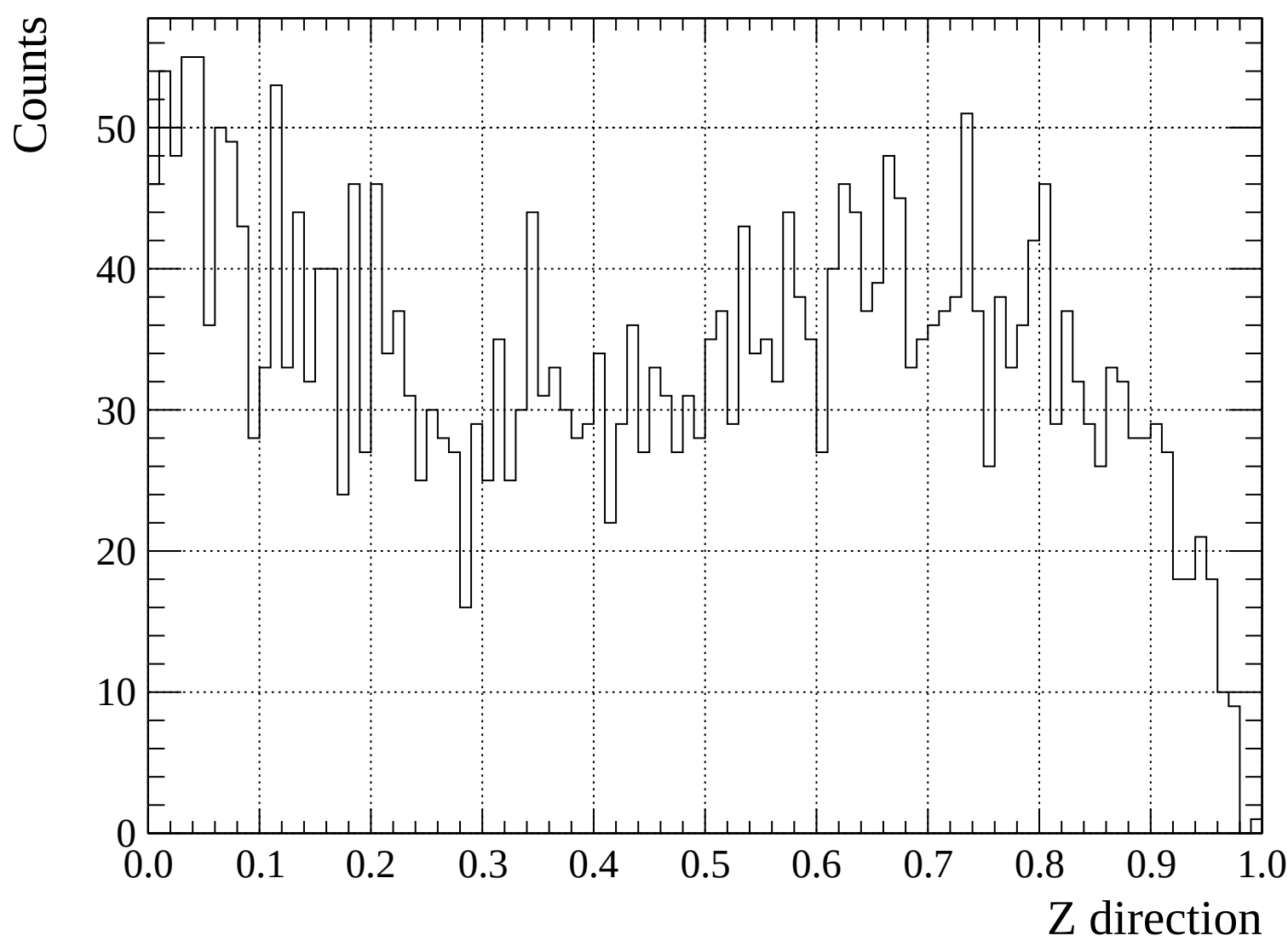


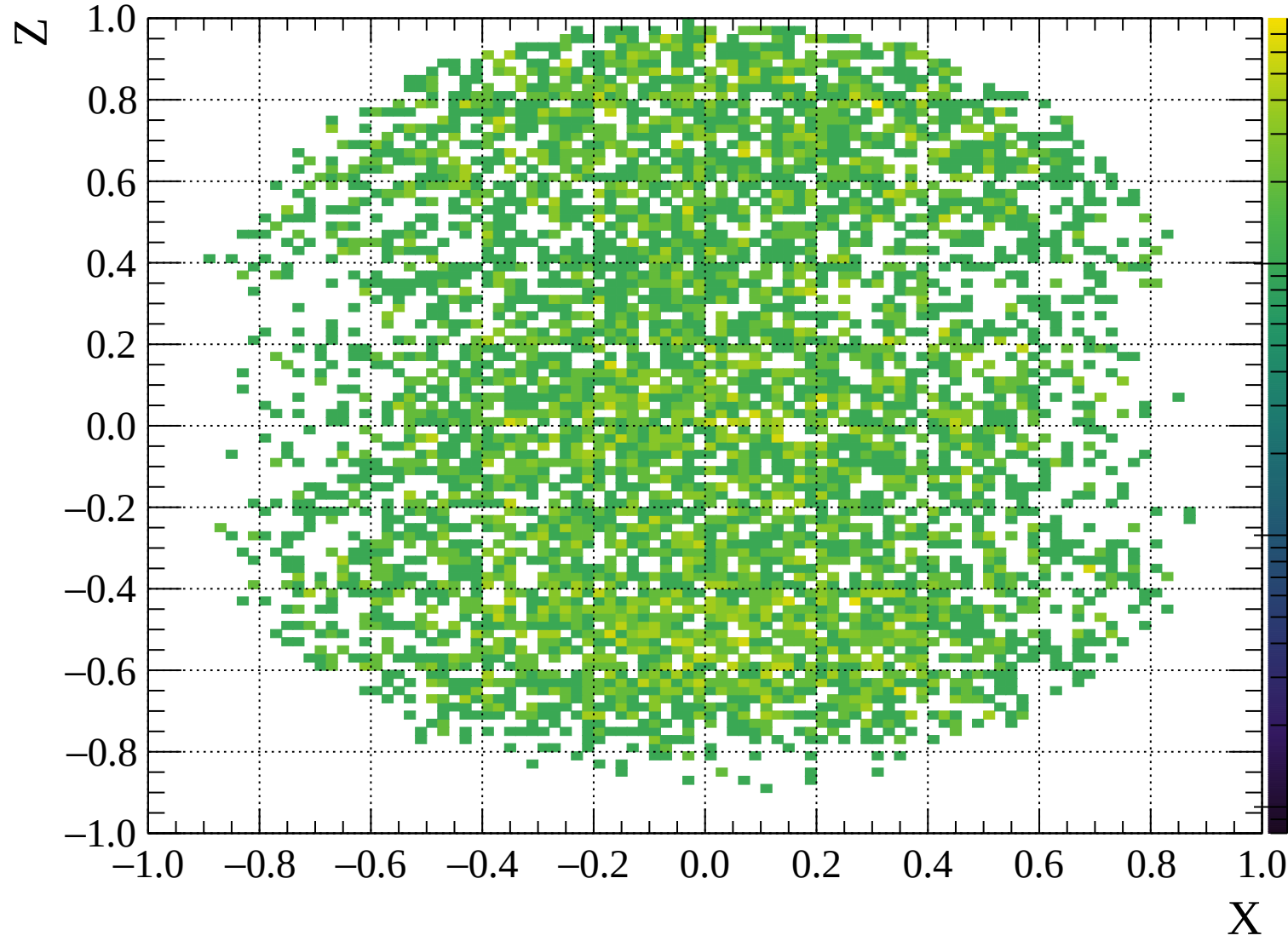


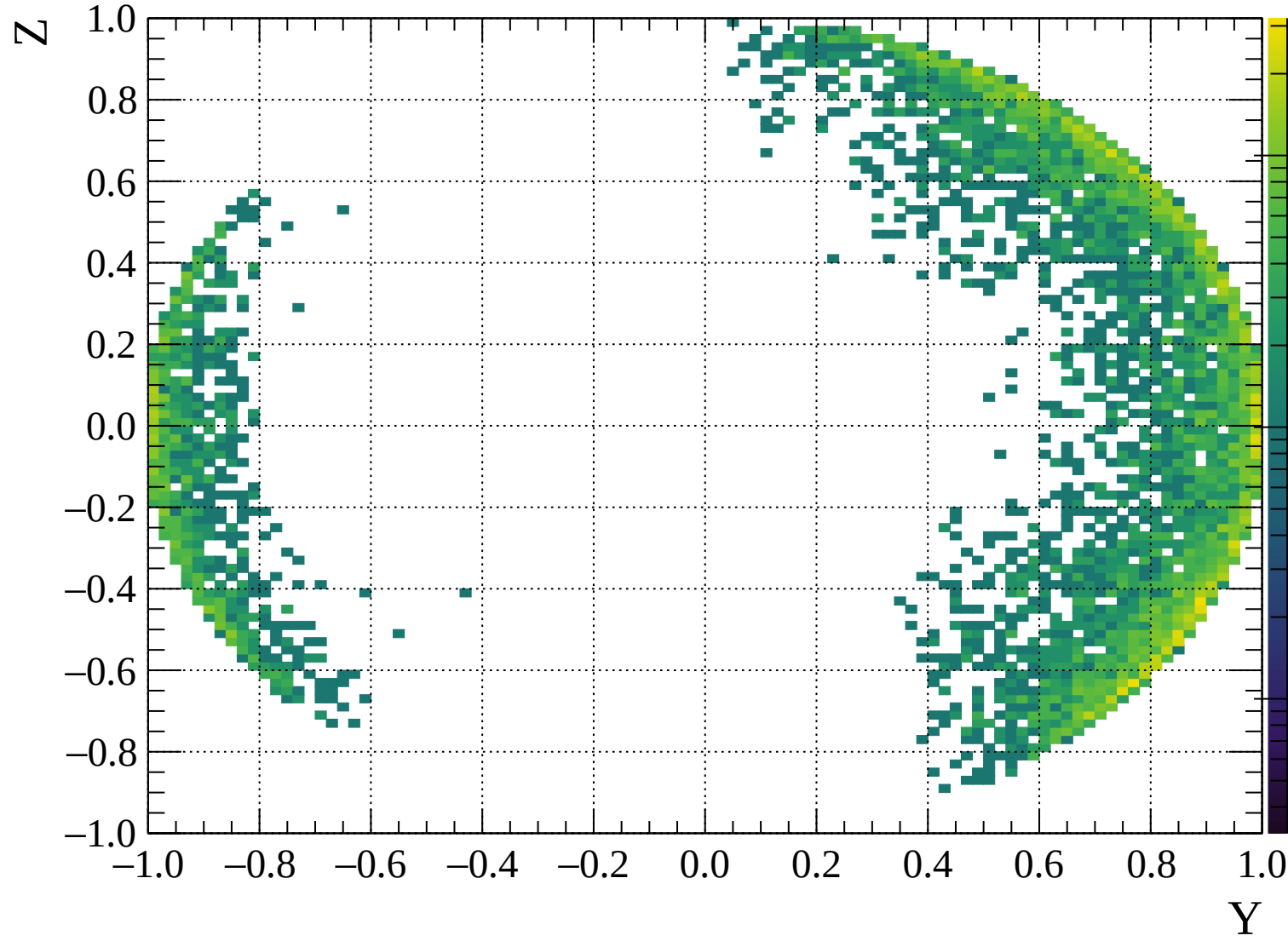


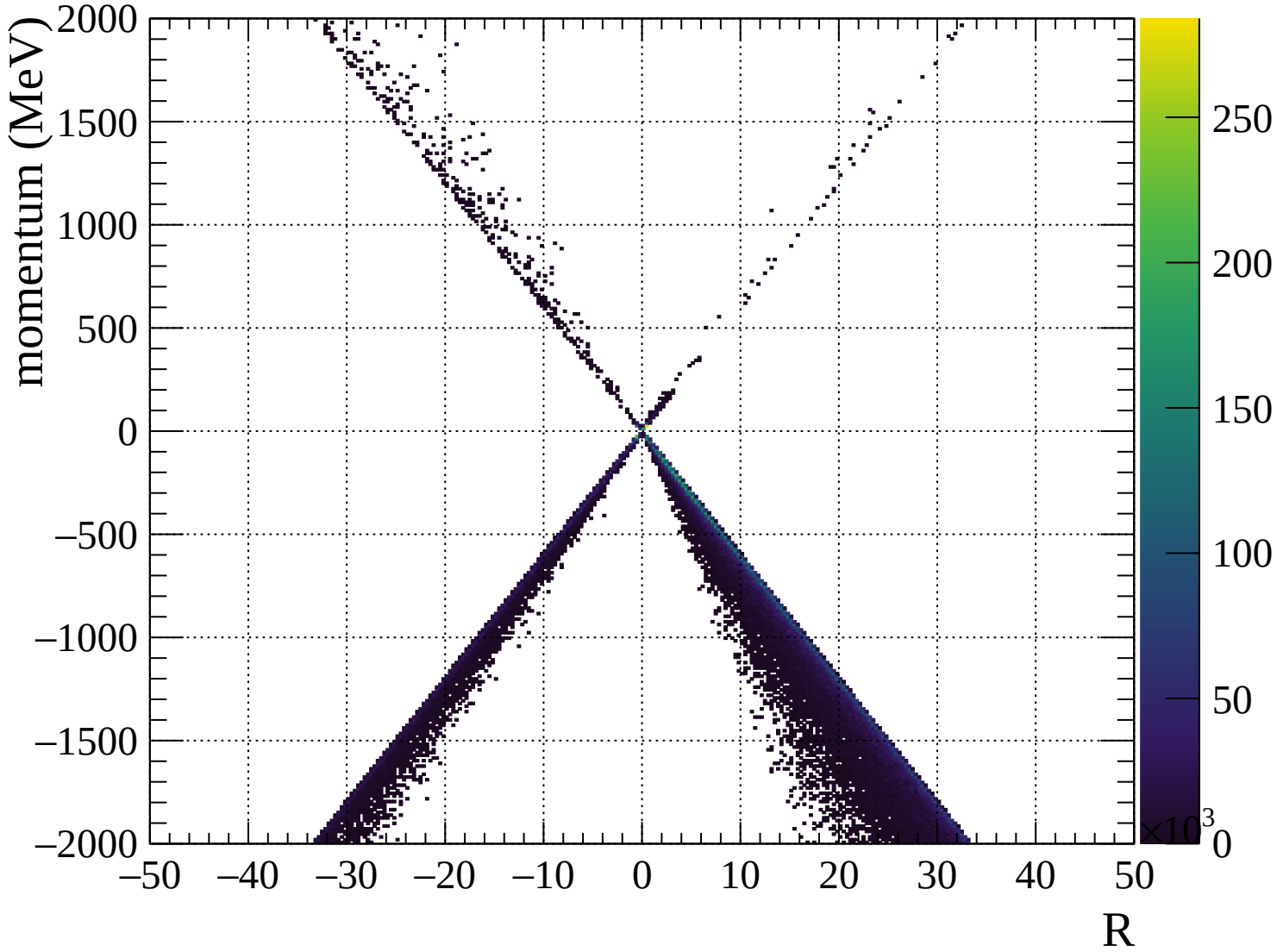


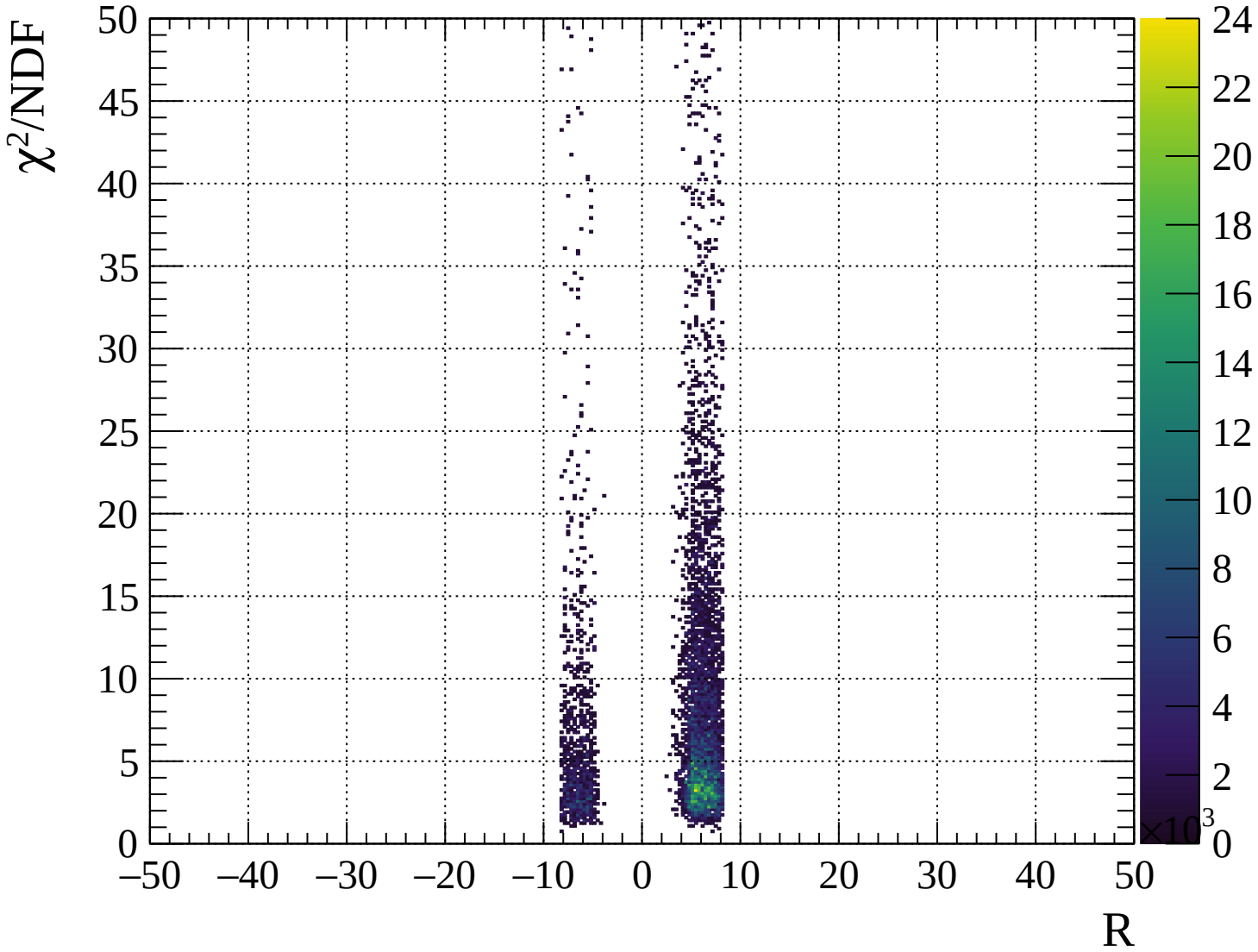






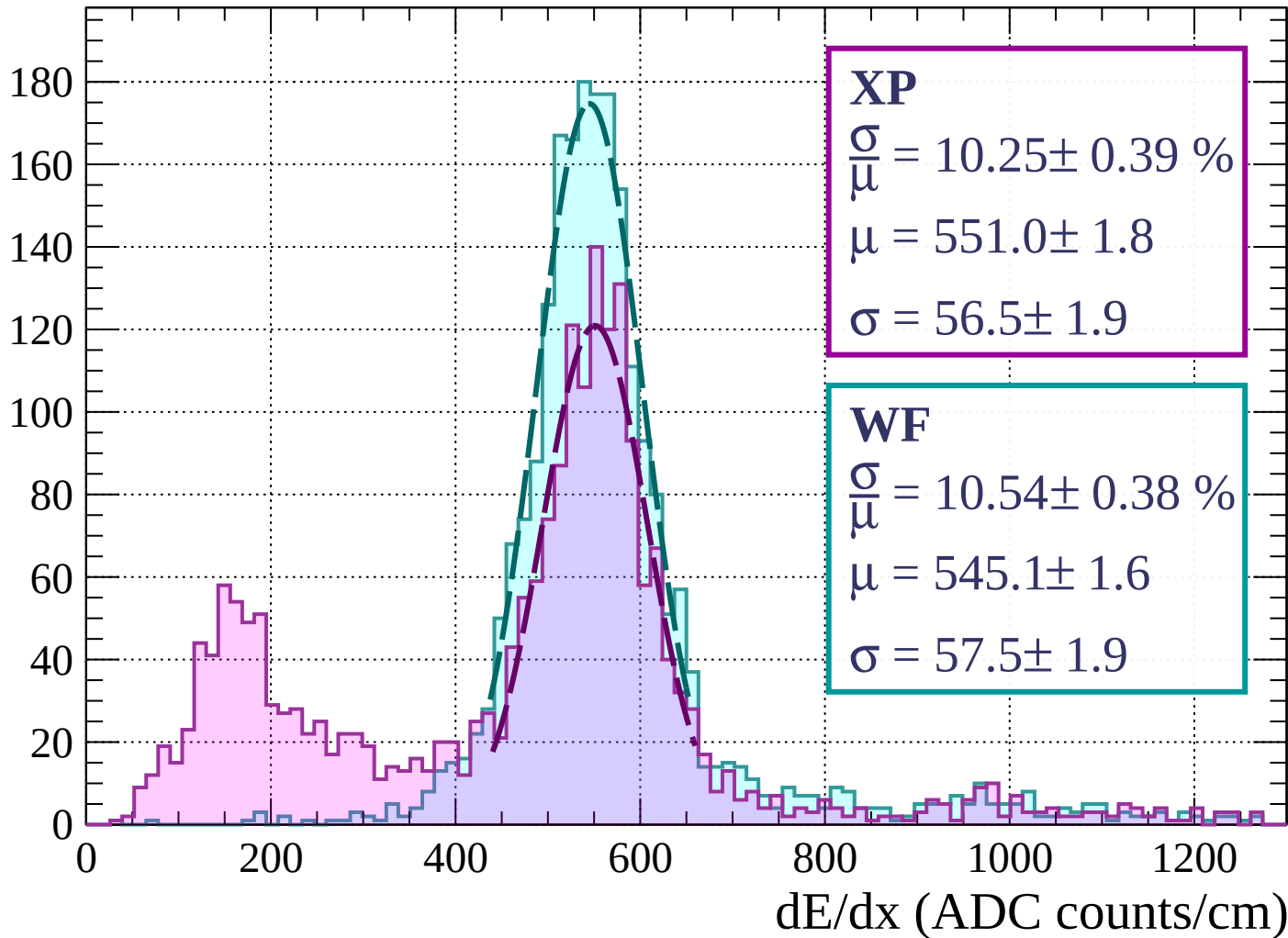




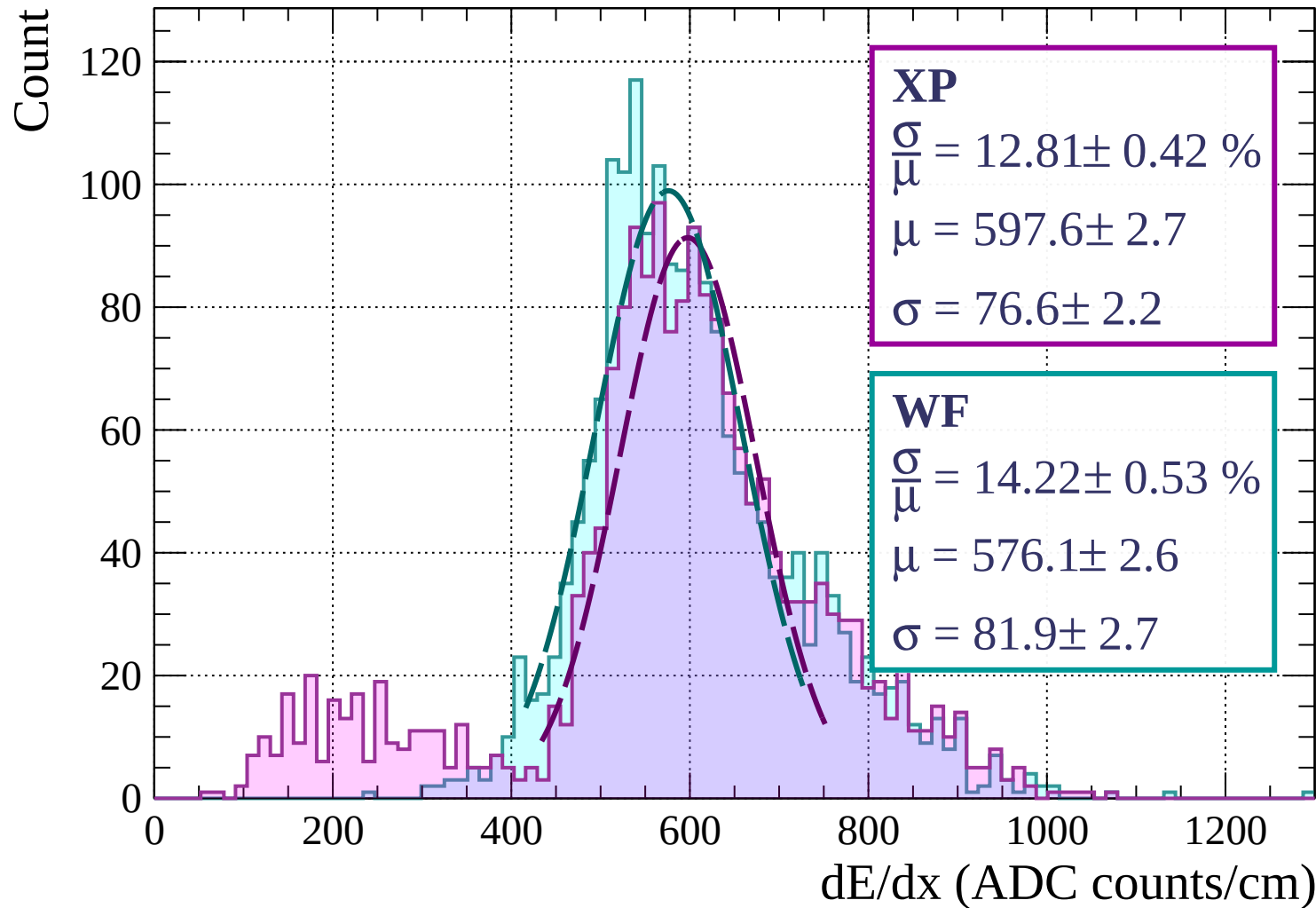


Energy loss | $0 < p < 80$

Count

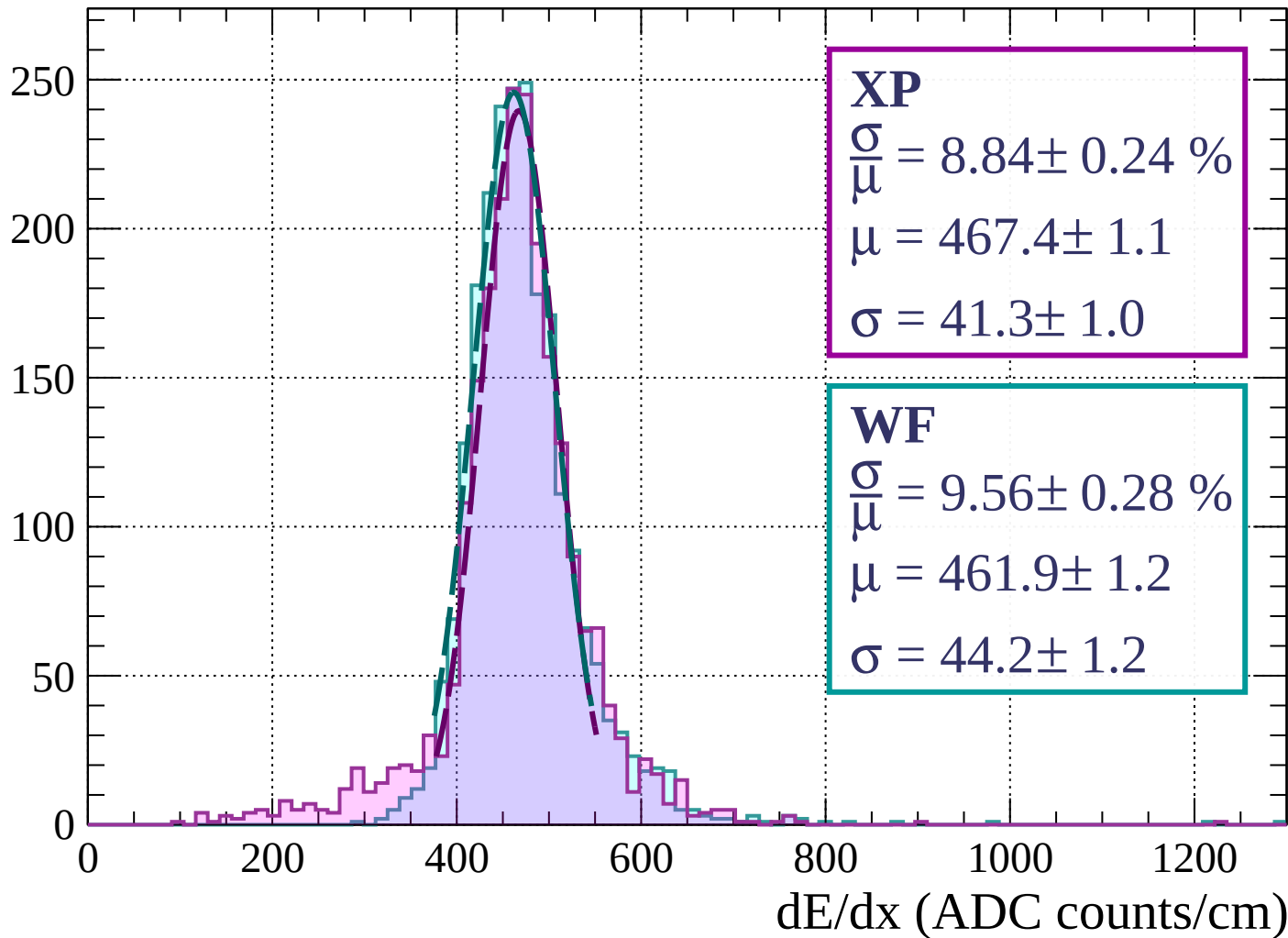


Energy loss | $80 < p < 160$



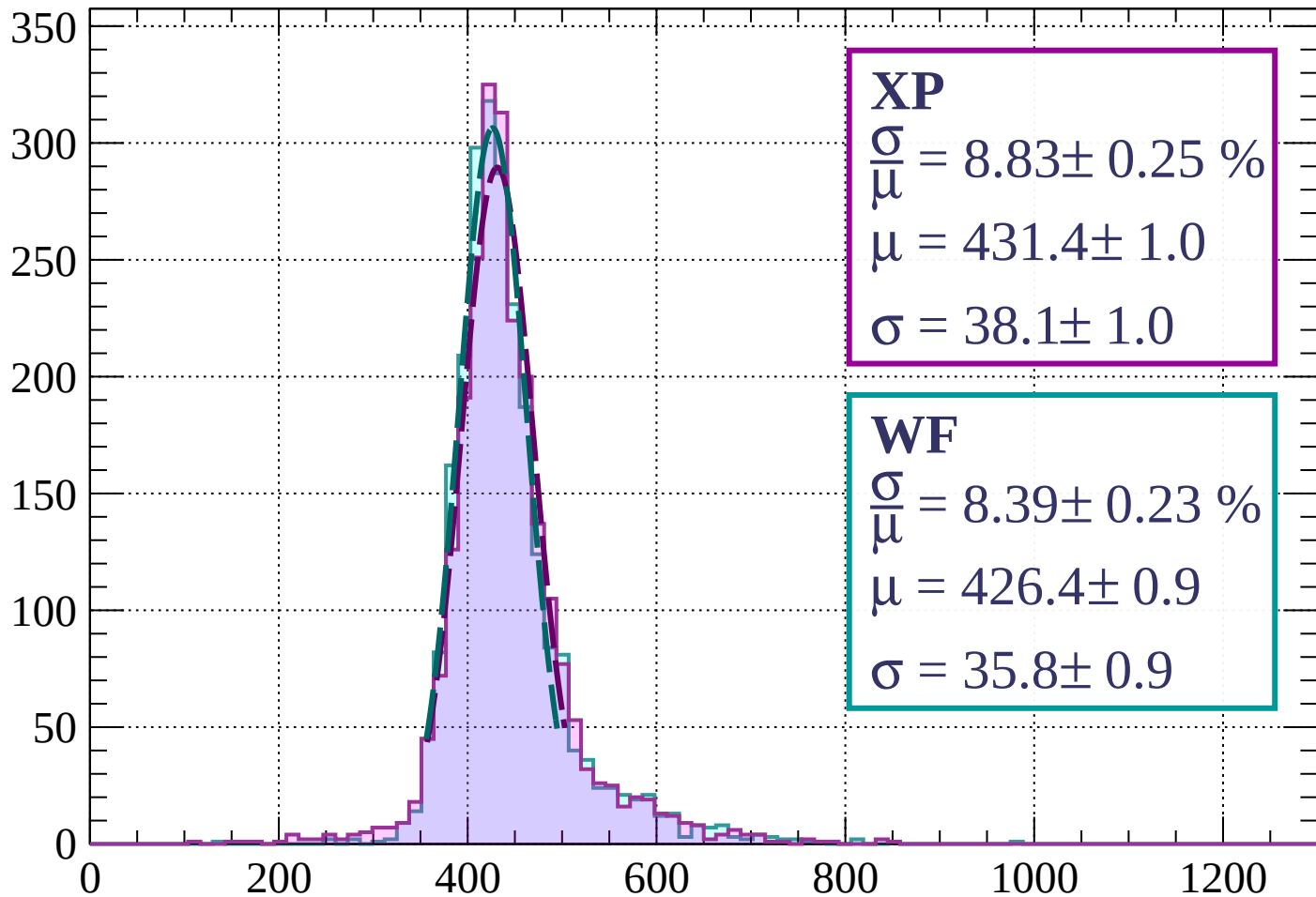
Energy loss | $160 < p < 240$

Count



Energy loss | $240 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 8.83 \pm 0.25 \%$$

$$\mu = 431.4 \pm 1.0$$

$$\sigma = 38.1 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.39 \pm 0.23 \%$$

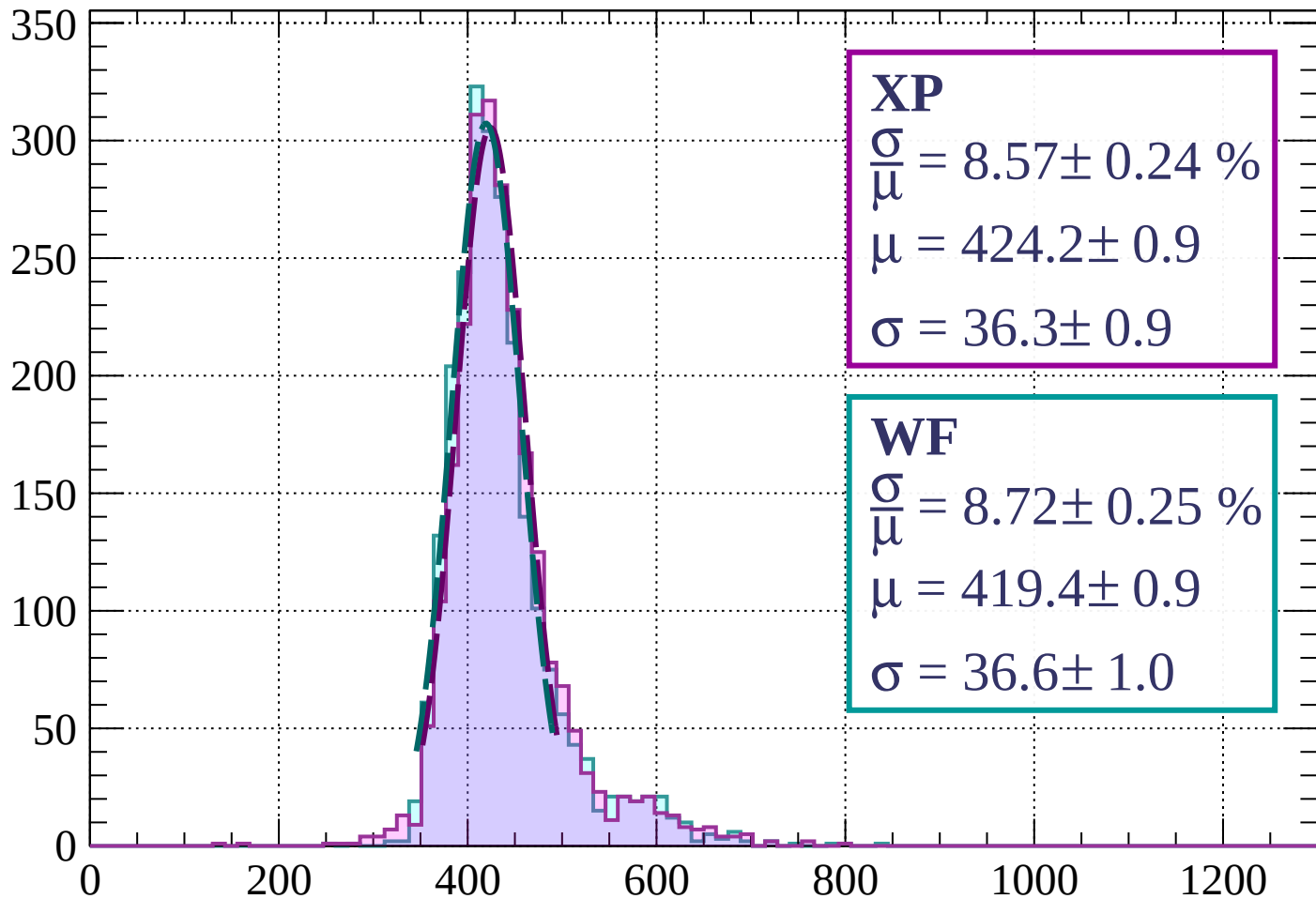
$$\mu = 426.4 \pm 0.9$$

$$\sigma = 35.8 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 400$

Count



XP

$$\frac{\sigma}{\mu} = 8.57 \pm 0.24 \%$$

$$\mu = 424.2 \pm 0.9$$

$$\sigma = 36.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.72 \pm 0.25 \%$$

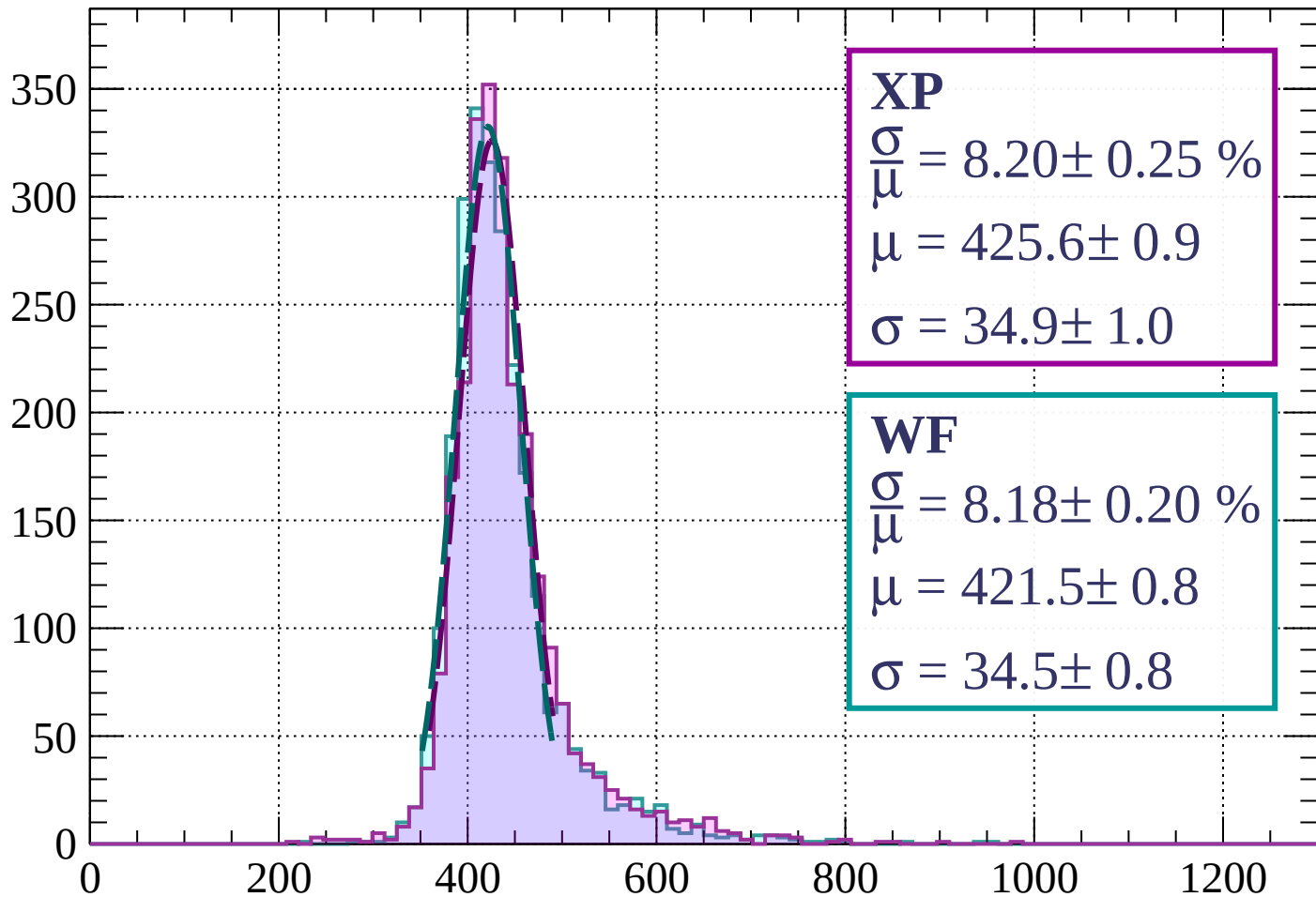
$$\mu = 419.4 \pm 0.9$$

$$\sigma = 36.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $400 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 8.20 \pm 0.25 \%$$

$$\mu = 425.6 \pm 0.9$$

$$\sigma = 34.9 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.18 \pm 0.20 \%$$

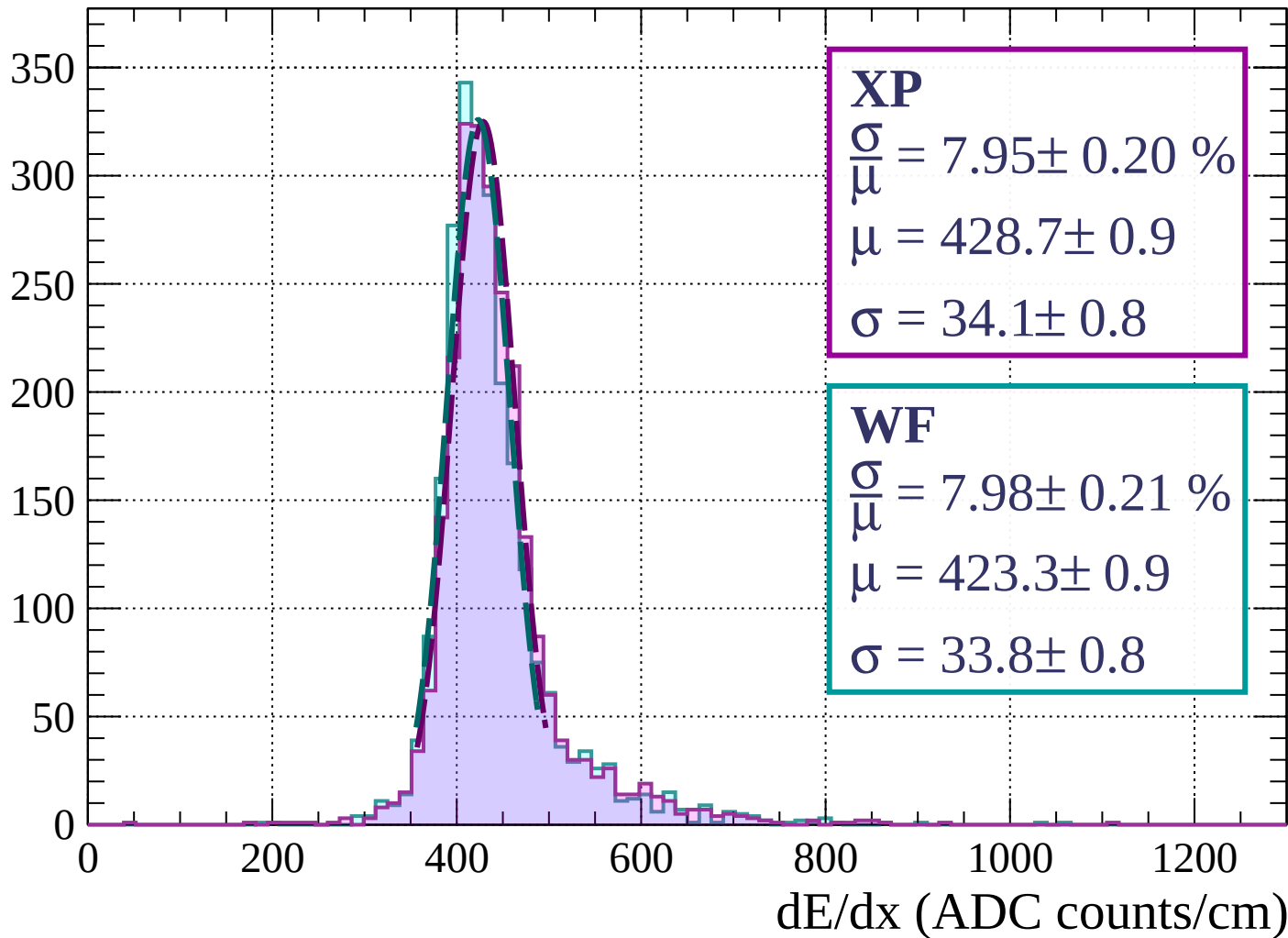
$$\mu = 421.5 \pm 0.8$$

$$\sigma = 34.5 \pm 0.8$$

dE/dx (ADC counts/cm)

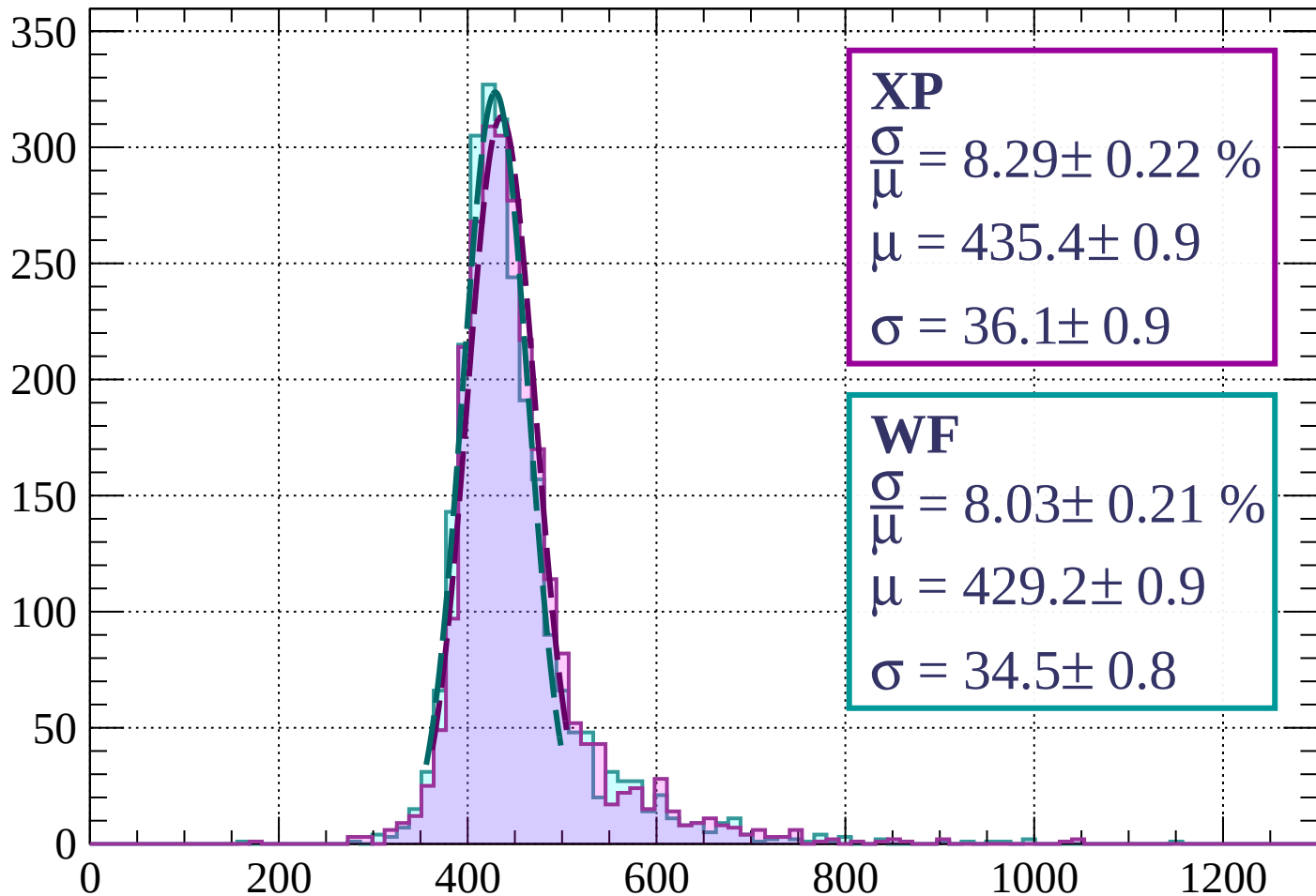
Energy loss | $480 < p < 560$

Count



Energy loss | $560 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 8.29 \pm 0.22 \%$$

$$\mu = 435.4 \pm 0.9$$

$$\sigma = 36.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.03 \pm 0.21 \%$$

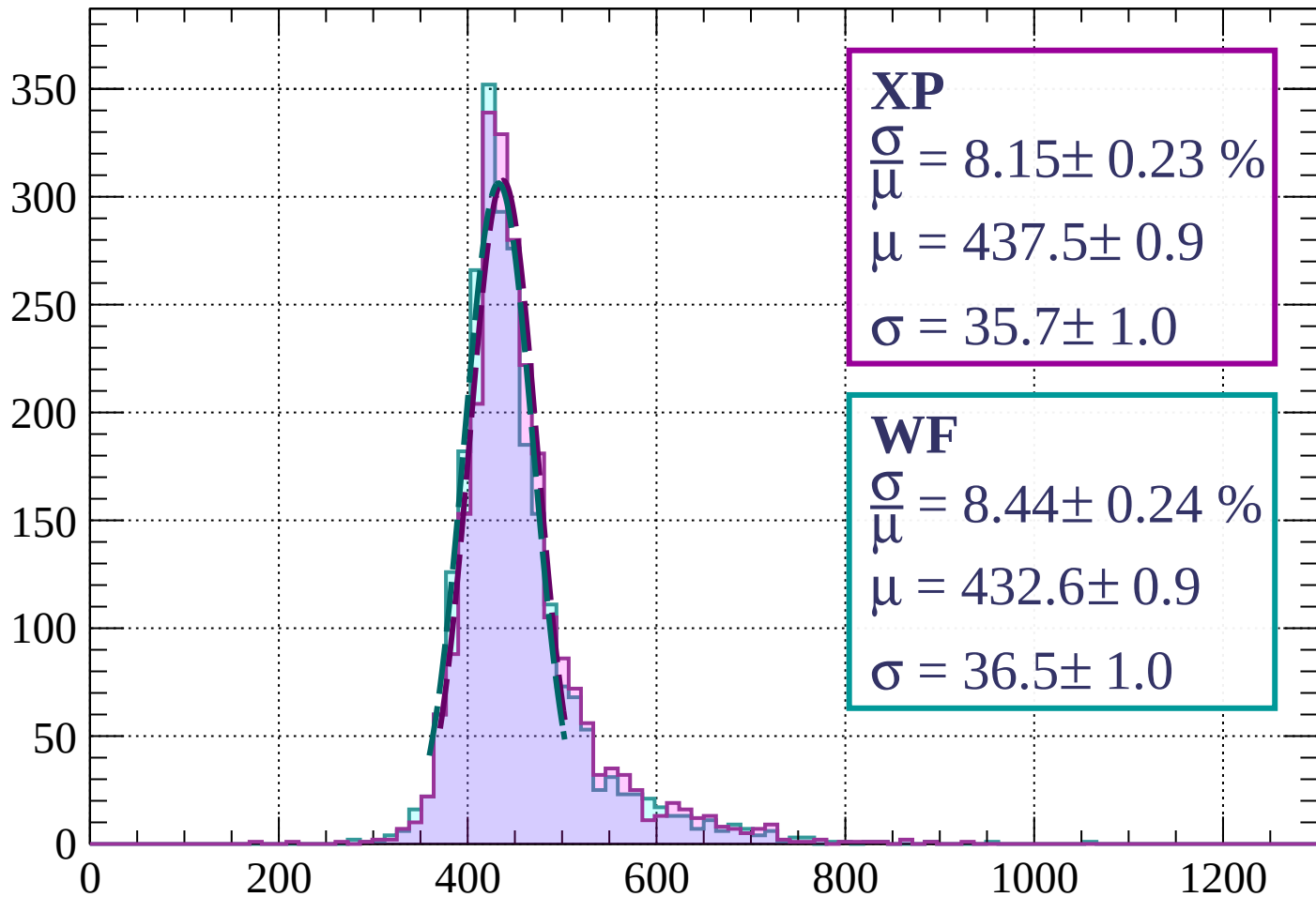
$$\mu = 429.2 \pm 0.9$$

$$\sigma = 34.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 8.15 \pm 0.23 \%$$

$$\mu = 437.5 \pm 0.9$$

$$\sigma = 35.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.44 \pm 0.24 \%$$

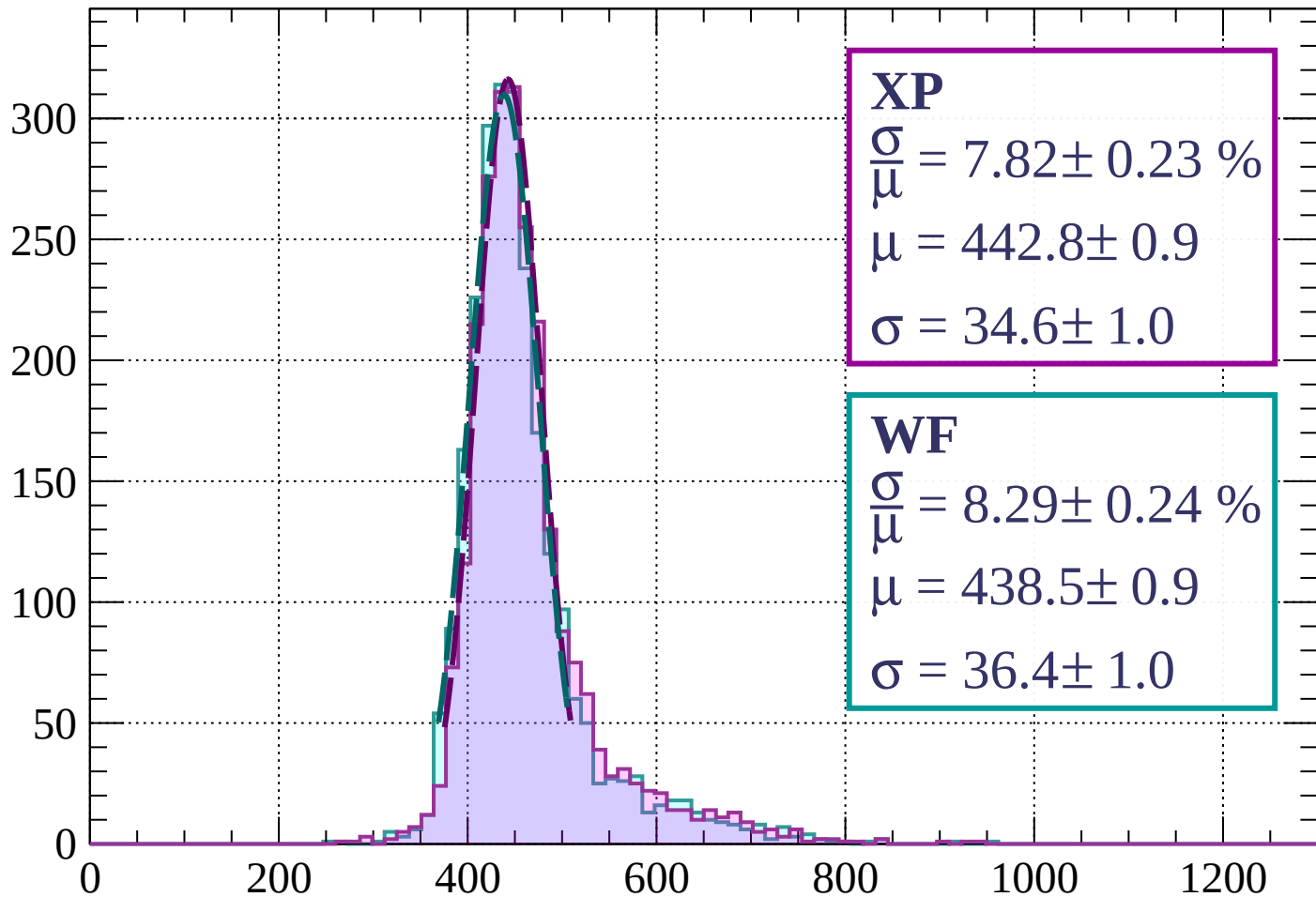
$$\mu = 432.6 \pm 0.9$$

$$\sigma = 36.5 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 7.82 \pm 0.23 \%$$

$$\mu = 442.8 \pm 0.9$$

$$\sigma = 34.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.29 \pm 0.24 \%$$

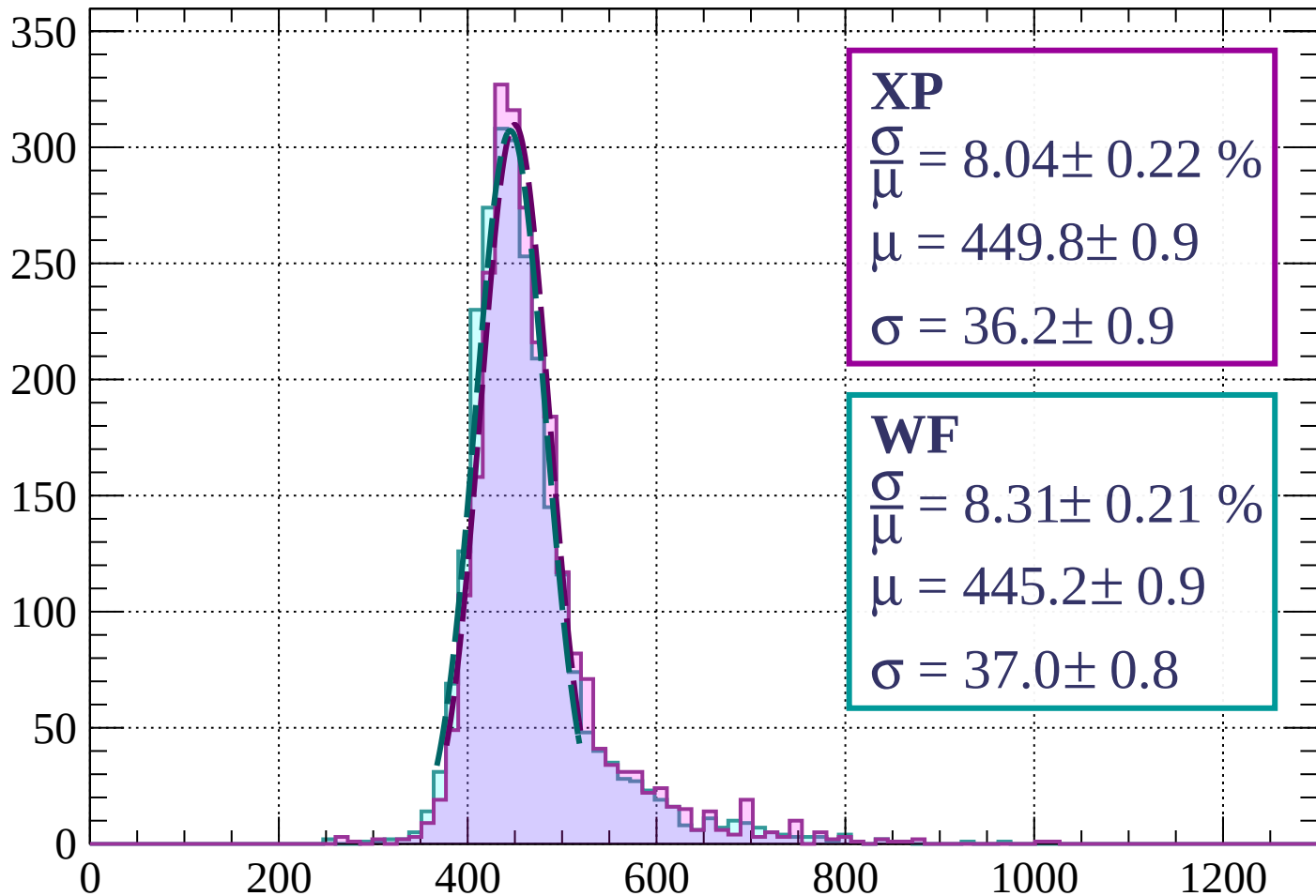
$$\mu = 438.5 \pm 0.9$$

$$\sigma = 36.4 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 880$

Count



XP

$$\frac{\sigma}{\mu} = 8.04 \pm 0.22 \%$$

$$\mu = 449.8 \pm 0.9$$

$$\sigma = 36.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.31 \pm 0.21 \%$$

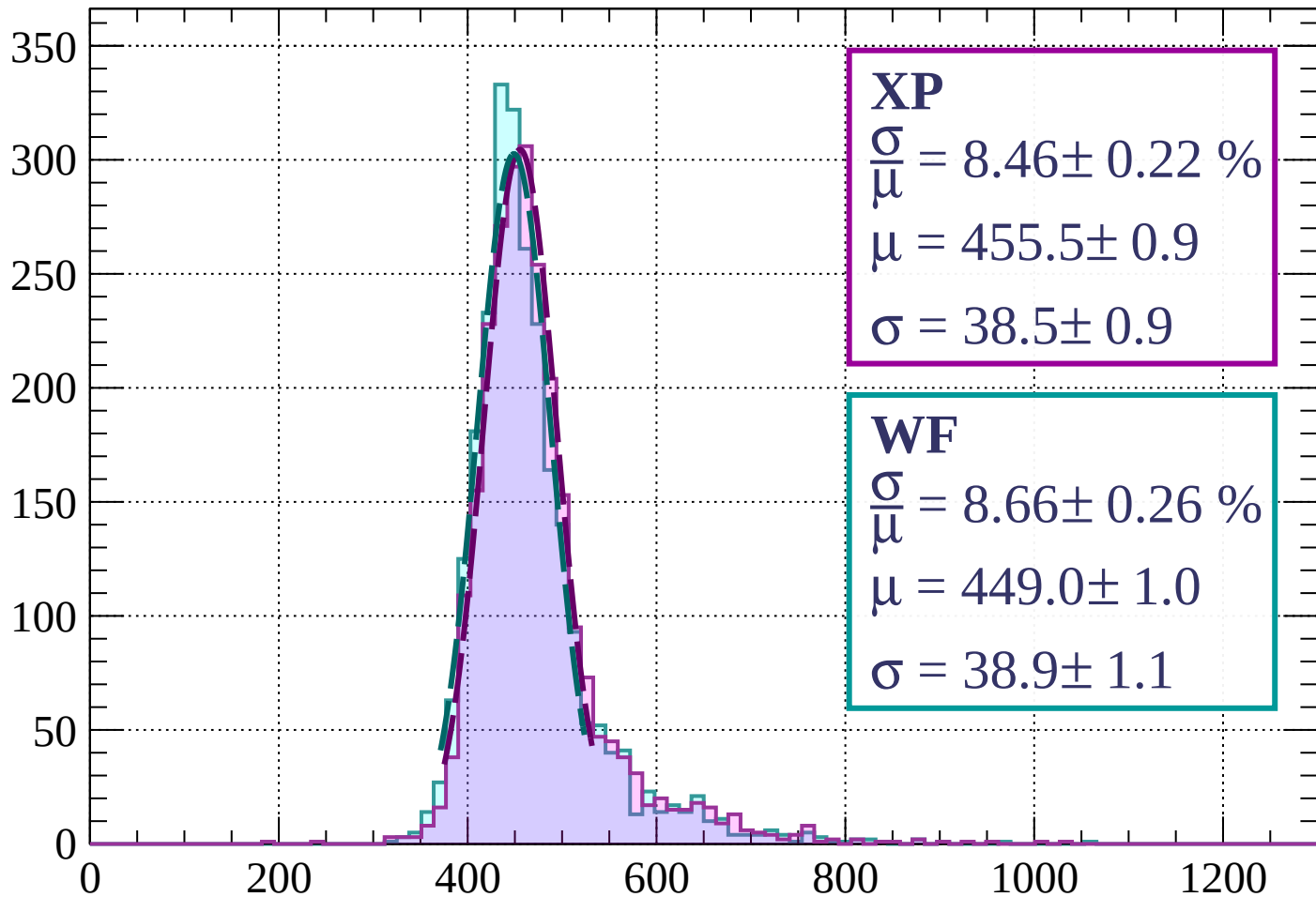
$$\mu = 445.2 \pm 0.9$$

$$\sigma = 37.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $880 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 8.46 \pm 0.22 \%$$

$$\mu = 455.5 \pm 0.9$$

$$\sigma = 38.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.66 \pm 0.26 \%$$

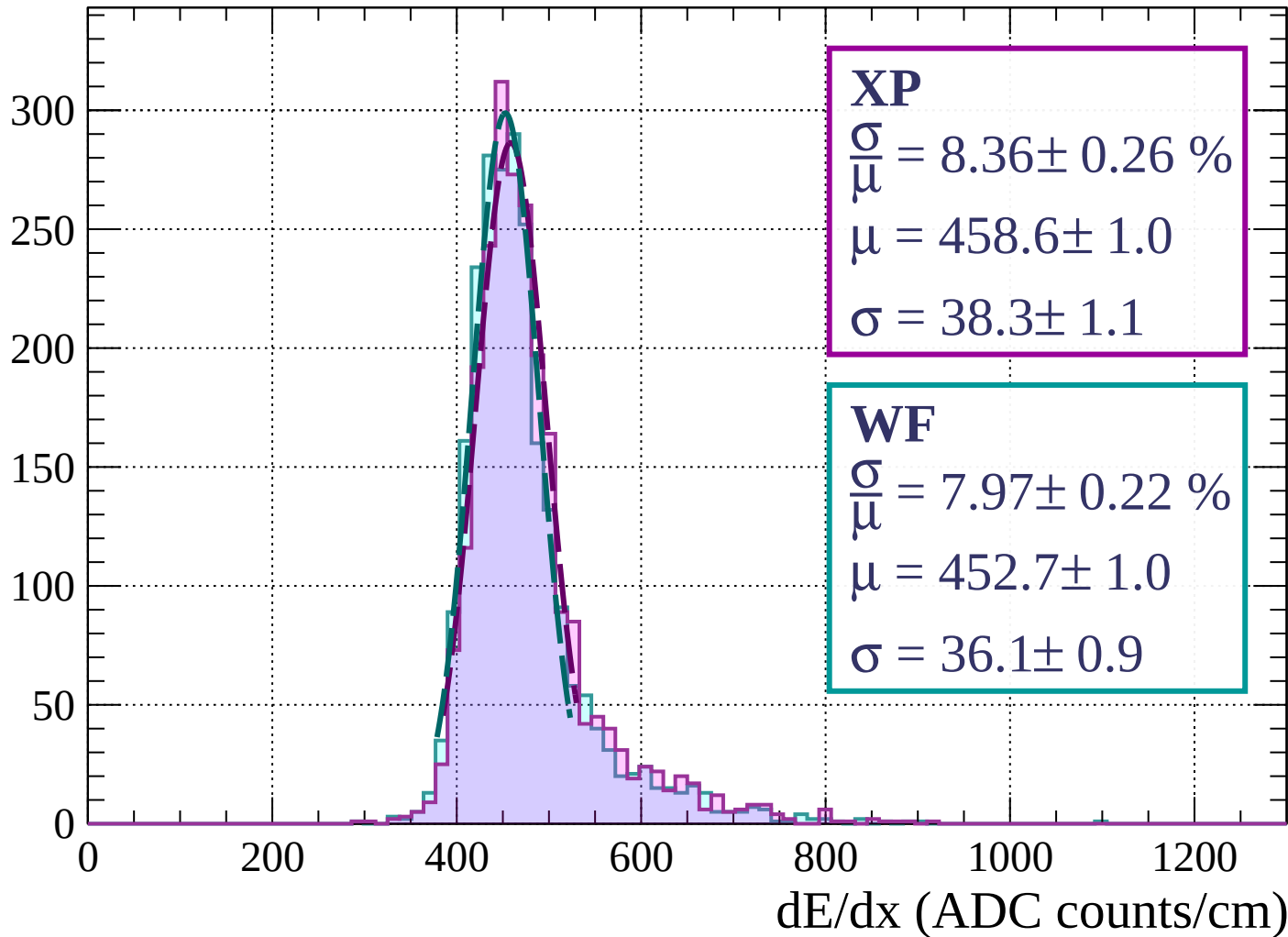
$$\mu = 449.0 \pm 1.0$$

$$\sigma = 38.9 \pm 1.1$$

dE/dx (ADC counts/cm)

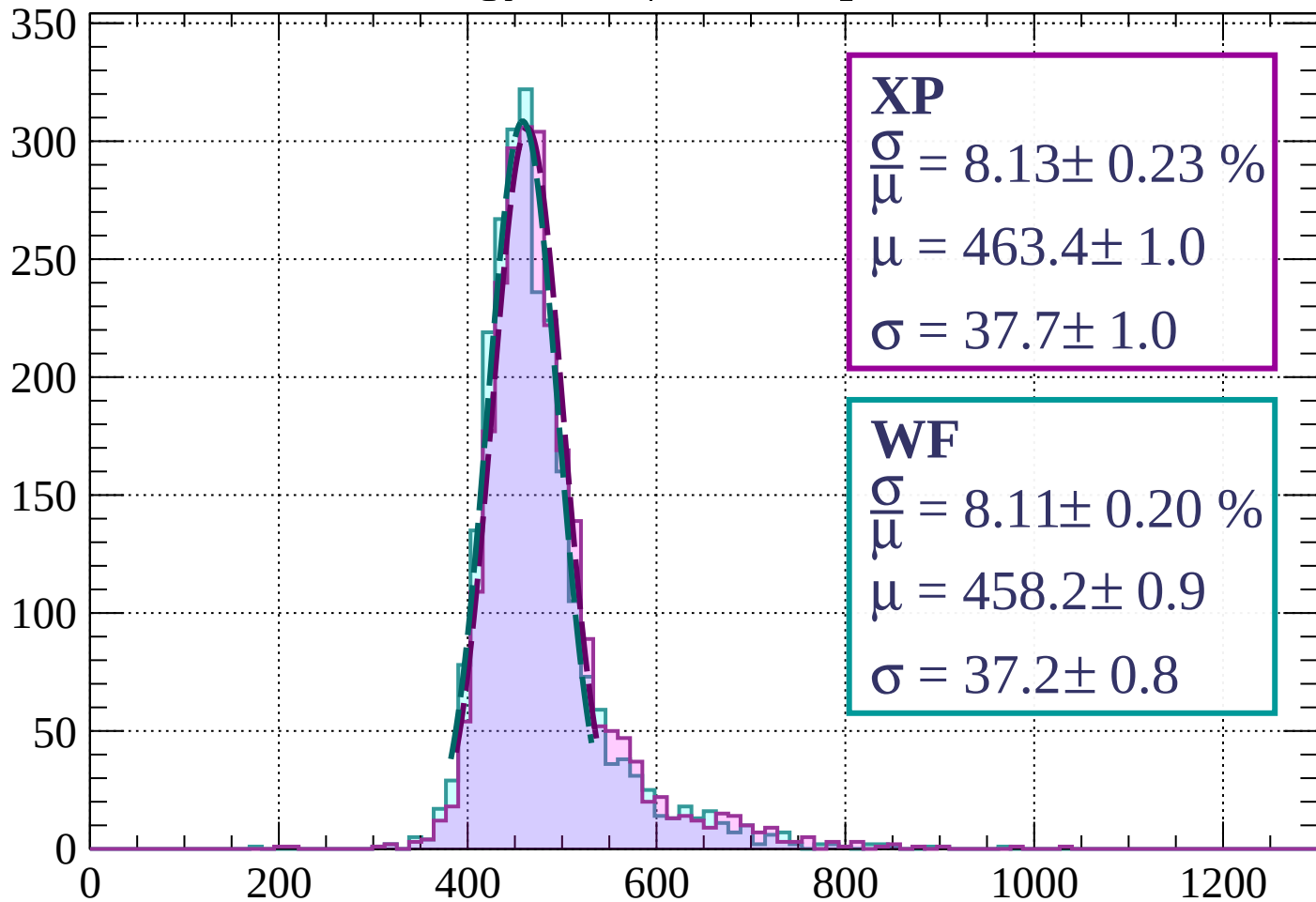
Energy loss | $960 < p < 1040$

Count



Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 8.13 \pm 0.23 \%$$

$$\mu = 463.4 \pm 1.0$$

$$\sigma = 37.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.11 \pm 0.20 \%$$

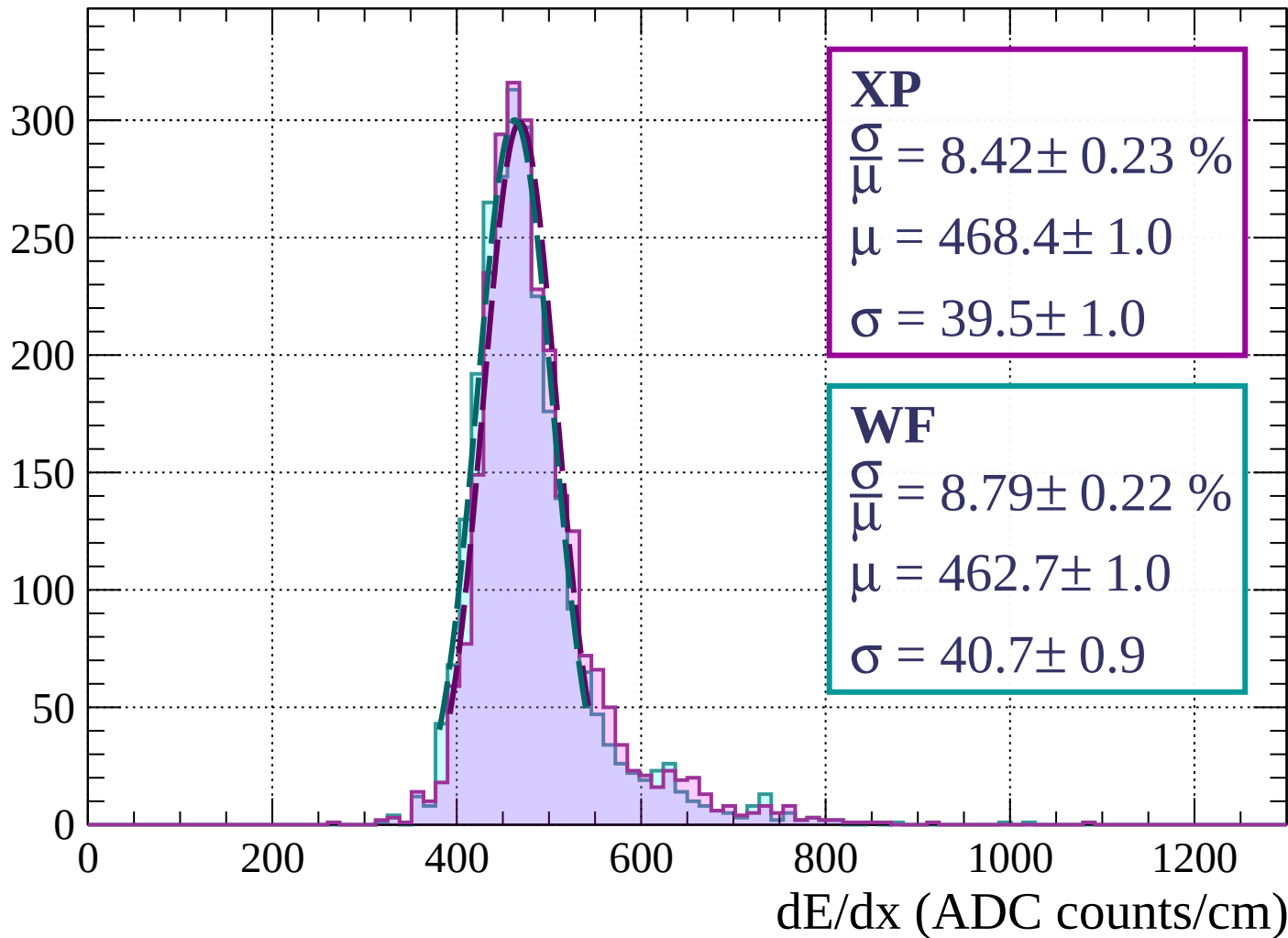
$$\mu = 458.2 \pm 0.9$$

$$\sigma = 37.2 \pm 0.8$$

dE/dx (ADC counts/cm)

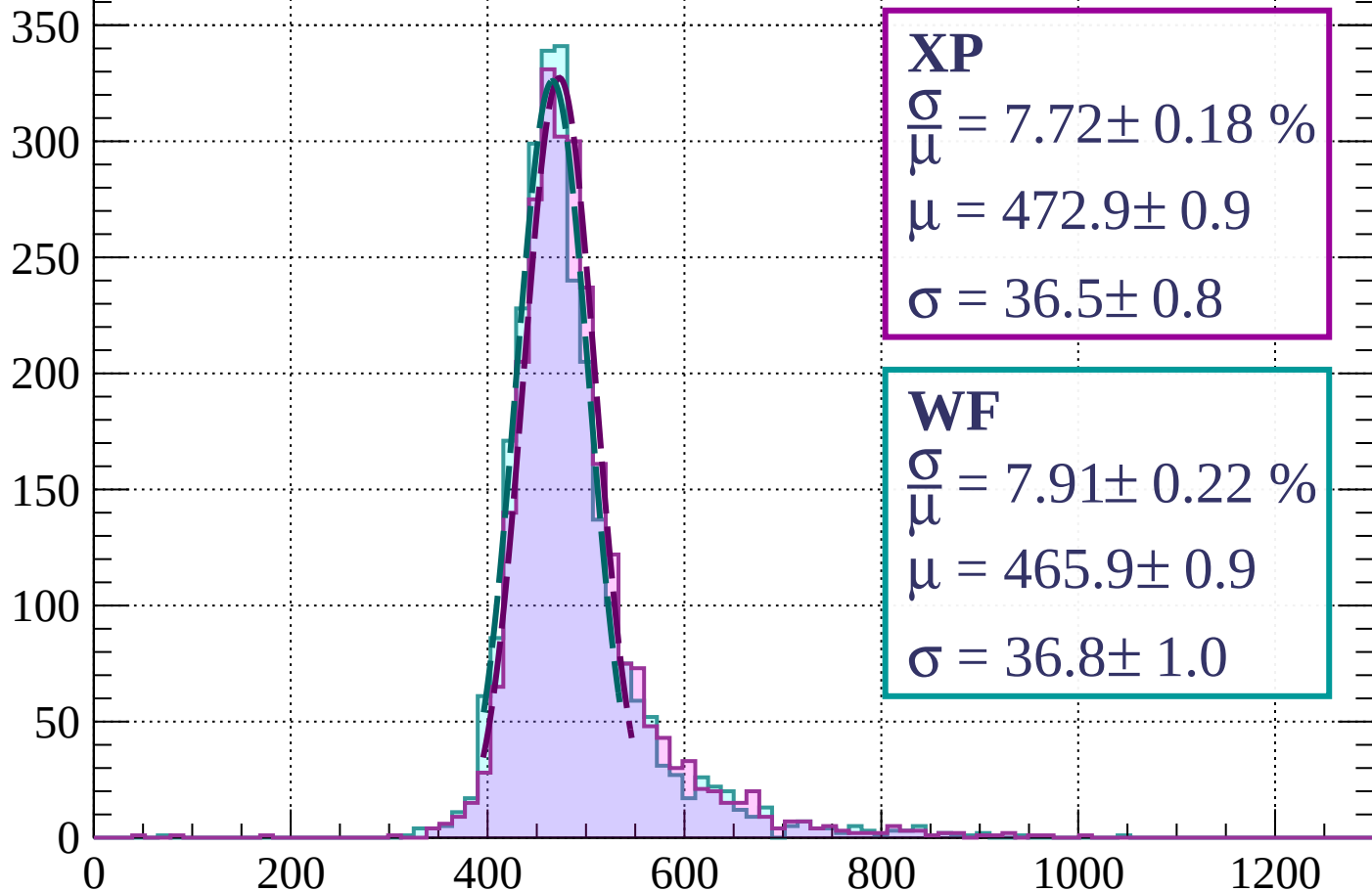
Energy loss | $1120 < p < 1200$

Count



Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 7.72 \pm 0.18 \%$$

$$\mu = 472.9 \pm 0.9$$

$$\sigma = 36.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.91 \pm 0.22 \%$$

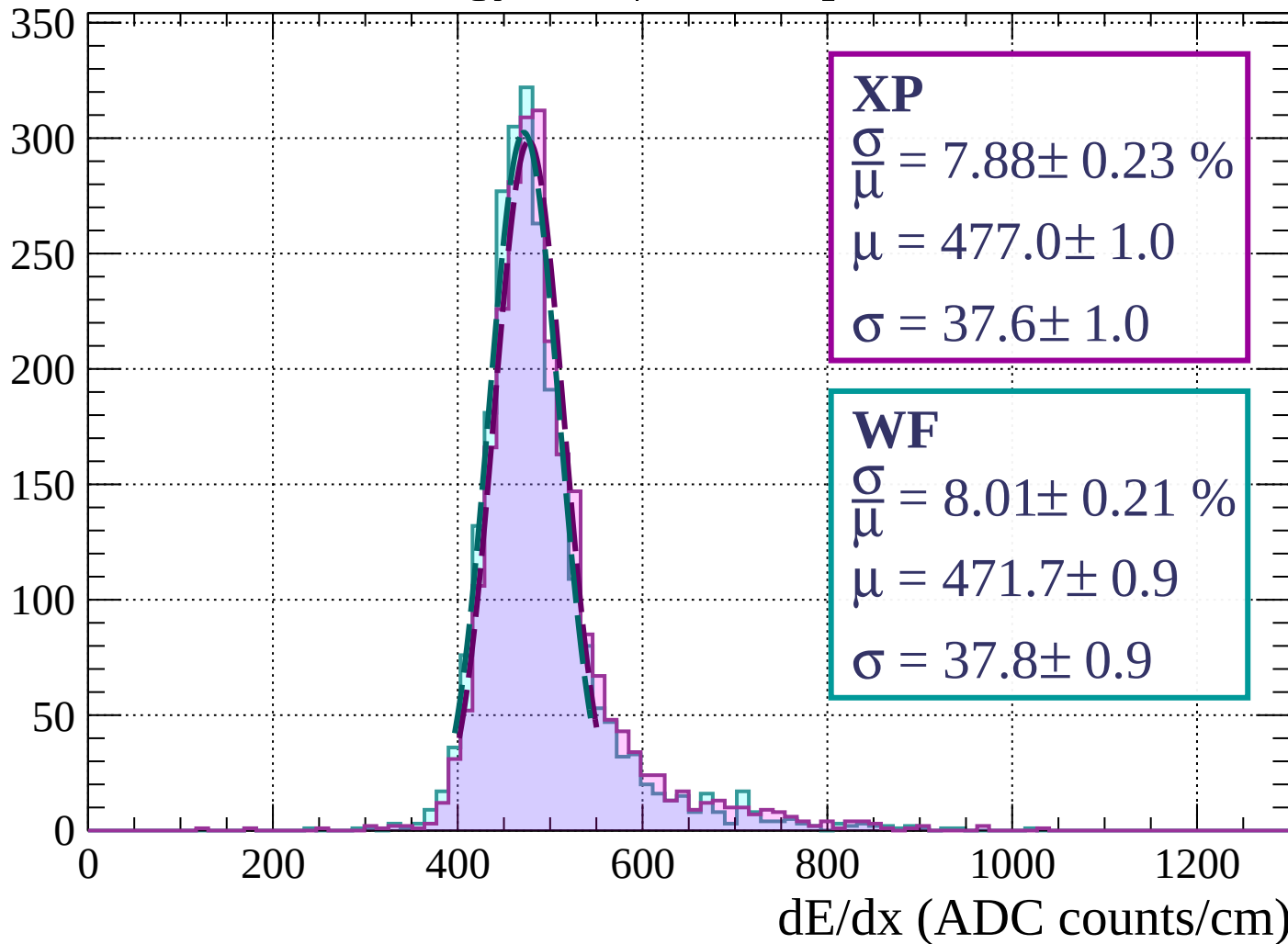
$$\mu = 465.9 \pm 0.9$$

$$\sigma = 36.8 \pm 1.0$$

dE/dx (ADC counts/cm)

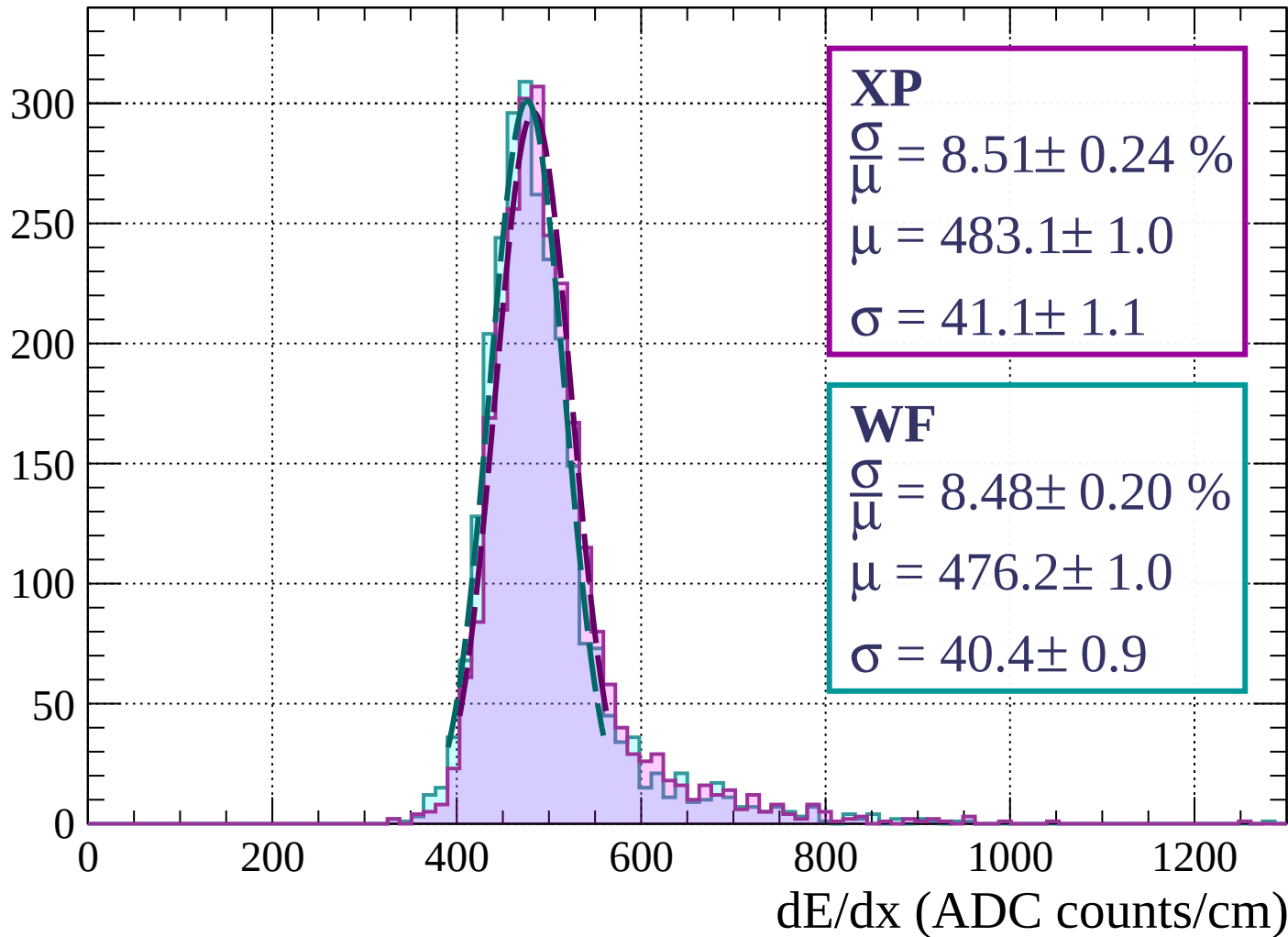
Energy loss | $1280 < p < 1360$

Count



Energy loss | $1360 < p < 1440$

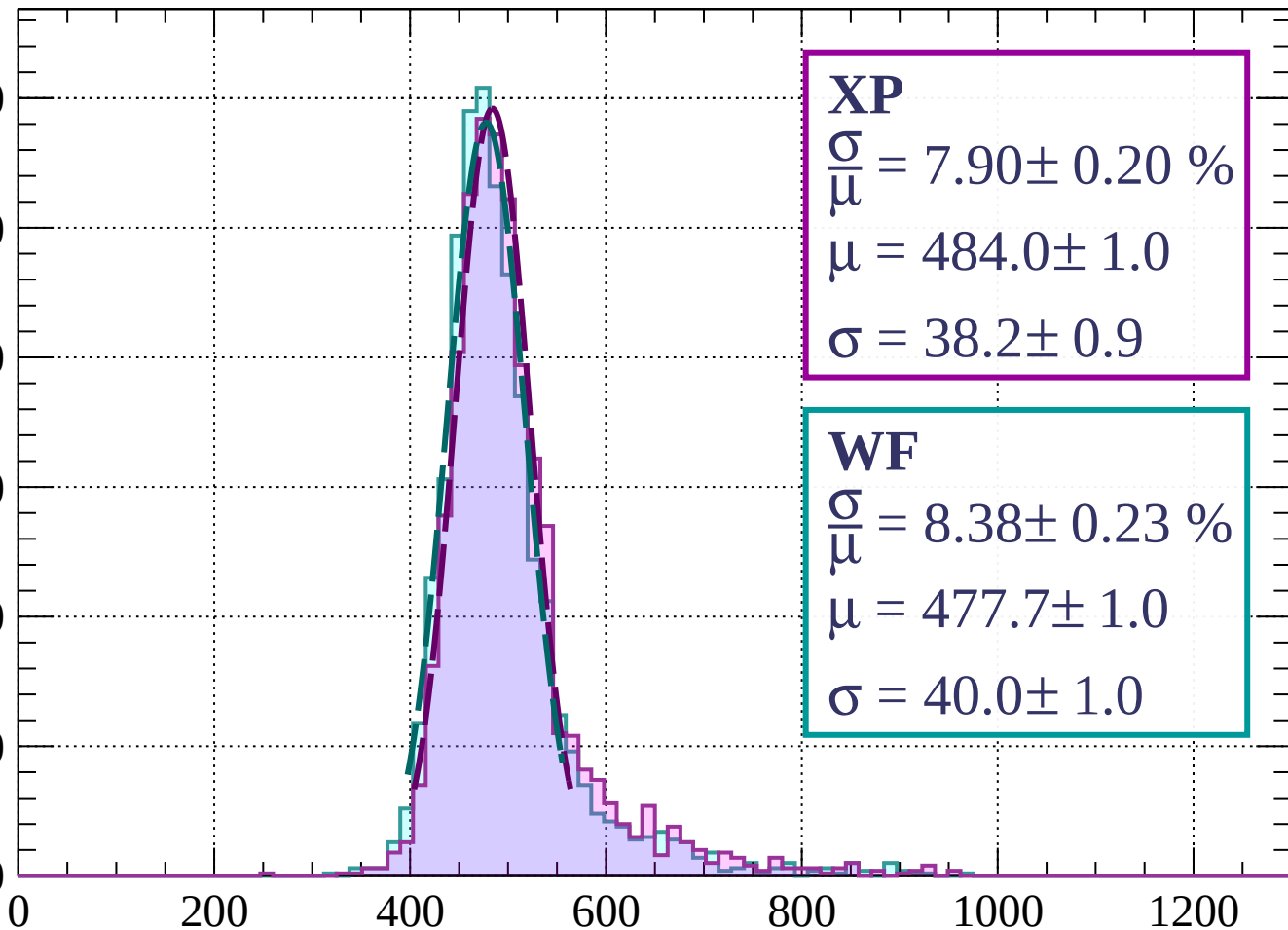
Count



Energy loss | $1440 < p < 1520$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 7.90 \pm 0.20 \%$$

$$\mu = 484.0 \pm 1.0$$

$$\sigma = 38.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.38 \pm 0.23 \%$$

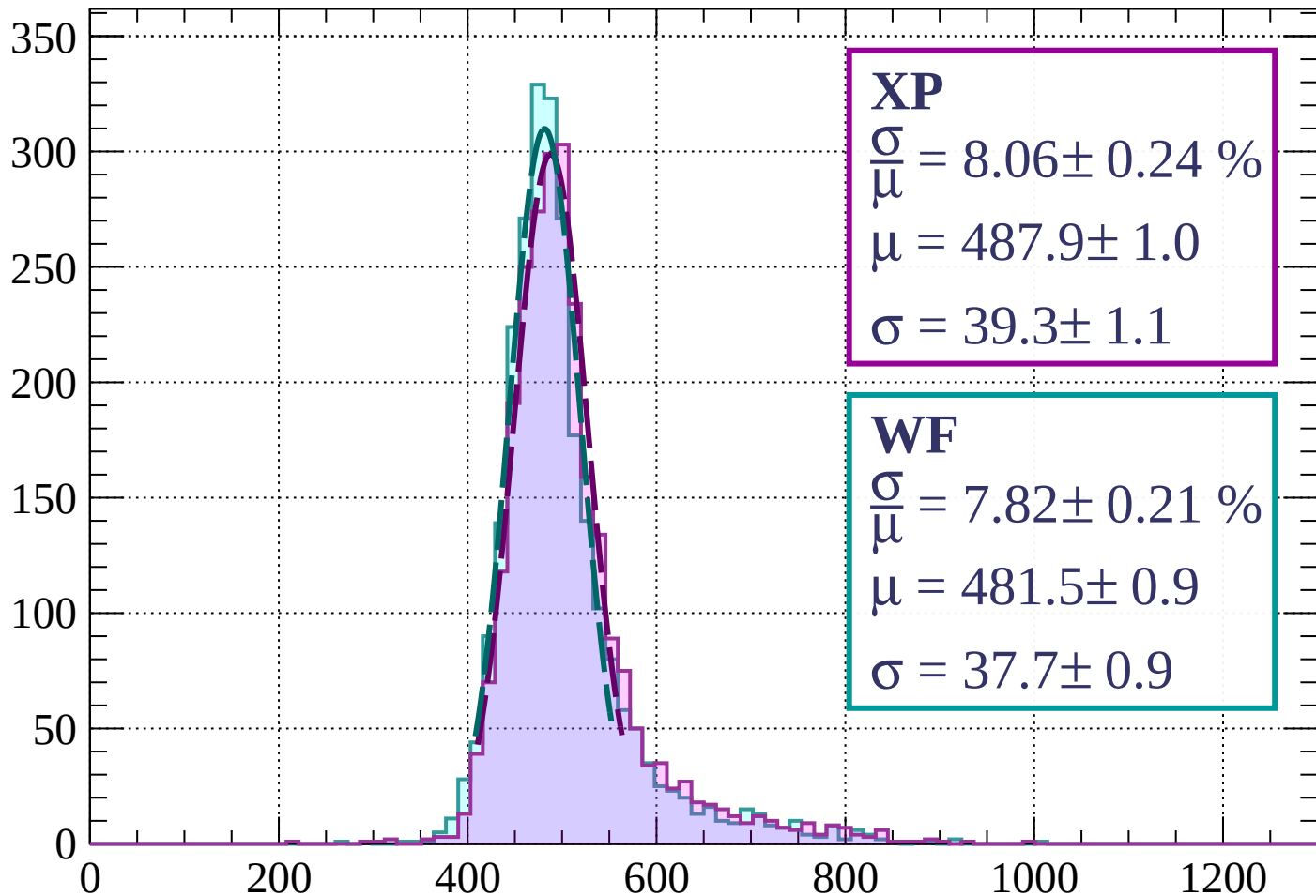
$$\mu = 477.7 \pm 1.0$$

$$\sigma = 40.0 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 8.06 \pm 0.24 \%$$

$$\mu = 487.9 \pm 1.0$$

$$\sigma = 39.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.82 \pm 0.21 \%$$

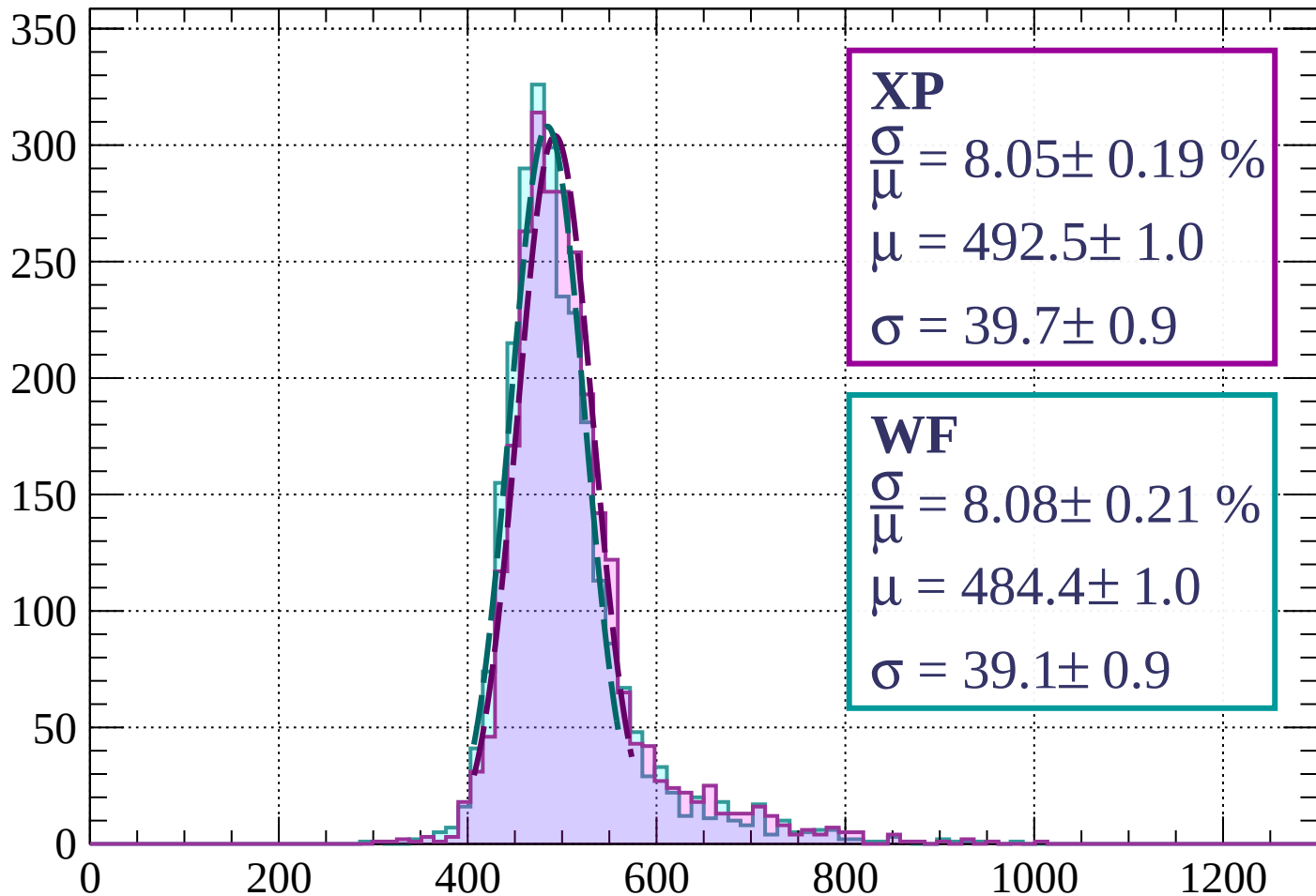
$$\mu = 481.5 \pm 0.9$$

$$\sigma = 37.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 8.05 \pm 0.19 \%$$

$$\mu = 492.5 \pm 1.0$$

$$\sigma = 39.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.08 \pm 0.21 \%$$

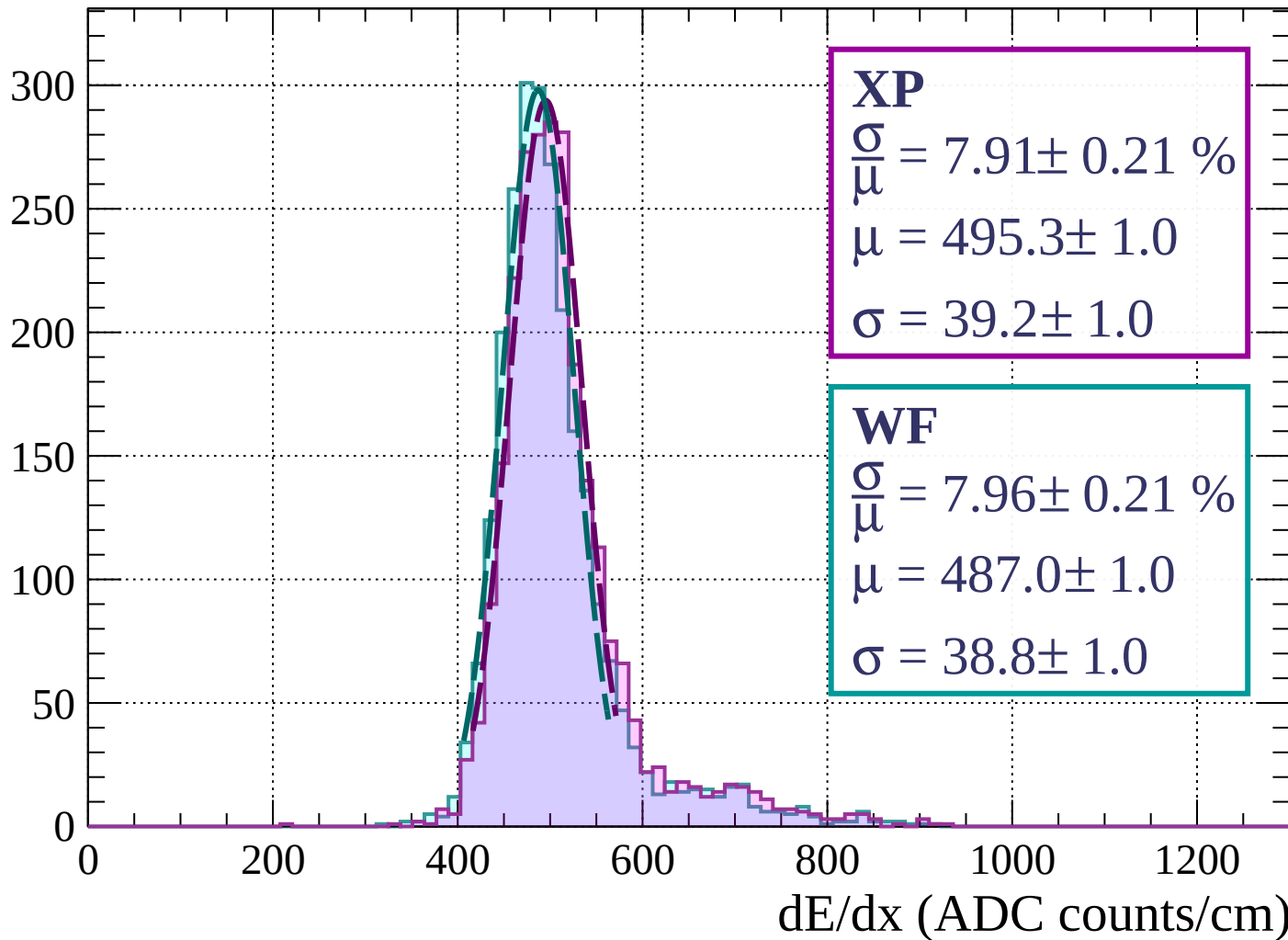
$$\mu = 484.4 \pm 1.0$$

$$\sigma = 39.1 \pm 0.9$$

dE/dx (ADC counts/cm)

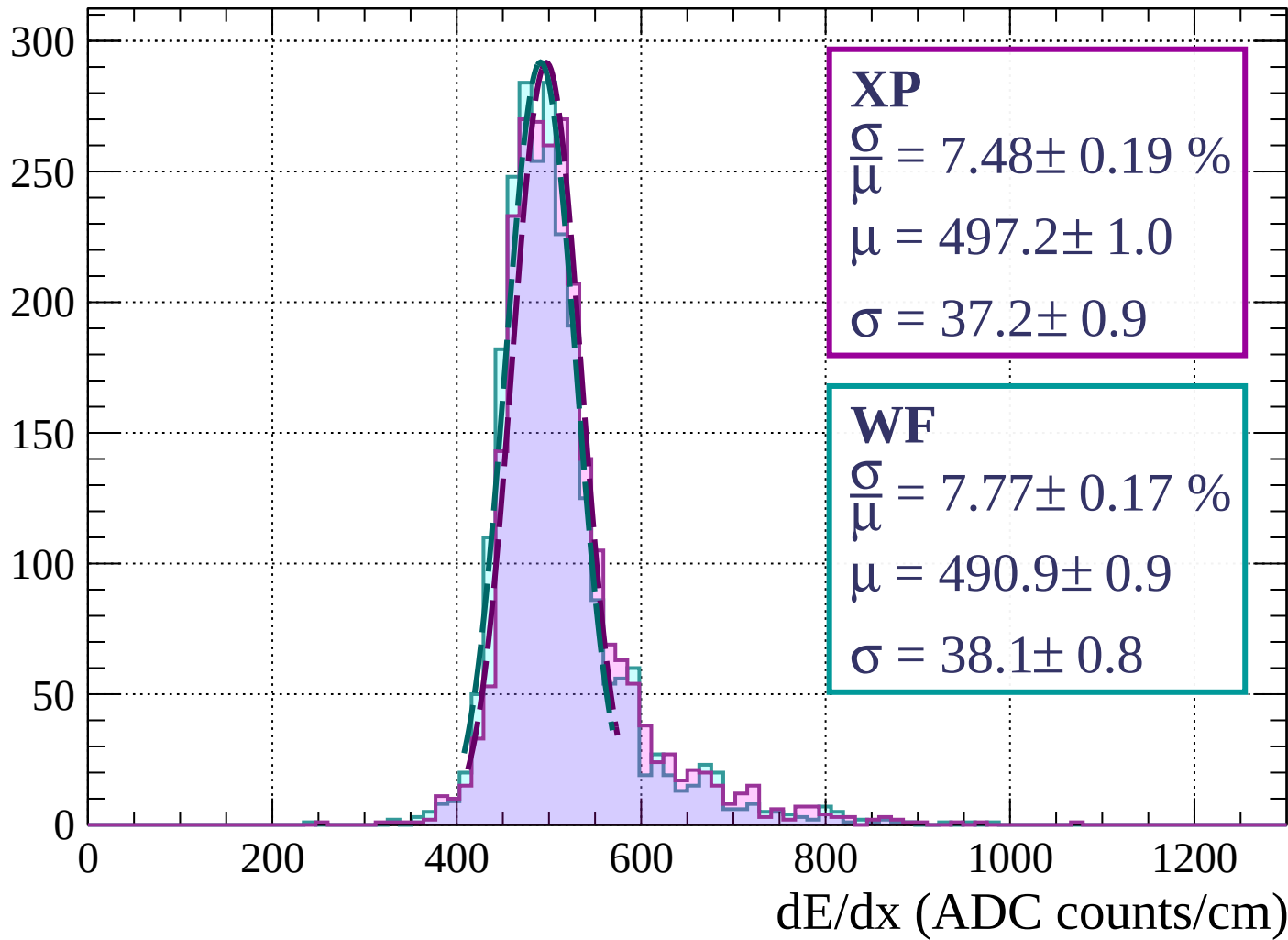
Energy loss | $1680 < p < 1760$

Count



Energy loss | $1760 < p < 1840$

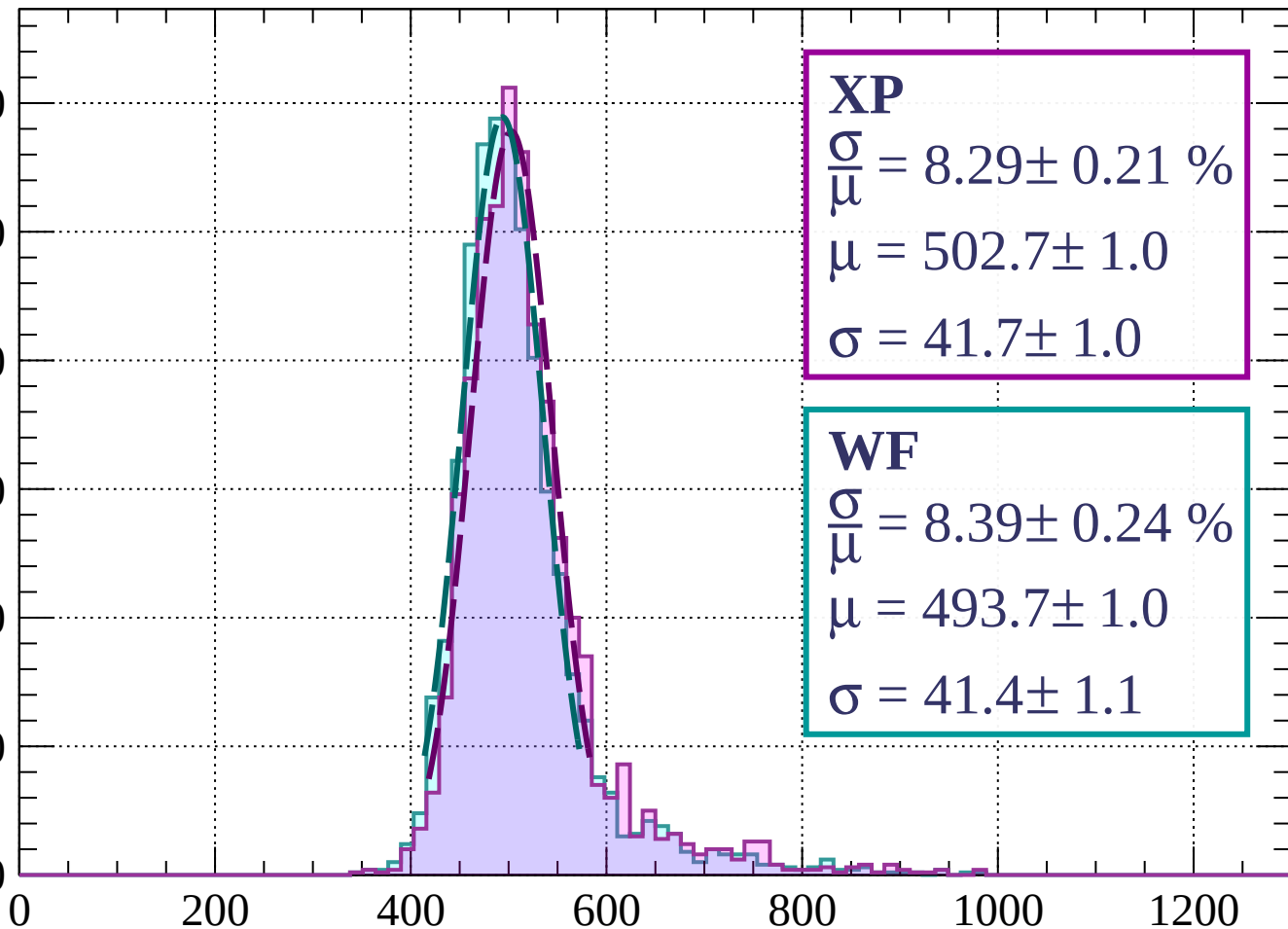
Count



Energy loss | $1840 < p < 1920$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 8.29 \pm 0.21 \%$$

$$\mu = 502.7 \pm 1.0$$

$$\sigma = 41.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.39 \pm 0.24 \%$$

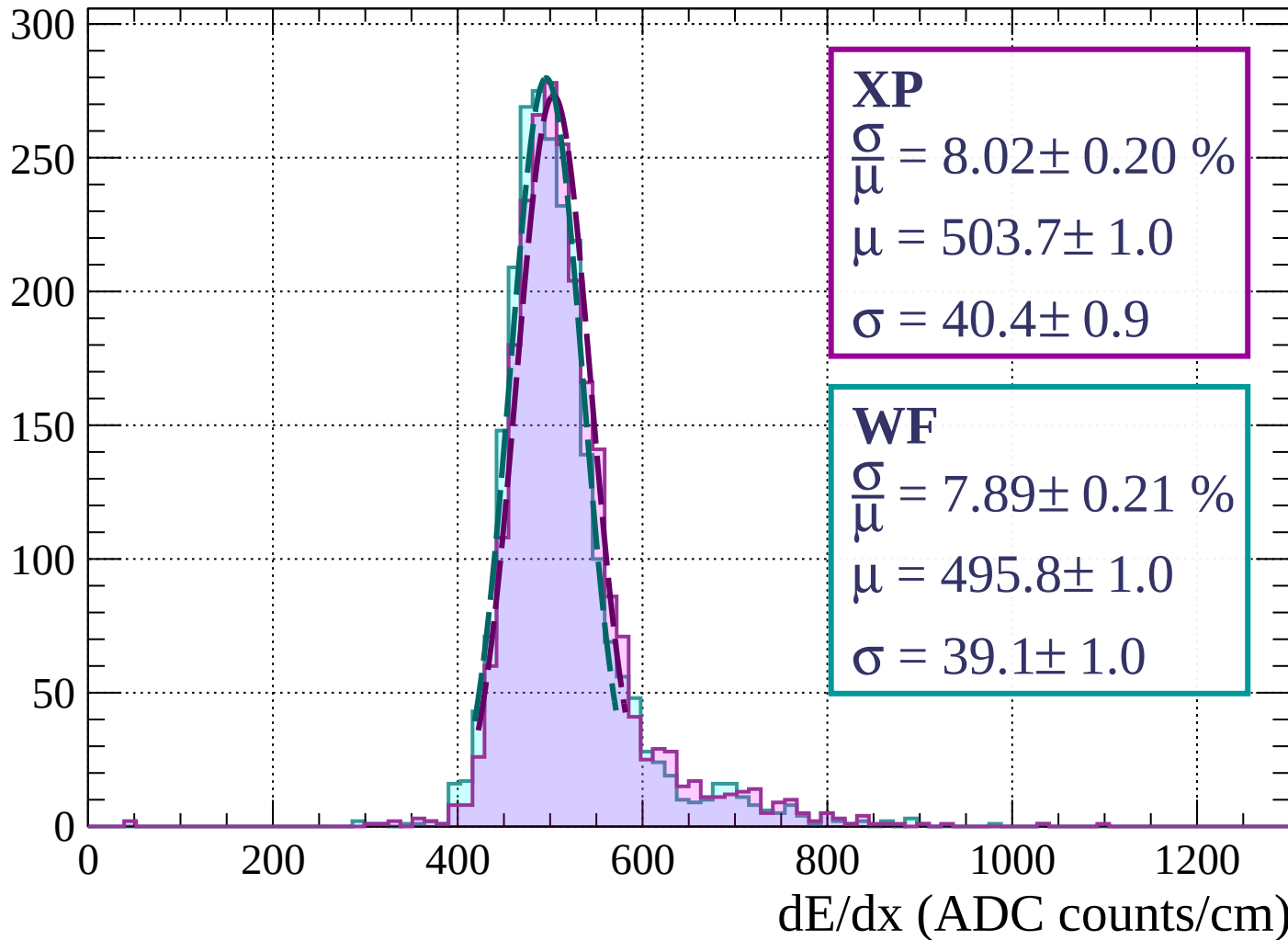
$$\mu = 493.7 \pm 1.0$$

$$\sigma = 41.4 \pm 1.1$$

dE/dx (ADC counts/cm)

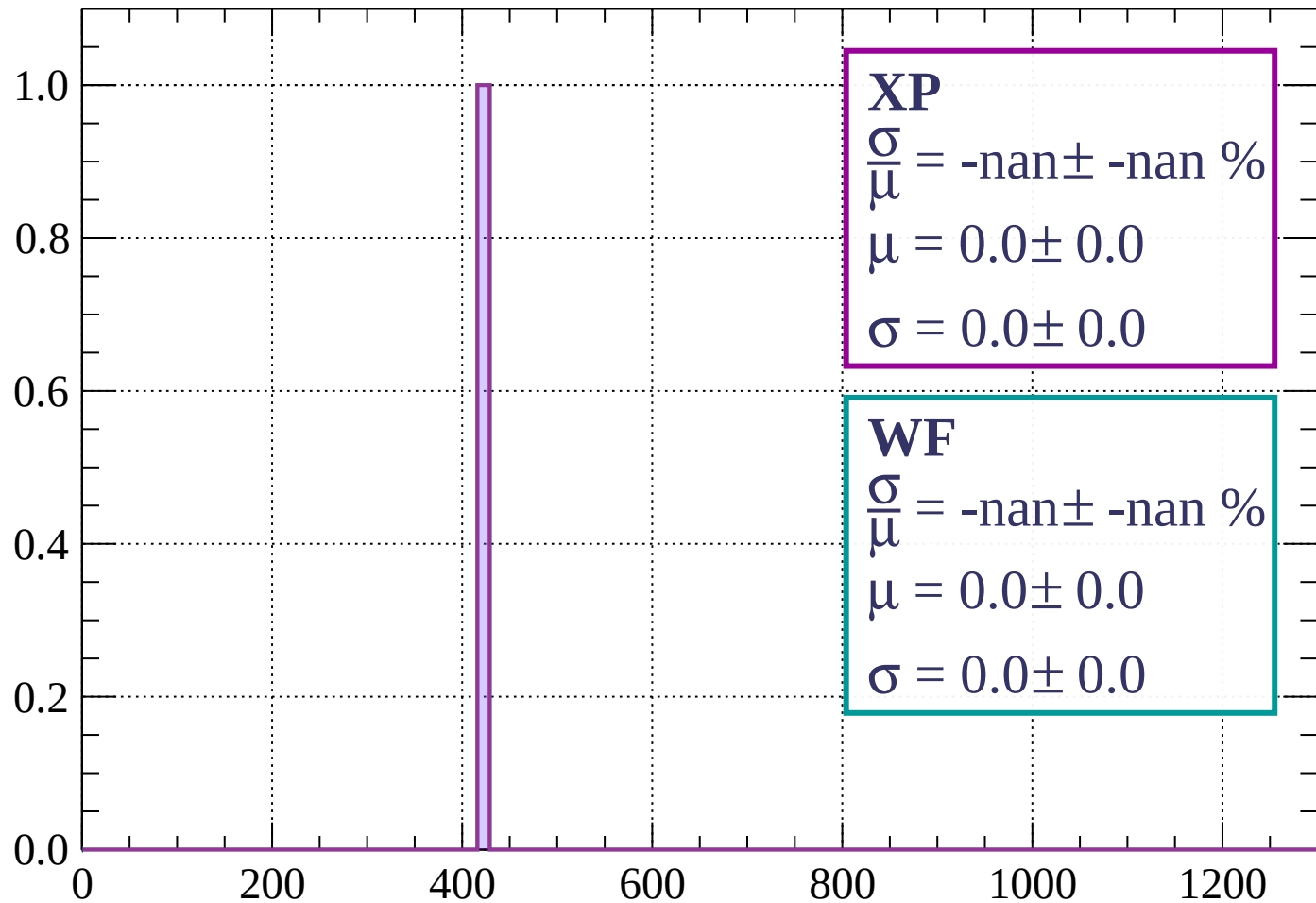
Energy loss | $1920 < p < 2000$

Count



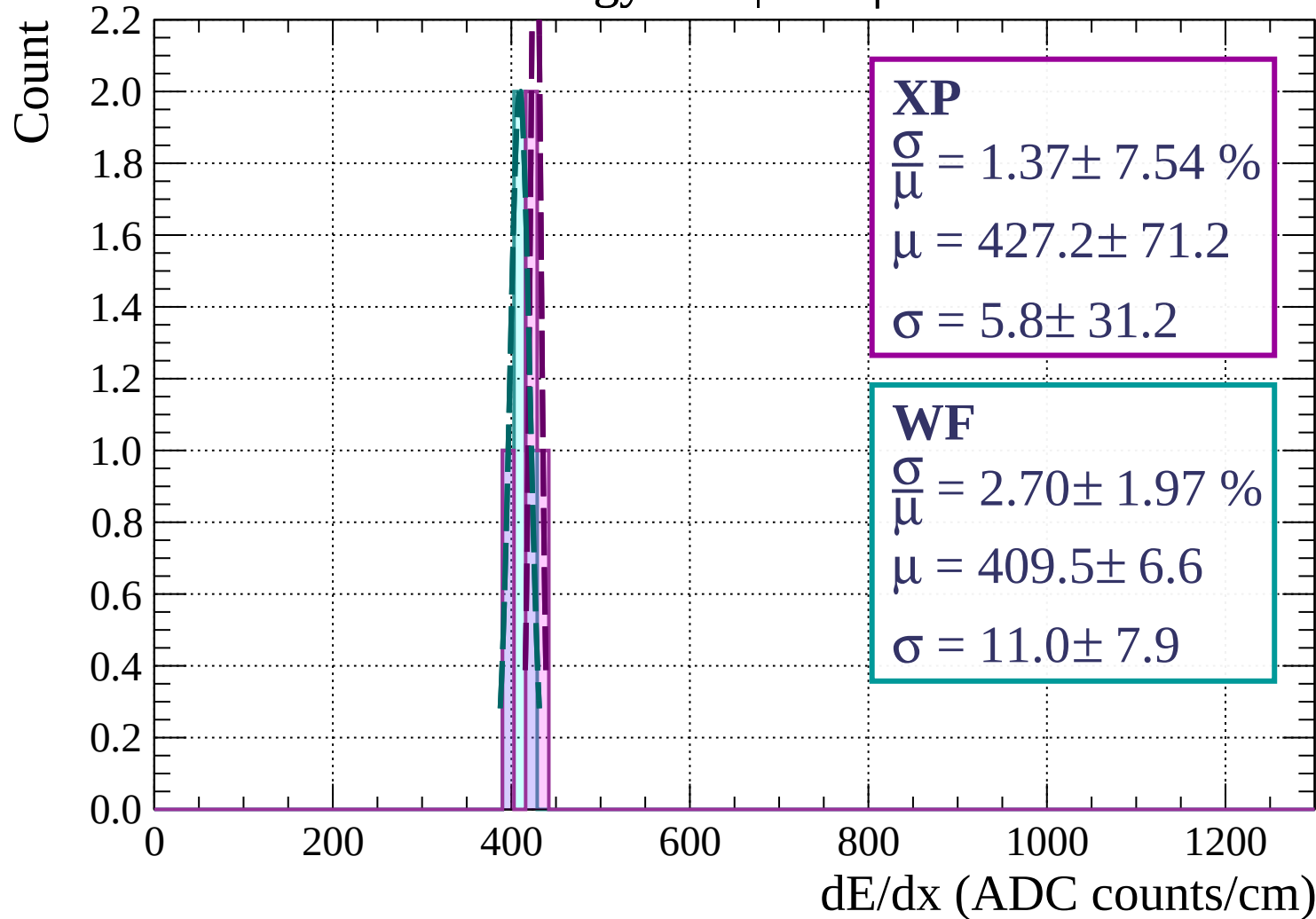
Energy loss | $0 < \phi < 3$

Count



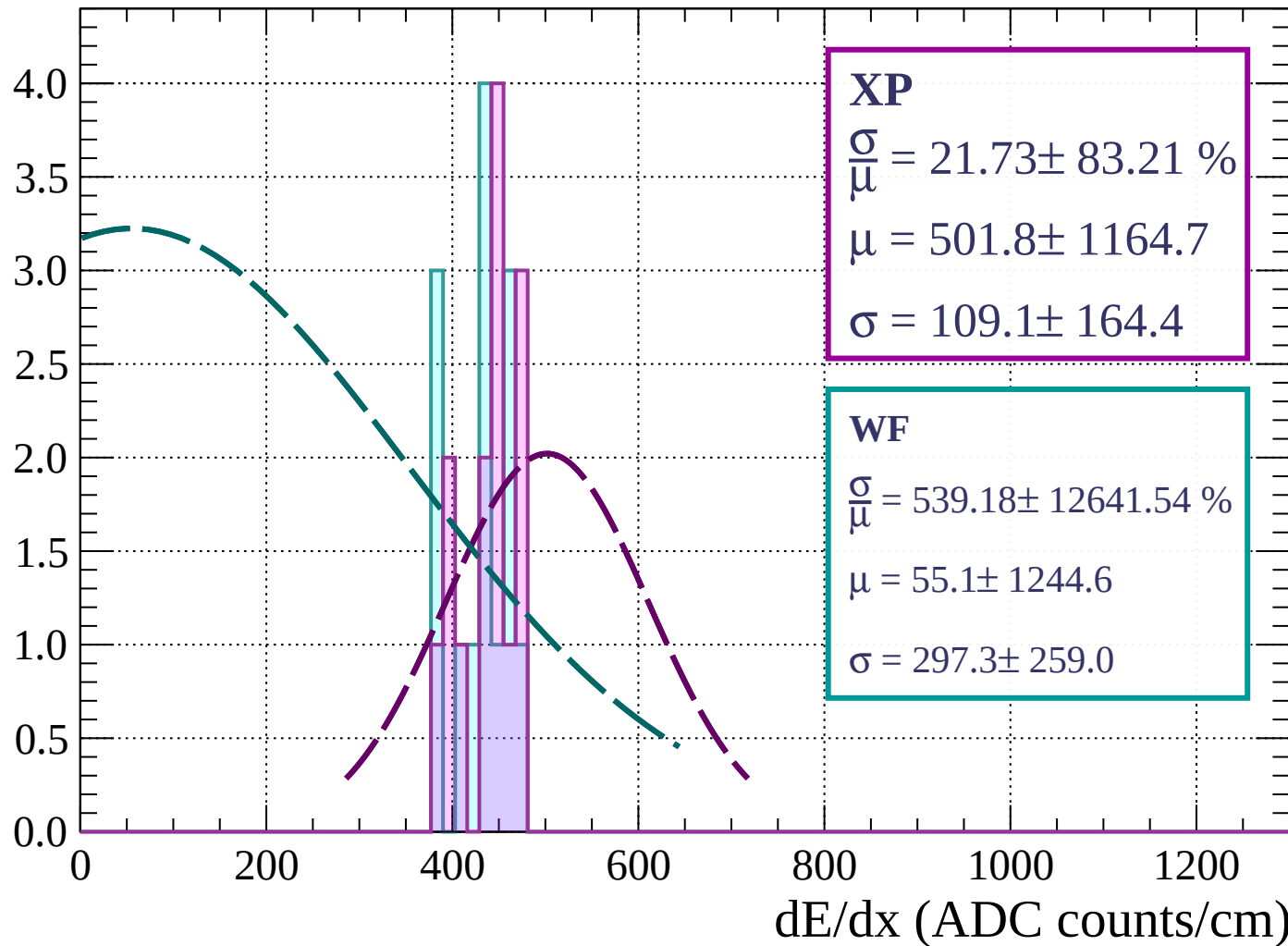
dE/dx (ADC counts/cm)

Energy loss | $3 < \phi < 6$



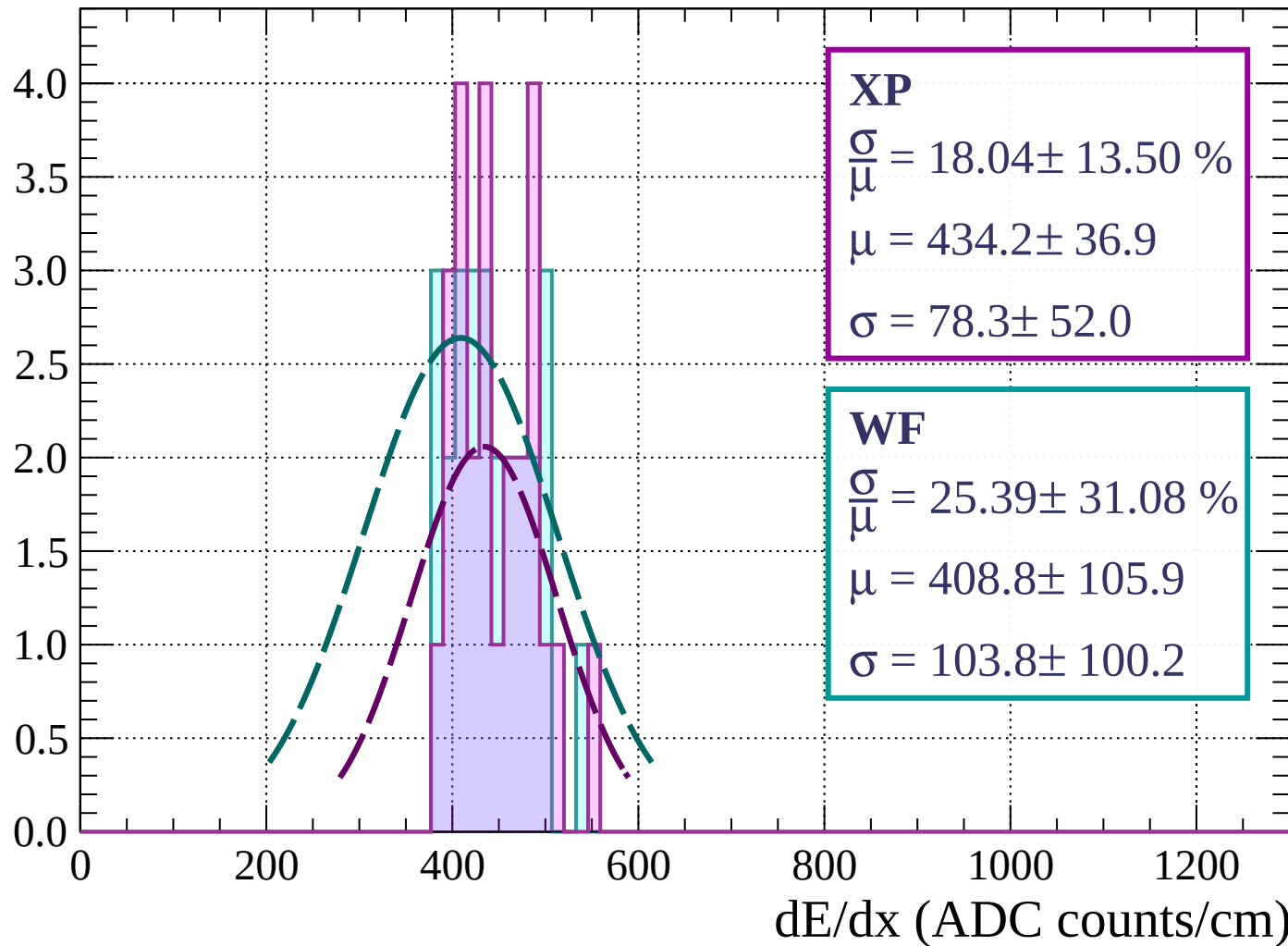
Energy loss | $6 < \phi < 9$

Count



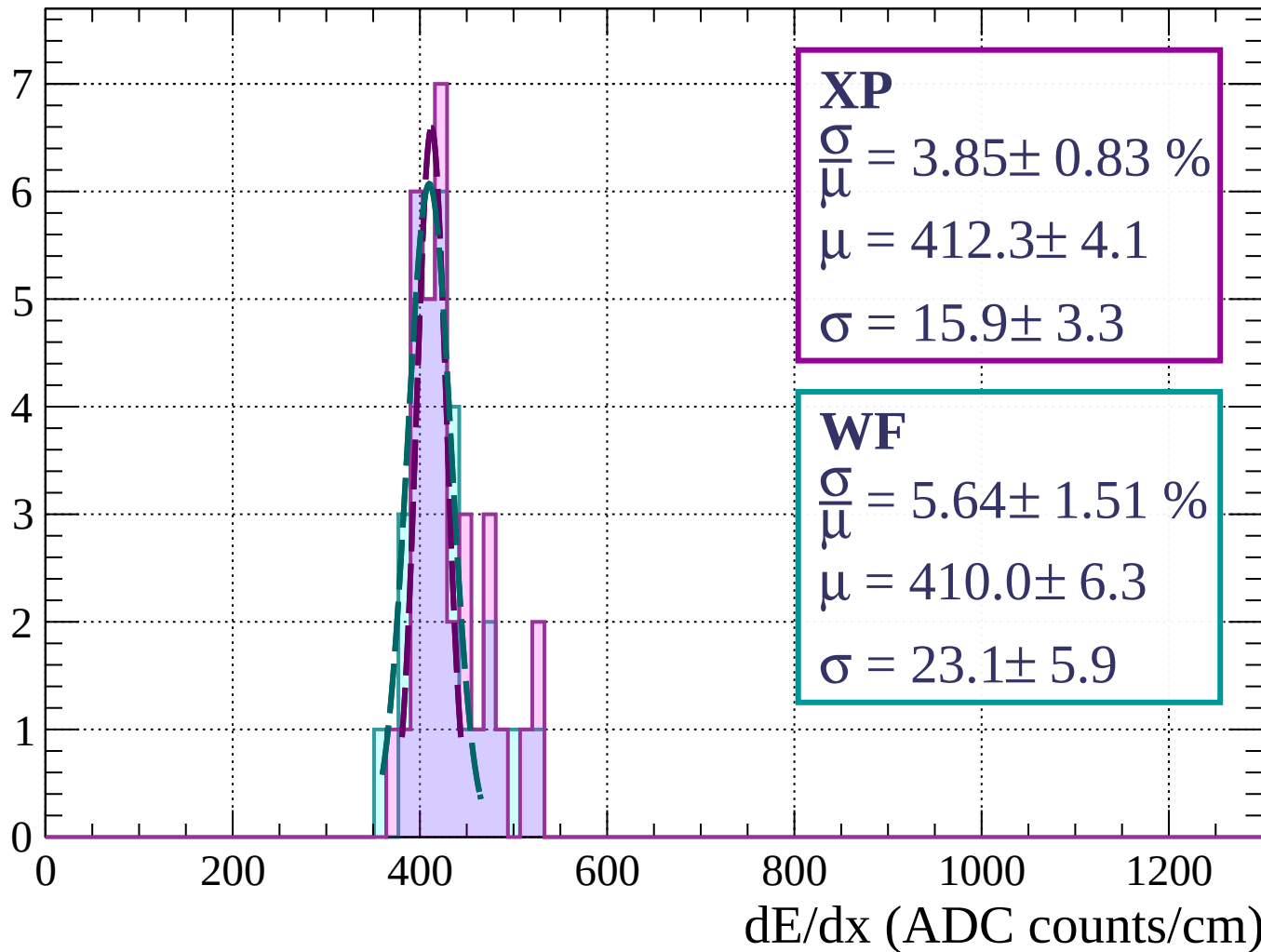
Energy loss | $9 < \phi < 12$

Count



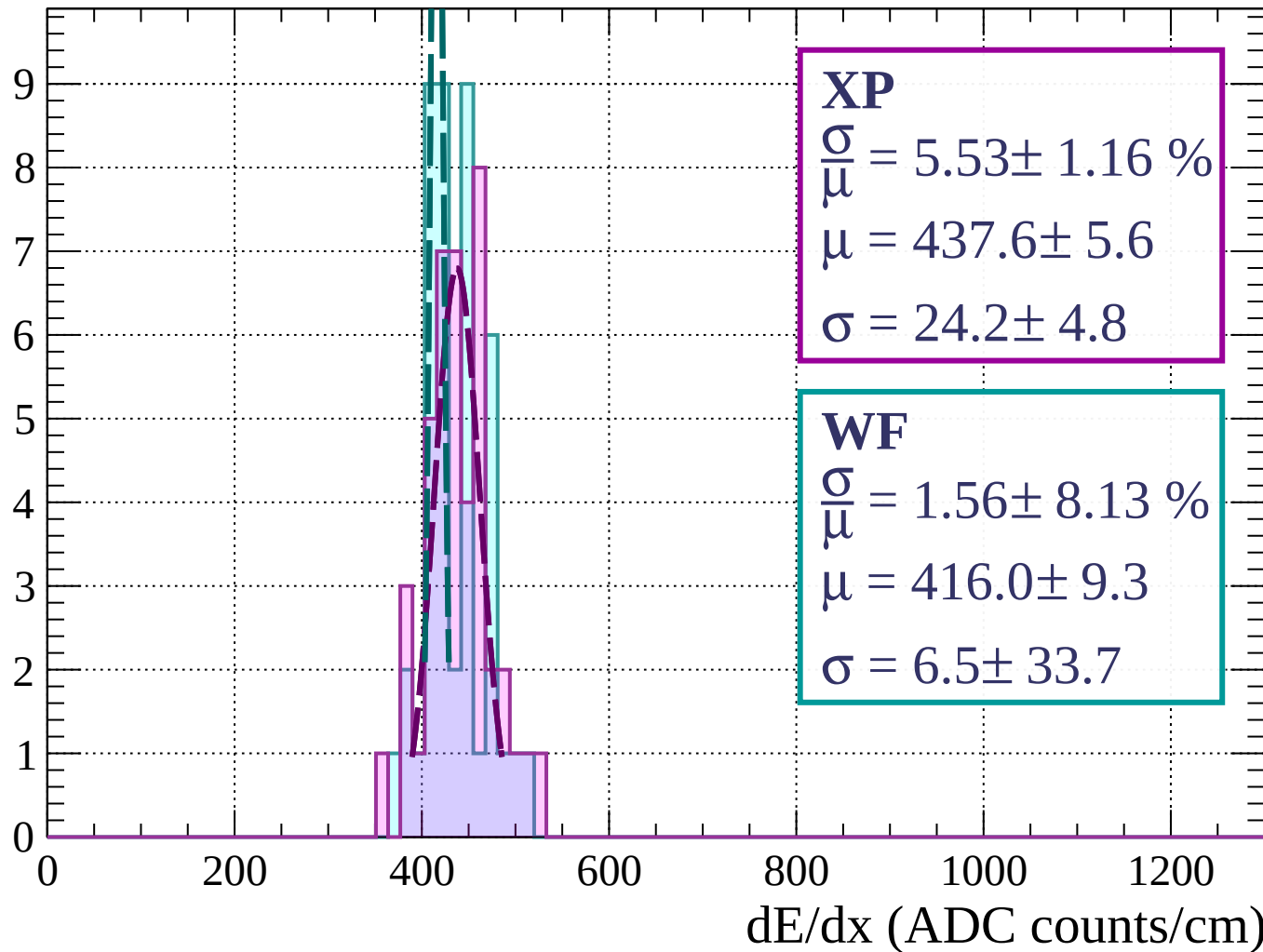
Energy loss | $12 < \phi < 15$

Count



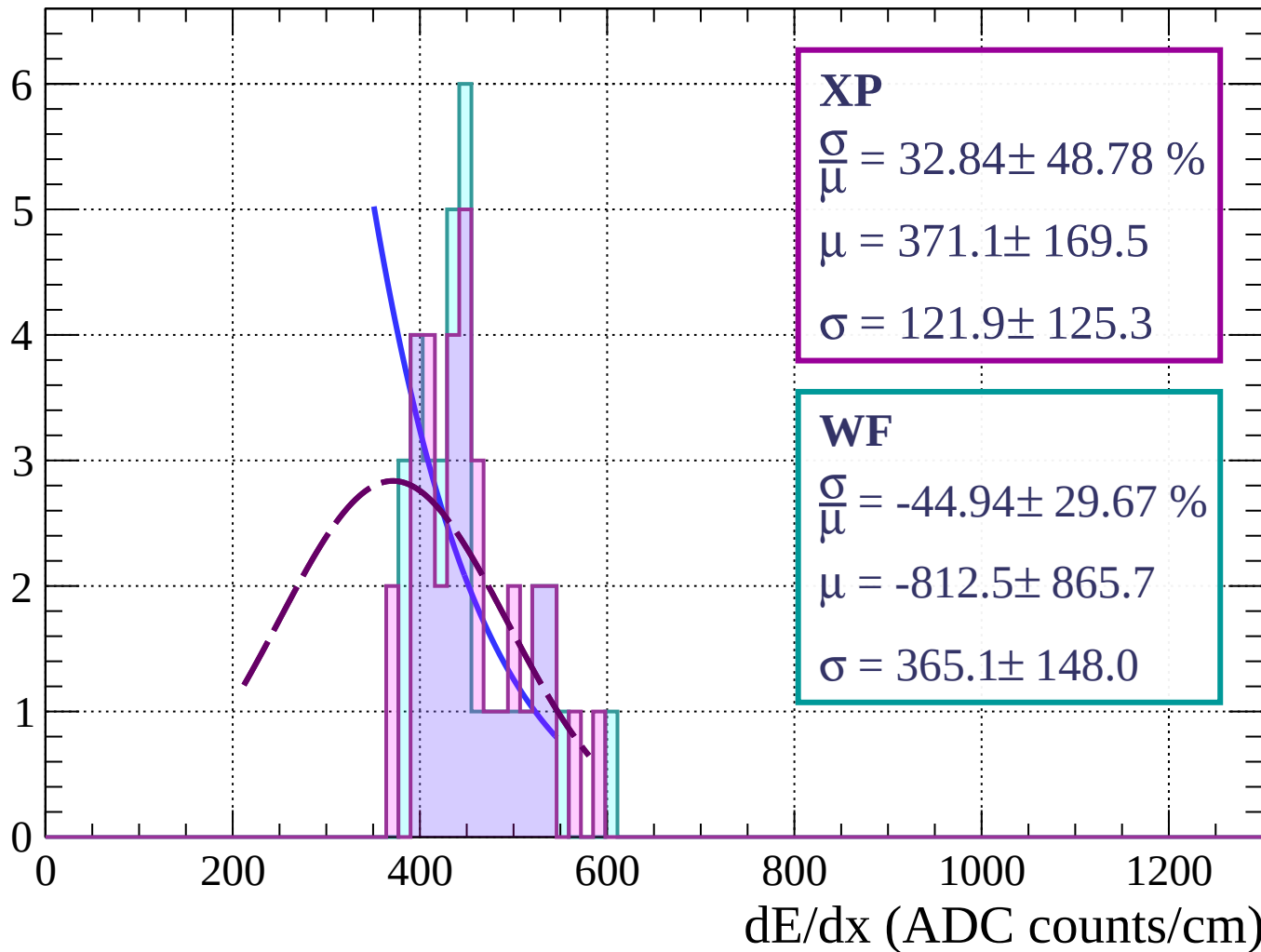
Energy loss | $15 < \phi < 18$

Count



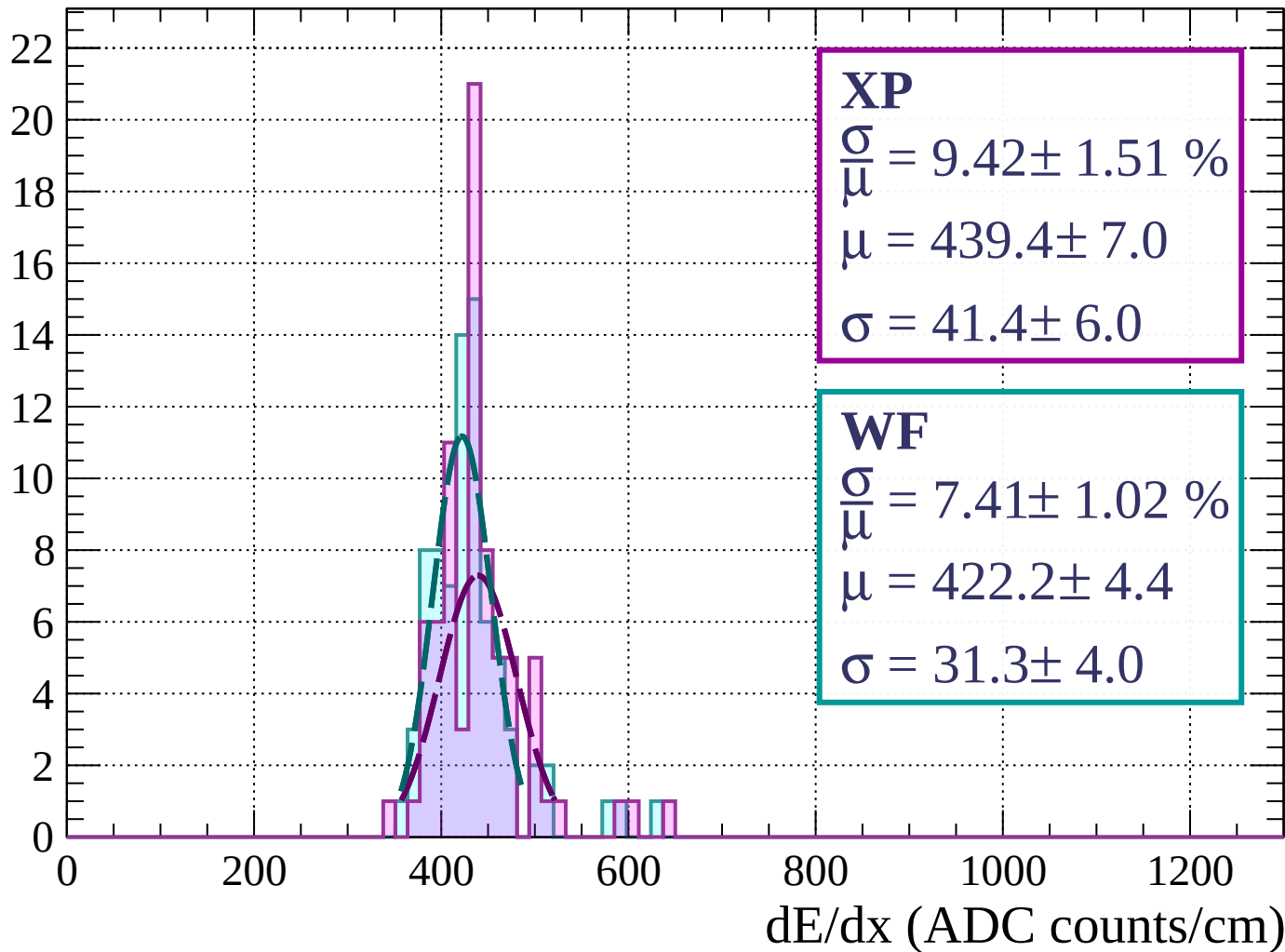
Energy loss | $18 < \phi < 21$

Count



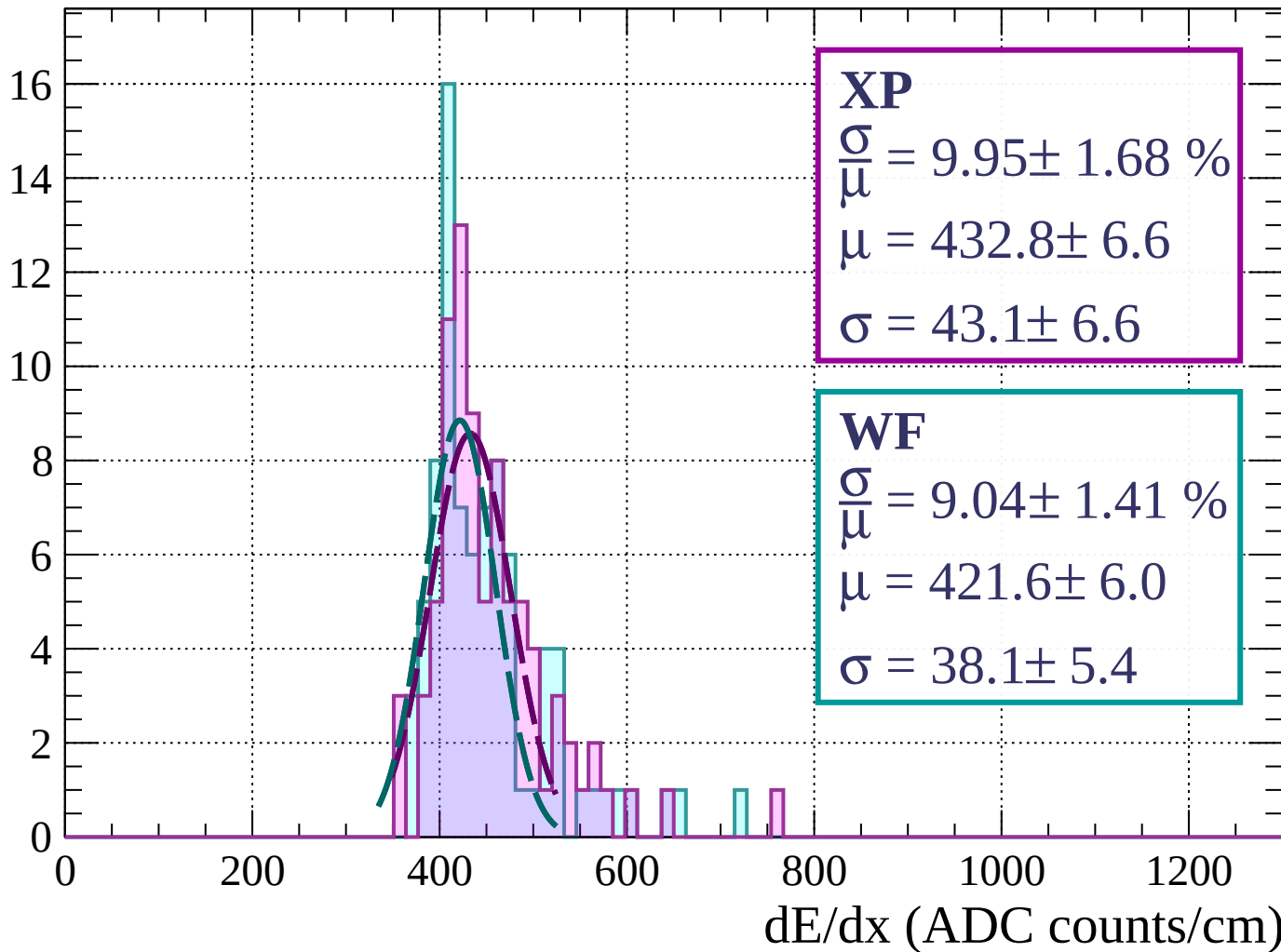
Energy loss | $21 < \phi < 24$

Count



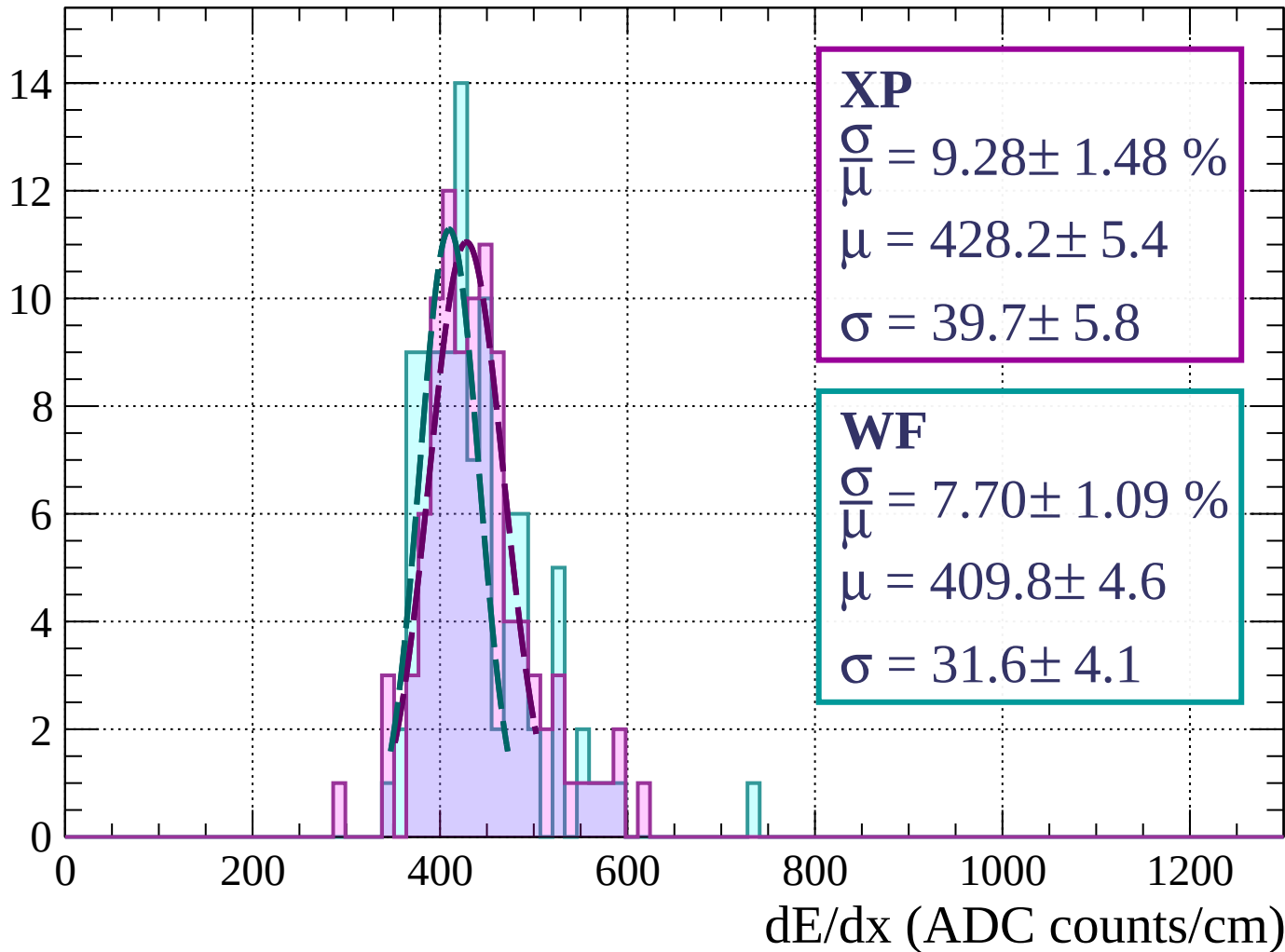
Energy loss | $24 < \phi < 27$

Count



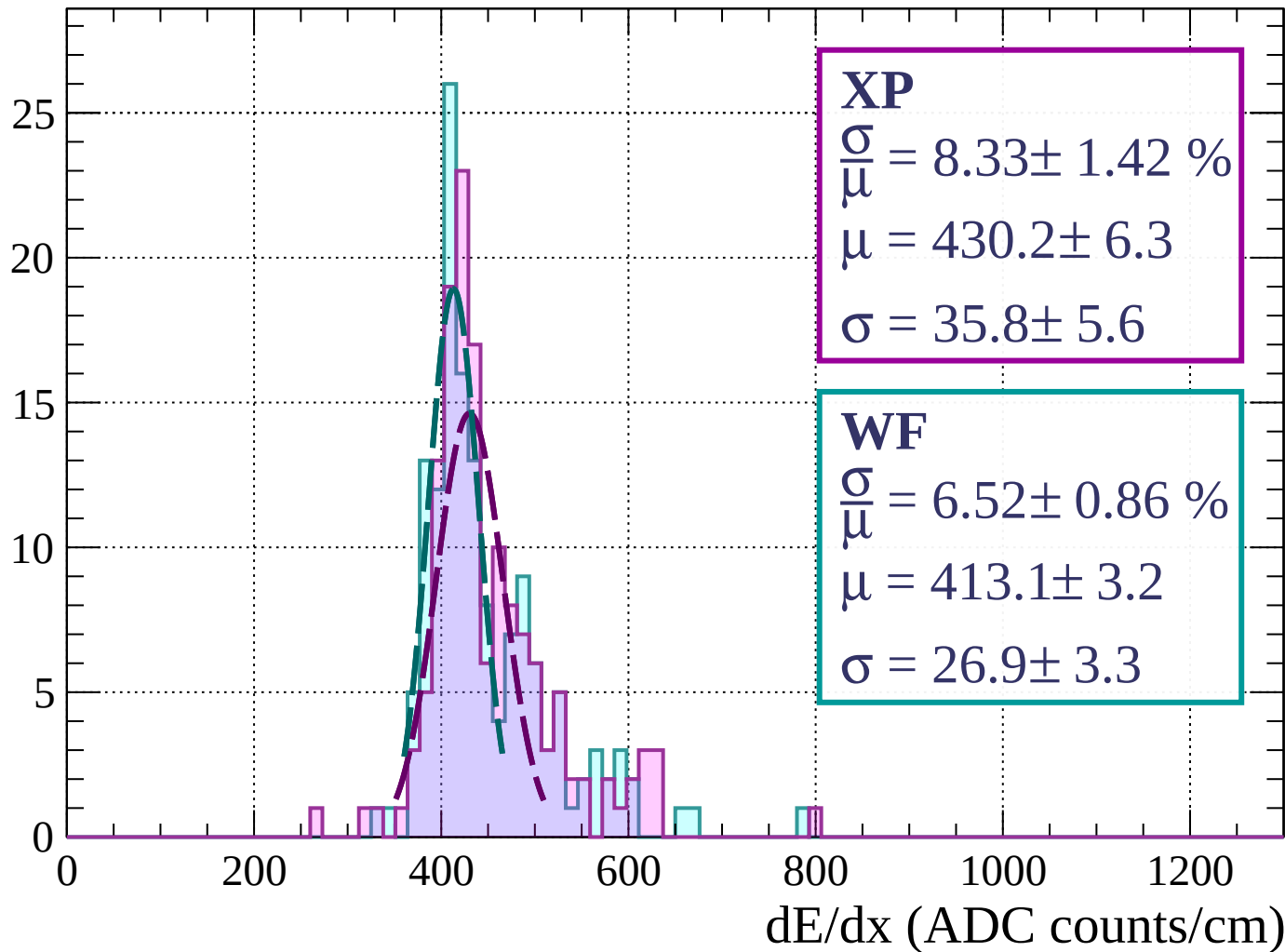
Energy loss | $27 < \phi < 30$

Count



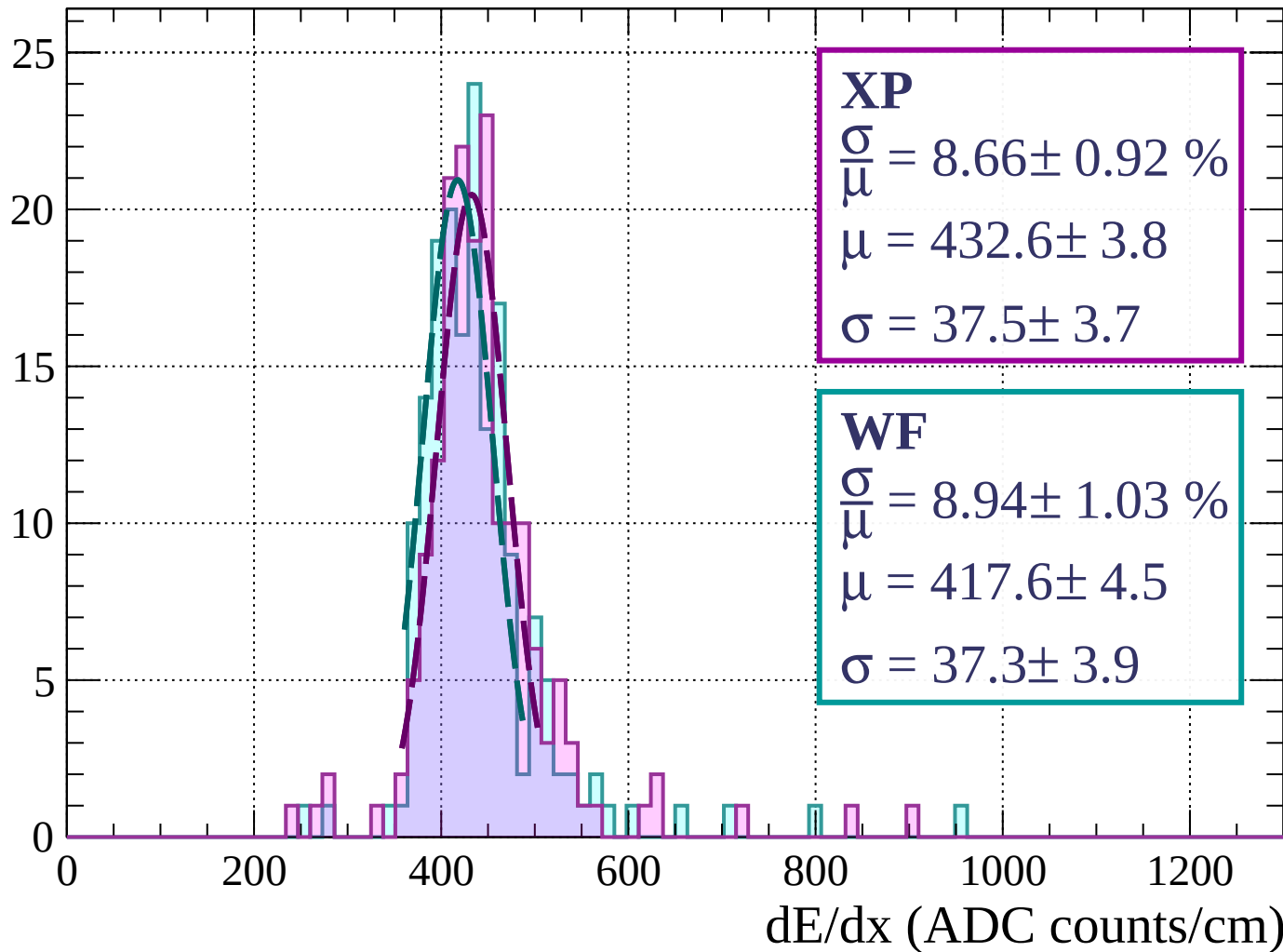
Energy loss | $30 < \phi < 33$

Count



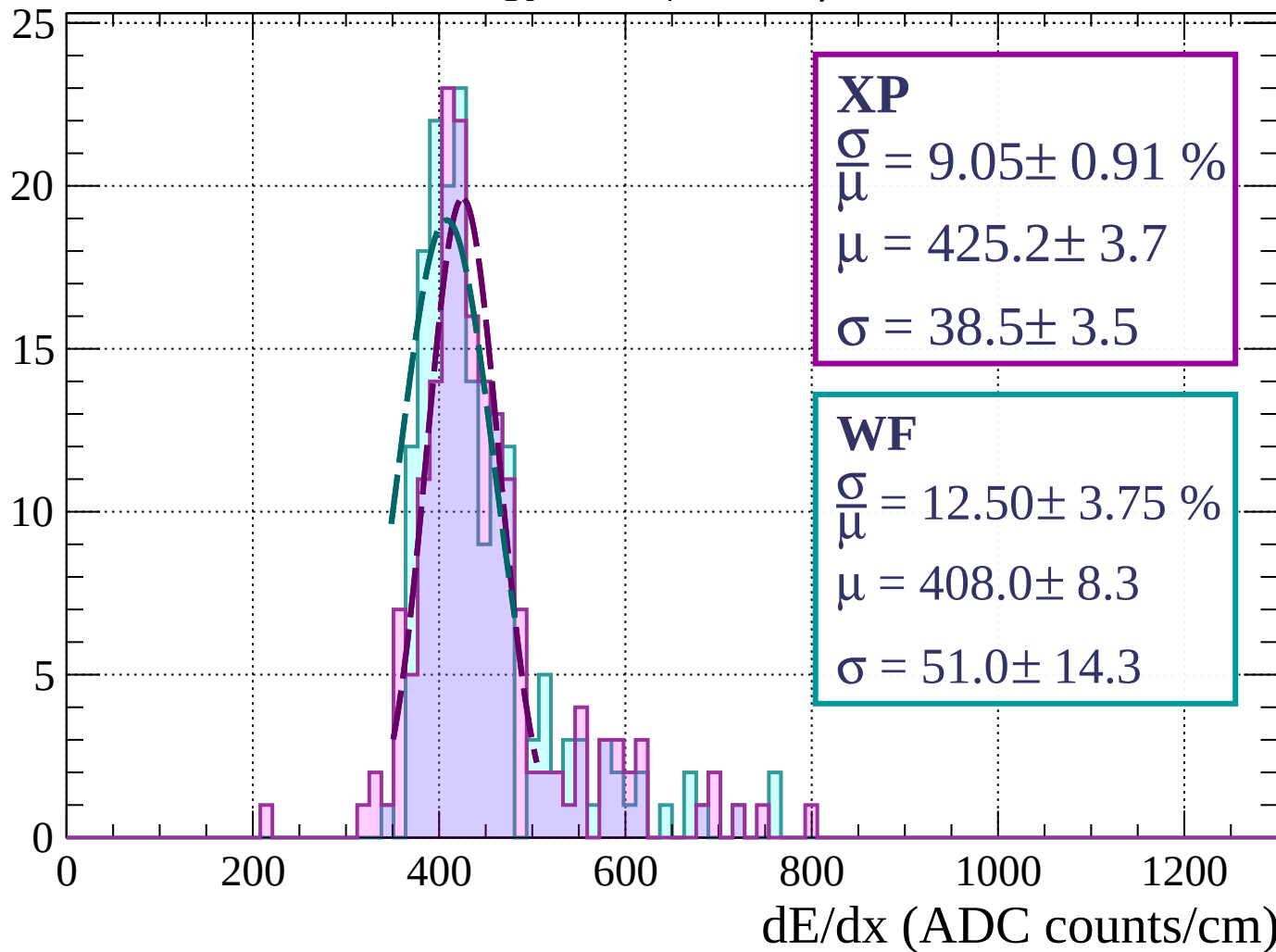
Energy loss | $33 < \phi < 36$

Count



Energy loss | $36 < \phi < 39$

Count



Energy loss | $39 < \phi < 42$

Count

XP

$$\frac{\sigma}{\mu} = 9.44 \pm 1.15 \%$$

$$\mu = 423.7 \pm 3.9$$

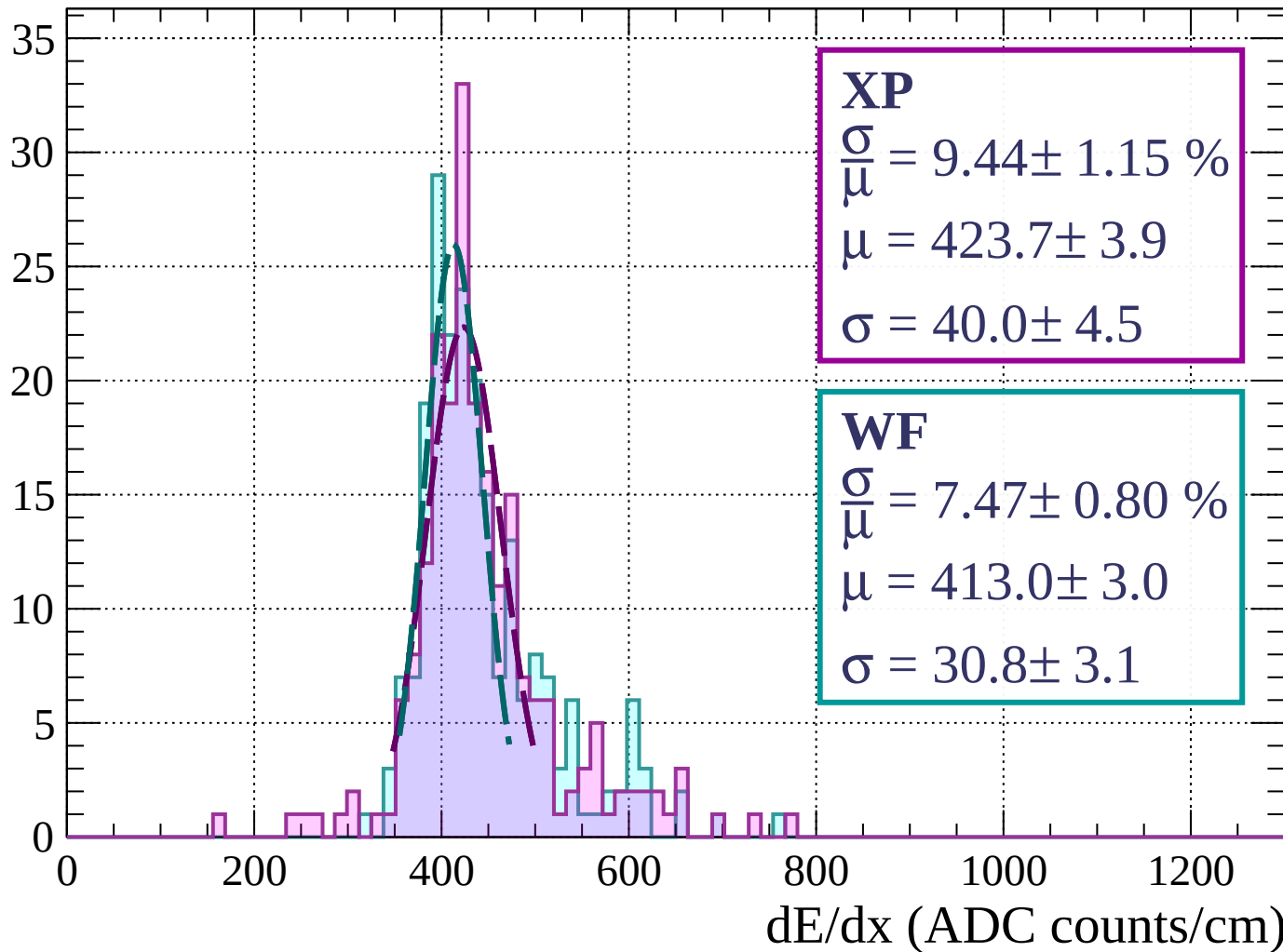
$$\sigma = 40.0 \pm 4.5$$

WF

$$\frac{\sigma}{\mu} = 7.47 \pm 0.80 \%$$

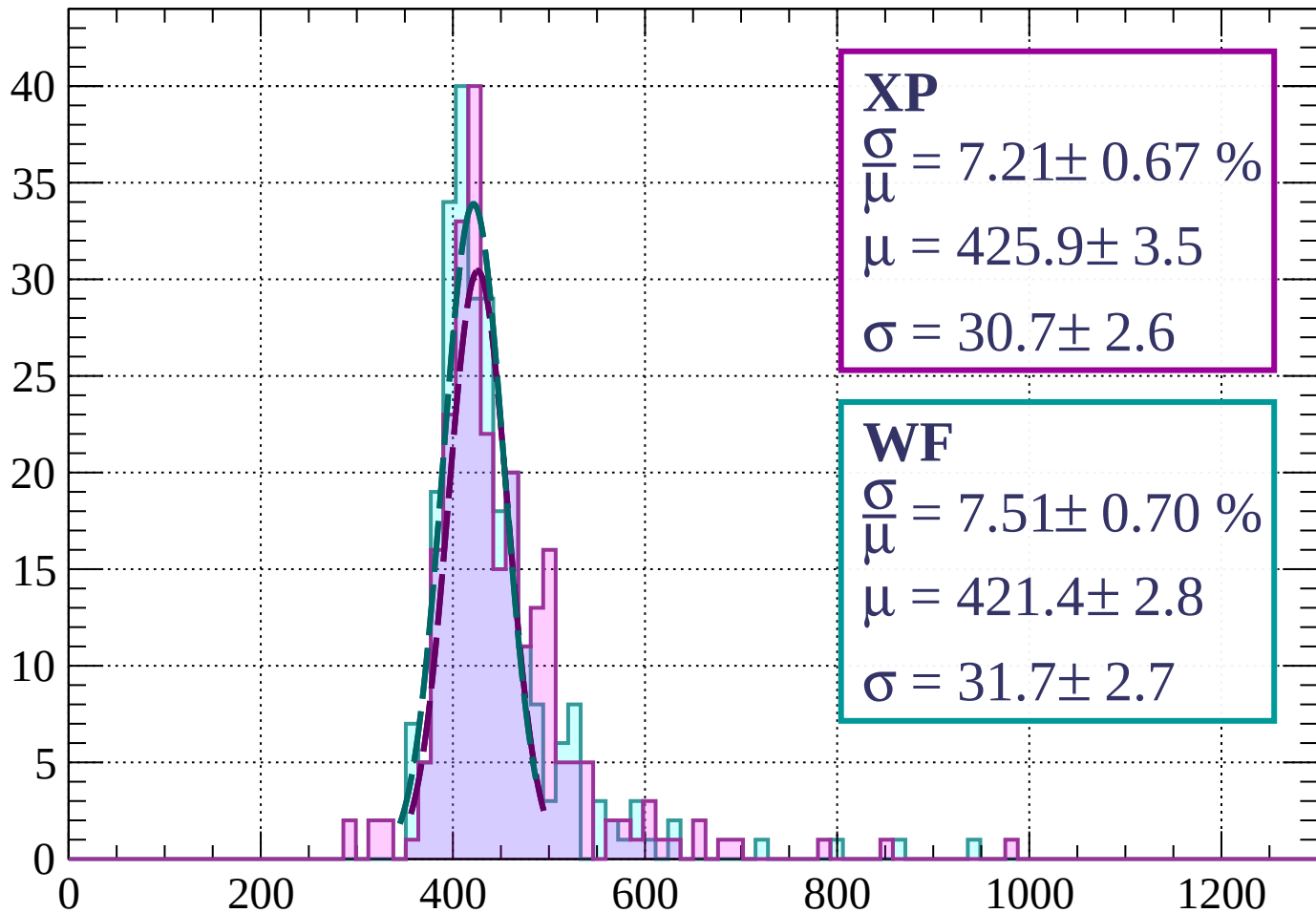
$$\mu = 413.0 \pm 3.0$$

$$\sigma = 30.8 \pm 3.1$$



Energy loss | $42 < \phi < 45$

Count



XP

$$\frac{\sigma}{\mu} = 7.21 \pm 0.67 \%$$

$$\mu = 425.9 \pm 3.5$$

$$\sigma = 30.7 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 7.51 \pm 0.70 \%$$

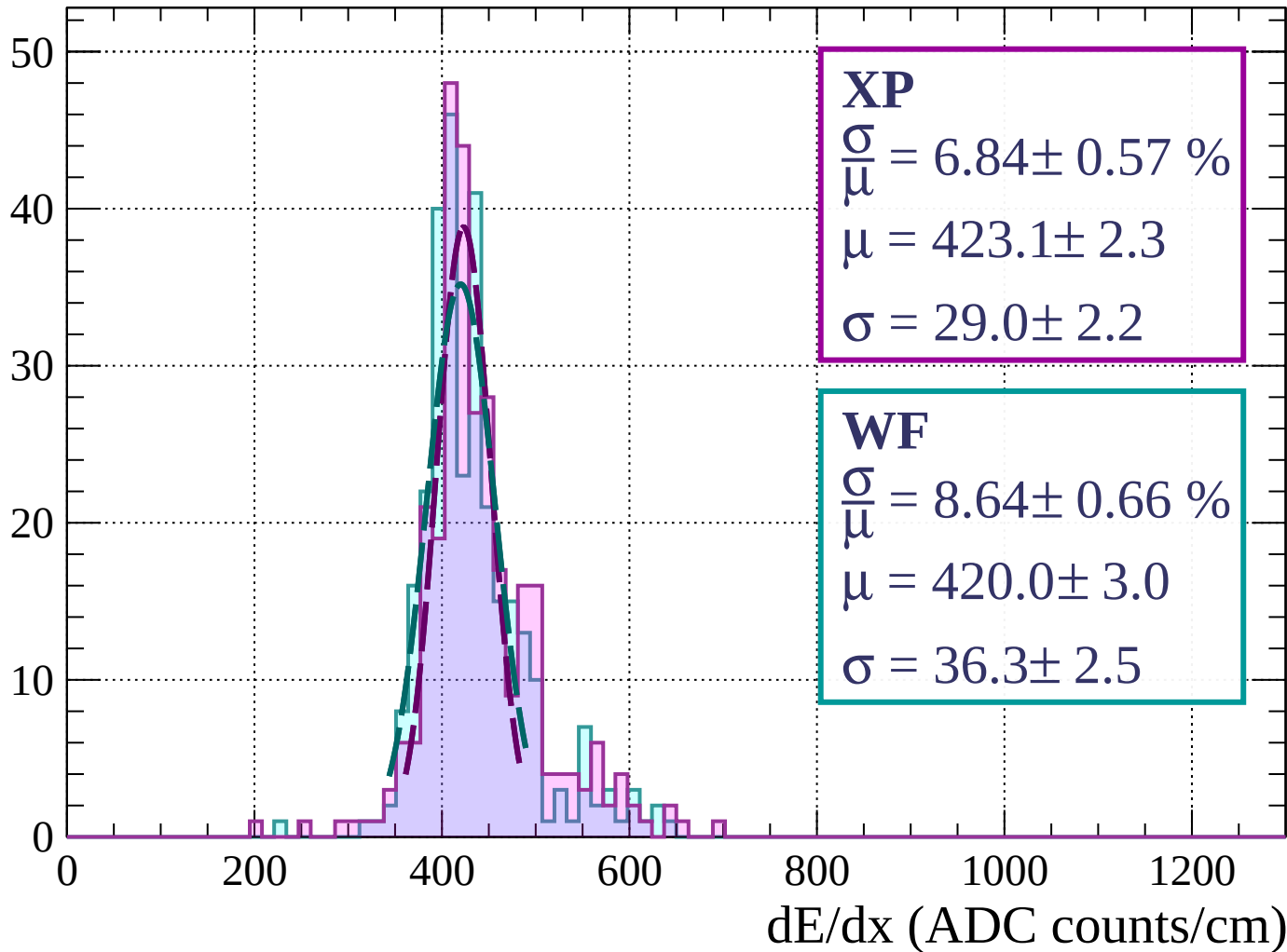
$$\mu = 421.4 \pm 2.8$$

$$\sigma = 31.7 \pm 2.7$$

dE/dx (ADC counts/cm)

Energy loss | $45 < \phi < 48$

Count



Energy loss | $48 < \phi < 51$

Count

XP

$$\frac{\sigma}{\mu} = 7.60 \pm 0.79 \%$$

$$\mu = 424.0 \pm 2.5$$

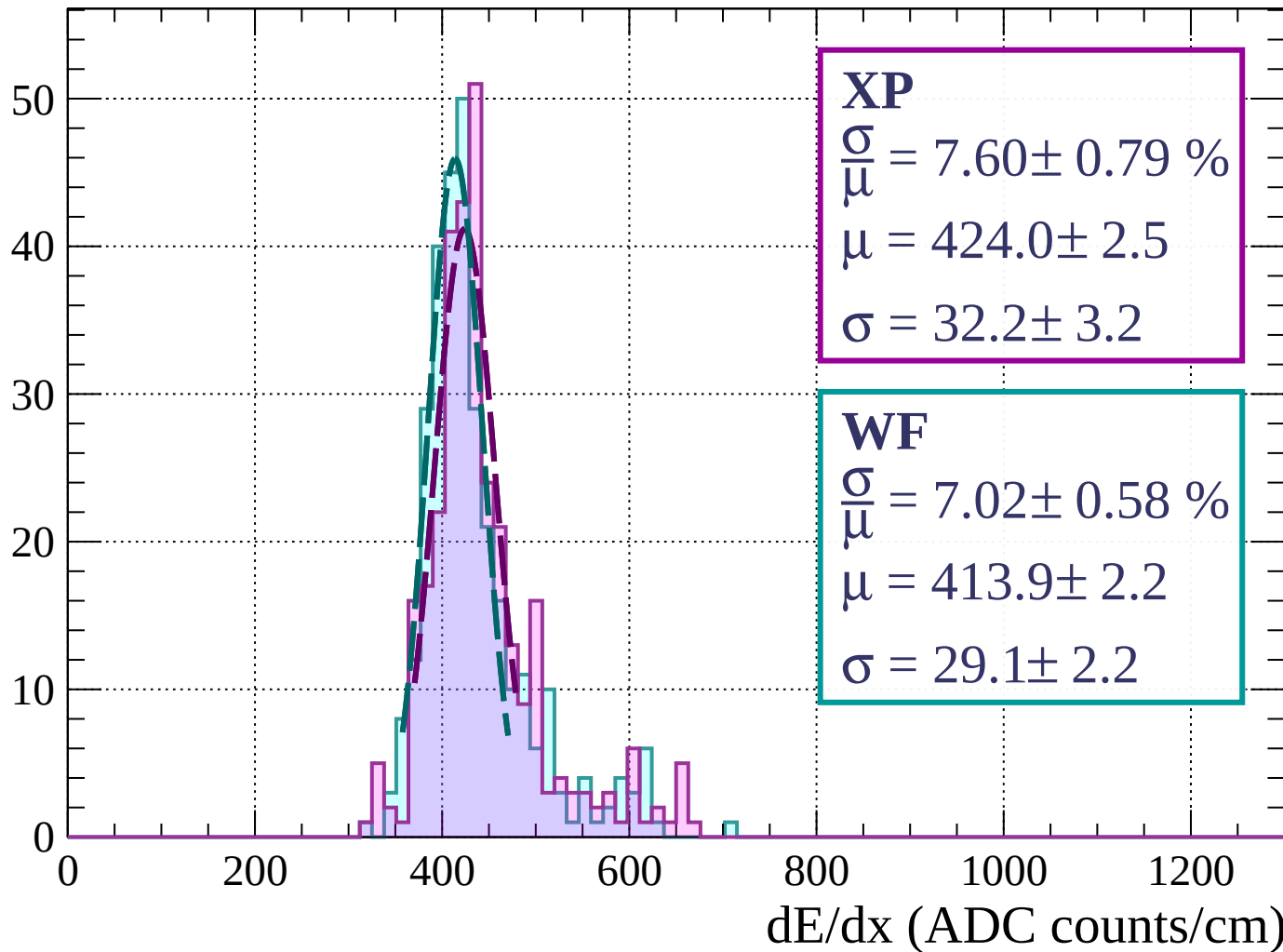
$$\sigma = 32.2 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 7.02 \pm 0.58 \%$$

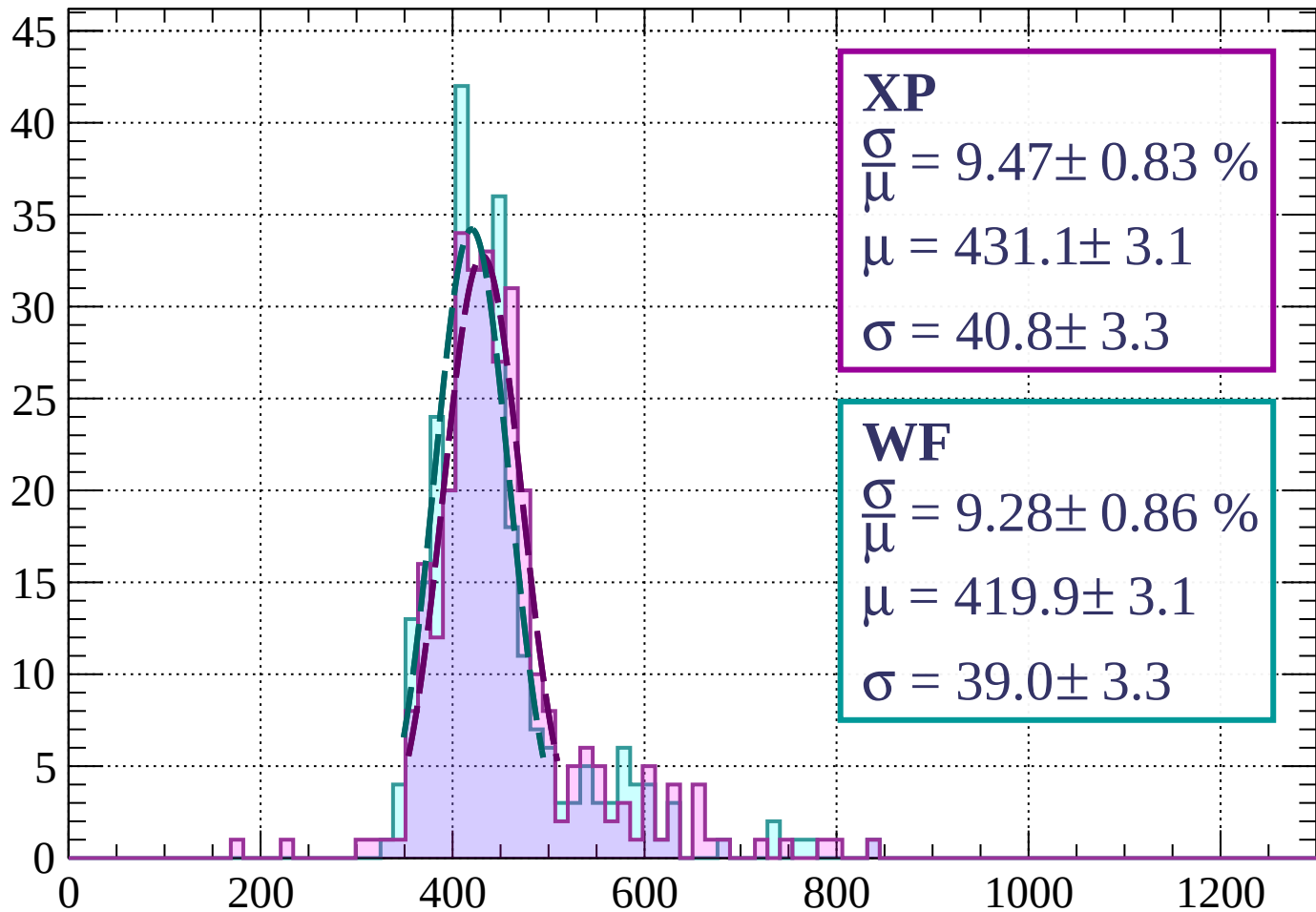
$$\mu = 413.9 \pm 2.2$$

$$\sigma = 29.1 \pm 2.2$$



Energy loss | $51 < \phi < 54$

Count



XP

$$\frac{\sigma}{\mu} = 9.47 \pm 0.83 \%$$

$$\mu = 431.1 \pm 3.1$$

$$\sigma = 40.8 \pm 3.3$$

WF

$$\frac{\sigma}{\mu} = 9.28 \pm 0.86 \%$$

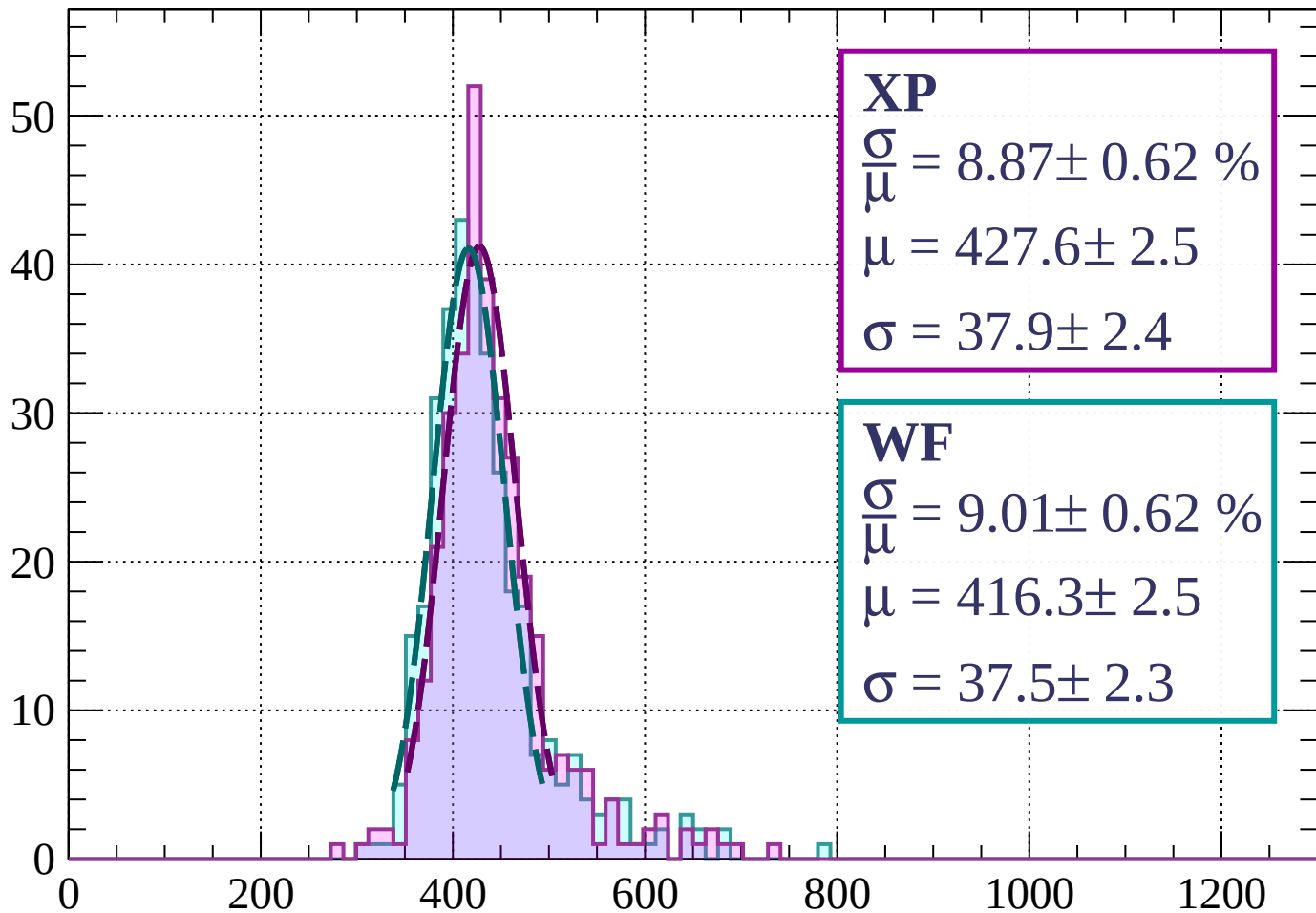
$$\mu = 419.9 \pm 3.1$$

$$\sigma = 39.0 \pm 3.3$$

dE/dx (ADC counts/cm)

Energy loss | $54 < \phi < 57$

Count



XP

$$\frac{\sigma}{\mu} = 8.87 \pm 0.62 \%$$

$$\mu = 427.6 \pm 2.5$$

$$\sigma = 37.9 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 9.01 \pm 0.62 \%$$

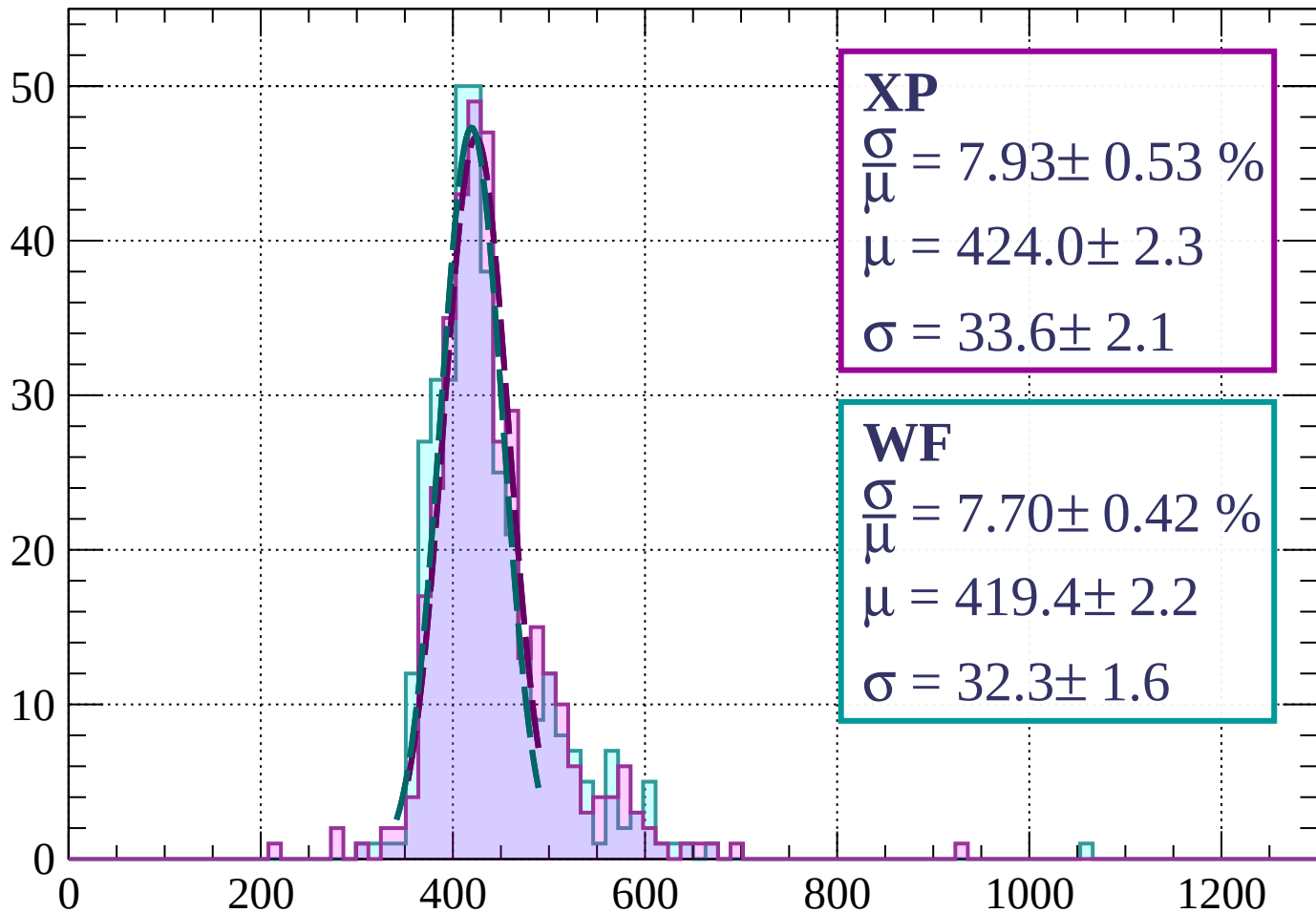
$$\mu = 416.3 \pm 2.5$$

$$\sigma = 37.5 \pm 2.3$$

dE/dx (ADC counts/cm)

Energy loss | $57 < \phi < 60$

Count



XP

$$\frac{\sigma}{\mu} = 7.93 \pm 0.53 \%$$

$$\mu = 424.0 \pm 2.3$$

$$\sigma = 33.6 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 7.70 \pm 0.42 \%$$

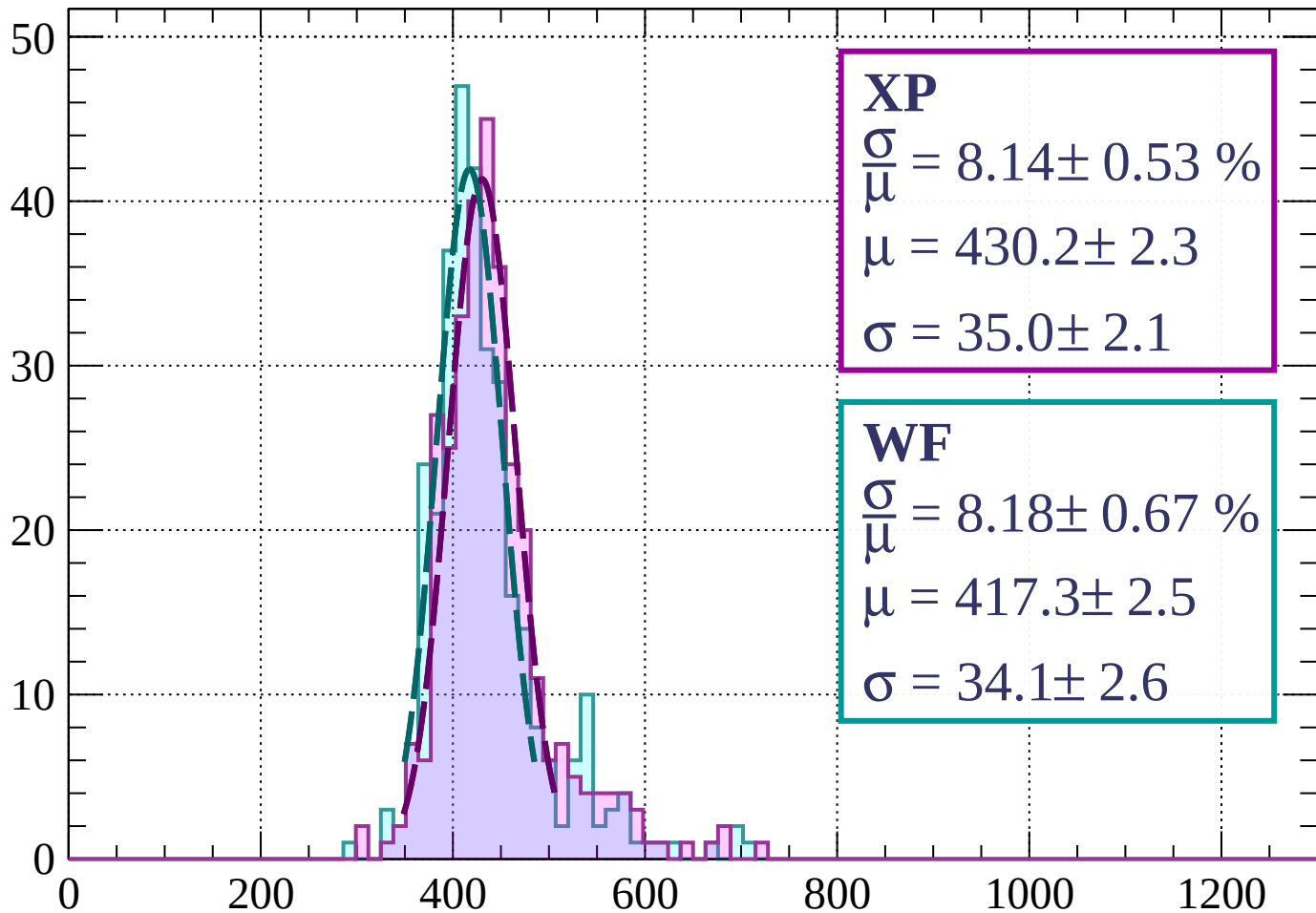
$$\mu = 419.4 \pm 2.2$$

$$\sigma = 32.3 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $60 < \phi < 63$

Count



XP

$$\frac{\sigma}{\mu} = 8.14 \pm 0.53 \%$$

$$\mu = 430.2 \pm 2.3$$

$$\sigma = 35.0 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 8.18 \pm 0.67 \%$$

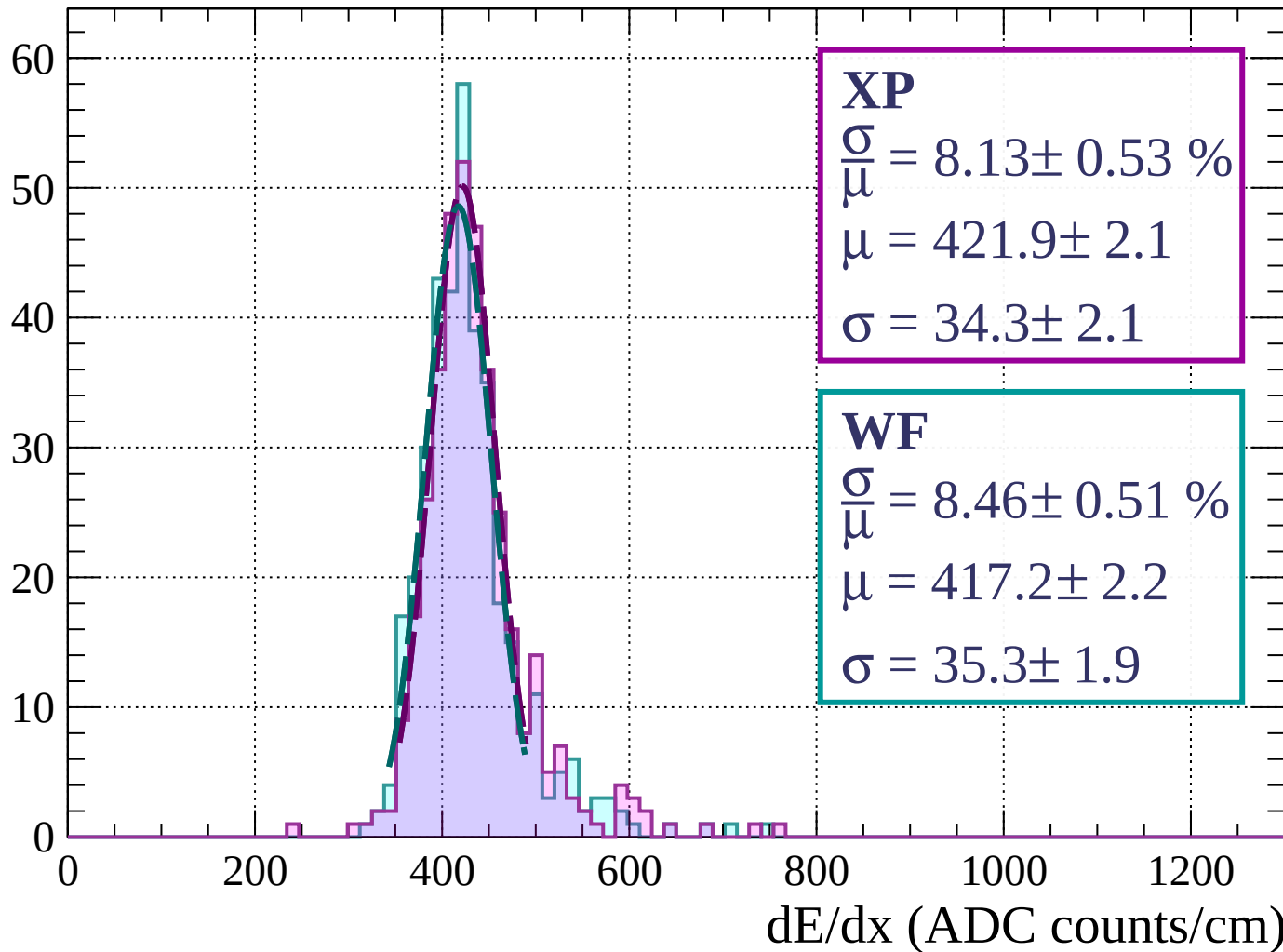
$$\mu = 417.3 \pm 2.5$$

$$\sigma = 34.1 \pm 2.6$$

dE/dx (ADC counts/cm)

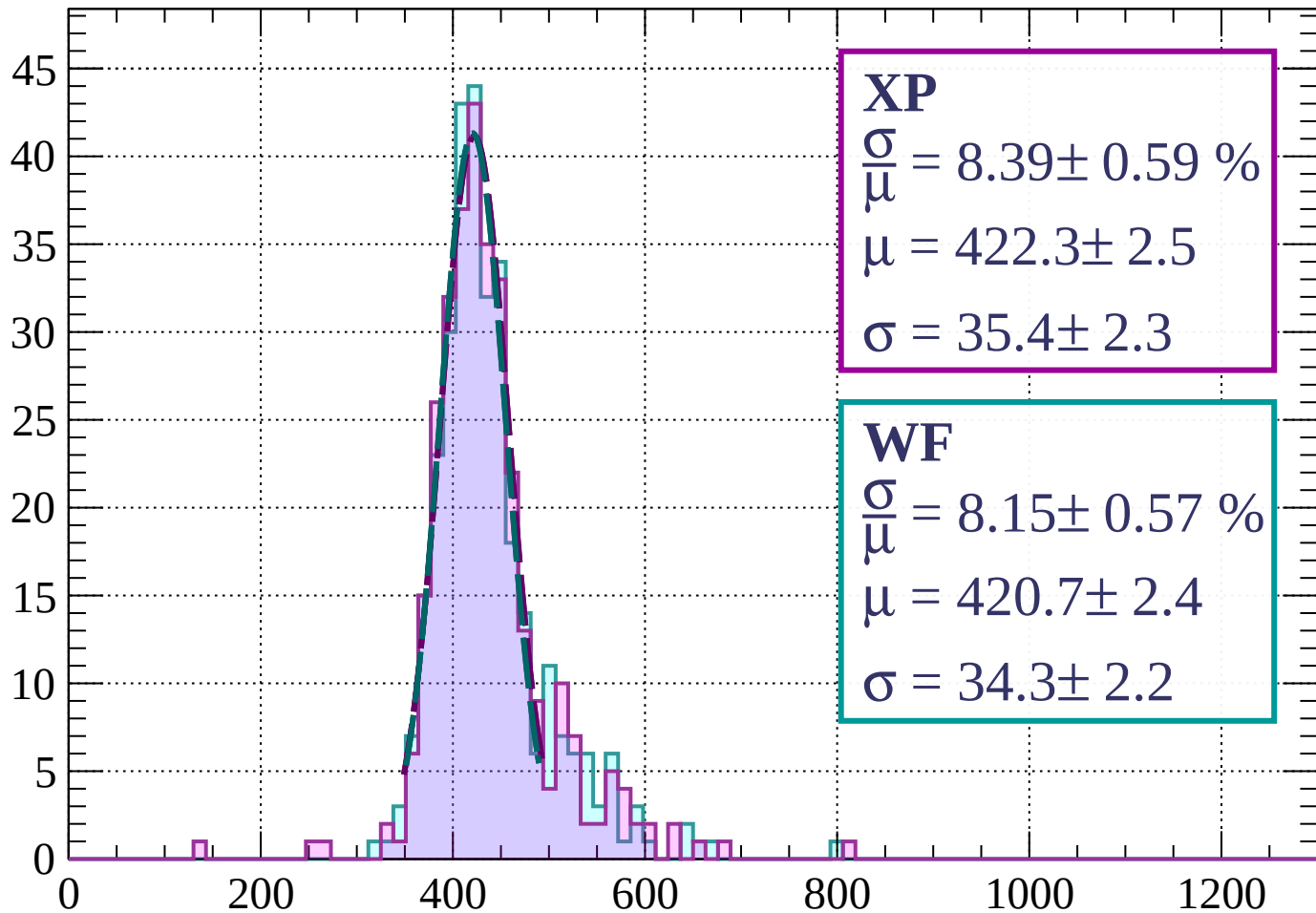
Energy loss | $63 < \phi < 66$

Count



Energy loss | $66 < \phi < 69$

Count



Energy loss | $69 < \phi < 72$

Count

XP

$$\frac{\sigma}{\mu} = 6.59 \pm 0.52 \%$$

$$\mu = 426.0 \pm 2.1$$

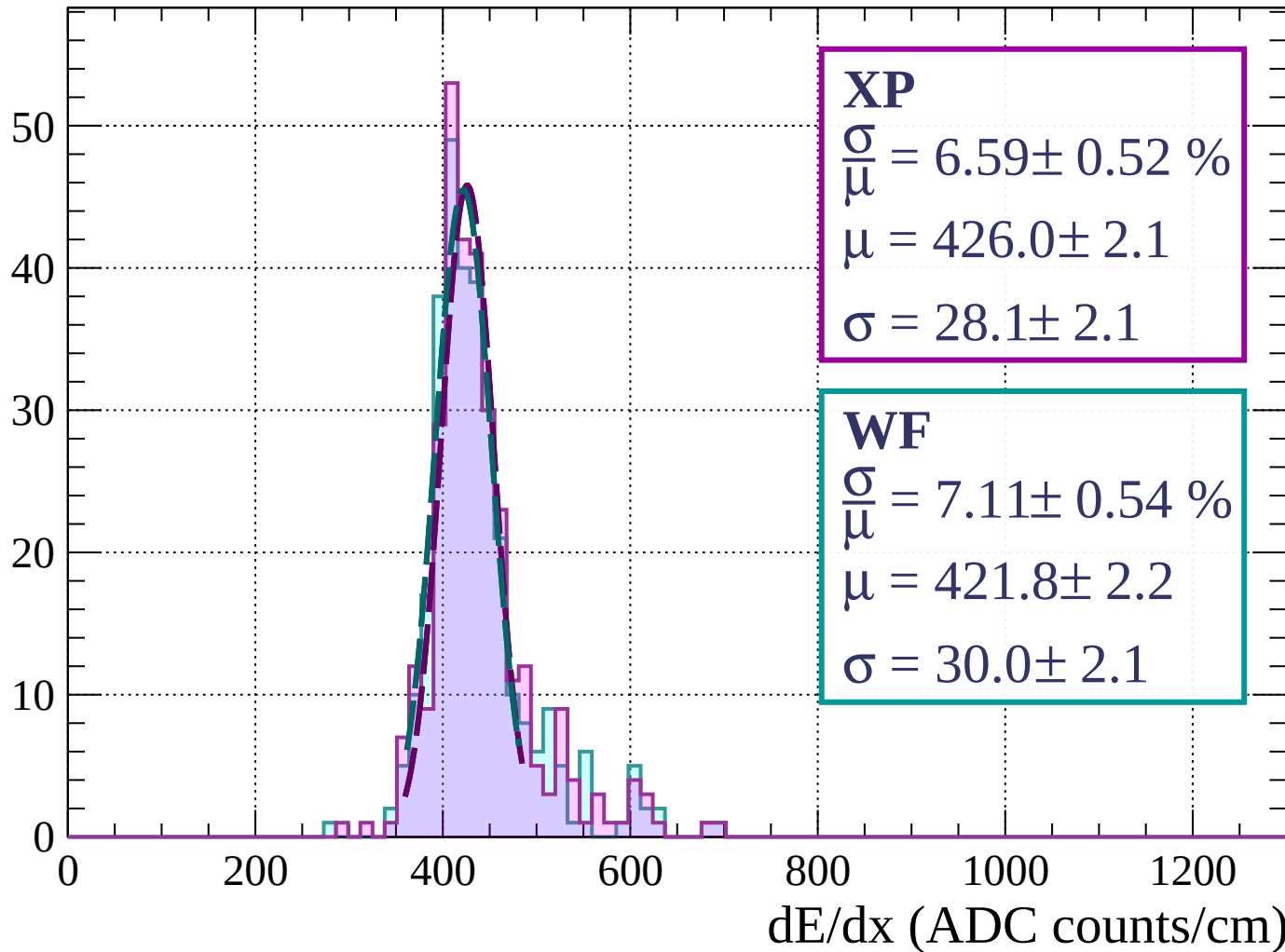
$$\sigma = 28.1 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 7.11 \pm 0.54 \%$$

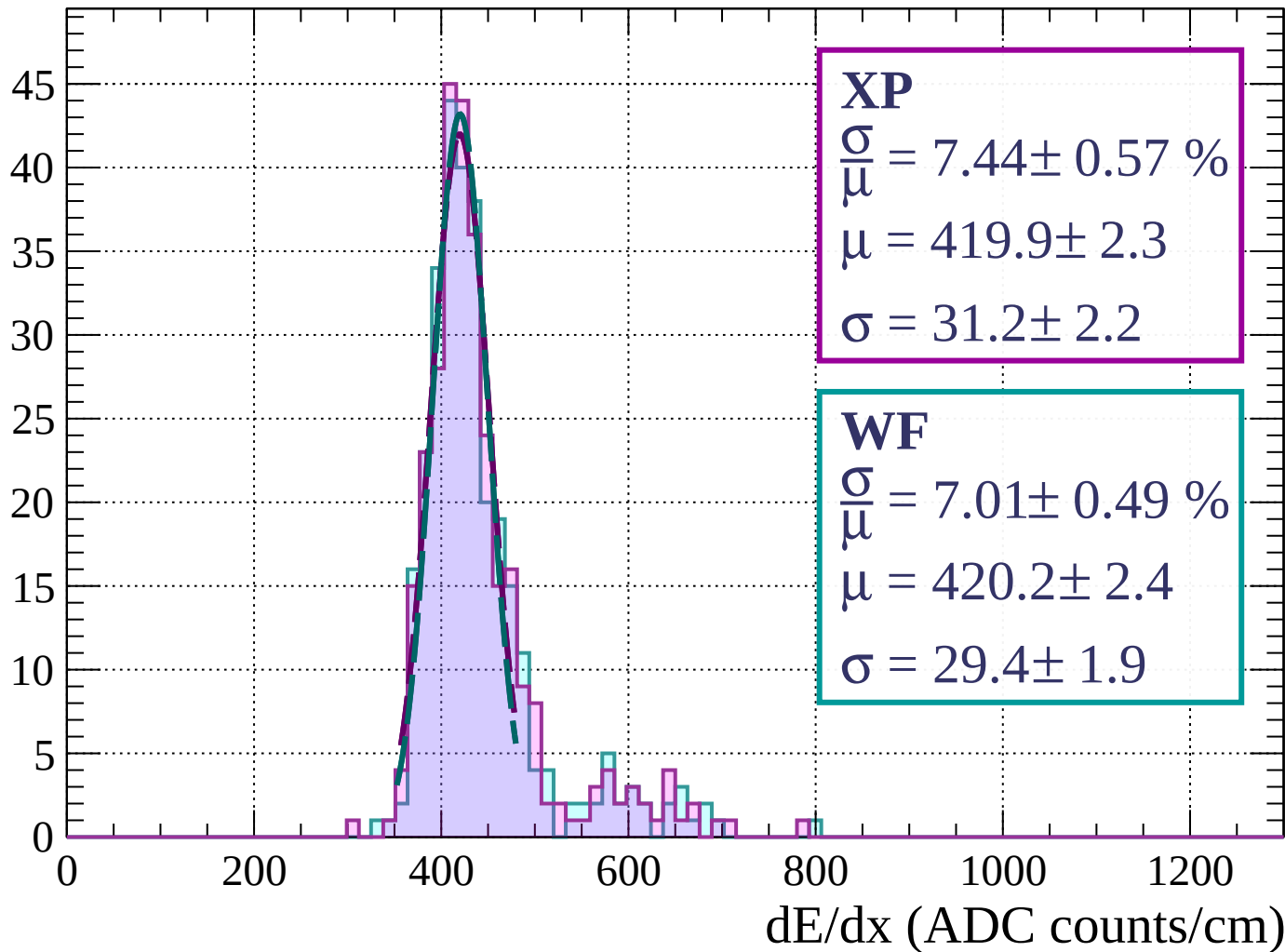
$$\mu = 421.8 \pm 2.2$$

$$\sigma = 30.0 \pm 2.1$$



Energy loss | $72 < \phi < 75$

Count



Energy loss | $75 < \phi < 78$

Count

XP

$$\frac{\sigma}{\mu} = 7.53 \pm 0.52 \%$$

$$\mu = 424.7 \pm 2.1$$

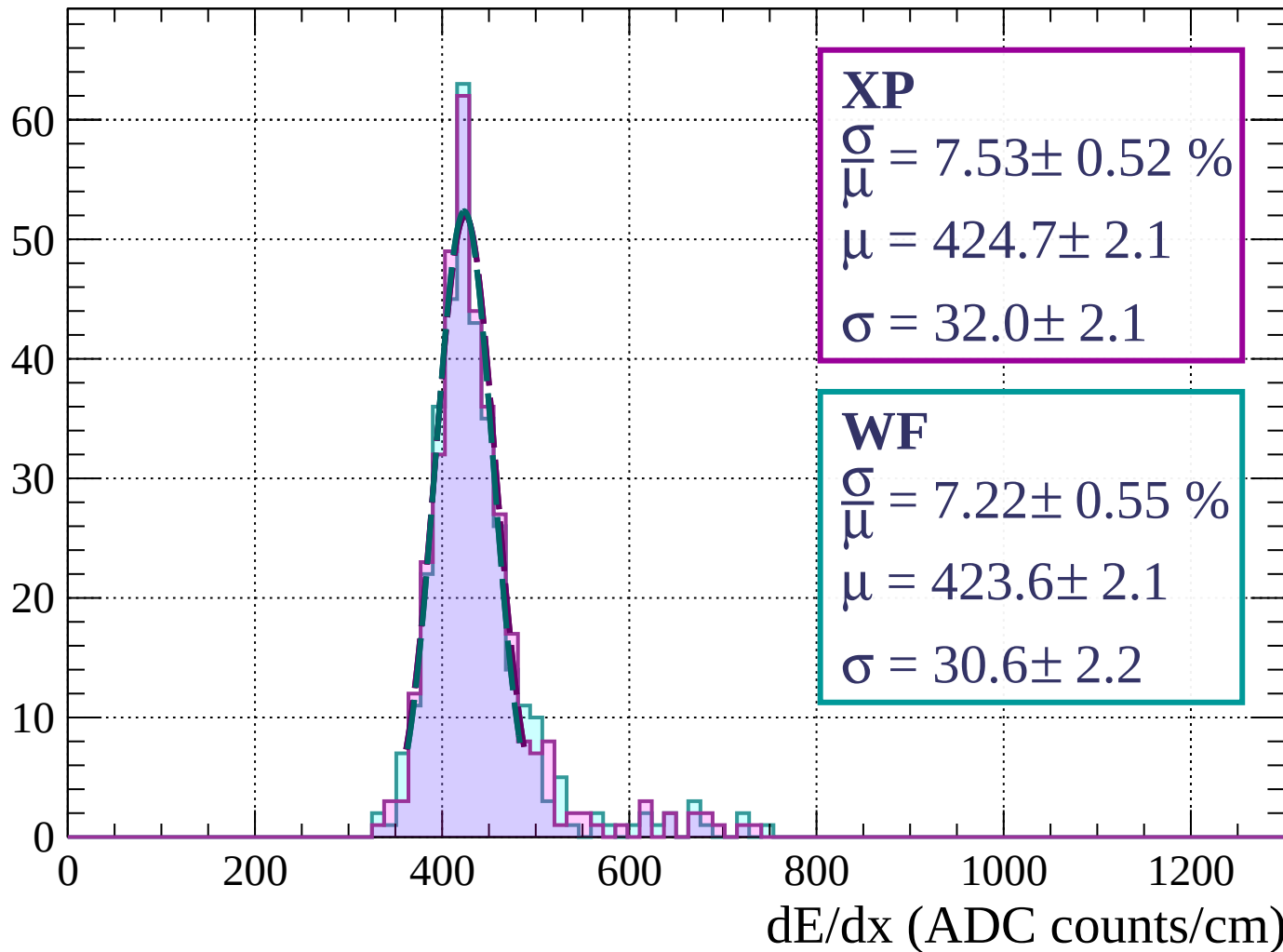
$$\sigma = 32.0 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 7.22 \pm 0.55 \%$$

$$\mu = 423.6 \pm 2.1$$

$$\sigma = 30.6 \pm 2.2$$



Energy loss | $78 < \phi < 81$

Count

XP

$$\frac{\sigma}{\mu} = 8.04 \pm 0.60 \%$$

$$\mu = 424.5 \pm 2.3$$

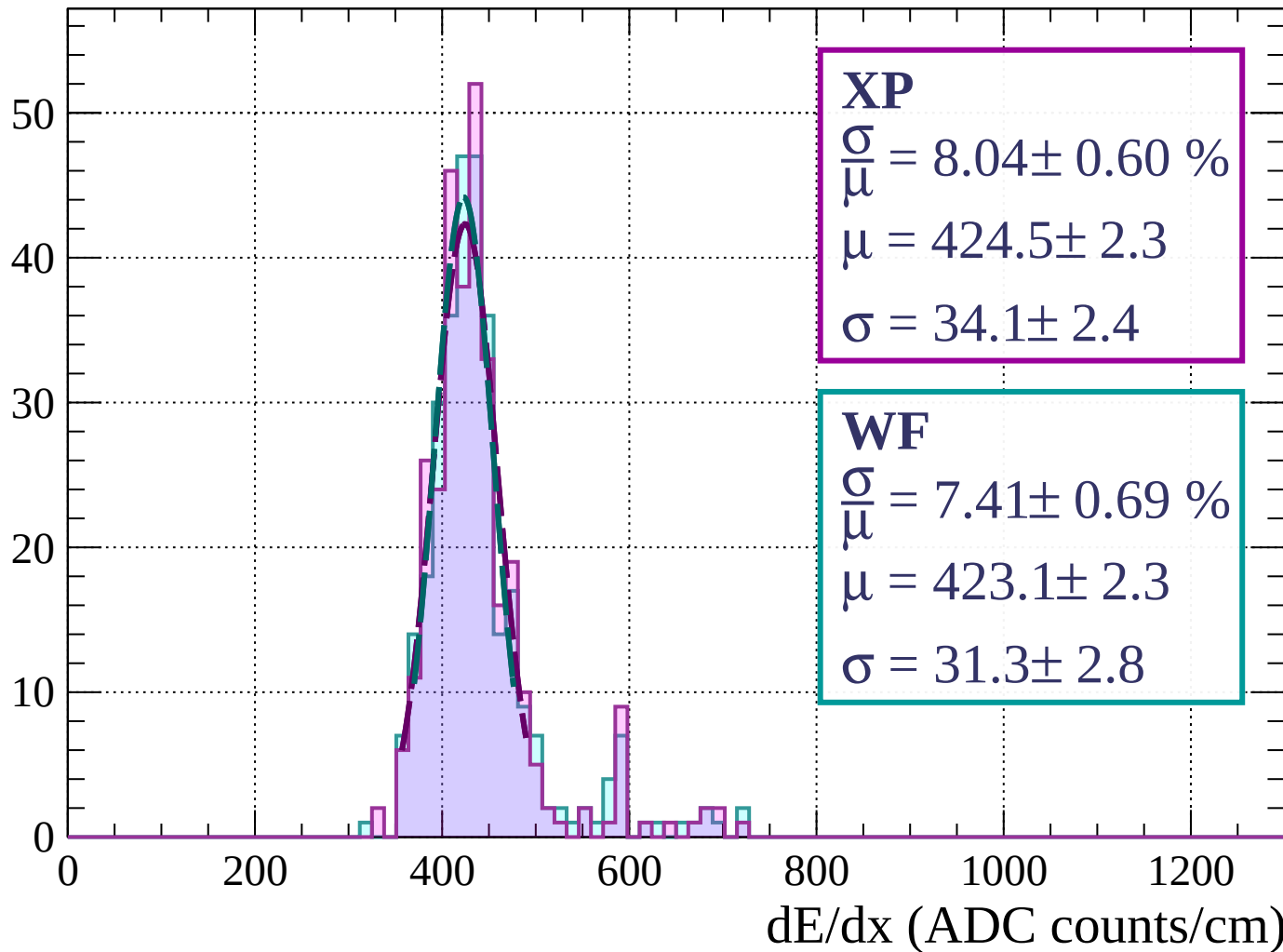
$$\sigma = 34.1 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 7.41 \pm 0.69 \%$$

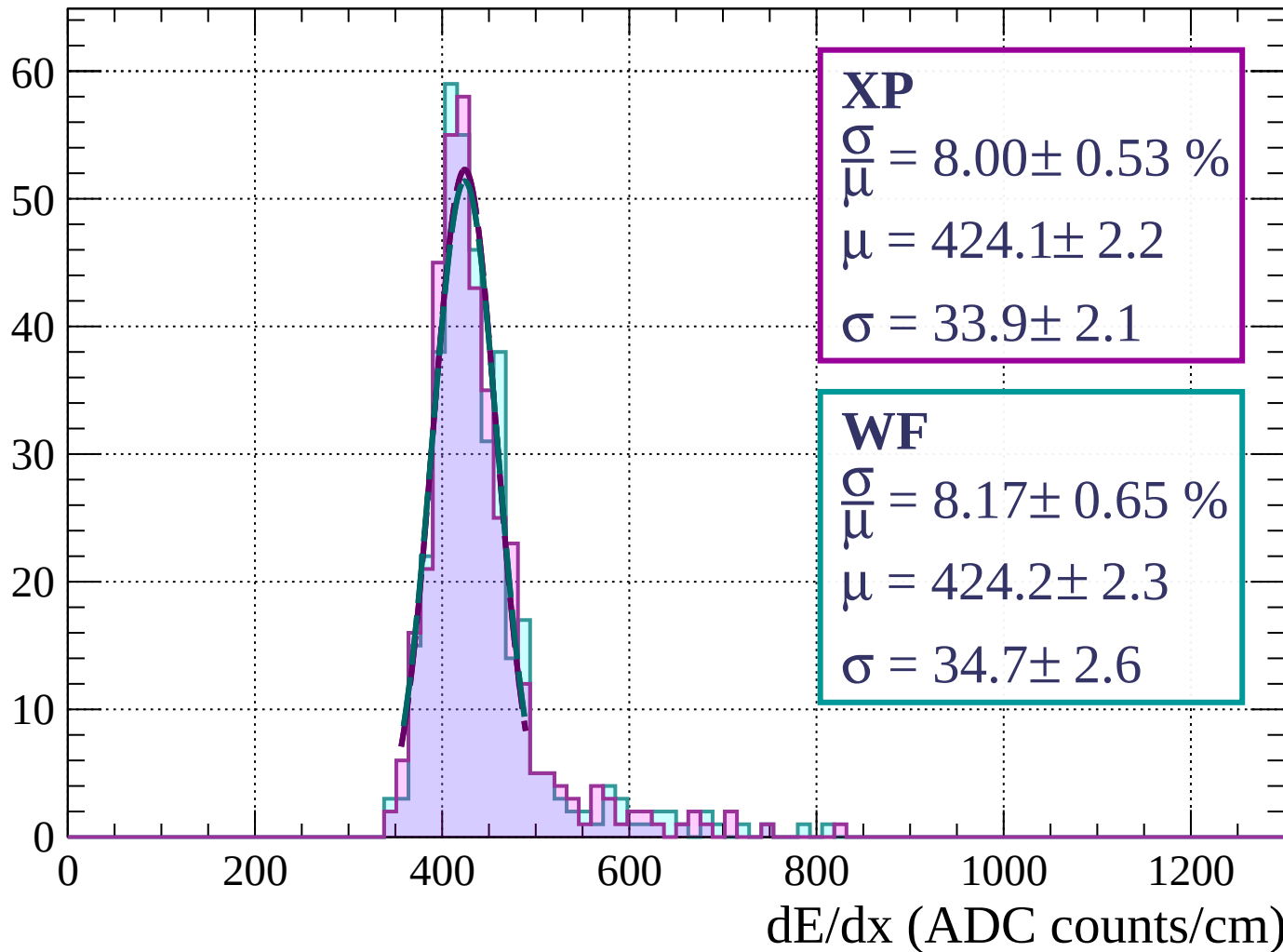
$$\mu = 423.1 \pm 2.3$$

$$\sigma = 31.3 \pm 2.8$$



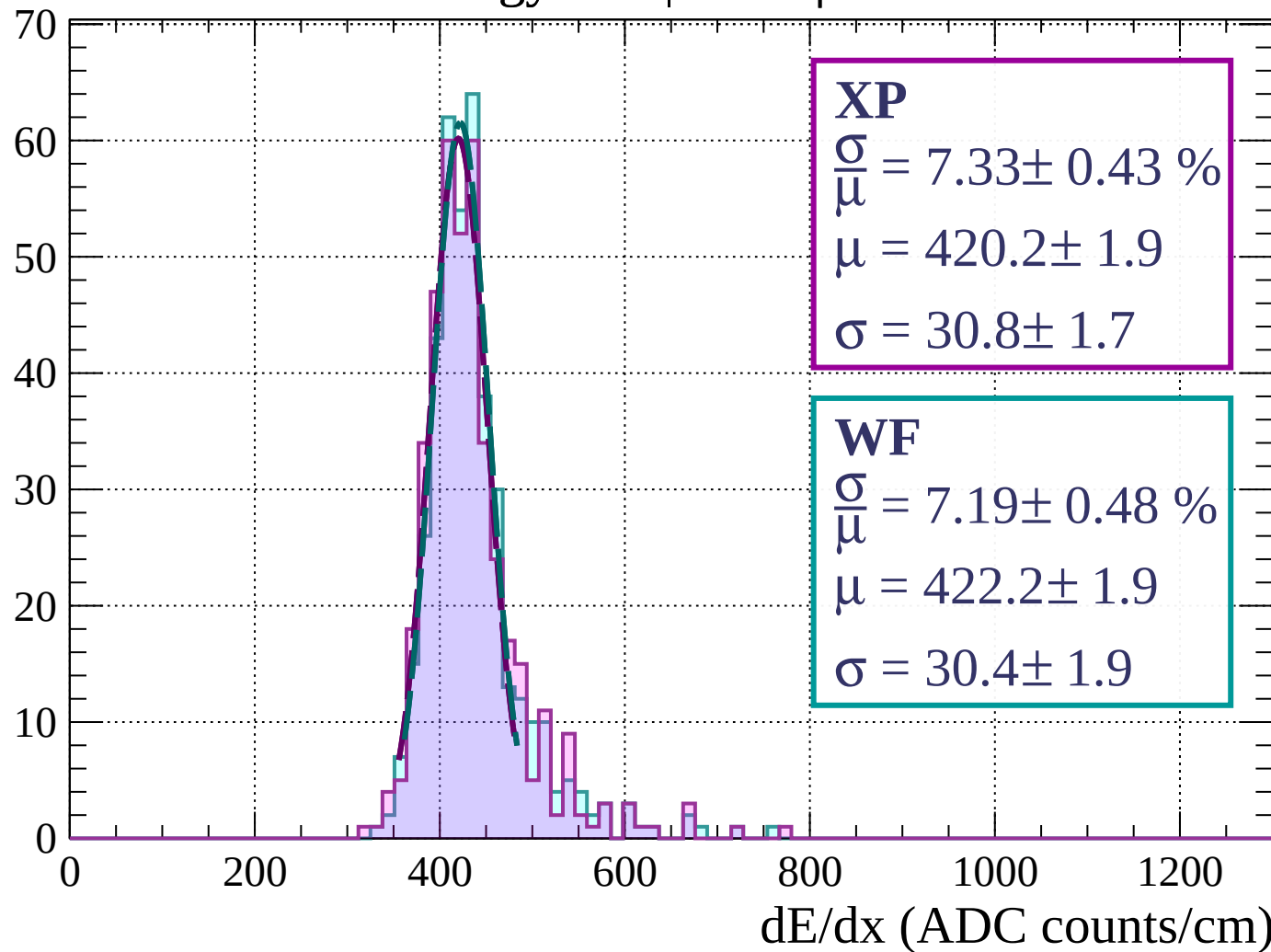
Energy loss | $81 < \phi < 84$

Count



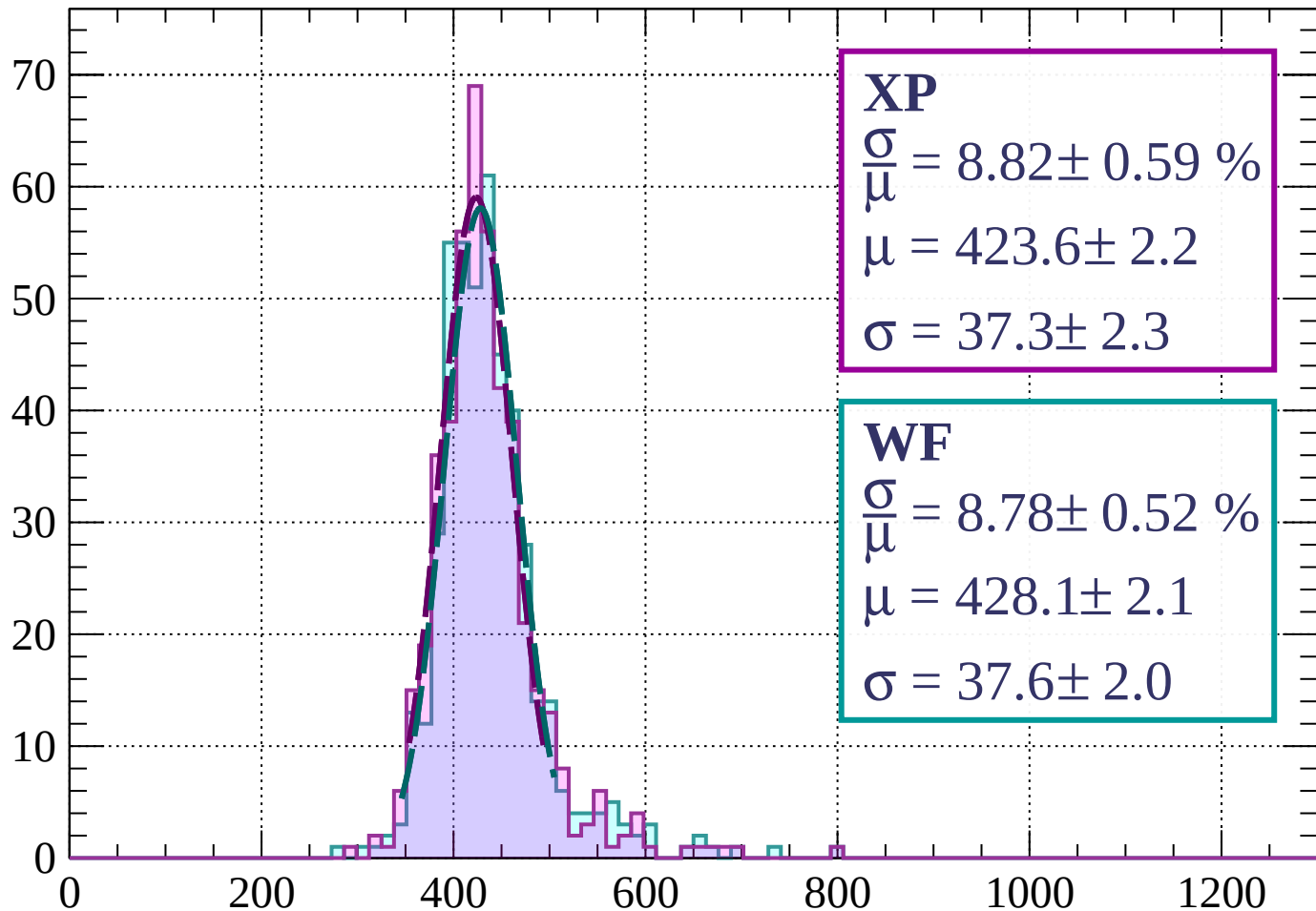
Energy loss | $84 < \phi < 87$

Count



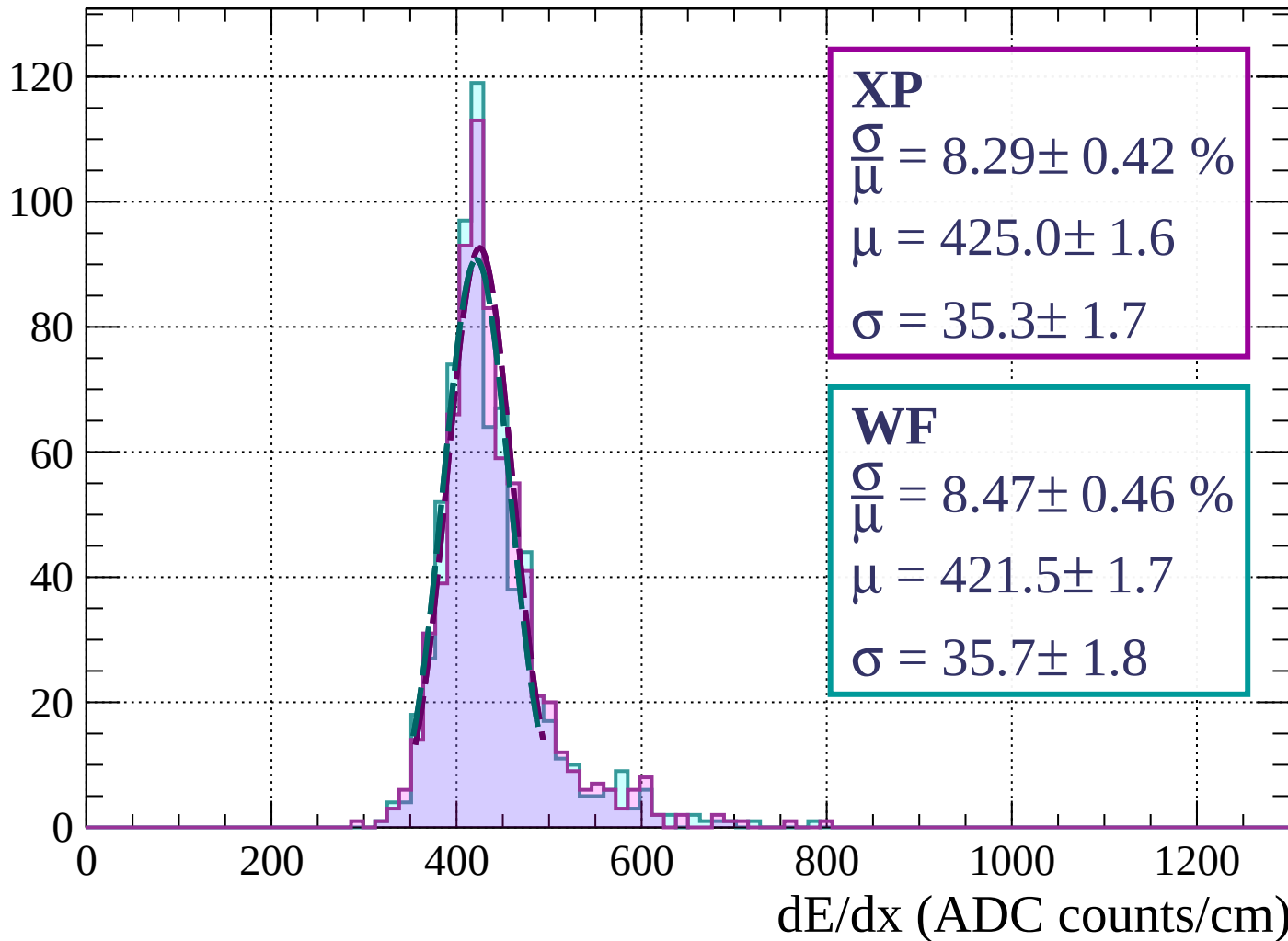
Energy loss | $87 < \phi < 90$

Count

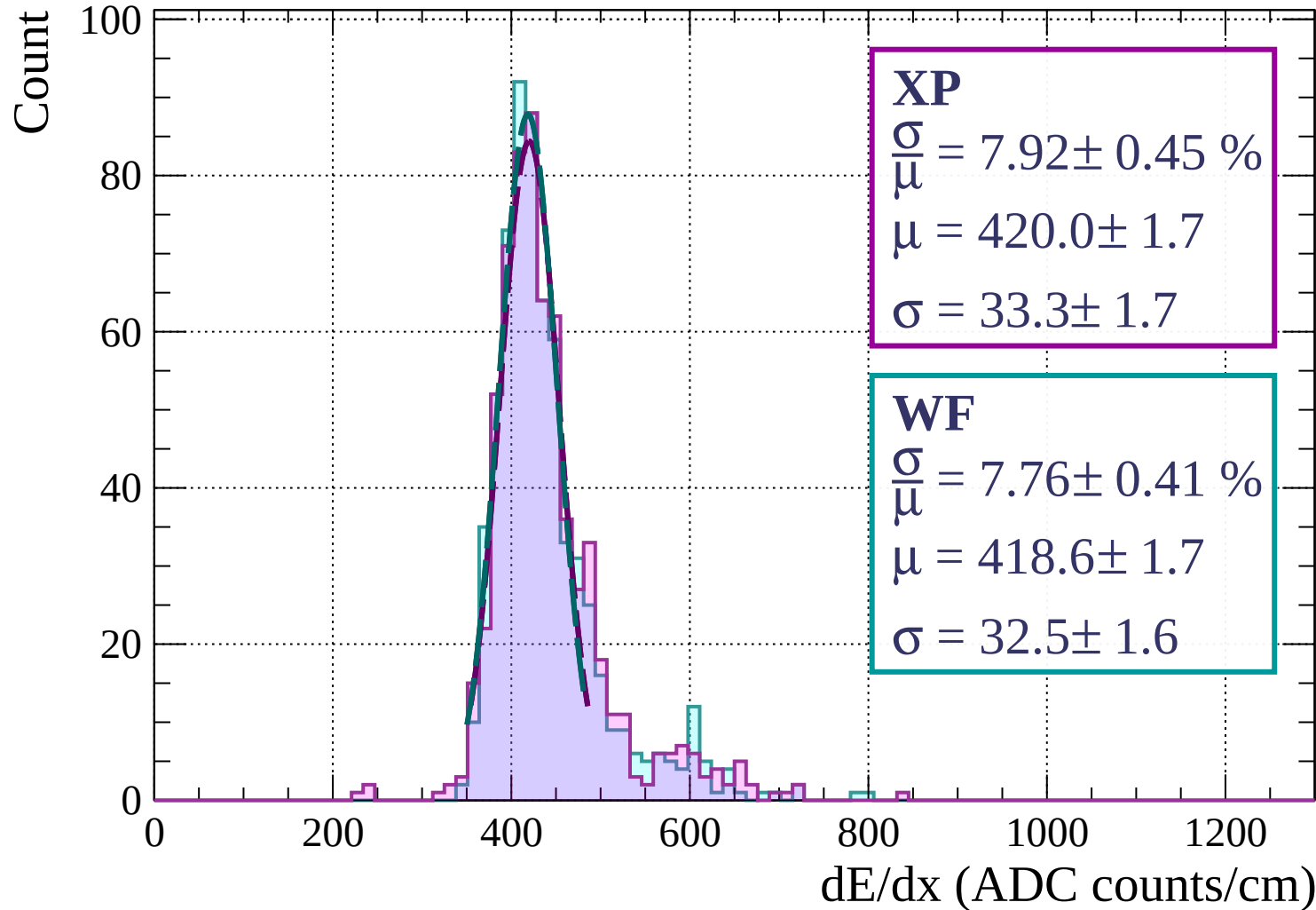


Energy loss | $0 < \theta < 3$

Count

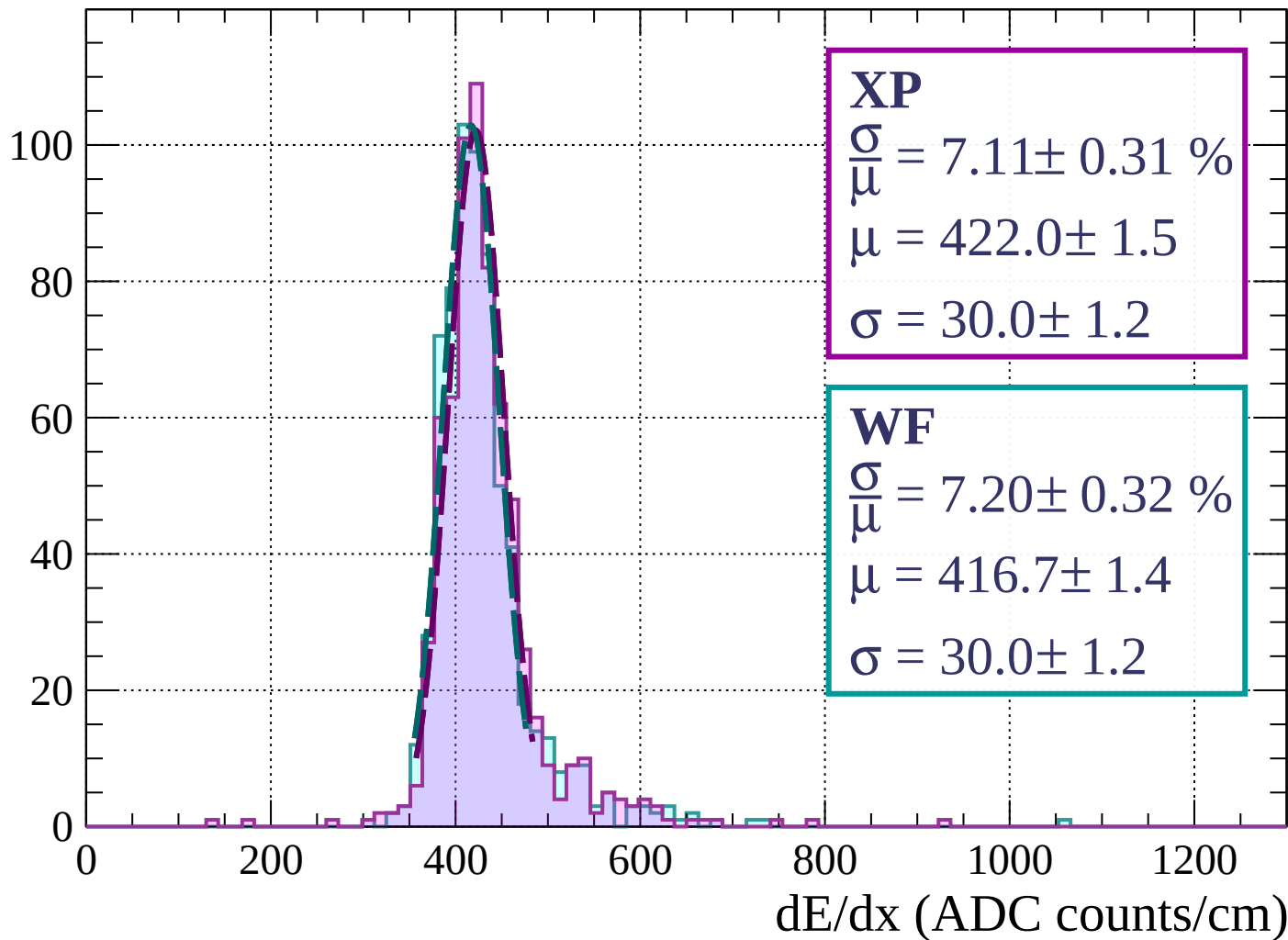


Energy loss | $3 < \theta < 6$



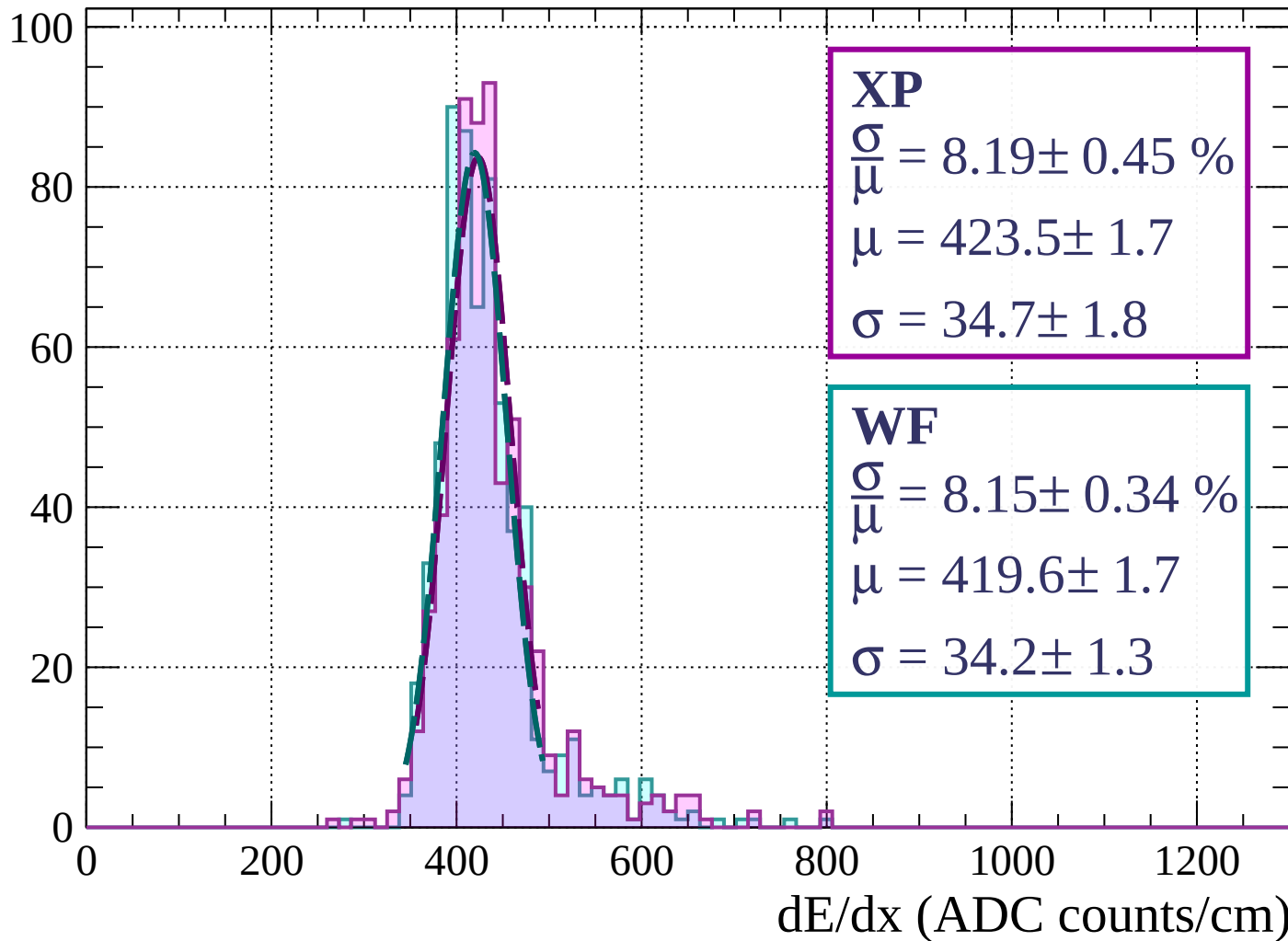
Energy loss | $6 < \theta < 9$

Count

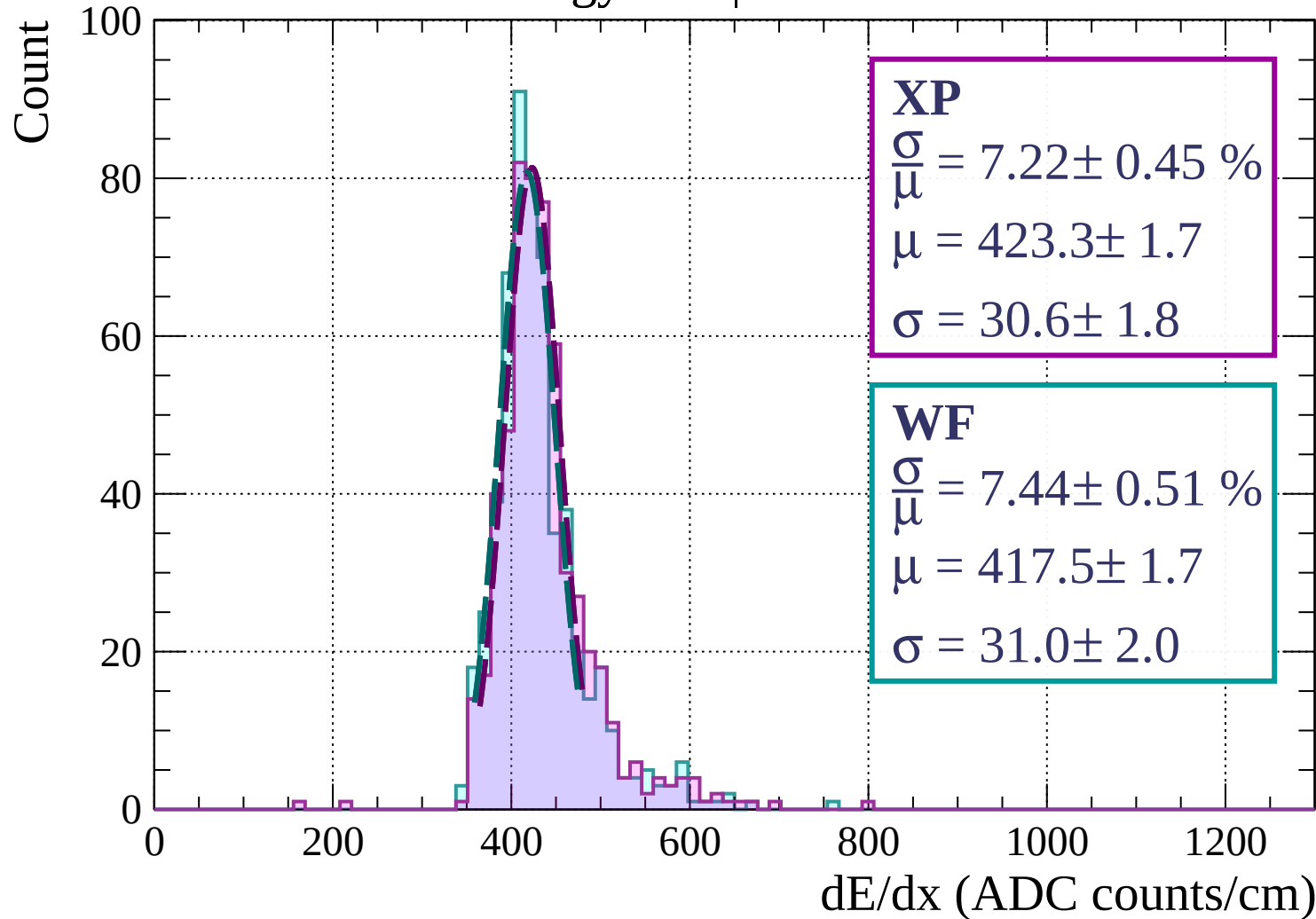


Energy loss | $9 < \theta < 12$

Count



Energy loss | $12 < \theta < 15$



Energy loss | $15 < \theta < 18$

Count

XP

$$\frac{\sigma}{\mu} = 8.30 \pm 0.47 \%$$

$$\mu = 423.7 \pm 2.0$$

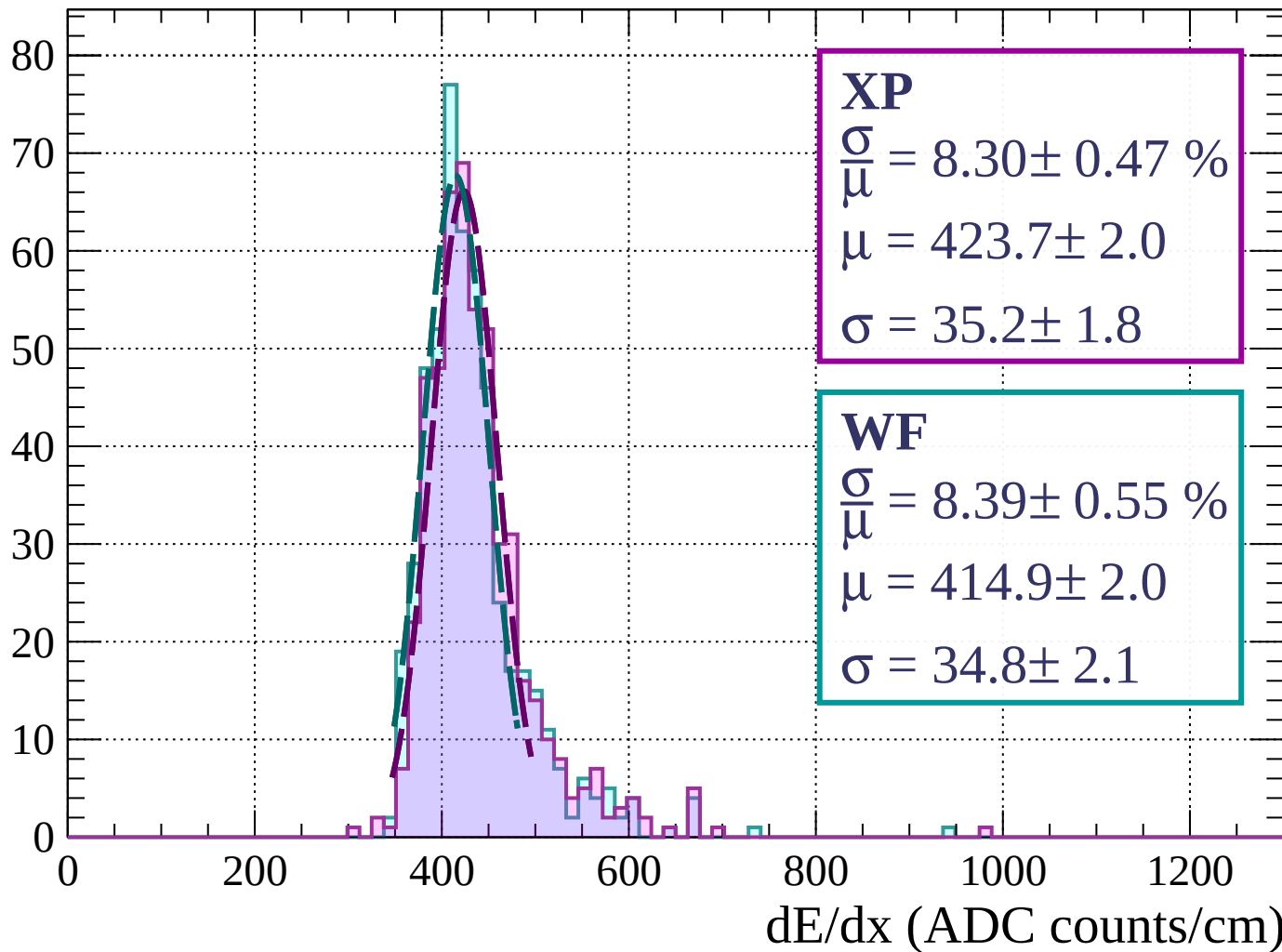
$$\sigma = 35.2 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 8.39 \pm 0.55 \%$$

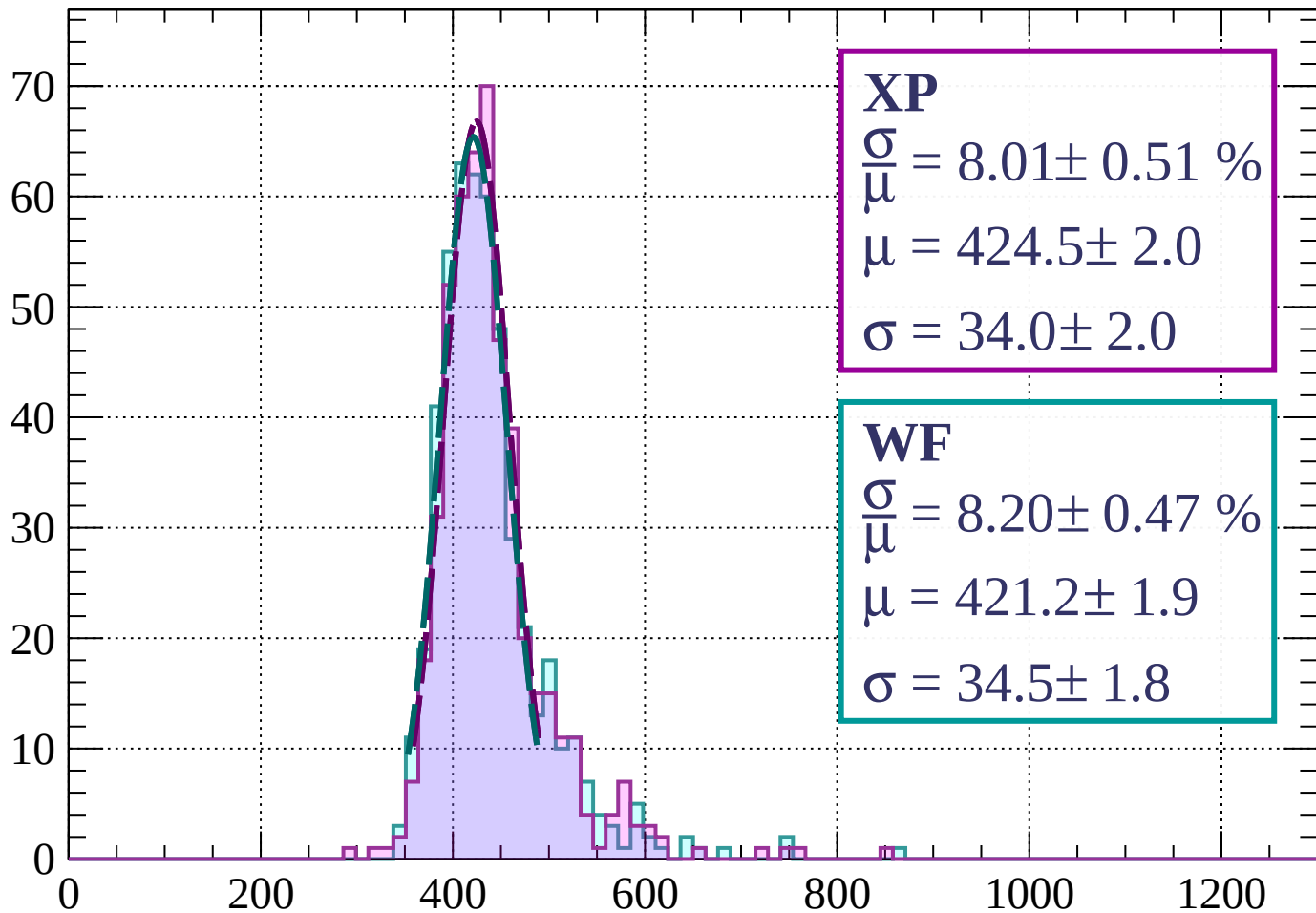
$$\mu = 414.9 \pm 2.0$$

$$\sigma = 34.8 \pm 2.1$$



Energy loss | $18 < \theta < 21$

Count



XP

$$\frac{\sigma}{\mu} = 8.01 \pm 0.51 \%$$

$$\mu = 424.5 \pm 2.0$$

$$\sigma = 34.0 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 8.20 \pm 0.47 \%$$

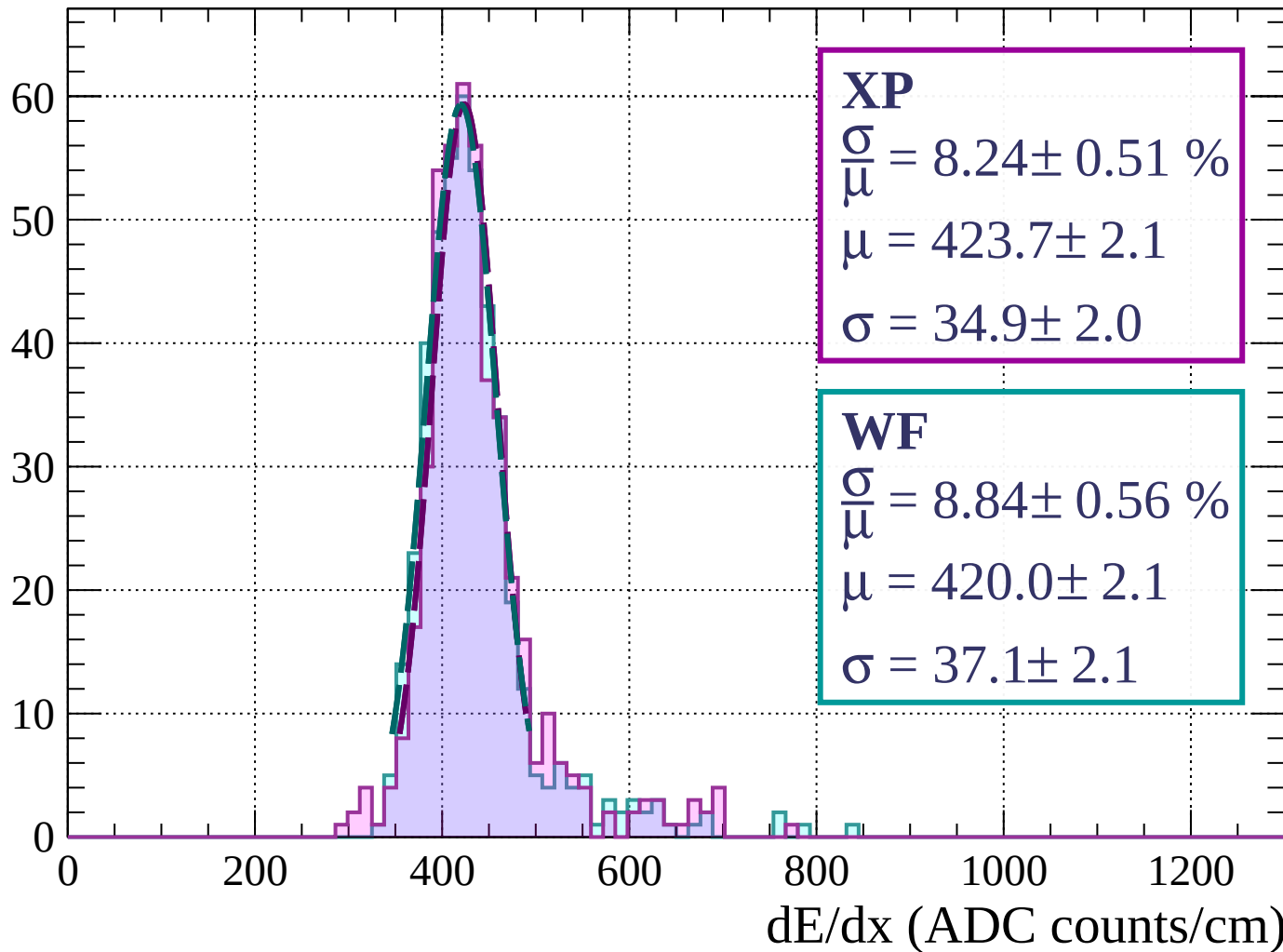
$$\mu = 421.2 \pm 1.9$$

$$\sigma = 34.5 \pm 1.8$$

dE/dx (ADC counts/cm)

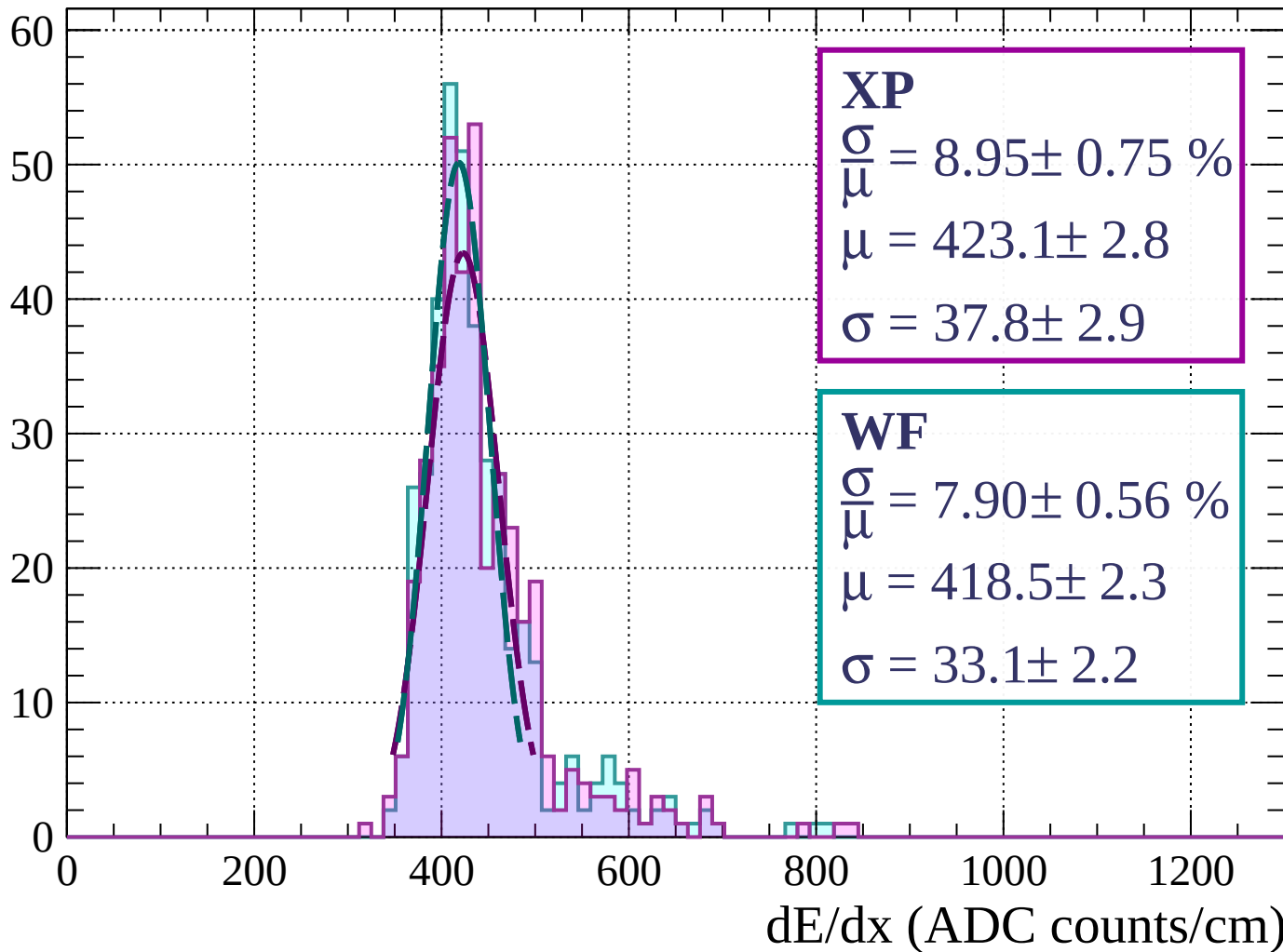
Energy loss | $21 < \theta < 24$

Count



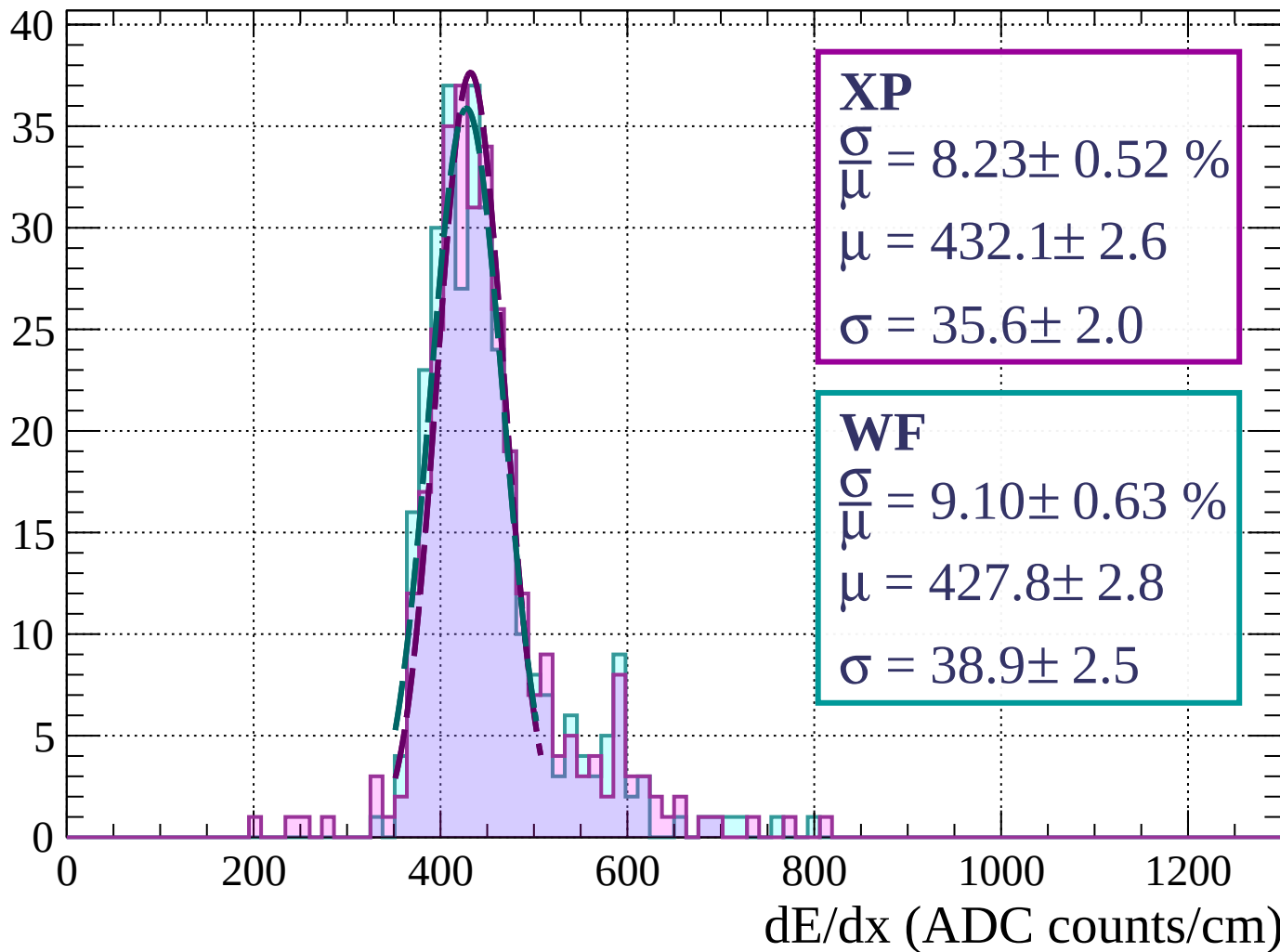
Energy loss | $24 < \theta < 27$

Count



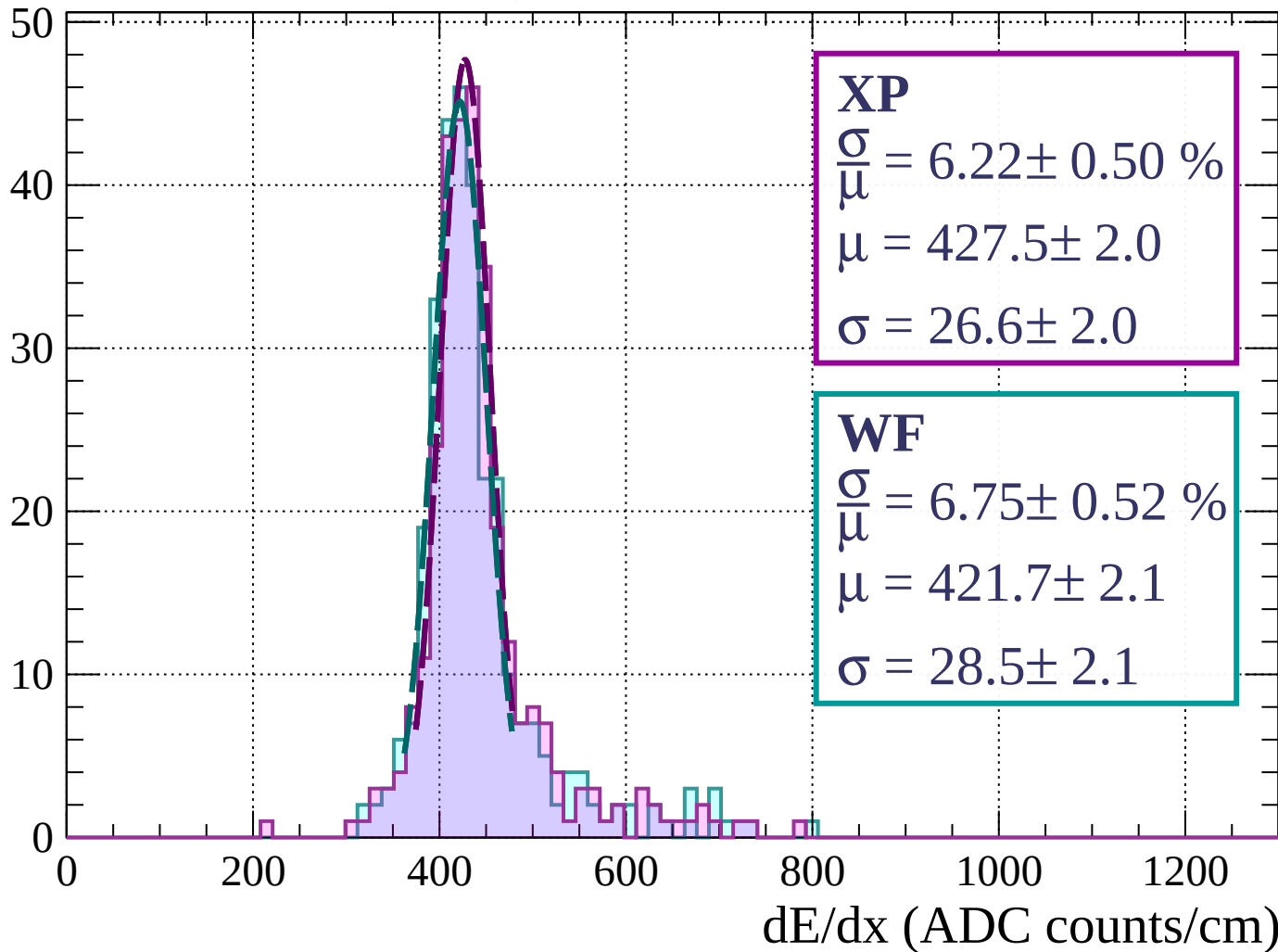
Energy loss | $27 < \theta < 30$

Count



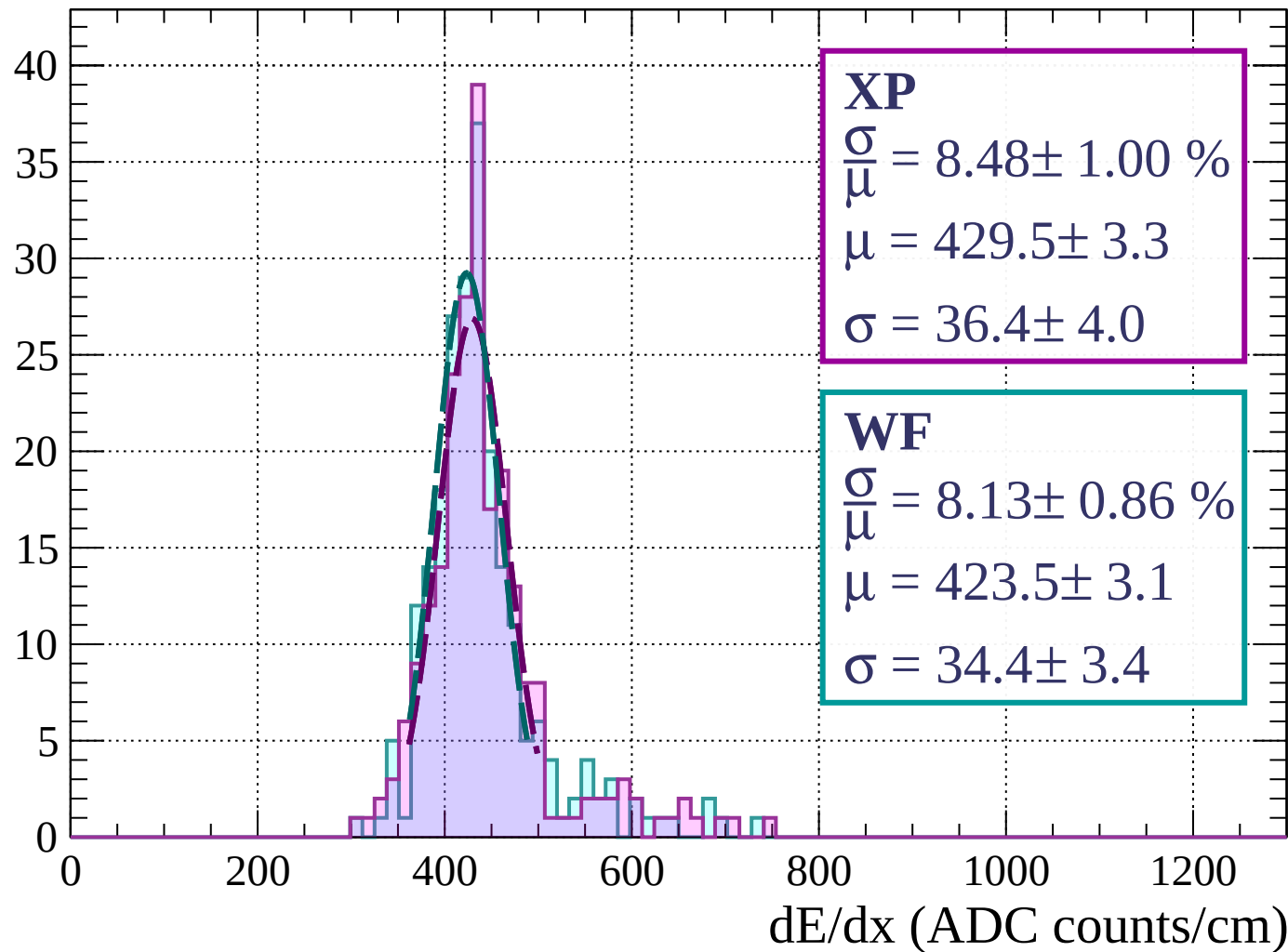
Energy loss | $30 < \theta < 33$

Count



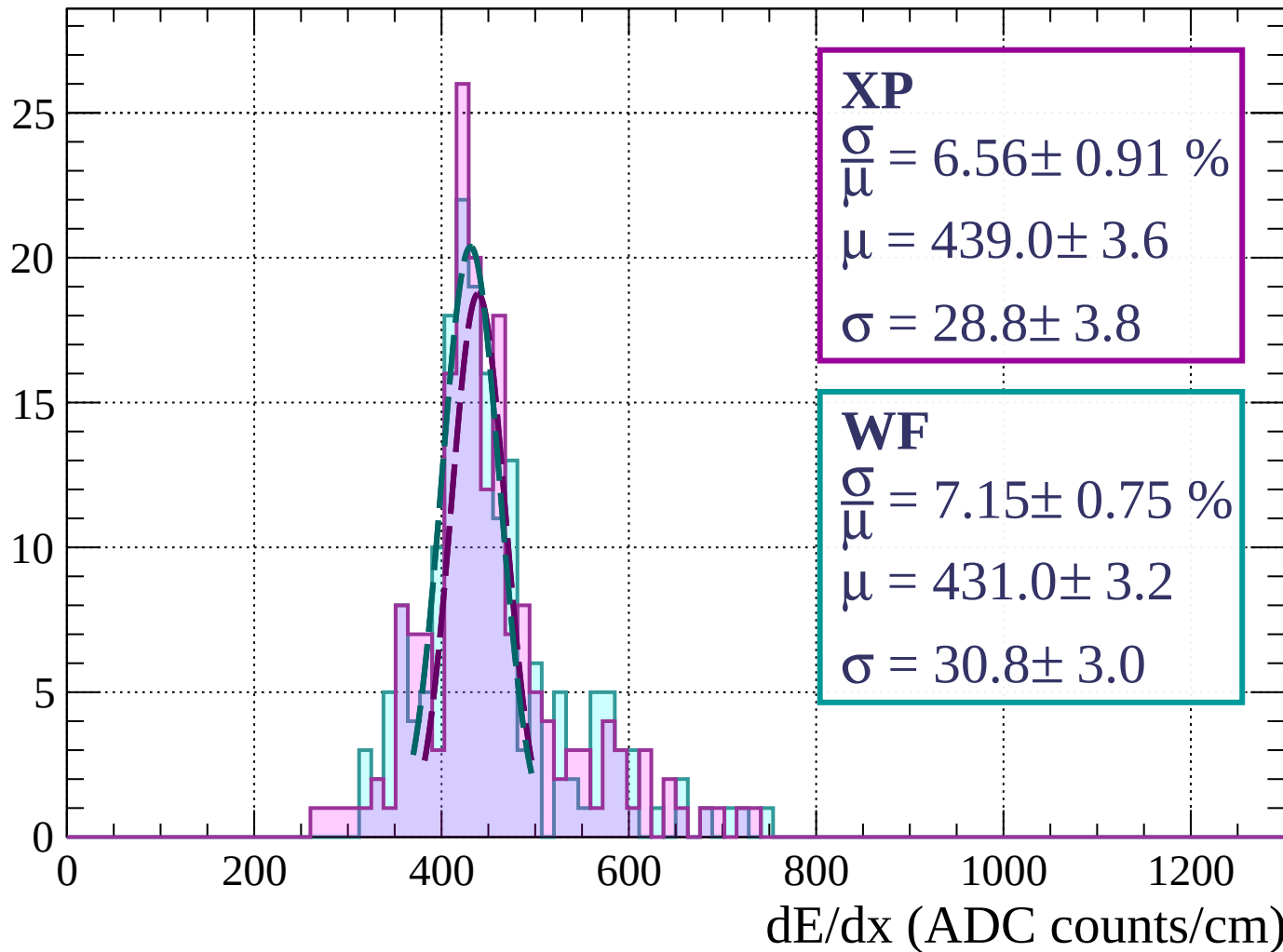
Energy loss | $33 < \theta < 36$

Count



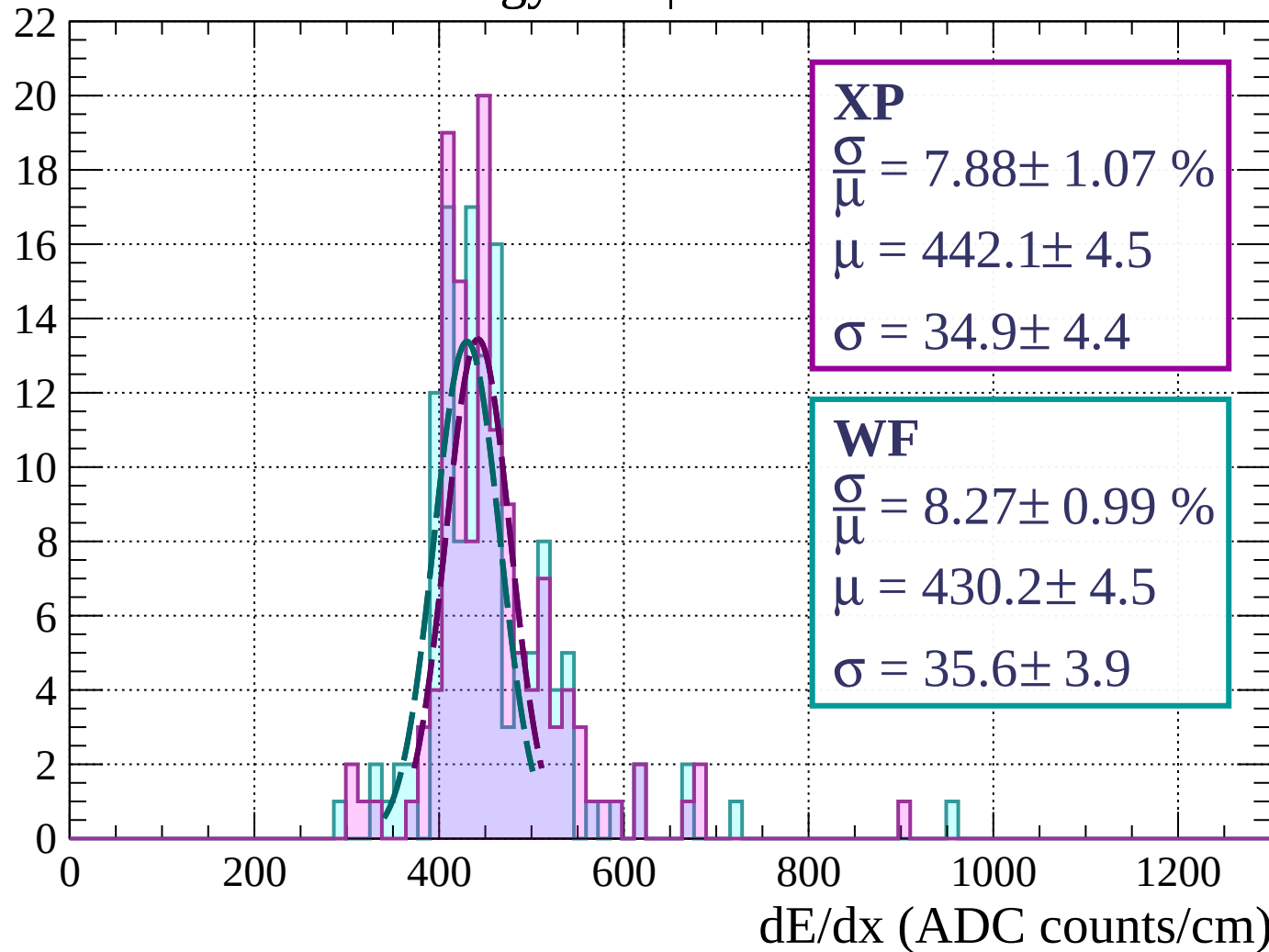
Energy loss | $36 < \theta < 39$

Count



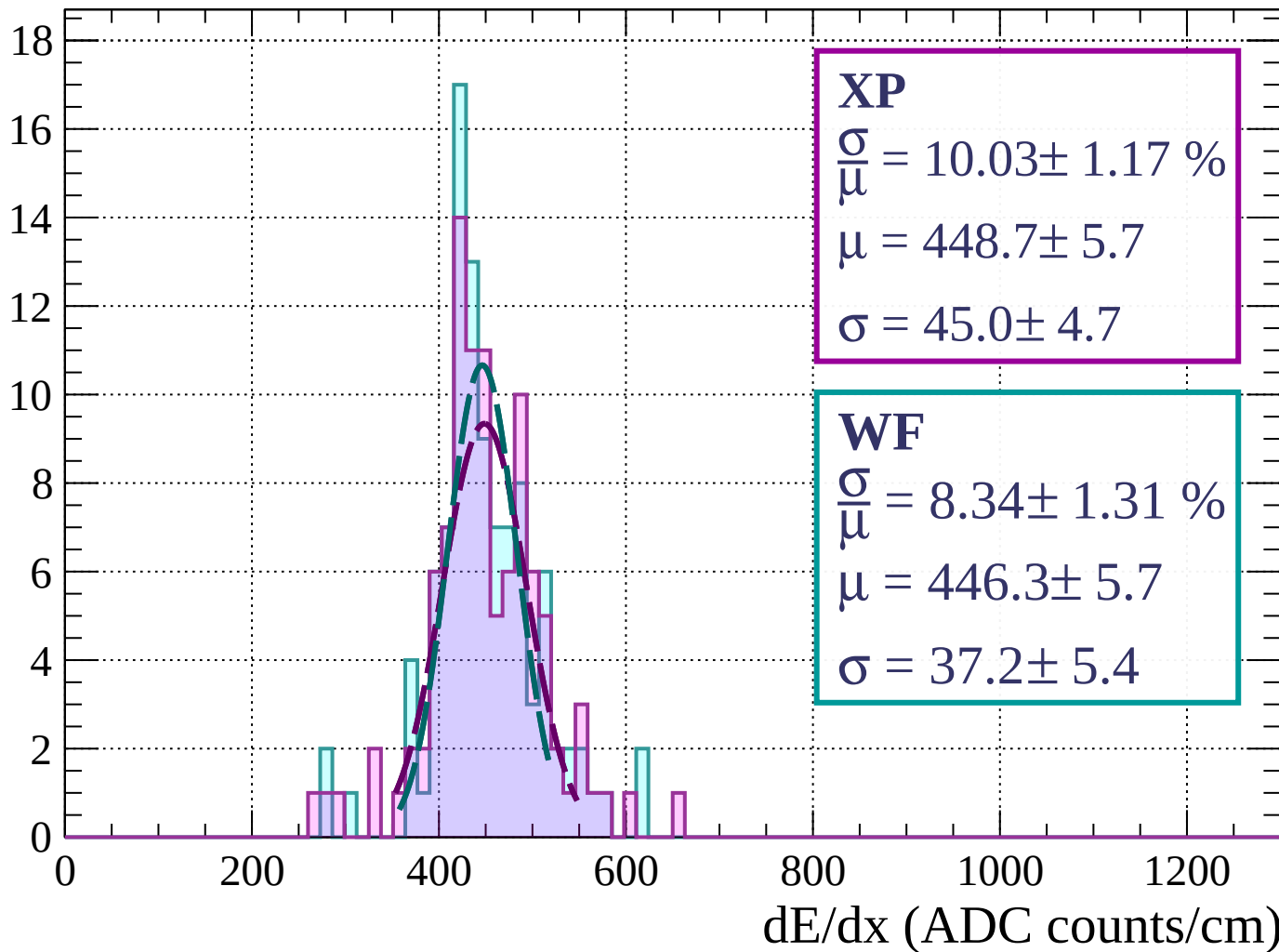
Energy loss | $39 < \theta < 42$

Count



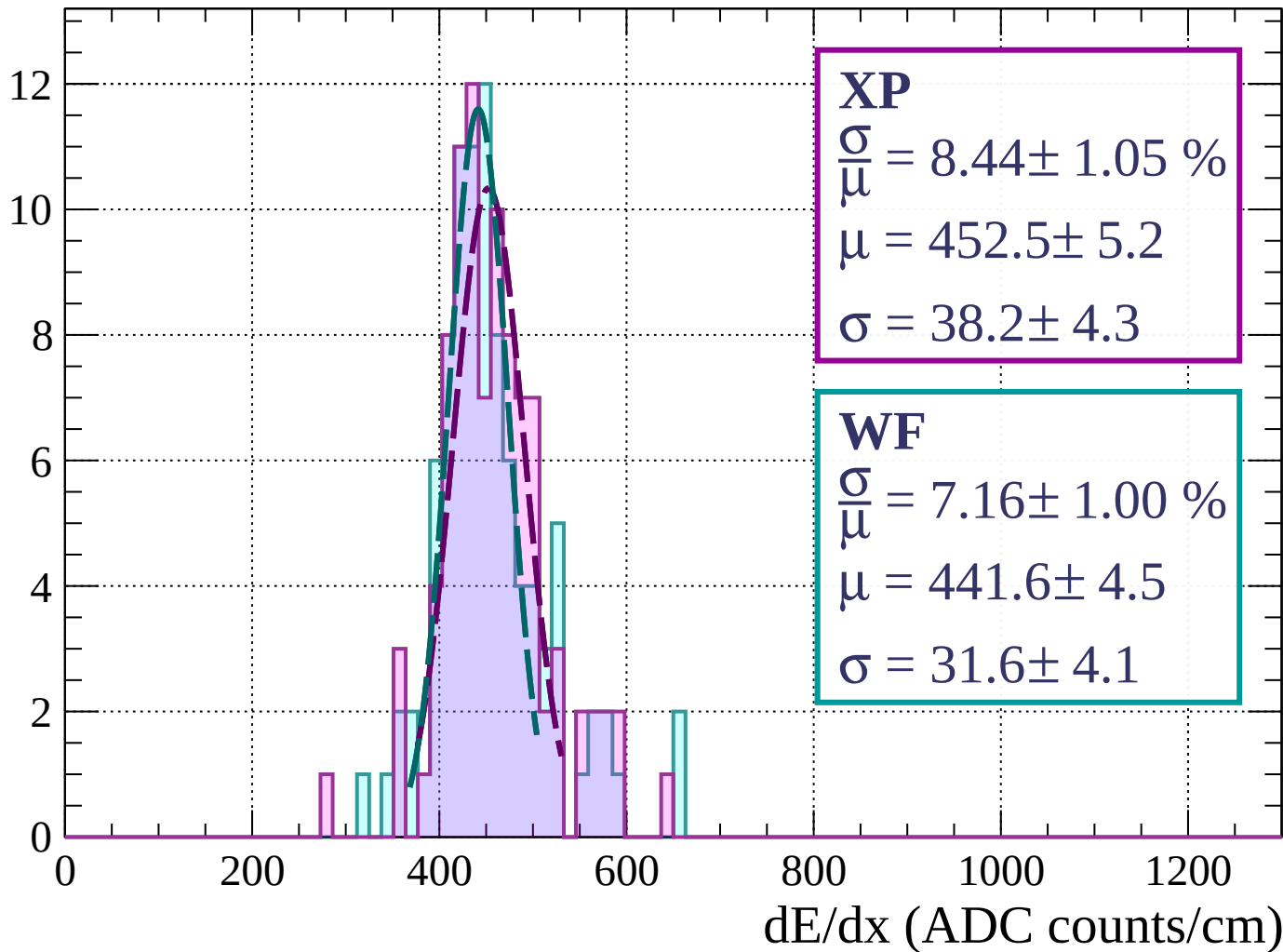
Energy loss | $42 < \theta < 45$

Count



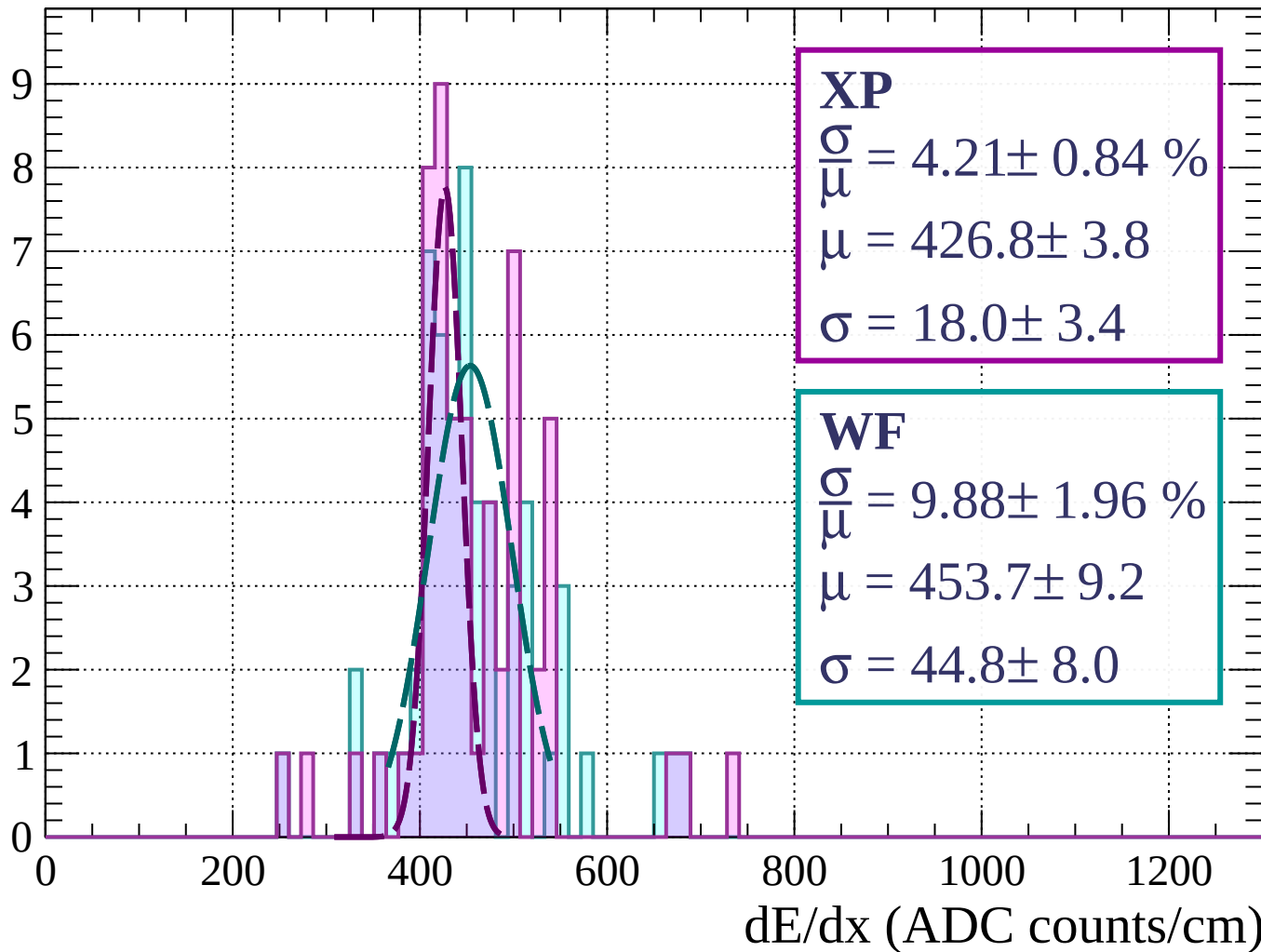
Energy loss | $45 < \theta < 48$

Count



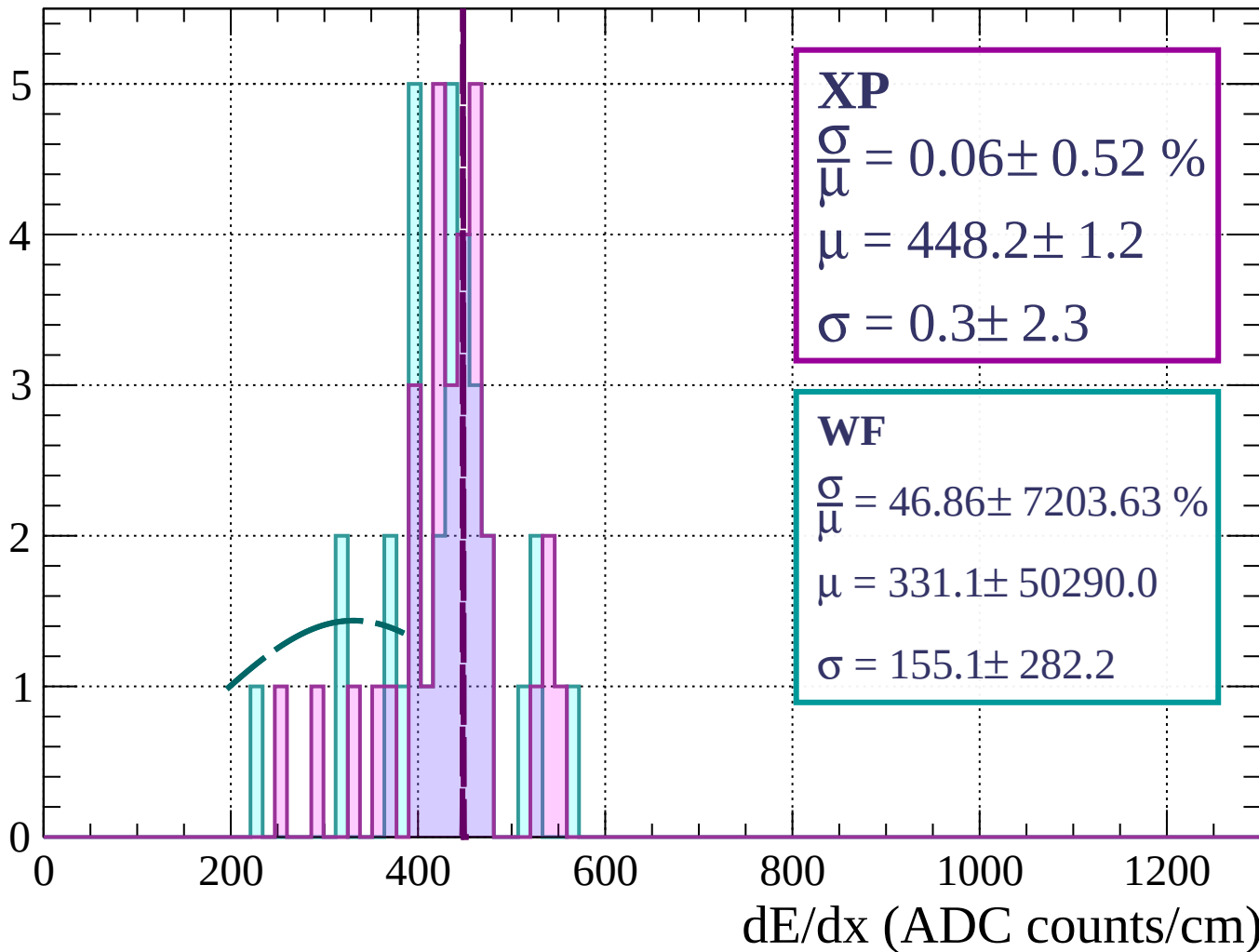
Energy loss | $48 < \theta < 51$

Count



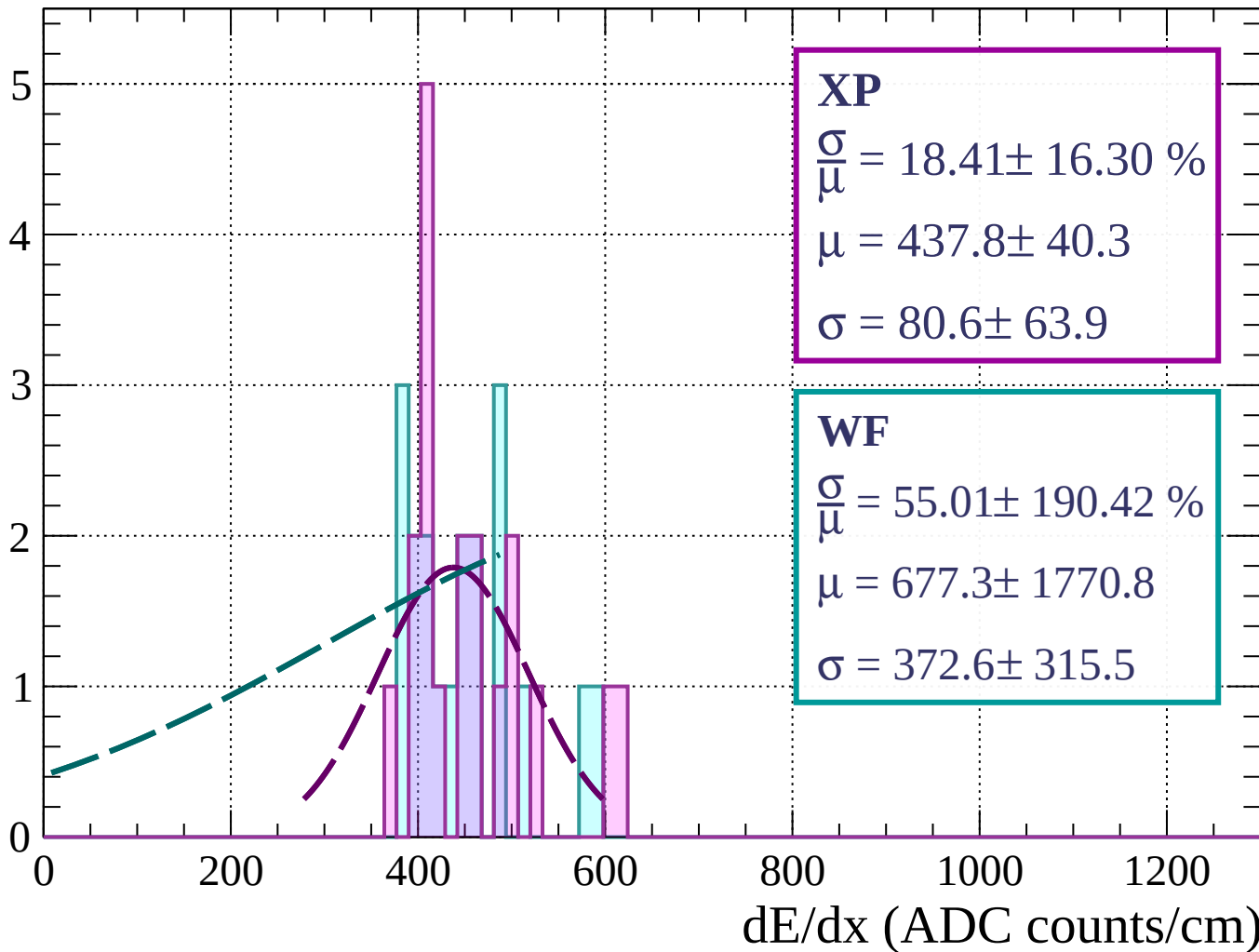
Energy loss | $51 < \theta < 54$

Count



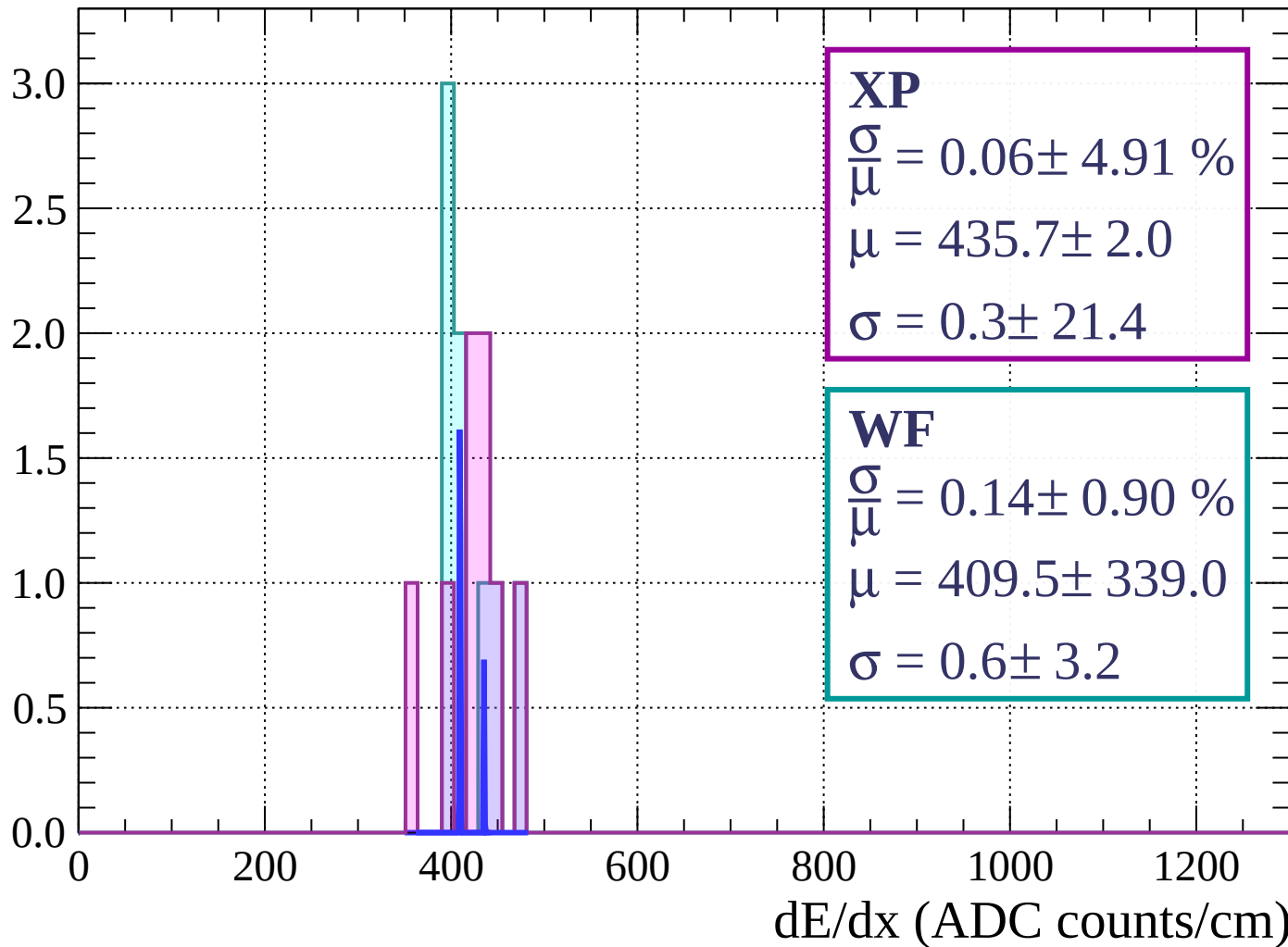
Energy loss | $54 < \theta < 57$

Count



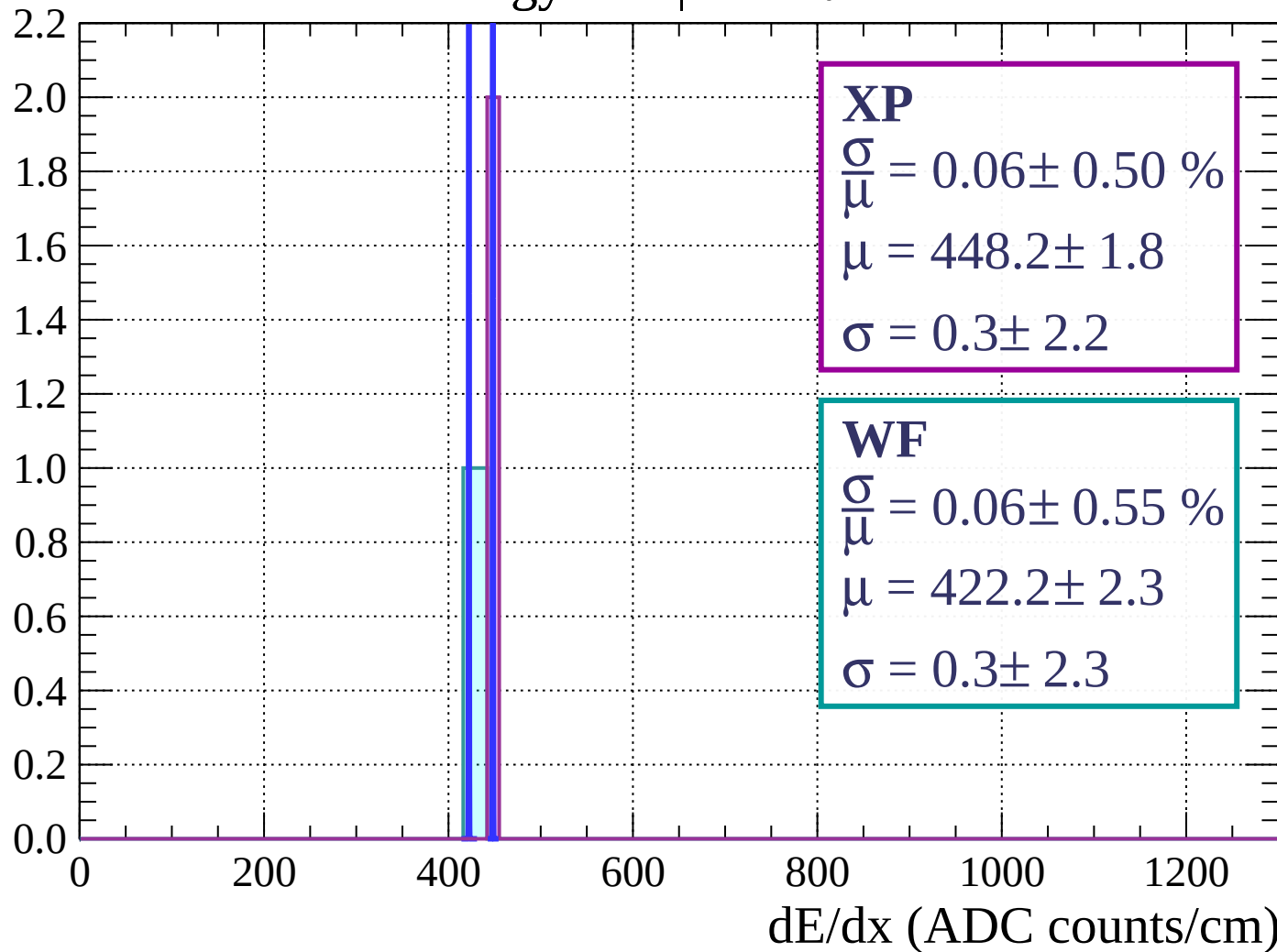
Energy loss | $57 < \theta < 60$

Count



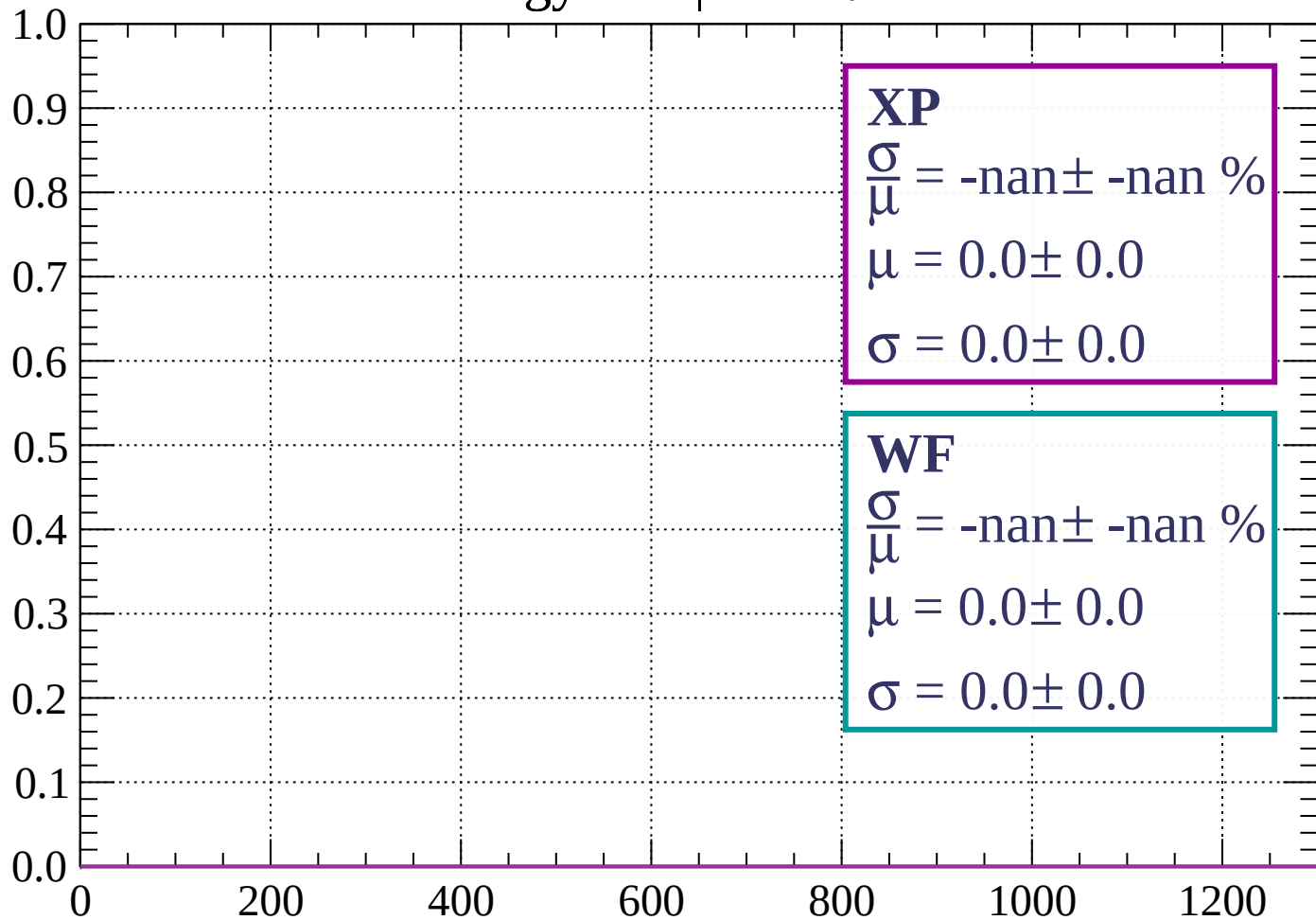
Energy loss | $60 < \theta < 63$

Count



Energy loss | $63 < \theta < 66$

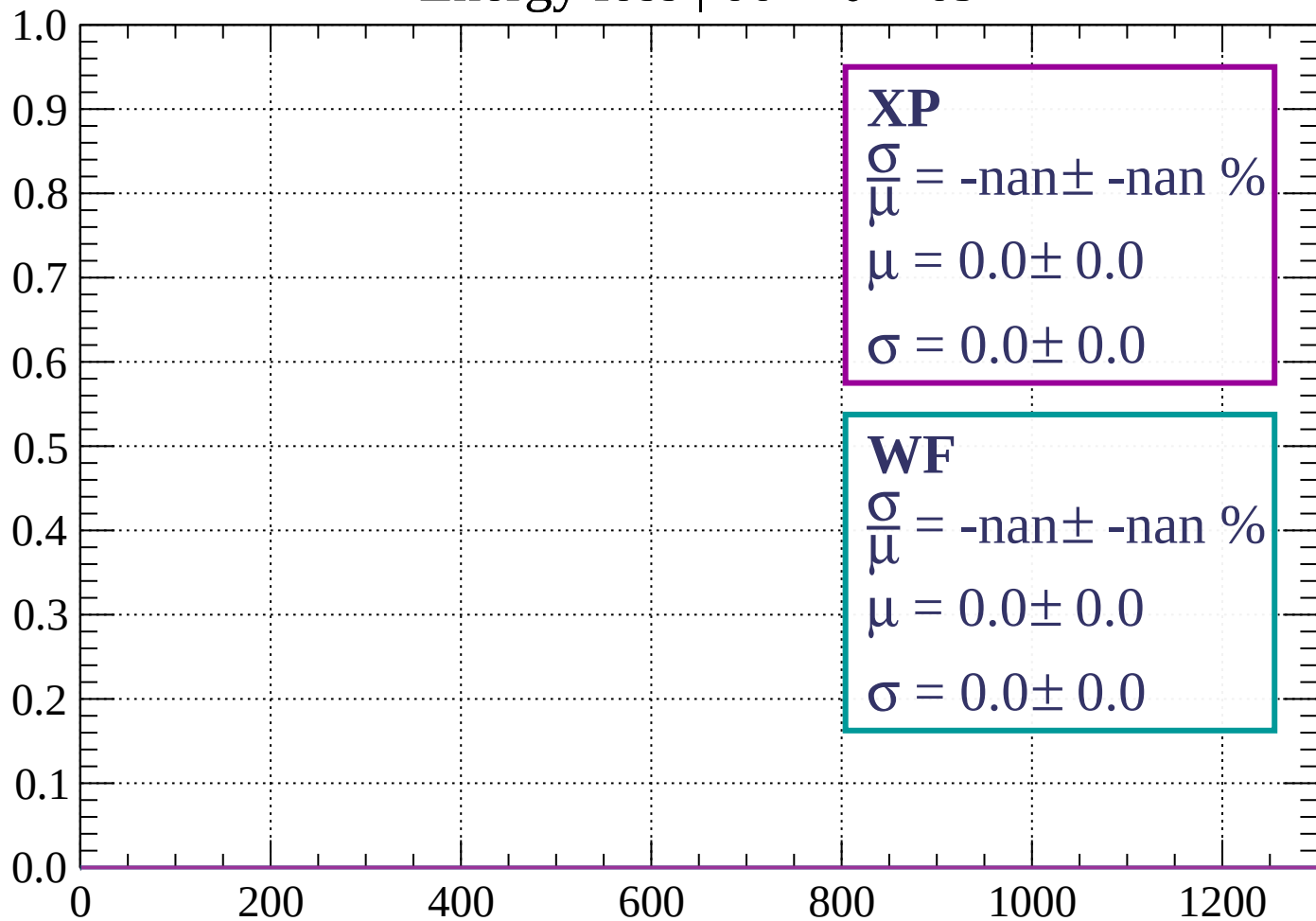
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 69$

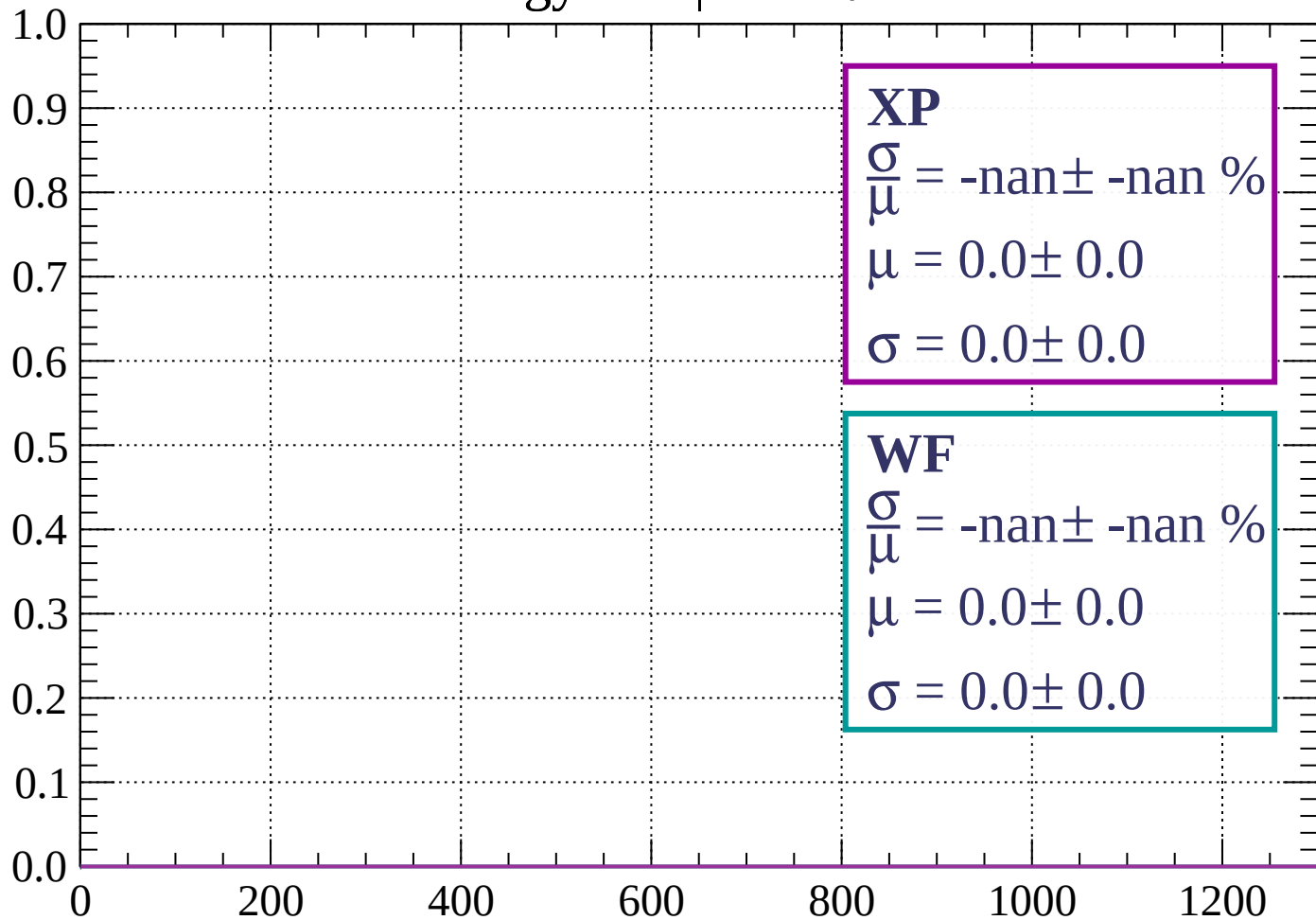
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \theta < 72$

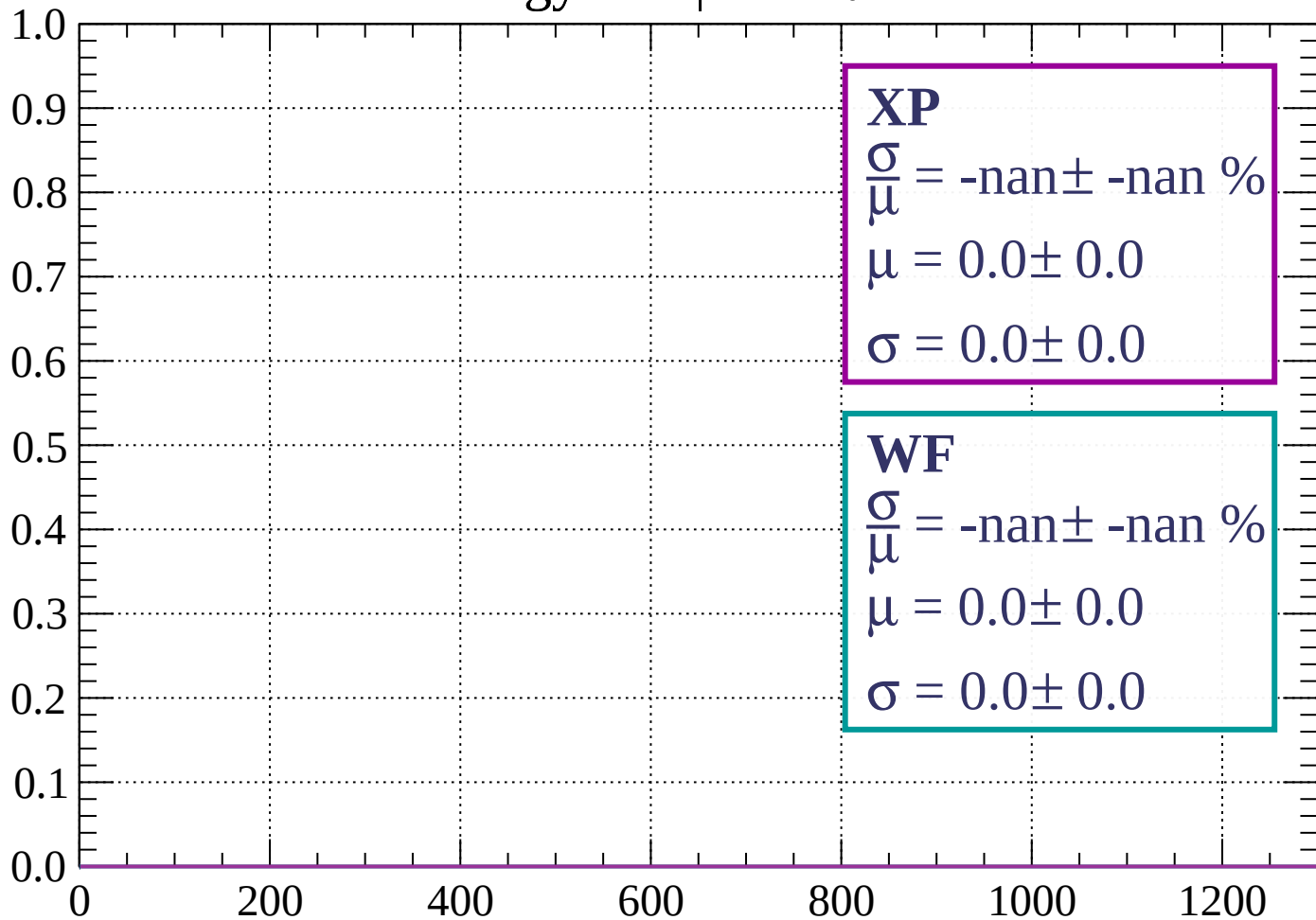
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 75$

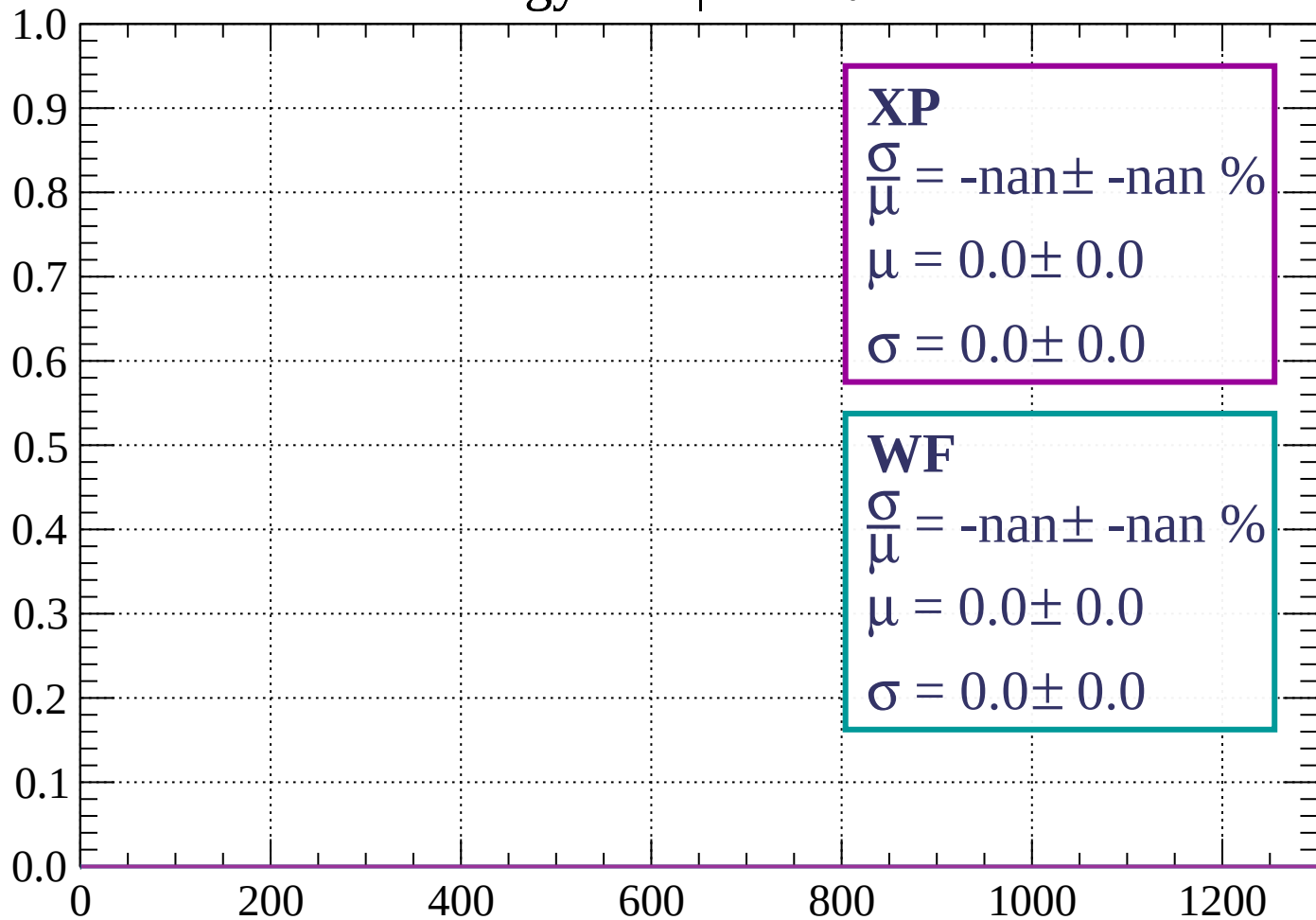
Count



dE/dx (ADC counts/cm)

Energy loss | $75 < \theta < 78$

Count



XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

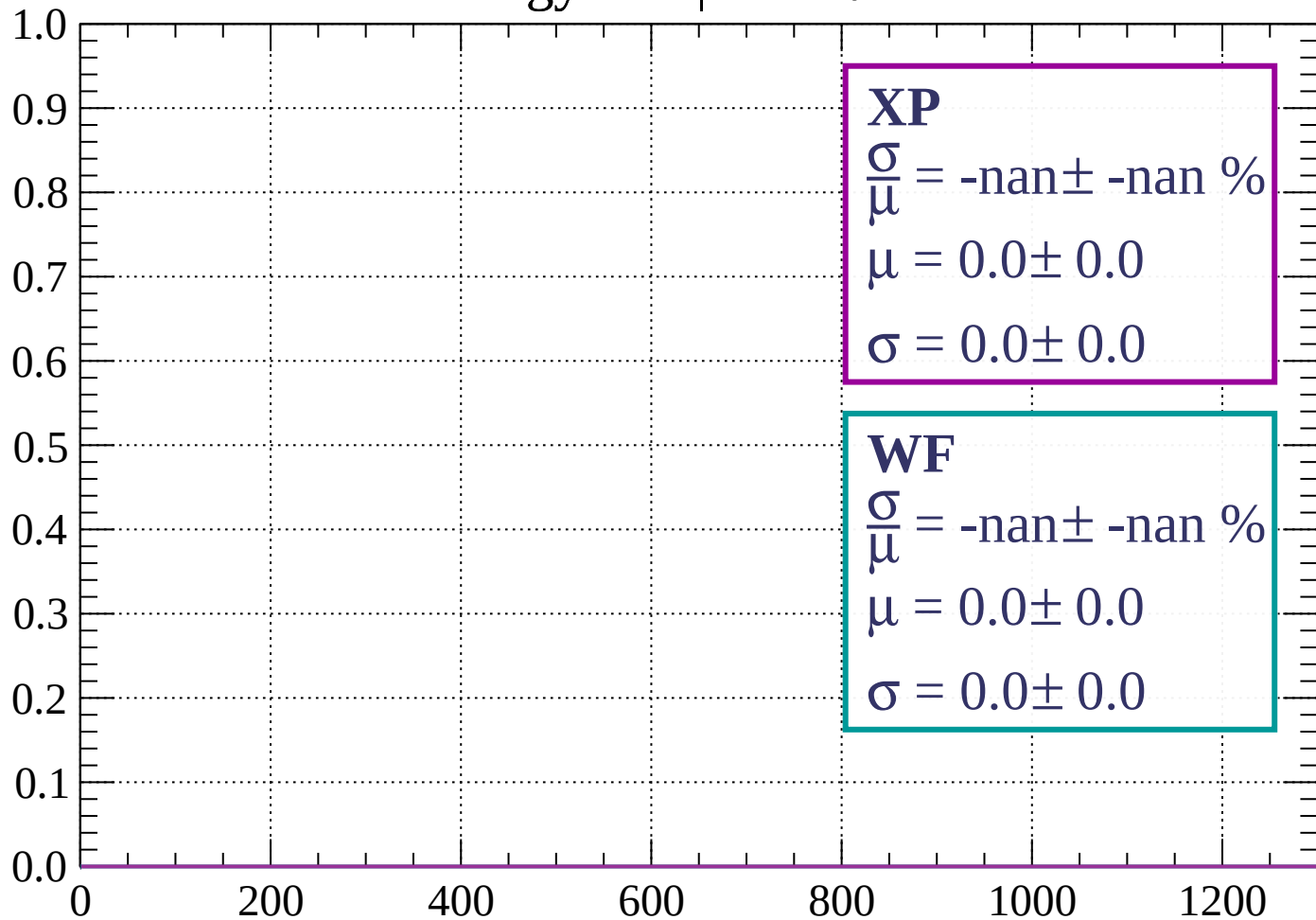
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

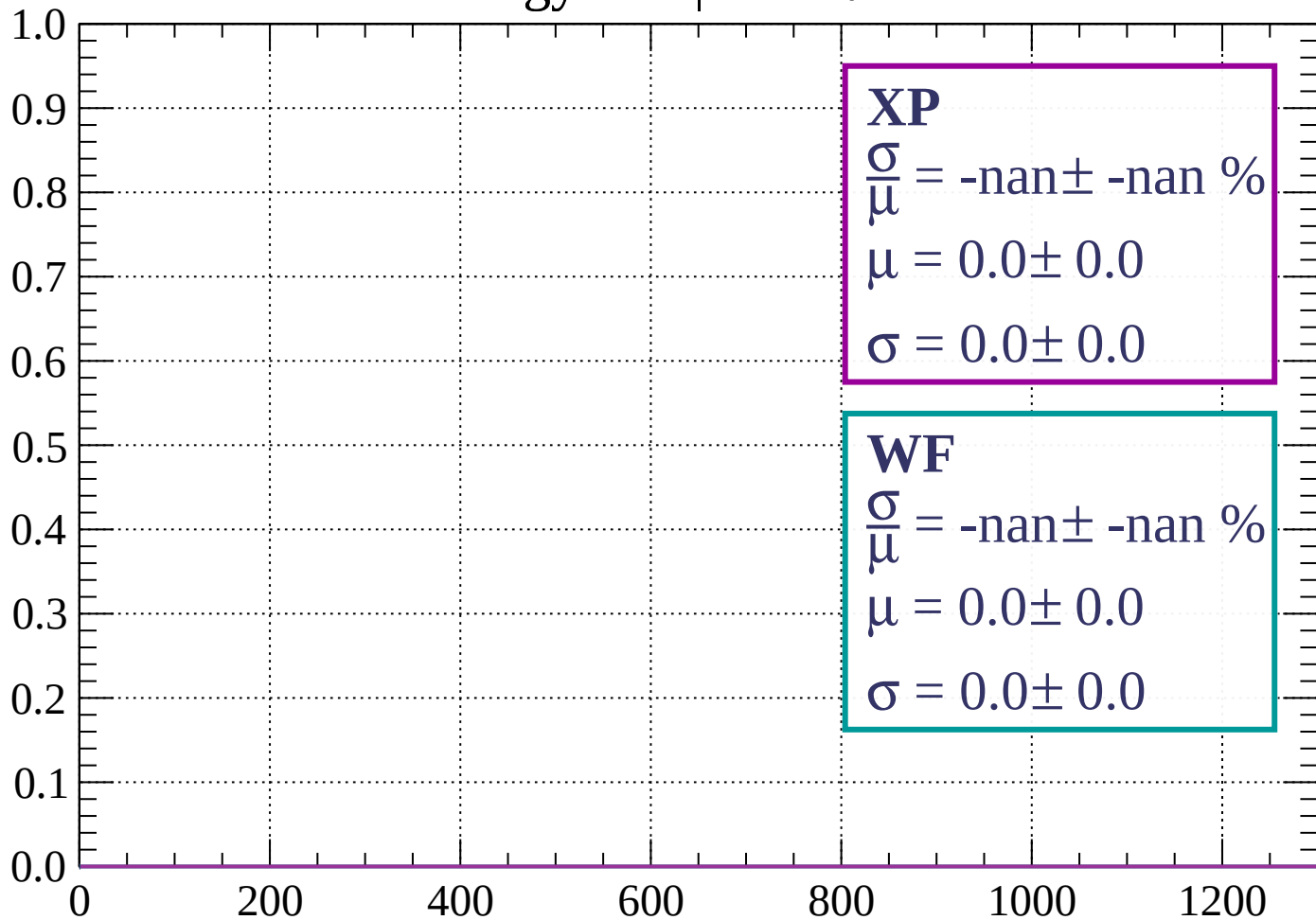
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

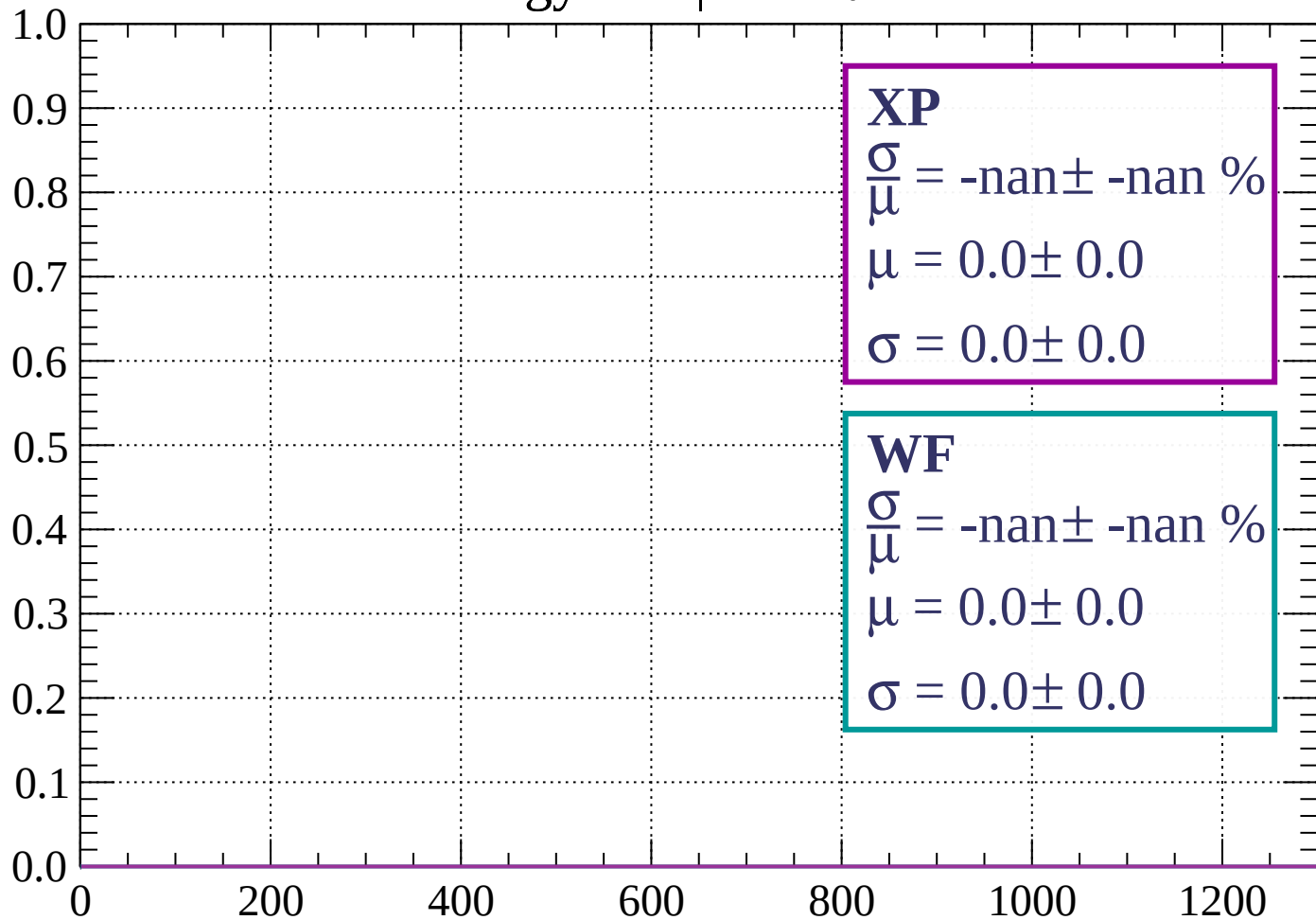
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

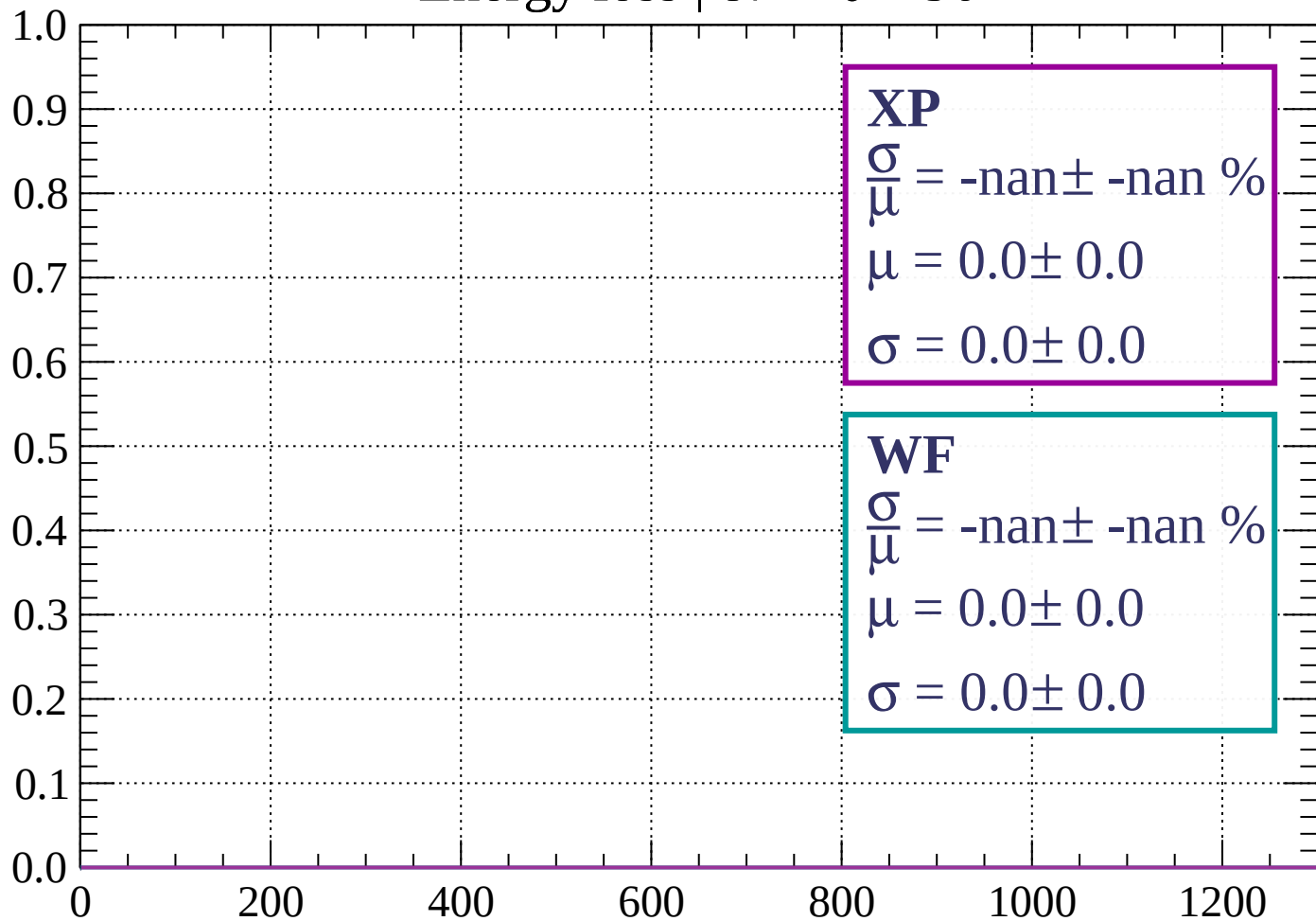
Count



dE/dx (ADC counts/cm)

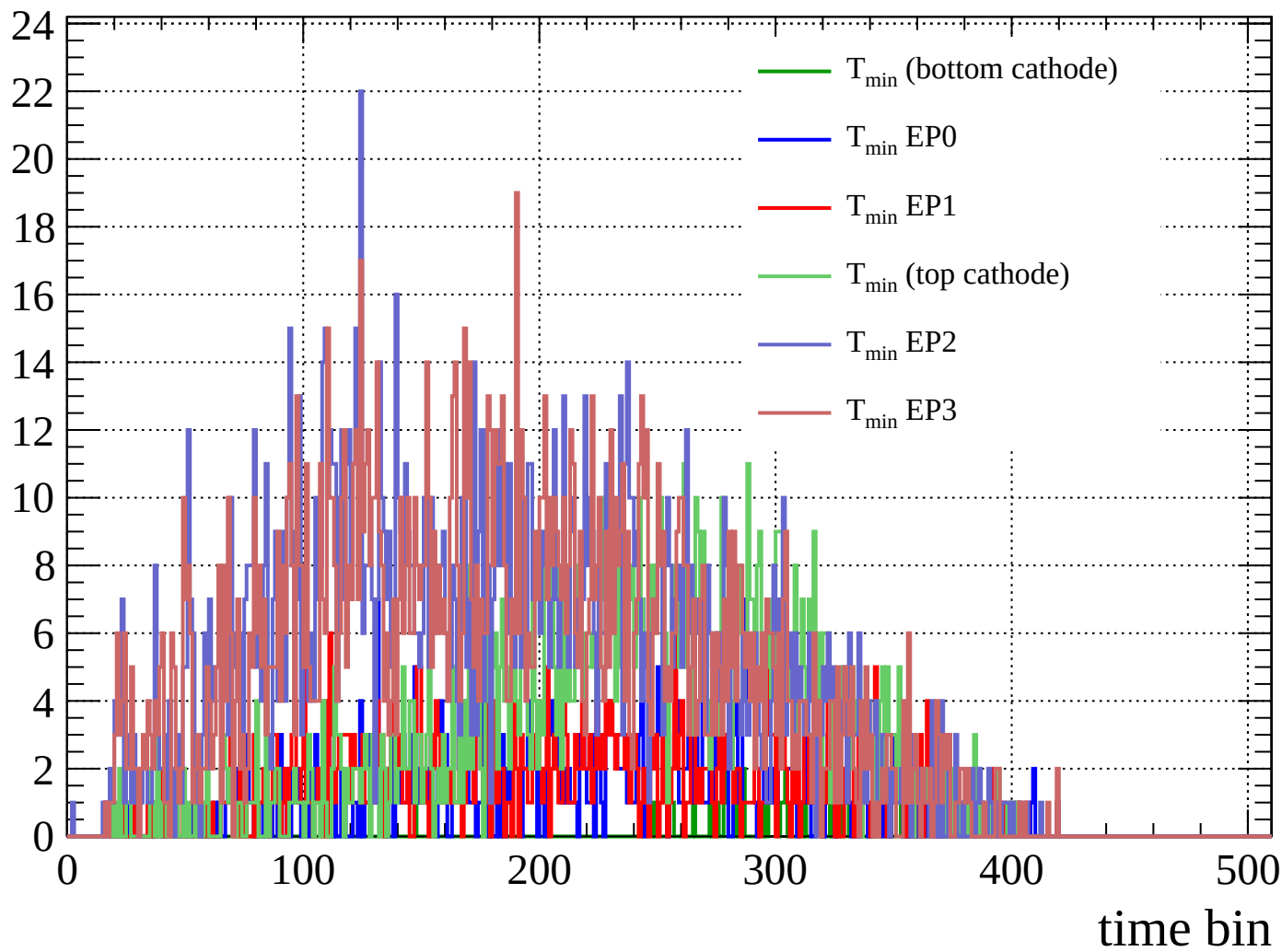
Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)

Count



Count

